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THE SOYBEAN CROP SIMULATOR GLYCIM:

MODEL DOCUMENTATION 1982

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Acknowledgements

The development of this model was largely funded by the AgRISTARS yield modeling project and the USDA-DOE project on crop response to increased atmospheric carbon dioxide concentration (Interagency Agreement No. DEA I-01-81 ER 60001 from the DOE, Carbon Dioxide Research Division, to the USDA). The authors have benefitted greatly over the last seven years from their involvement in the S-107 regional soybean modeling project. Their colleagues in that project have freely discussed their results and conceptual models of how soybeans grow.

Executive Summary

Man is interested in making predictions because they give him better control of his environment. Sometimes event A always follows event B and there is no need to know the reason for this, to make a reliable prediction. Usually, however, the events do not always occur together and, to be able to make reliable predictions, we need to know what mechanism links A and B and what factors influence the mechanism. The relationship between crop yield and the weather is an example of this latter type of unreliable association. We can state in broad terms how the sun, rain, CO_2, etc. influence crops but it is very difficult to predict how a crop will respond to given levels of these factors in combination, especially when they vary daily through the growing season. Our only hope is to understand the mechanisms involved.

GLYCIM is a model (a means for making predictions) about the soybean crop. It consists of a set of mathematical equations written in FORTRAN 77 so that the calculations can be done quickly and accurately by a computer. The equations describe most of the mechanisms involved in the physical and physiological processes going on in the plant and its environment. These are processes like light interception, carbon and nitrogen fixation, organ initiation, growth and death, flows of water, nutrients, heat and oxygen in the soil and many more. All of the important factors known to influence these processes are included in the equations along with information about how the factors interact.

The model is divided into sections called subroutines which deal with single processes or groups of related processes. This makes it easier to

improve the model by inserting subroutines dealing with new processes, e.g., insect attack or by replacing subroutines when better information about the mechanisms become available. Unfortunately, there are several plant physiological processes where the mechanisms are unknown and at present we lack the experimental techniques to discover them. For instance, the initiation of branches seems to depend on adequate supplies of carbon and nitrogen and the age of the node on the plant. Stand density probably affects branching via carbon and nitrogen availability. Beyond that nothing is known. In such instances, mechanisms have been proposed and built into the model that are plausible to soybean crop physiologists, do not conflict with any of the data in the literature and give correct predictions when the model is compared with the growth of real plants. The authors of the model consider this better than putting in an empirical (experimentally observed) relationship such as that between branch number and row spacing. A rigid relationship of that type would not allow the model plant to respond to the extra carbon fixed in a high CO_2 concentration.

Because it handles processes mechanistically, GLYCIM can be used to make predictions beyond the range of the data base used to build it and it can be used to test hypotheses (new ideas). It should be able to simulate the growth of any variety of soybean on any soil and at any location and time of year. Because it is comprehensive and includes most of the processes in the plant, air and soil, GLYCIM makes available to non-experts the knowledge of soybean crop physiologists, soil physicists and agro-climatologists in a usable form. It has the potential for enabling (a) farmers to optimize the use of their resources, (b) government agencies to make better yield predictions including the prediction of yields in a future world with a high CO_2 concentration and a changed climate, (c) administrators to manage research more effectively, (d) plant breeders and genetic engineers to select desirable traits for combining, and (e) teachers to instruct students in a new and exciting way. To that end, the model has recently been made "user-friendly". That means it asks for the information needed to initiate a run by putting up questions on the screen of the terminal and reading the answers that are typed in. This most recent version of the model is available from the authors but has not yet been documented.

The development of GLYCIM started in earnest in late 1978. The literature review took one year, and coding in FORTRAN took another. During all that time, the conceptual models for the process mechanisms were being formulated, tested and reformulated. In 1981, the individual subroutines were verified, assembled into a complete model and debugged. Also in 1981, the model was chosen for use in the USDA-DOE Project on crop response to CO_2. This improved the quality and quantity of data available for developing the model and it increased the number of researchers who constructively criticized the model. The rate of development accelerated. By the end of 1982, the model had been validated (tested) against several field data sets and the deficiencies revealed had been corrected. This publication encapsulates GLYCIM as it was at the end of 1982. It is the documentation for GLYCIM 82A. Improved versions will be issued in the future and any scientist is welcome to contribute improvements with full acknowledgement guaranteed.

In the first part of this documentation an attempt has been made to present the salient features of the model in simple terms. This has necessarily resulted in oversimplification. Physiological processes are highly complex. Plants have evolved homeostatic mechanisms which allow them to adapt, survive and reproduce even in extreme environments. A functional balance between the various organs must be maintained at all times and this requires tight coordination. For instance, roots have a different environment from shoots and they do respond directly to that environment. However, over-riding that response is a process that controls root growth so that root water uptake approximately equals water transpired by the shoot. Numerous other examples could be cited; each process in the plant seems to influence every other process. To put all of the connections between plant processes into one diagram is to overwhelm the uninitiated; to exclude some is to oversimplify.

Most of this documentation consists of details about the mechanisms in the model, the rationale behind them and the sources of ideas, data and parameter values. Whenever possible this has been done by citing published work. This makes the information available to the reader without reproducing it in these pages. The level of detail in the model is such that this documentation would be unmanageably large if all the information used in the model was reproduced here. For instance the notes on the program THERMS which occupy 2 pages here are based on arguments presented in 25 pages in a specialist book on the plant environment.

Finally, at the end of the documentation, there is an example of a validation and there is a discussion of the strengths and weaknesses of the model. Essentially, GLYCIM 82A gives good simulations of soybean crop growth and development, except for the final senescence. The leaves either fall prematurely or stay on too long. The mechanism controlling leaf fall is obviously wrong and no plausible alternative had been proposed by the end of 1982.

General Introduction

GLYCIM is a dynamic simulator of soybean crop growth which is mechanistic at the level of the physical and physiological processes involved in the transfer of materials in the soil, plant and atmosphere. It comes from the same laboratories as GOSSYM, the cotton crop simulator (Baker et al., 1983) and has much in common with that model. It also contains parts of the rhizophere simulator RHIZOS by Lambert and Baker (1984). Like its antecedents, it is organized into subroutines dealing with individual processes or groups of related processes. The model runs in time steps equal to one tenth (or more) of the photoperiod, mainly so that it can simulate diurnal variation in plant water potential. A full simulated season takes about 8 minutes to run on a VAX 11-750 computer.

This documentation is for

GLYCIM 82A

the version in use at the end of 1982. This version was initially written in FORTRAN 66 with amendments in FORTRAN 77 and it occupies nearly 4000 lines of code. It has been validated against field data from Mississippi, Florida, and North Carolina.

GLYCIM is not offered as a finished product but as a vehicle which other scientists may be able to use to test their own submodels and so improve the main model. The authors hope it will achieve an identity of its own independent of them. You are invited to send constructive criticism, alternative hypotheses for testing or rewritten subroutines, all with full acknowledgement guaranteed.

A Brief Introduction to Crop Modeling for the "Scientific Layman"

Man, in common with all higher animals, is a modeler. We observe things going on around us and try to discern some pattern which will enable us to adapt our behavior in order to survive and even thrive. One of the most important of mans' early achievements was the development of a model of the changing seasons. The first gropings after truth were conceptual models: they existed in the form of ideas or words only. Next, considerable resources were devoted to building physical models which could predict the advent of each season. These are the numerous circles of standing stones, like Stonehenge, worldwide. The development of mathematics made it possible to write mathematical models which were just symbols on paper and, therefore, much easier to manipulate and much more portable than the physical models. Improvements in instrumentation and techniques yielded better data which in turn made better models possible. Most recently, calculation aids like computers have enabled us to predict events with this model to a fraction of a second. The model is, of course, the calendar.

Crop modeling has gone through all of these stages and even today most crop models are developed by following the historic pattern. Everyone who works with crops has a conceptual model of their growth and development and many people never attempt to develop the model beyond that stage. They have no reason to do so even if they had the necessary skills. The commonest physical model of a crop has always been an isolated plant growing in a pot. It was this model that enabled van Helmont in the 17th century to demonstrate that plants are not primarily composed of materials taken up from the soil. Conceptual models and physical models have a long and honorable history in crop science.

Probably the earliest mathematical model of crop growth was the one proposed by Briggs, Kidd and West (1920) and used extensively in the form of a system of equations for "Growth Analysis". These equations are:

Relative Growth Rate, $R_w = (1/W)\ (dW/dt)$

Net Assimilation Rate, $E_a = (1/A)\ (dW/dt)$

Leaf Area Ratio, $F_a = A/W$

Therefore, $R_w = E_a * F_a$

where W = plant dry weight, A = plant leaf area and * = multiply. The equations are written in a form to solve for the parameters but we can rearrange them to give a model capable of prediction as follows:

Rate of dry weight gain, $dW/dt = A*E_a$

Plant leaf area, $A = W*F_a$

Therefore, $dW/dt = W*F_a*E_a = W*R_w$

Unfortunately, the model assumes that growth is exponential, which is only true of the early growth of seedlings before mutual shading by leaves on the same or adjacent plants occurs. (The model is often used outside the range of its validity.) It is depicted in Figure 1 in a Forrester diagram (named after J. W. Forrester who first described the box and arrow symbols in 1961). In the diagram, an irregular cloud shape represents an infinite source or sink of material, a rectangular box is a finite quantity of material and a solid arrow is a flow of material. A cross or diabolo placed over a solid arrow represents a valve and things linked to that cross, exercise control over the flow of material represented by the arrow. A dashed arrow is a flow of information. Thus, the model does not allow for the fact that CO_2, H_2O or minerals may be available in limited amounts.

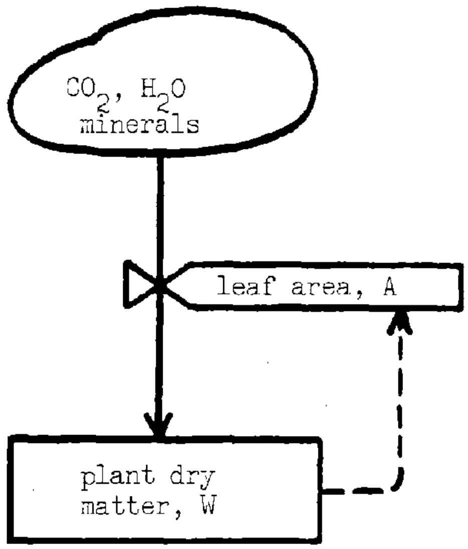

Fig. 1. A Forrester diagram of the "growth analysis" model proposed by Briggs, Kidd and West (1920).

Leaf area is a function of plant dry weight, and the rate of accretion of CO_2, H_2O and minerals to make more dry weight is controlled only by leaf area. Because there is a closed control loop in the model, the rate of dry weight gain can also be expressed as a function of the current plant dry weight.

We realize that this model, useful though it has been, is a gross oversimplification. The model parameters vary according to the environmental conditions in which the crop is grown. If we look at just the uptake of CO_2 and its incorporation into the dry matter in the plant, we can produce a Forrester diagram like the one shown in Figure 2. Here we have used some of our knowledge about the processes involved in carbon fixation. The model is now a process-level model or an organ-level model. The classification can be based either on the level of activity modeled (behavior, process, biochemistry) or on the size of the crop subunit modeled (crop, plant, organ, cell, organelle, molecule). It would, of course, be possible to take one of the processes named in Figure 2 and write a biochemical or molecular model describing it. However, the current conventional wisdom is that it is not desirable to build our models more than one, or at most two, levels below the one at which we wish to make predictions. This is because (a) activities at lower levels run faster and to simulate the variation it is necessary to use shorter time steps in the model which increases the running time, (b) there is usually insufficient information to allow the whole model to be built at the same level of detail and (c) with the introduction of extra detail it becomes increasingly difficult to find faults in the model. Since we wish to predict crop behavior, a process-level, organ-level model is appropriate.

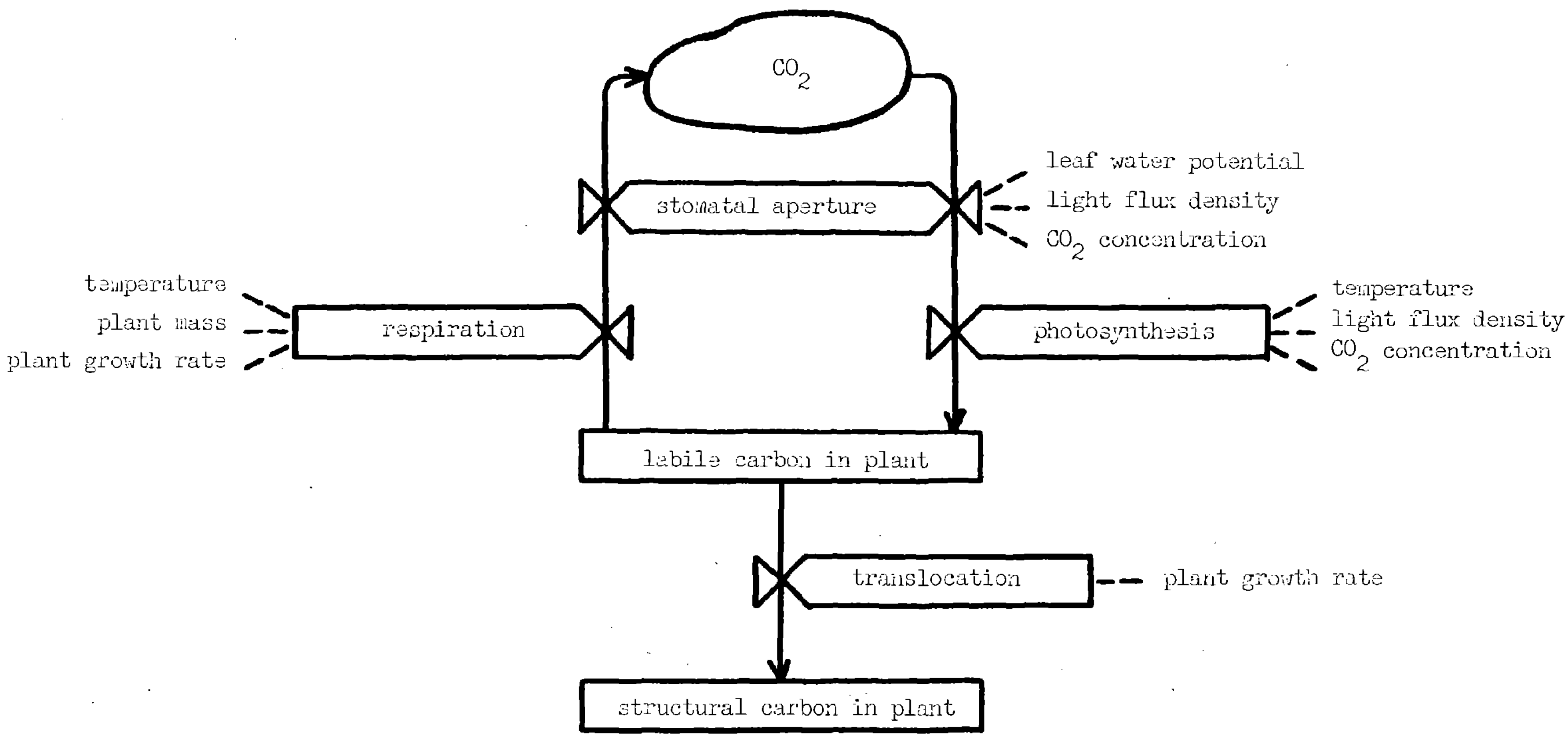

Fig. 2. A Forrester diagram of a model for carbon fixation.

Another important pair of adjectives used to describe models is, empirical and mechanistic. The model in Figure 2 is an attempt to predict carbon fixation by simulating the mechanisms involved. It is, therefore, mechanistic. We might equally well have gathered data on carbon fixation and some or all of the factors known to influence it, put all the data into the computer and found, by trail and error, the equation that explained most of the variation in the data. This would have been an empirical model. It would have summarized our results and the relationship found between factors but it would not have improved our understanding of why the results were obtained. In general, empirical models can only be used to make predictions over the range of conditions observed; they are notoriously bad for extrapolating beyond the data base. Despite these criticisms of empiricism, it must be admitted that all models are empirical at some level. To the modeler working at the cell level, the model in Figure 2 is empirical. Therefore, to use this pair of adjectives in a meaningful way, we should say of Figure 2 that it is a model of carbon fixation that is mechanistic at (or down to) the process level.

The "Growth Analysis" model in Figure 1 is a static model. Given an initial plant dry weight, it will predict a rate of increase in dry weight and then stop. By adding an iterative loop and an equation to increment the plant dry weight, it could be made a dynamic model. It would then calculate a rate and a new state on each iteration with the next rate being calculated from the updated state. There is, of course, no point in actually doing this with such a simple model, because we can make predictions for any day in the future with a single set of calculations. It is merely used to illustrate the difference between static and dynamic models.

In Figure 2, we modeled the processes involved in the fixation of carbon: just one of several materials needed in plant dry matter. To continue our expansion of Figure 1, we must add the processes involved in water and mineral uptake, the partitioning of dry matter and leaf area expansion. When all this is done, we have a diagram that looks like a ball of wool after the kitten has played with it. This is not surprising to the crop scientist, because plants have many feedback mechanisms to maintain homeostasis, but the sheer complexity of the diagram discourages people. However, the reader, when he encounters such a diagram in a few pages, will know that it just consists of elements like the one in Figure 2 joined together. Undaunted, he will be able to trace out the pathway of water through the plant, the movement of carbon from the atmosphere to the leaves, etc.

The Philosophy Behind GLYCIM

GLYCIM is intended for practical application; it is not just an academic toy. The objectives in building it were to provide a tool for (a) optimizing the use of farm resources (b) making yield predictions in all environmental conditions including those of a future would with a high CO_2 concentration (c) managing research (d) directing the efforts of plant breeders and genetic engineers (e) instructing students and (f) testing hypotheses not directly testable by experimentation.

Most members of the team that built GLYCIM are agronomists or horticulturists with a solid training in the basic sciences. They are committed to understanding the operation of the whole plant in its normal environment. Thus, all parts of the plant and all processes have been dealt with at approximately the same level of detail. The idea that one can sum transpiration or photosynthesis over the season, multiply by a factor and predict yield is unrealistic.

The team that built GLYCIM is multi-disciplinary and the members have discussed the concepts in the model at conferences and privately with other experts. For these reasons, the model should represent the current state of knowledge within the scientific community. Undoubtedly there are experts who have not been consulted and undoubtedly our knowledge will be advanced by further experimentation. We expect to improve the model and we would be very happy to have other scientists suggest or make improvements even to the extent that the model is completely replaced piece by piece and assumes an identity of its own, independent of the team that gave it birth. To facilitate this process, GLYCIM is written as a string of subroutines, each dealing with various processes. Each can be removed and replaced without affecting the others so long as the necessary output data are generated.

GLYCIM is mechanistic at the level of the physical and physiological processes involved in the transfer of materials in the soil, plant and atmosphere. These are processes such as light interception, photosynthesis, water uptake and leaf expansion. Below that level, i.e., at the biochemical level, GLYCIM is empirical. Since it handles processes mechanistically, it can contain present and future knowledge about processes, it can be used to make predictions beyond the range of its data base and it can be used to test hypotheses. Empirical models can only be used to make predictions within the range of their data base. They cannot take advantage of knowledge about processes and cannot be gradually improved, although they are cheaper and easier to build. During the development of the conceptual models for GLYCIM, considerable effort was expended in proposing plausible mechanisms for processes where no mechanisms had been proposed before or where those proposed were clearly inadequate. The novel hypotheses are described in greater detail in the documentation that follows in order to justify their use. Generally-accepted hypotheses are briefly referenced to their source in the literature. This may give the impression the GLYCIM consists of mainly controversial new hypotheses but this is false.

A very real problem with this type of model is the lack of data for building it. For most modelers, this limits the model they can produce. For the team that built GLYCIM, it just means that they must do experiments to get the data. The team operates a group of ten daylit controlled-environment chambers similar to those described by Phene et al.

(1978). These chambers are used to measure photosynthesis, respiration, transpiration and light interception by plants arranged in rows like a field crop. The growth in number and size of organs on the shoot are estimated from non-destructive measurements and root growth is measured on the glass face of the soil bin. Temperature, CO_2 concentration and watering are controlled by a computer which also continuously records all environmental variables of importance. Some of the data used in the model come from experiments in these chambers.

Many modelers have claimed that models with process-level mechanism are impractical because they require computers with too much memory and they take too long to run. It is, of course, true that models like GLYCIM cannot conveniently be run on the present generation of personal microcomputers. It would probably take about 2 hours to make a run on the IBM PC with an 8087 arithmatic co-processor. However, the cost of memory continues to fall rapidly and the speed of operation continues to increase. Improvements already being effected ensure that, in no more than 5 years, personal microcomputers able to run GLYCIM in a few minutes will be available.

The History of GLYCIM

Members of the modeling team based at the USDA, ARS, Crop Simulation Research Unit and MAFES Agronomy Department, Mississippi State University have recognized the need for a dynamic simulator of soybean crop growth for some time. They have participated in the S-107 Regional Soybean Modeling Project since its inception in 1976. Along with other scientists involved in that project, they discussed the form that such a model should take and even wrote some flow charts. However, no one had enough time to direct a major effort on soybeans.

Late in 1978, Basil Acock joined the team to lead the work on the soybean model. He was appointed through a cooperative agreement with the Agronomy Department at Mississippi State University. He spent 1979 reading the available literature on soybean growth and developing the concepts necessary for the model. That same year there were also field and controlled-environment experiments with soybean. The field experiment was to obtain a validation data set and the controlled-environment experiment was to obtain data on the temperature response of soybeans.

In 1980 coding of the model started and the development of concepts continued. There was no experimental work on soybeans in that year and by the end of the year the first draft of all the code for the model had been written. During the first few months of 1981, the individual subroutines were verified to ensure that they performed calculations as expected. They were then assembled into a complete model program called GLYCIM.

At this point, the Department of Energy became interested in using GLYCIM for its assessment of crop response to increased atmospheric carbon dioxide. The first controlled-environment experiments on the response of soybeans to CO_2 were done in that year. Our involvement in the USDA-DOE project on crop response to CO_2 had several effects on our work. First it put us under tremendous pressure to develop the model rapidly. The cotton model GOSSYM took about 12 years to develop. Essentially, we attempted to halve that development time. There were also a number of conferences, meetings to attend and reports to write. While this was all very stimulating, it also took time. However, the contact with the Department of Energy enabled us to upgrade our facilities so that we are now collecting much better data than we were before we became involved in the project. Also, colleagues at other universities and research centers who were involved in the project contributed their facilities and time to gather data for use in the development of the model. We moved rapidly from the position where the limitation of data was the block to development of the model to the position where the limitation was the rate at which this data could be incorporated in the model.

Once the subroutines had been assembled into the complete model, GLYCIM, we spent the rest of 1981 debugging the model. It is said that professional computer programmers write an average of about 10 lines of debugged code each day. Our experience confirms this figure. By the end of 1981, the model was running through a complete simulated season without aborting. At the request of the Department of Energy, we hastily assembled some preliminary documentation and did some sensitivity testing.

During 1982, we moved both our offices and controlled-environment facilities to a new site. We then performed the experiments on soybean response CO_2 that had been commissioned by the DOE. In the little time that remained, the model was streamlined to decrease the length of time it took to run and several mechanisms in the model were investigated. In particular, we were concerned about the mechanisms controlling the following: 1) Nodule initiation and growth, 2) Pod and seed retention or abortion, 3) Leaf abscission. This also involved our investigating the control of photosynthetic rate by sink strength.

Throughout the period of its development, the ideas in the model have been disseminated orally. We have presented our concepts at those conferences and workshops, where we felt we would benefit most from the ensuing discussion. This was considered to be more appropriate than formal publication. As a result of this policy, our work is well known and has been exposed to public criticism.

This publication encapsulates GLYCIM as it was at the end of 1982. Since then, further changes have been made and these are described below. These later versions of the model are available from the modeling team but documentation has not yet been prepared.

During the initial development of the model, we became aware that it is not sufficient to build mechanistic models that are scientifically sound. It is also necessary to make them easy to use. We find that the choice of model by the end-user is determined more by its ease of use than by its scientific content or applicability to the situation under investigation. Therefore, we have made GLYCIM user-friendly. The model program now starts with an interactive section in which the user is asked, through questions that appear on the screen, to choose a soil type, a climate file, and provide information about the crop that is to be simulated. The user also has the option of stopping the model at any time during the simulation, storing the results and either resuming the simulation on some other day or making multiple runs forward in time from the stopping point using different weather scenerios. The user also has the possibility of making small corrections to a few of the most important state variables in the model. This feature is intended for the use of farmers and extension agents and other people who may not be able to correct the model by finding and changing the relevant parameters.

There has been further work on the mechanisms in GLYCIM and this work will probably continue for several years. The mechanisms of concern are those which cannot easily be investigated experimentally. It is therefore necessary to propose hypotheses, put them in mathematical form and test them in the model to see if they give results like those observed in the field.

GLYCIM and the USDA/DOE Project on

Crop Response to Increase Atmospheric CO_2 Concentration

Plants grown in greenhouses always respond to increased CO_2 concentrations with higher rates of photosynthesis and dry weight gain. However, it was realized at the start of the project that field crops might not be able to take advantage of extra CO_2 if they were limited by temperature or water or nutrients. To fully evaluate the impact of increased CO_2 on crops, it would be necessary to look at the interactions between CO_2 and other factors. To do this experimentally, would require large numbers of controlled environments over long periods of time; an expensive proposition. Therefore, a decision was made to use computer models to simulate crop response to CO_2 in various soil types and climatic conditions.

At the time this decision was made, GLYCIM had just been coded and the subroutines verified but the model had not been debugged. It was chosen for use in the project for the following reasons: 1) All the participants in the project had an interest in soybeans. 2) It was the most completely mechanistic model available. 3) CO_2 is an explicit variable in the model. 4) The effect of CO_2 on nitrogen fixation by legumes is of special interest. Experiments were planned and performed, which provided data for developing, parameterizing or validating various parts of the model.

In GLYCIM, CO_2 concentration has a direct effect on gross photosynthetic rate and on photorespiration rate. Through its effect on carbon availability, if affects the expansion and dry weight gain of all the organs on the plant. Through its effect on root growth, it influences water uptake, plant water relations and stomatal conductance. Ultimately,

CO_2 concentration, both in the real plant and the model, has the potential to affect every aspect of plant growth and development. For this reason, and also because we wish to use the model to examine interactions between CO_2 and other factors, all subroutines in the model were brought to a similar level of mechanistic detail. None of the processes that were included could be considered sub-critical and modeled empirically.

General Description of GLYCIM

GLYCIM consists of a collection of subroutines coded in standard FORTRAN 77 which describe related sets of physical or physiological processes. It describes the growth of an average plant in a uniform crop that is free of pests and diseases. Materials' balances are kept of individual leaves and petioles on the plant to facilitate interfacing with pest and disease models. Materials' balances are kept for other plant organs by type (i.e. stems, flowers, seeds) and for the soil by cells (Figure 3). These cells are rectangular blocks of soil with a thickness of one centimeter in the direction of the plant row and a width which is a submultiple of the row spacing. Cell depth must be specified. Root activity and soil processes are assumed to be symmetrical about the plant row, so materials' fluxes are only calculated for a representative set of cells from the row to the mid-row and to the depth specified. Fluxes of water, heat, nitrate and oxygen are simulated for the soil while fluxes of carbon, nitrogen and other structural dry matter are simulated for the plant.

In the soil, the fluxes of materials are calculated from equations analogous to Ohm's law. The planes vertically below the plant row and the mid-row are assumed to be planes of symmetry and they are modelled mathematically as impervious boundaries (Figure 4). Since they are planes of symmetry, any material flowing across the boundary in one direction will be balanced by an identical flow in the other direction. The treatment of upper and lower boundaries depends on the material under consideration. For water, the lower boundary is a sieve; water lost from the bottom of the profile is never recovered. This is easily changed to fit other known

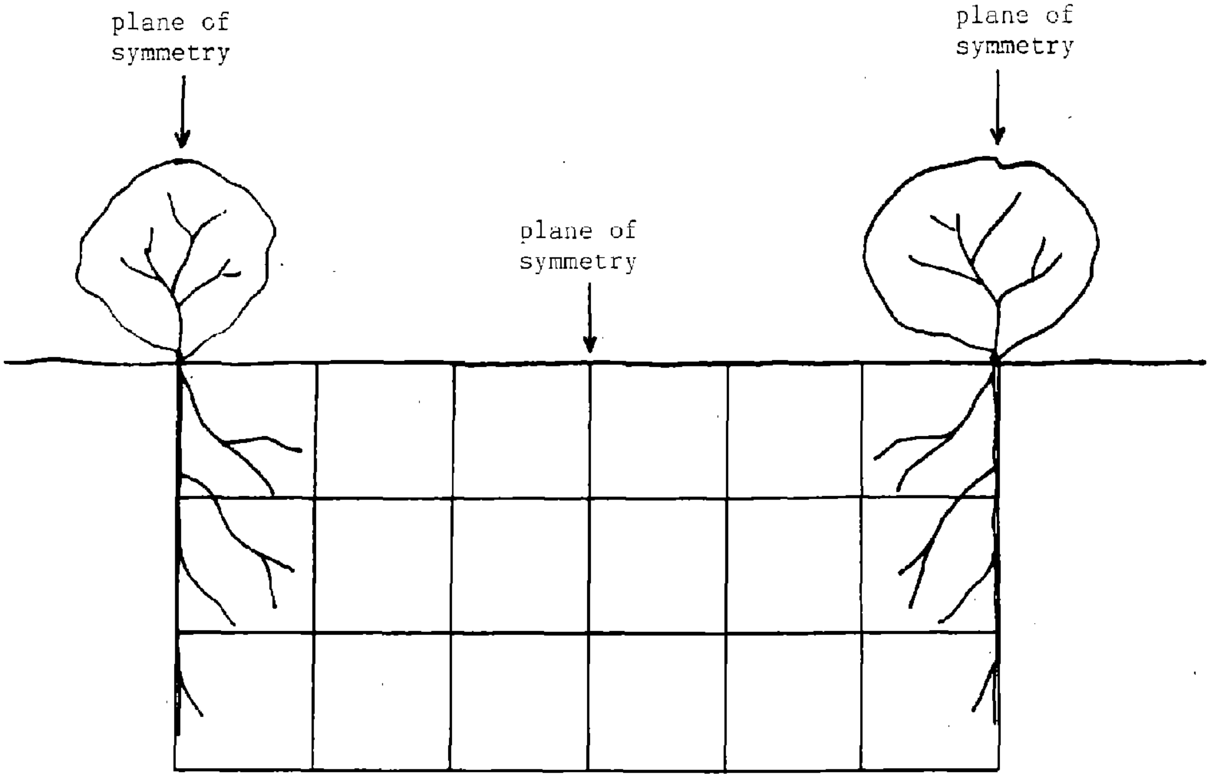

Fig. 3. In GLYCIM the soil profile between the plant rows is divided into a grid of rectangluar cells each 1 cm thick.

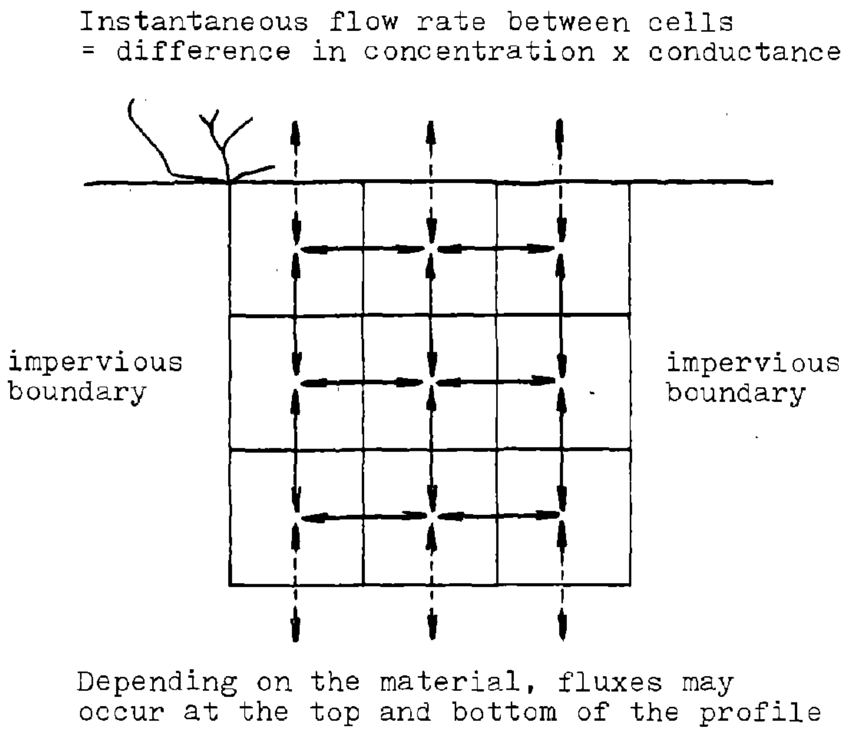

Fig. 4. Diagram to illustrate how materials' fluxes in the soil profile are handled in GLYCIM.

bottom boundary conditions, e.g., a water table.

Materials' fluxes are calculated for each pair of adjacent cells, the materials are moved and new concentrations of the materials are calculated. Depending on the material being considered, it is first moved in one direction (say vertical) and then in the other (horizontal). For calculating vertical movement, the fluxes and new concentrations for the cells in layers 1 and 2 of column 1 are computed, then those for the cells in layers 2 and 3 of column 1 and so on down each column of soil cells. After trying many different methods, we have found this to be the most efficient computationally and the least prone to instability.

The flux equations allow for the fact that as flow occurs the gradient driving the flow decreases. Thus, over time, the material concentration in the two cells tends towards the mean (Figure 5). This avoids the instability resulting for the assumption that the gradient remains constant.

The model has been designed to simulate the growth of any maturity group on any soil and at any location and time of year. A maturity group is a group of cultivars with a similar length of time between sowing and harvest. It seems unlikely that maturity group number will be an adequate description of differences between cultivars but it is the best we can do at present. All soil processes in the model are mechanistic and soil characteristics by horizon are required for all locations simulated. Date and latitude fix daylength, solar altitude and light interception.

Simulations are initiated at the cotyledon stage with appropriate data on the number, size and weight of organs on the plant. Thereafter the only inputs are daily weather data of the type collected by class A stations

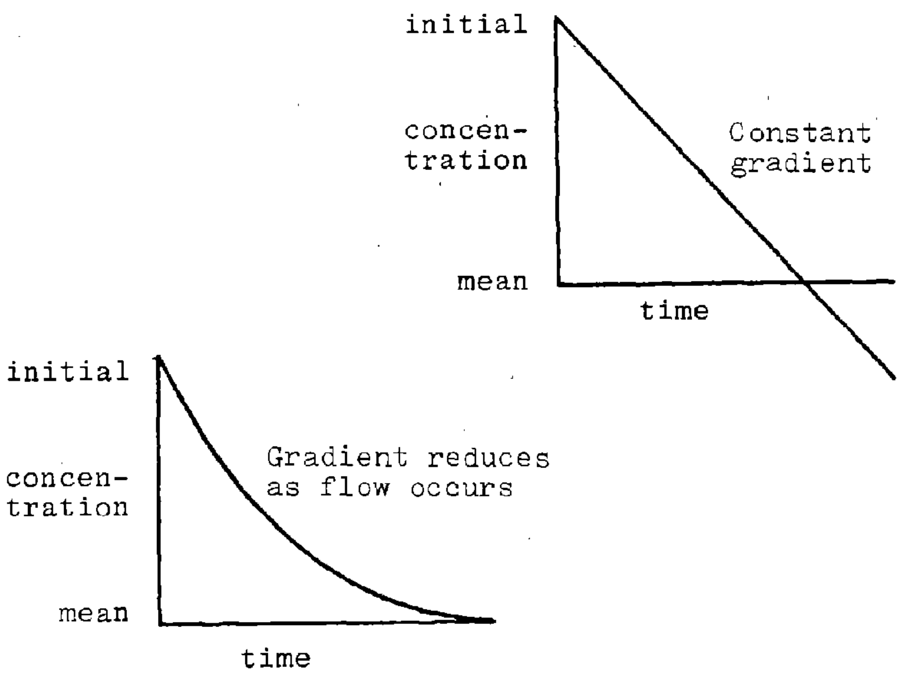

Fig. 5. Comparison of changes in material concentration in soil cells over time either assuming the material gradient is constant or that it is reduced as flow occurs.

with the vital addition of daily total solar radiation. The growth in size of the plant and its advance through the phenological stages are all predicted by the model. In the course of the simulation, the model provides predicted values for most of the physiological variables that can be measured. This feature is intended to facilitate the improvement of the model.

A simplified diagram of carbon partitioning in GLYCIM is shown in Figure 6. Seeds have the highest priority with pods next. Depending on the nitrogen supply/demand ratio, carbon may be sent to the nodules where it is used to fix nitrogen or to grow nodules or initiate new nodules. The carbon left over is used to grow vegetative organs on the plant. If the shoot is turgid, the carbon goes to grow roots or it goes into the root storage pool. The full complexity of the partitioning processes are described in the subroutines.

In order to simulate mechanistically the daytime reduction in shoot growth and the mid-day stomatal closure seen during water stress, the model runs in time steps of one-tenth of a daylength. A further step carries it through the night.

For each of the ten daytime calculation periods, process rates are calculated for the mid-point of the current period based on states at the end of the previous period. For most processes, these rates are assumed to be constant over the period and are merely multiplied by the length of the period to determine fluxes and hence new states. However, soil water content is considered sufficiently important that the rates in the middle of one period are used to predict the state in the middle of the next period assuming that diurnal root water uptake can be described by a half

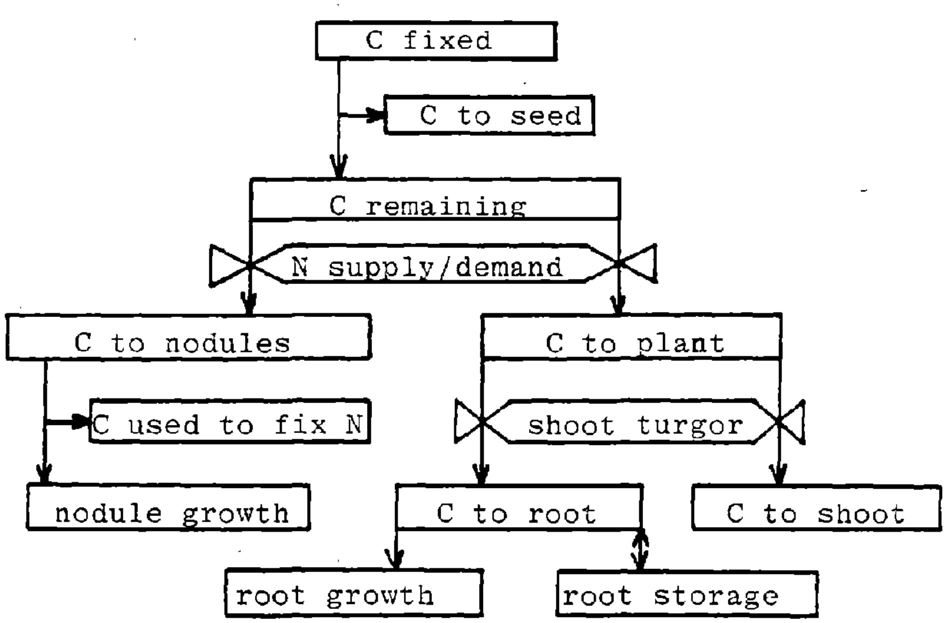

Fig. 6. Simplified diagram of carbon partitioning in GLYCIM.

sine wave. Calculations for the overnight period are treated similarly but instead of calculating rates at midnight they are calculated at the time when air temperature has reached its overnight mean value.

Most of the processes and materials' fluxes simulated in GLYCIM are shown diagrammatically in Figure 7. Their approximate spatial distribution in the soil-plant-air continuum is indicated but their interactions are not. Some of the most important of these interactions are evident in the Forrester diagram in Figure 8. This diagram shows the interactions typically incorporated into models built at Mississippi State but is not specific to any of them.

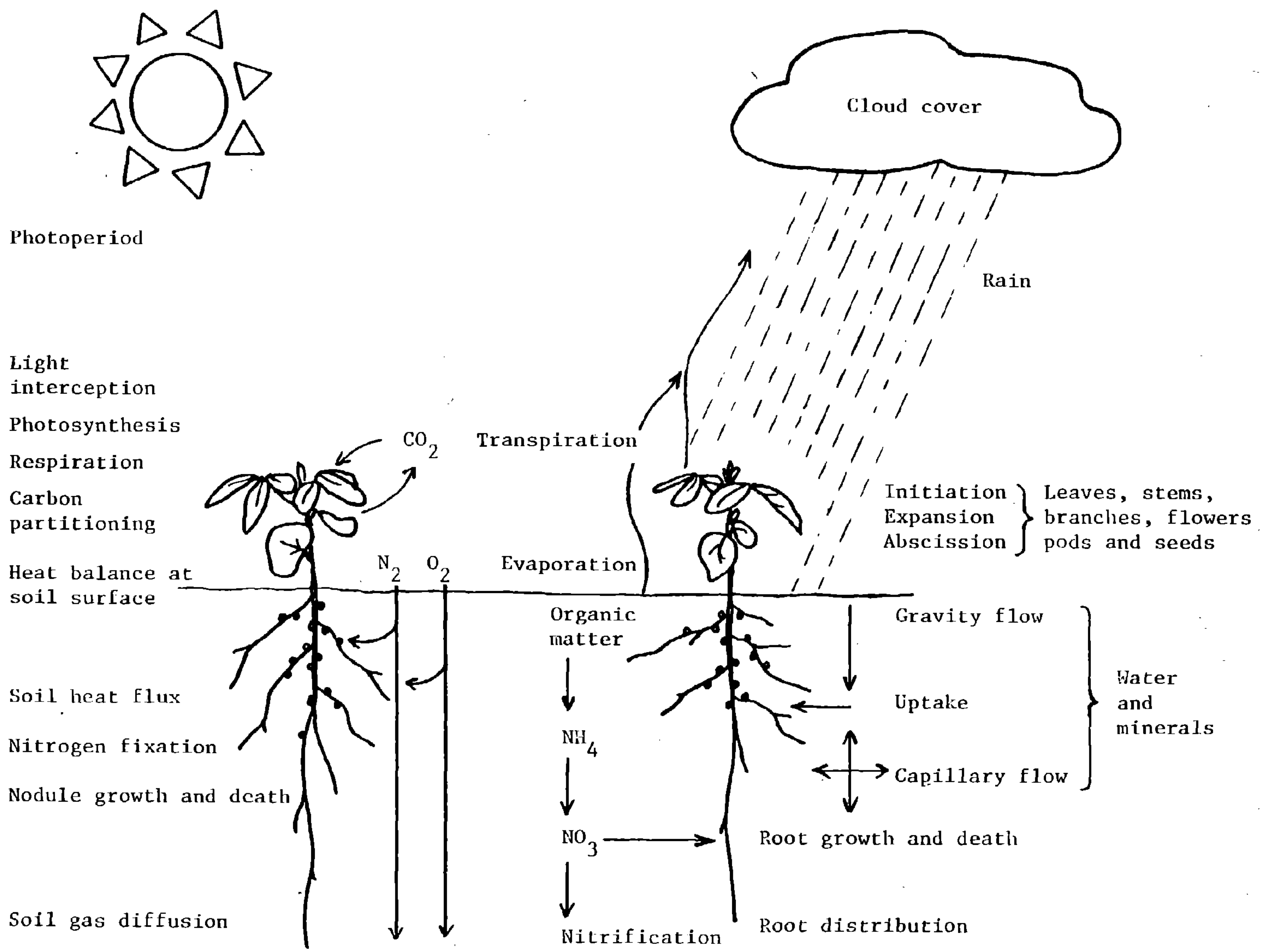

Fig. 7. Diagram of the influence of environmental factors on the processes and flows of materials in a soybean crop.

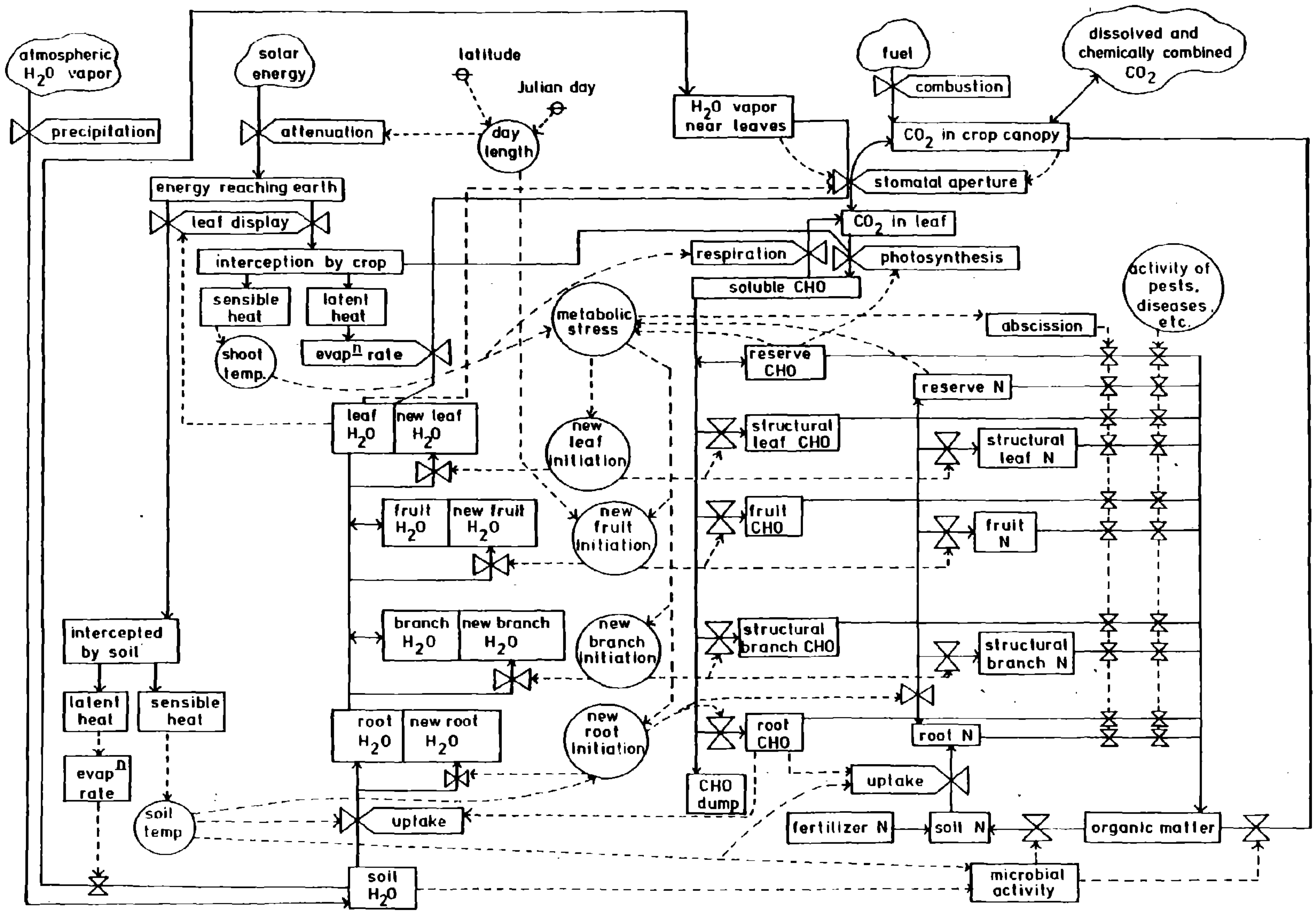

Fig. 8. Forrester diagram of the processes and flows of materials considered in models built at Mississippi State.

The Structure of GLYCIM

As far as practicable, GLYCIM has been organized into a set of subroutines each dealing with a process or related set of processes. This is to facilitate the testing of alternative hypotheses about the control of a process and the addition of code describing new processes that have been recognized as important. These subroutines are all called from MAIN, the control subroutine, or from NIGHT, which controls overnight calculations. See Figure 9.

The names of the subroutines in GLYCIM and a brief description of their functions are given below in the order in which they are normally called.

BLK:D (Block Data) Initializes the values of some variables and variable arrays.

MAIN Controls the order and frequency of call of the other subroutines. Controls the frequency and format of output.

NIGHT Calculates all overnight activity in the soil-plant-atmosphere continuum. Calls other subroutines as needed.

CLYMAT Calculates daylength, effective photoperiod, cloud cover, and mean day and night air temperature. For ten or fewer equally-spaced times each day it estimates air temperature and the interception of diffuse and direct light by "opaque" crop rows. These diurnal patterns of temperature and light interception are generated using measured maximum and minimum temperature and measured daily total solar radiation.

EVTR Calculates potential water evaporation rates from the crop and soil and estimates actual water loss from exposed surface soil cells as limited by hydraulic diffusivity. It also estimates the amount of

Fig. 9. Call sequence for subroutines in GLYCIM

```
------------------------------------------------------------------------
Control routines     Called from MAIN      Called from NIGHT
----------------     ----------------      -----------------
BLK:D (block data)
MAIN
for each day of the crop growing season, do the following:
                     CALL CLYMAT

  for each tenth (or other fraction) of the photoperiod,
     do the following:
                     CALL EVTR
                     CALL TMPSOL
                     CALL PNET
                     CALL SHTGRO
                     CALL RUTGRO
                     CALL NITRO
                     CALL TRADE
                     CALL GROWTH
                     CALL PLTMAP
                     CALL NODULE

      if this is the last tenth of the photoperiod, jump down
                     CALL UPTAKE

      if there has been no rain or irrigation, jump down
                     CALL GRAFLO
                     CALL CAPFLO

                     CALL NIGHT
                                           CALL UPTAKE
                     if there has been no rain or irrigation,
                        jump down
                                           CALL GRAFLO
                                           CALL CAPFLO
                                           CALL EVTR
                                           CALL TMPSOL
                                           CALL PNET
                                           CALL SHTGRO
                                           CALL RUTGRO
                                           CALL NITRO
                                           CALL TRADE
                                           CALL GROWTH
                                           CALL PLTMAP
                                           CALL NODULE
                                           CALL UPTAKE
                     if there has been no rain or irrigation,
                        jump down
                                           CALL GRAFLO
                                           CALL CAPFLO
                                           CALL NITRIF
                                           CALL SOLOXY
   if printed output is not required today, jump to end of loop
                     CALL SOYPLT
```

radiation available for soil heating.

TMPSOL Calculates a heat flux balance for the exposed soil surface and hence a heat flux into the soil. It also estimates soil temperature in all parts of the soil profile.

PNET Uses single-leaf photosynthetic parameters to calculate crop canopy parameters and canopy gross photosynthesis rate. Respiration rates from various causes are calculated and subtracted to give net photosynthesis rate which is corrected for stomatal conductance (determined in TRADE). From this are calculated the net carbon fixation rate and the rate of carbon translocation out of the leaf.

SHTGRO Calculates potential growth rate for organs on the shoot assuming that temperature is the only limiting factor.

RUTGRO Calculates potential root growth in each soil cell assuming that carbon and nitrogen are not limiting but allowing for limitations of soil temperature, soil water potential, soil oxygen content and soil root impedence. It "kills" roots in soil cells with unfavorable conditions.

NITGRO Calculates the supply of and the demand for nitrogen in the whole plant and determines a stress factor which is used to partition carbon to the nodules. Controls leaf and petiole abscission.

TRADE Maintains a functional balance between root and shoot by growing root as necessary to meet transpiration demand. Calculates potential root water uptake for a number of key shoot water potentials and compares these with the potential transpiration rate to estimate shoot water potential. Depending on shoot trugidity, the shoot or roots or both may grow. Root growth occurs in the soil cells where conditions are most favorable and is shared with adjacent cells. Stomatal conductance is a function of shoot turgidity.

GROWTH Calculates the actual growth in size and dry weight of organs on the shoot. Initiates branches, pods and seeds according to the plant's ability to support them.

PLTMAP Calculates the vegetative and reproductive stages of growth and limits the growth of determinate plants.

NODULE Calculates nitrogen fixing activity, growth and death of nodules in each soil cell.

UPTAKE Calculates uptake of water and nitrate by roots from each soil cell.

GRAFLO Calculates flow of water and nitrate down the soil profile after rain or irrigation.

CAPFLO Calculates the capillary flow of water and nitrate in all directions in the soil profile. Also calculates soil water potential in each soil cell.

NITRIF Calculates the nitrification of organic matter and ammonia in the soil.

SOLOXY Calculates oxygen concentration in all parts of the soil profile.

All of the subroutines contain the FORTRAN statement INCLUDE COMMON.FOR. The effect of this is to insert the entire contents of the COMMON.FOR file into the subroutine at the point occupied by the statement. This file contains all of the REAL, INTEGER and COMMON statements. The COMMON statements are lists of all variables passed between subroutines.

The relationships between the subroutines and the principal flows of information are shown diagrammatically in Figure 10. The diagram greatly

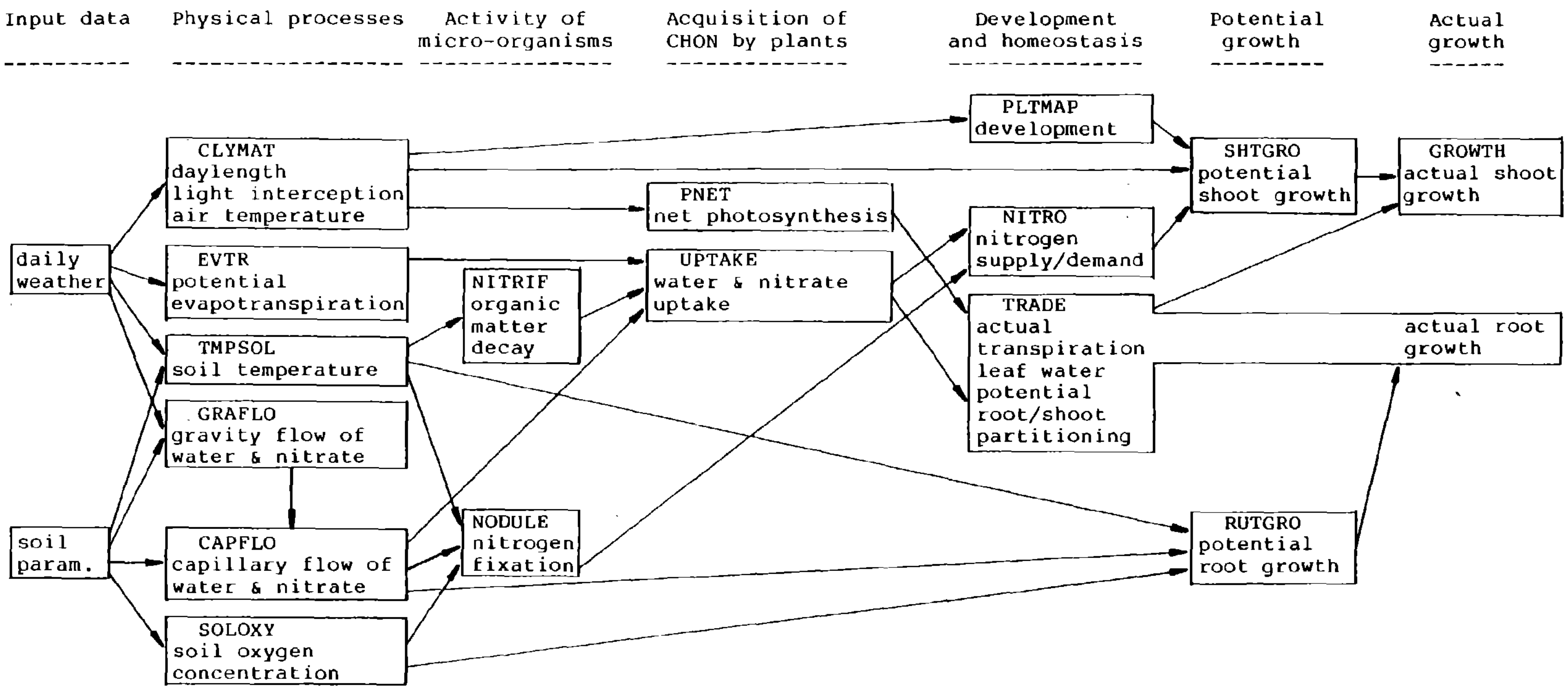

Fig. 10 Simplified diagram to show the principal functions of the routines in GLYCIM and the principal flows of information between them.

simplifies the true relationships and none of the feedback loops of the homeostatic mechanisms are represented. Nevertheless, the reader may find it helpful when studying the individual subroutines.

Input Needed and Output Provided by GLYCIM

In this section, the data needed to initialize and run the model are listed. The list is not very demanding except in the soil data which are normally available for experimental plots or may often be obtained from soil surveys. Since GLYCIM is initialized at the cotyledon stage (VC), the size, dry weight, and spatial disposition of all organs on the plant are also needed as input. These plant data will probably not change very much from one soybean cultivar to another and it should be possible to use a standard set of values. A typical set of input data is listed in BLK:D later in this publication.

Data Essential to Initialize the Program

Variety or maturity group number

Whether determinate or indeterminate

Place or latitude of planting

Date of emergence

Row spacing

Plant population within the row

Row orientation

Carbon dioxide concentration (if other than ambient)

Mean windspeed for site (if daily windspeed unavailable)

Depths of major soil horizons

Particle size analysis, bulk density and water content - pressure head curve for each horizon

Residual nitrogen in soil at start of season

Fertilizer nitrogen applied this season (if any)

Organic matter content of soil

Deep soil temperature (at a depth where it is constant all year)

Daily Meteorological Data Needed to Run the Program

Daily total solar radiation

Maximum and minimum temperature

Rainfall

Wind run (if available)

Wet and dry bulb temperature or other measure of moisture content of air (if available)

Routine Output at Chosen Time Intervals

Day of the year

Dry weights of leaves, petioles, stems, pods, seeds and roots

Numbers of pods and seeds

Leaf area

Vegetative and reproductive stages of growth (Fehr and Caviness, 1977)

Supply/demand ratio for nitrogen

Number of the lowest leaf present on mainstem

Number of branches

Plant height

Stored carbon (i.e. starch) in the shoot and root

Leaf water potential and evaporation rate at 1/10 day intervals

Dry weights and heights of mainstem and branches 1 through 8

Dry weights of leaf and petiole, leaf area and petiole length for each node on mainstem and branches 1 through 8

Two-dimensional soil matrices showing the distribution of:

Root dry weight

Soil moisture

Soil temperature

Soil oxygen concentration

Dry weight of nodules capable of fixing nitrogen

Additional Ouput Available

Atmospheric transmission coefficient

Cloud cover

Diffuse and direct radiation and interception of each by the crop

Air temperature at any time of day

Photoperiod for soybeans

Heat flux into soil at the surface

Rates of photorespiration and maintenance respiration

Gross photosynthetic rate as limited by light, carbon dioxide concentration, leaf nitrogen and leaf age

Carbon translocation rate

Potential (temperature-limited) and actual (substrate-limited) rates of expansion and dry weight gain for individual organs on the plant

Nitrate reductase activity for each leaf on the plant

Leaf osmotic potential and turgor pressure

Stomatal conductance

Documentation for Individual Subroutines in GLYCIM and for Ancillary Programs

The documentation for each subroutine and program consists of (a) a verbal description of what it does, (b) the pseudo-code which summarizes its logic structure, (c) the FORTRAN code itself and (d) notes on the sources of the relationships and parameter values used in the code. The notes referred to in (d) are keyed to the relevant lines in the code and are liberally cross-referenced to the notes and code for other subroutines.

The first subroutines of the model to be coded were written in FORTRAN 66. From the start, a conscious effort was made to list the statements and equations in the order that they would be used. This effort was greatly facilitated by the decision in early 1982 to start writing in FORTRAN 77. The BLOCK IF statements make it possible to eliminate many of the GO TO statements which otherwise leapfrog readers around the code. Because of this, the code mostly reads from top to bottom with certain parts skipped according to circumstances. Thus flow diagrams of the subroutines are unnecessary. In fact, the use of flow diagrams in program documentation is declining rapidly. Instead of flow diagrams, we present here the pseudo-code that is replacing them.

From the pseudo-code the reader can see the various paths through the code, the calculations performed and the decisions made in each subroutine. The words used to describe the calculations are identical with those used as comments in the code. Headings for major subsections of the subroutines are delineated with dashed lines. Conditions which cause parts of the code to be skipped are described in lower case letters. If the condition, which is described, exists, then the arrow must be followed. The variables which control the number of interactions in a DO loop are also described in lower case letters and the range of each DO is denoted with a square bracket.

Because of the physiological detail in the model it is impracticable to describe it all and give all the relevant references in this publication. To do that would be to put textbooks on soybean physiology, agro-climatology and soil physics between a single set of covers. Rather, it is our intention here to establish the pedigree of the model and present enough of the arguments that those interested can read the references and reproduce our equations for themselves. The authors will be pleased to provide further details on request.

For processes that are well understood or have been modeled by others, only a reference to the source of the equation is given. For processes that are not well understood or are being treated here in a novel way, the sources of ideas and the ways they have been used, are discussed.

In the notes, the variable names have the same meaning as in the code unless they are redefined. This has been done in some places to simplify the explanation. Also, equations in the notes use FORTRAN notation for the arithmetic operators. These differ from the common notation in that * = multiply and ** = exponentiate.

If readers find errors in the code and dictionary, we would like to be advised of these, especially if they are logical errors. You are also invited to suggest alternatives to the code or even alternative hypotheses for coding, especially if you have good supporting data.

BLK:D

The subroutine BLK:D is used to initialize the values of certain variables at the start of a run.

```
      BLOCK DATA

C     ---------------------------( BLK:D )------------------------------
C      THIS ROUTINE SETS THE VALUES OF CERTAIN VARIABLES AT THE START
C     OF THE PROGRAM (I. E., AT COTYLEDON STAGE VC ).
C     ------------------------------------------------------------------
      INCLUDE 'COMMON.FOR'
C     ------------------------------------------------------------------
      DATA ABSCIS/0/,ABSLPN/0./,ABSPON/0./,ADRL/200*0./,ADWR/200*0./,
     &      AFWUP/200*0./,AFWUPS/0./,AIRDR/5*0./,ASCRAT/11*0./,
     &      ASGT/11*0./,AVP/0./,AWR/200*0./,AWUP/200*0./,AWUPS/0./,
     &      AWUPSS/0./
      DATA BD/5*0./,BDC/200*0./,BETA/5*0./,BLAREA/400*0./,
     &      BLEAFW/400*0./,BPETL/400*0./,BPETW/400*0./,BRNCHH/20*0./,
     &      BRNCHW/20*0./
      DATA CEC/.55/,CLAY/5*0./,CLIMAT/7*0./,CLOUD/0./,
     &      COND/200*0./,CONRES/0./,CONVN/0.51/,CONVPO/0./,
     &      CONVSE/0./,CXT/200*0./
      DATA DAYBR/20*0/,DEADRN/0./,DIFF0/5*0./,DIFIMG/0./,
     &      DRAIN/0./,DROP/0/
      DATA EO/11*0./,EORS/0./,EORSCS/0./,ETA/5*0./,EWU/11*0./,EWUS/0./
      DATA FC/20*0./,FCI/5*0./,FILDAY/0/,FIXC/0./,FIXCS/0./,
     &      FIXDAY/0./,FLWRWT/0./,FWUP/200*0./
      DATA GAMMA/0./
      DATA IDAY/0/,ILOW/0/,ILOWB/20*1/,INITBR/20*0/,IPERD/10/
      DATA KLJ/0/
      DATA LAREAB/0./,LAREAT/7.33/,LCAI/0./,LEAFWT/.03/,LKH/0/,
     &      LKR/0/,LKRP1/0/,LLUE/.012/,LTC/.17/
      DATA MATURE/0/,MLAREA/25*0./,MLEAFW/25*0./,MPETL/25*0./,
     &      MPETW/25*0./,MSTEMH/.5/,MSTEMW/0./
      DATA NBRNCH/0/,NBTPLF/20*0/,NCD/0/,NCPOOL/0./,NFCF/200*0./,
     &      NFIXNU/200*0./,NFIXWT/200*0./,NFLWRS/0/,NJ/20*0/,
     &      NKH/0/,NKN/0/,NLN/1/,NNEWWT/2000*0./,
     &      NODES/0/,NODIE/200*0./,NODN/0./,NODNEW/2000*0./,
     &      NODNUM/200*0./,NODWAR/200*0./,NODWT/0./,NPRM/0/,
     &      NRATIO/1.0/,NTOPLF/0/,NYTE/0/
      DATA OSMREG/-0.25/
      DATA PADWR/200*0./,PAR/10*0./,PARINT/10*0./,PCRPAR/200*0/,
     &      PCRS/0./,PCST/0./,PDBRW/20*0./,PDFW/0./,PDLAB/400*0./,
     &      PDLAM/3*0./,PDMH/0./,PDMW/0./,PDNODC/0.0/,PDNW/200*0./,
     &      PDPLB/400*0./,
     &      PDPLM/3*0./,PDPODC/0./,PDPODW/0./,PDPWB/400*0./,PDPWM/3*0./,
     &      PDR/0./,PDSC/0./,PDSW/0./,PDV/0./,PDWR/200*0./,PERIOD/0./,
     &      PERTWT/200*0./,PETWT/.001/,PHOTO/0./,PILD/-9.0/,PILOSM/-9.0/,
     &      PLANTN/0.0065/,PNLLFS/0./,PODS/0./,PODWT/0./,PPDRL/200*0./,
     &      PRM/99*0./,PSIL/10*0./,PSILD/-2.0/,PSIS/200*0./,
     &      PSISAT/5*0.0/,PSISM/0./,PWUP/200*0./
      DATA Q/0./,QCST/0./
      DATA RADINT/10*0./,RAIN/0./,RCPOOL/0./,REMNO3/0./,RESPS/200*0./,
     &      RGCF/200*0./,RHS/10*0./,ROOTWT/.035/,ROUGH/0./,RRRM/8.0E3/,
     &      RRRY/1500/,RSTAGE/-1.0/,RTMINW/0./,RTWL/9.0E-5/,
     &      RTWT/200*0./,RUTDEN/200*0./,RVR/200*0./,RVRL/17.5/
      DATA SAND/5*0./,SCF/1.0/,SCOND/5*0.0/,SCPOOL/.0124/,
     &      SCRATO/0./,SDEPTH/5*0./,
     &      SEEDS/0./,SEEDWT/0./,SGT/0./,SGTR/0./,SINK/0.0/,
     &      STEMWT/.002/,STPOOL/0.0/,SUPNO3/0./,
     &      SWR3/0/,SWR4/0/,SWV1/0/,SWV2/0/
      DATA TAIR/11*0./,TAUN/0./,TB/20*-1./,TBMAH/20*30./,TCADS/0./,
     &      TDSX/25*0./,THETAS/5*0./,THETA0/5*0./,TLAGE/0./,TLIFE/30./,
     &      TNAIRD/10*0./,TNL/0./,TNYT/0./,TOPORS/200*0./,TPL/0./,
     &      TPLD/5.0/,TRANSC/0./,TRIFMG/0./,TRNDAY/0.0/,TS/200*0./
      DATA ULAREA/3.33/,ULEAFW/.01/,UPETL/.17/,UPETW/.001/,USEDCS/0./
      DATA VCLAY/200*0./,VEGSRC/0./,VH2OC/200*0./,
     &      VLOST/0.0/,VMUCK/200*0./,VNC/200*0./,VNH4C/200*0./,
     &      VNO3C/200*0./,VSAND/200*0./,VSTAGE/0./,VSTMAH/30.0/
      DATA W/0./,WATTSM/10*0./,WATWK/0./,WIND/0./
      DATA XTRACS/0./
      DATA YRL/200*0./
      DATA Z/5.5/
      END
```

MAIN

The subroutine MAIN reads all the files needed to intialize a run of the model. It calculates the values of some variables from the input data It converts the arrays of data which describe soil characteristics by horizon to arrays which describe them by soil cell. It calls other subroutines in the model with the frequency and in the sequence desired. It calculates the values of a few variables that are adjusted once a day.

The subroutine writes a heading and copies of the input files into an output file. It adds to the file, model predictions of the type and frequency requested. It also prepares a data file for the subsequent comparison of model predictions and experimental observations. No processes are simulated in MAIN.

```
-------------------( MAIN )-----------------------------------------
THIS ROUTINE CONTROLS THE ORDER IN WHICH THE OTHER ROUTINES ARE
CALLED
--------------------------------------------------------------------
    WRITE HEADING AND SELECT DATA TO BE RECORDED

    READ IN DETAILS ABOUT CROP LOCATION, TIMING, SPACING, ETC.

    READ IN DETAILS ABOUT NUTRIENT STATUS OF SOIL AT PLANTING.

    READ DESCRIPTORS AND CONVERSION CONSTANTS FOR WEATHER DATA

    IF DAILY WINDSPEED DATA ARE UNAVAILABLE, PUT IN AN AVERAGE
       VALUE FOR SITE.

    CALCULATE THE DIMENSION OF THE "CLIMAT" ARRAY.

    FIND THE CORRECT DAY IN THE WEATHER FILE

    READ IN DETAILS ABOUT SOIL COMPOSITION, STRUCTURE, ETC.

    IF INITIAL SOIL WATER CONTENTS ARE AVAILABLE READ THEM

    READ IN THE INITIAL ROOT DISTRIBUTION

    CALCULATE SOIL CHARACTERISTICS IN EACH CELL FROM DATA ON
       HORIZONS.

    DISTRIBUTE FERTILISER AND ORGANIC MATTER THROUGH THE PLOW
       LAYER AND CALCULATE TOTAL SOIL POROSITY.

    INITIALIZE SOME SOIL VARIABLES IN THE REMAINING CELLS.

    CALCULATE THE THERMAL CONDUCTIVITY OF AIR-DRY SURFACE SOIL
       USING EQUATIONS FROM THERMS

    CALCULATE A CLOUD COVER FACTOR FOR THIS LATITUDE.

    SET VALUES FOR SOME ENVIRONMENT VARIABLES.

    SET VALUES FOR SOME PLANT VARIABLES.

    for each day of the crop growing season, do the following:
    |
    | CALL CLYMAT
    |
    | AVERAGE LIGHT FLUX DENSITY ON CROP FOR PAST WEEK
    |
    | THICKNESS OF NEW LEAVES
    |
    | NUMBER OF DAYS SINCE SEED FILL STARTED
    |
    A
```

```
A
 TIME SINCE MAINSTEM TERMINAL LEAF REACHED FULL EXPANSION

 NUMBER OF TOP LEAF ON MAIN STEM

 CALCULATE A CROP SURFACE ROUGHNESS PARAMETER

 ON STILL DAYS, CONVECTION CURRENTS AT CROP LEVEL ASSURE SOME
    AIR MOVEMENT.

 for each tenth (or other fraction) of the photoperiod,
    do the following:

   CALL EVTR
   CALL TMPSOL
   CALL PNET
   CALL SHTGRO
   CALL RUTGRO
   CALL NITRO
   CALL TRADE
   CALL GROWTH
   CALL PLTMAP
   CALL NODULE

   if this is the last tenth of the photoperiod, i.e. dusk,
      jump down

     CALL UPTAKE

       if there has been no rain or irrigation, jump down

       CALL GRAFLO
       CALL CAPFLO

 CALL NIGHT

 if printed output is not required today, jump to end of loop

   PRINT OUT THE MODEL PREDICTIONS IF TODAY IS A MULTIPLE OF
      THE REQUIRED FREQUENCY OR THE CROP HAS REACHED MATURITY

   CALL SOYPLT

STOP
```

```
      PROGRAM MAIN
C     -------------------------( MAIN )--------------------------------
C     THIS ROUTINE CONTROLS THE ORDER IN WHICH THE OTHER ROUTINES ARE
C     CALLED
C     -----------------------------------------------------------------
      INCLUDE 'COMMON.FOR'
C     ------------------------( FORMATS )------------------------------
    1 FORMAT(36X,52H********************** GLYCIM **********************
     &     ,/,36X,52H*     A DYNAMIC SIMULATOR FOR SOYBEAN CROPS CREATED*
     &     ,/,36X,52H*BY MEMBERS OF THE USDA CROP SIMULATION RESEARCH    *
     &     ,/,36X,52H*UNIT, THE MISSISSIPPI AGRICULTURAL AND FORESTRY    *
     &     ,/,36X,52H*EXPERIMENT STATION AND THE AGRONOMY DEPARTMENT AT *
     &     ,/,36X,52H*MISSISSIPPI STATE UNIVERSITY, WITH COLLEAGUES IN  *
     &     ,/,36X,52H*THE S107 REGIONAL SOYBEAN MODELING PROJECT.       *
     &     ,/,36X,52H*       FUNDED BY THE AGRISTARS YIELD MODELING     *
     &     ,/,36X,52H*PROJECT AND THE USDA/DOE PROJECT ON CROP RESPONSE *
     &     ,/,36X,52H*TO INCREASED ATMOSPHERIC CARBON DIOXIDE.          *
     &     ,/,36X,52H*      ADDRESS CORRESPONDENCE TO BASIL ACOCK,      *
     &     ,/,36X,52H*P.O.BOX 5367, MISS.STATE,MISSISSIPPI 39762.       *
     &     ,/,36X,52H****************************************************
     &     )
    9 FORMAT(1X,25HINSUFFICIENT WEATHER DATA)
   10 FORMAT(/,1X,4H MG=,I2,5H DET=,I1,8H LATUDE=,F5.2,8H JDFRST=,I3,
     &       8H JDLAST=,I3,7H ROWSP=,F5.1,8H POPROW=,F4.1,/,
     &       8H ROWANG=,F5.1,6H NFRQ=,I2,6H FILL=,F3.1,5H CO2=,F5.0,
     &       8H SDWMAX=,F5.3)
   12 FORMAT(1X,8H LYRSOL=,I1)
   13 FORMAT(/,8H HORIZON,1X,6HSDEPTH,2X,2HBD,4X,4HSAND,2X,4HCLAY,2X,
     &       6HPSISAT,1X,6HTHETAS,1X,3HFCI,4X,6HTHETA0,1X,5HAIRDR,2X,
     &       5HSCOND,1X,5HDIFF0,5X,4HBETA,2X,3HETA)
   14 FORMAT(8H   TOP   ,1X,F5.1,3X,3(F4.2,2X),F5.2,3(2X,F5.3),2X,F5.3,
     &        2X,F4.1,2X,E8.2,1X,F5.1,2X,F4.2)
   15 FORMAT(I5,4X,F5.1,3X,3(F4.2,2X),F5.2,3(2X,F5.3),2X,F5.3,
     &        2X,F4.1,2X,E8.2,1X,F5.1,2X,F4.2)
  160 FORMAT(/,8H HORIZON,3X,7HTDSX(1),3X,7HTDSX(2),3X,7HTDSX(3),3X,
     &        7HTDSX(4),3X,7HTDSX(5))
  161 FORMAT(8H   TOP   ,5(1X,E9.2))
  162 FORMAT(I5,3X,5(1X,E9.2))
   17 FORMAT(//,1X,27H SITUATION AT DAWN ON IDAY=,I3,5X,6H JDAY=,I3)
   18 FORMAT(1X,8H LEAFWT=,F6.3,2X,7H PETWT=,F6.3,2X,8H STEMWT=,F6.3,
     &        2X,7H PODWT=,F6.3,2X,8H SEEDWT=,F6.3,2X,8H ROOTWT=,F6.3,
     &        7H NODWT=,F6.4)
   19 FORMAT(1X,6H PODS=,F6.2,5X,7H SEEDS=,F6.2,5X,8H LAREAM=,F7.2,
     &5X,8H LAREAB=,F7.2,5X,8H LAREAT=,F7.2)
   20 FORMAT(1X,8H VSTAGE=,F4.1,5X,7HNTOPLF=,I2,5X,8H RSTAGE=,F4.1,5X,
     &        6HVLOST=,F4.1)
   21 FORMAT(1X,8H PDNODC=,F5.2,5X,8H NRATIO=,F5.2)
   22 FORMAT(1X,6H ILOW=,I2,5X,3H Z=,F5.1,5X,8H NBRNCH=,I2)
   23 FORMAT(1X,8H SCPOOL=,F6.3,2X,8H RCPOOL=,F6.3,2X,8H XTRACS=,F6.3,
     &7H FIXCS=,F6.3,8H USEDCS=,F6.3)
   24 FORMAT(/,10X,9HMAIN STEM,5X,8HBRANCH 1,5X,8HBRANCH 2,5X,
     &        8HBRANCH 3,5X,8HBRANCH 4,5X,8HBRANCH 5,5X,8HBRANCH 6,5X,
     &        8HBRANCH 7,5X,8HBRANCH 8)
   25 FORMAT(1X,I6,2X,F5.3,1X,F5.3)
   26 FORMAT(1X,I6,9(2X,F5.3,1X,F5.3))
   27 FORMAT(6X,1HU,2X,F5.3,1X,F5.3)
   28 FORMAT(7X,9(3X,4HPETL,1X,5HLAREA))
   29 FORMAT(1X,6HHEIGHT,5X,F5.1,8(8X,F5.1))
   30 FORMAT(1X,I6,9(2X,F5.1,1X,F5.1))
   31 FORMAT(/,1X,4HRTWT,64X,4HRGCF)
   32 FORMAT(1X,5(F5.3,2X),35X,5(F5.3,2X))
   33 FORMAT(/,1X,6H VH2OC,62X,2HTS)
   34 FORMAT(1X,6H EORS=,F6.1,2X,8H EORSCS=,F6.1,2X,8H AWUPSS=,F6.1,2X,
```

```
     &8H AFWUPS=,F6.1,2X,6H EWUS=,F6.1,2X,7H DRAIN=,F6.1)
   35 FORMAT(1X,I6,9X,F4.1,2X,F5.1)
   37 FORMAT(1X,5(F5.2,2X),35X,5(F5.2,2X))
   38 FORMAT(6X,1HU,2X,F5.1,1X,F5.1)
   39 FORMAT(/,1X,6H   CXT,62X,6HNFIXWT )
   40 FORMAT(1X,18HRADINT(ITIME=1,10),10(F6.2))
   41 FORMAT(5X,14HEO(ITIME=1,11),11(F6.1))
   42 FORMAT(3X,16HPSIL(ITIME=1,10),10(F6.1),2X,6HPSILD=,F6.1,2X,5HPILD=
     &        ,F6.1,7H PSISM=,F6.1)
   43 FORMAT(2X,5HSTEMW,5X,F5.1,8(8X,F5.1))
   44 FORMAT(7X,9(3X,4HPETW,1X,5HLEAFW))
   45 FORMAT(4X,15HSGT(ITIME=1,11),11(F6.3))
   46 FORMAT(1X,18HSCRATO(ITIME=1,11),11(F6.3))
   47 FORMAT(/,1X,4HLINE,1X,6HTRANSC,4X,4HFIXC,6X,6HSCPOOL,4X,3HSEN,
     &  3X,4HREDN,2X,3HSCF,2X,6HRCPOOL,4X,2HQZ,3X,6HSCRATO,1X,4HSGTR,
     &  2X,6HXTRACS,4X,6HUSEDCS,4X,5HFIXCS,5X,4HTAUN,7X,4HSINK)
   48 FORMAT(' ENTER RUN IDENTIFICATION')
   49 FORMAT(18A4)
   50 FORMAT(/,1X,18A4)
   51 FORMAT(' ENTER 1 UNDER THE FIRST LETTER OF EACH PLOT YOU WISH TO',
     &' SEE.'/' ROOTS RGCF VH2OC TS CXT NFIXWT TOPS PNDT LW LA.')
   52 FORMAT(I1,5X,I1,4X,I1,5X,I1,2X,I1,3X,I1,6X,I1,4X,I1,4X,I1,2X,I1)
   53 FORMAT(I2,5X,I1,4X,I1,5X,I1,2X,I1,3X,I1,6X,I1,4X,I1,4X,I1,2X,I1)
   54 FORMAT(/,1X,8H ULEAFW=,F5.3,7H UPETW=,F5.3,8H ULAREA=,F6.4,
     &         7HUPETL=,F5.3,8H MSTEMH=,F5.3,3H Z=,F6.4,8H LEAFWT=,F4.2,
     &         7H PETWT=,F6.4,8H STEMWT=,F6.4,1X,8H ROOTWT=,F5.3,/,
     &         8H LAREAT=,F6.3,5H CEC=,F5.3,6H LLUE=,F5.3,5H LTC=,F5.3,
     &         8H SCPOOL=,F6.4,6H RRRM=,F6.1,6H RRRY=,F6.1,6H RVRL=,
     &         F5.2,7H TLIFE=,F4.1,6H RTWL=,E8.1)
   57 FORMAT(1X,5HLAYER,I2,11H RTWT(1,K)=,15(F6.4,2X))
   58 FORMAT(/,1X,7H RNNO3=,F7.3,7H RNNH4=,F7.4,
     &6H FERN=,F7.4,6H FNO3=,F7.3,6H FNH4=,F7.3,5H OMA=,F8.1)
   59 FORMAT(/)
   60 FORMAT(1X,F5.1,F7.1,F6.1,8F6.2,F6.1,F6.2)
C     ----------------------------------------------------------------
C      *** WRITE HEADING AND SELECT DATA TO BE RECORDED ***
C
       WRITE(2,1)
C
C       WRITE(10,48)
       READ(*,49)AA,AB,AC,AD,AE,AF,AG,AH,AI,AJ,AK,AL,AM,AN,AO,AP,AQ,AR
       WRITE(2,50)AA,AB,AC,AD,AE,AF,AG,AH,AI,AJ,AK,AL,AM,AN,AO,AP,AQ,AR
C       WRITE(10,51)
       READ(*,52)LRTS,LRGF,LVHC,LSTP,LOXT,NFX,LTOPS,LPNDT,LW,LA
C
       MSTEMW=STEMWT
       SHTWT = LEAFWT+PETWT+STEMWT
       LAREAM = LAREAT
C
       WRITE(2,54)ULEAFW,UPETW, ULAREA,UPETL,MSTEMH,Z,LEAFWT,PETWT,
     &            STEMWT,ROOTWT,LAREAT,CEC,LLUE,LTC,SCPOOL,RRRM,RRRY
     &            ,RVRL,TLIFE,RTWL
C
C      *** READ IN DETAILS ABOUT CROP LOCATION, TIMING, SPACING, ETC. ***
C
       READ(4,*) MG,DET,LATUDE,JDFRST,JDLAST,ROWSP,POPROW,ROWANG,
     &  NFRQ,FILL,CO2,SDWMAX
       WRITE(2,10) MG,DET,LATUDE,JDFRST,JDLAST,ROWSP,POPROW,ROWANG,
     &            NFRQ,FILL,CO2,SDWMAX
        NEXT = NFRQ - 1
       WRITE(2,18) LEAFWT, PETWT, STEMWT,PODWT,SEEDWT,ROOTWT,NODWT
       WRITE(2,19) PODS,SEEDS,LAREAM,LAREAB,LAREAT
C
```

```
C      *** READ IN DETAILS ABOUT NUTRIENT STATUS OF SOIL AT PLANTING. ***
C
       READ(4,*) RNNO3,RNNH4,FERN,FNO3,FNH4,OMA
       WRITE(2,58) RNNO3,RNNH4,FERN,FNO3,FNH4,OMA
C
C      *** READ DESCRIPTORS AND CONVERSION CONSTANTS FOR WEATHER DATA ***
C
       READ(1,*)MSW1,MSW2,A,B,C,E,F
C
C      *** IF DAILY WINDSPEED DATA IS UNAVAILABLE, PUT IN AN AVERAGE
C          VALUE FOR SITE. ***
C
       IF( MSW2.EQ.1 ) GOTO 7773
         READ(1,*) WIND
C
C      *** CALCULATE THE DIMENSION OF THE "CLIMAT" ARRAY. ***
C
 7773. NCD=7
       IF( MSW1.EQ.0 ) NCD=NCD-2
       IF( MSW2.EQ.0 ) NCD=NCD-1
C
C      *** FIND THE CORRECT DAY IN THE WEATHER FILE ***
C
 9998 READ(1,*)JDAY,(CLIMAT(I),I=1,NCD)
       IF(JDAY.LT.JDFRST)GO TO 9998
       IF (JDAY.GT.JDFRST)THEN
       WRITE (2,9)
       STOP
       END IF
       RI = CLIMAT(1)*A
       TA = (CLIMAT(3)-C)/B
C
C      *** READ IN DETAILS ABOUT SOIL COMPOSITION, STRUCTURE, ETC. ***
C

       READ(3,*) LYRSOL,SITE,DATA,NL,NK,D,NLP,DIMPL,TC,MSW3
       W = ROWSP/NK
       DO 1003 I=1,LYRSOL
 1003    READ(3,*) SDEPTH(I),BD(I),SAND(I),CLAY(I),PSISAT(I),THETAS(I),
      &        FCI(I),THETA0(I),AIRDR(I),SCOND(I),DIFF0(I),BETA(I),ETA(I),
      &        (TDSX(M,I),M=1,5)
       WRITE(2,12) LYRSOL
       WRITE(2,13)
       WRITE(2,14)  SDEPTH(1),BD(1),SAND(1),CLAY(1),PSISAT(1),THETAS(1),
      &        FCI(1),THETA0(1),AIRDR(1),SCOND(1),DIFF0(1),BETA(1),ETA(1)
       IF(LYRSOL.GT.1) THEN
       DO 125 I=2,LYRSOL
  125 WRITE(2,15) I,SDEPTH(I),BD(I),SAND(I),CLAY(I),PSISAT(I),THETAS(I),
      &        FCI(I),THETA0(I),AIRDR(I),SCOND(I),DIFF0(I),BETA(I),ETA(I)
       END IF
       WRITE(2,160)
       WRITE(2,161) (TDSX(M,1),M=1,5)
       IF(LYRSOL.GT.1) THEN
       DO 135 I=2,LYRSOL
  135 WRITE(2,162) I,(TDSX(M,I),M=1,5)
       END IF
       IF (SITE.EQ.2.AND.DATA.EQ.1)READ(3,*) T0,THETAI,DUMB,DUMC
C
C  *** IF INITIAL SOIL WATER CONTENTS ARE AVAILABLE READ THEM ***
C
        IF(MSW3.EQ.0) GO TO 7776
        READ(3,*)(VH2OI(L),L=1,NL)
        WRITE(2,*)(VH2OI(L),L=1,NL)
```

```
C
C      *** READ IN THE INITIAL ROOT DISTRIBUTION ***
C
1.01    7776 READ(3,*) NLR,NKR,((RTWT(L,K),L=1,NLR),K=1,NKR)
             WRITE(2,59)
             DO 1039 L = 1,NLR
             WRITE(2,57) L,(RTWT(L,K),K=1,NKR)
        1039 CONTINUE
             WRITE(2,59)
             READ(5,*) NPRM,(PRM(I),I=1,NPRM)
             WRITE(2,*) NPRM,(PRM(I),I=1,NPRM)
C
C      *** CALCULATE INITIAL NODULE DISTRIBUTION. ***
C
             NLN = NLR
             NKN = NKR
             DO 1025 L = 1,NLN
             DO 1028 K = 1,NKN
             NFIXWT(L,K) = RTWT(L,K)/100.
             NFIXNU(L,K) = NFIXWT(L,K)/CONVN/1.58E-4
             NODNUM(L,K) = NFIXNU(L,K)
        1028 CONTINUE
        1025 CONTINUE
C
C      *** CALCULATE SOIL CHARACTERISTICS IN EACH CELL FROM DATA ON
C          HORIZONS. ***
C
             NKH=NK/2
             LKH = NL * NKH
             J=1
1.02         DO 1010 L=1,NL
               X200=(L-0.5)*D
         200   CONTINUE
               IF( X200.LE.SDEPTH(J) ) GOTO 205
                 J=J+1
                 IF( J.LT.LYRSOL ) GOTO 200
         205     NJ(L) = J
1.03             TSI=TA*EXP(ALOG(TC/TA)/NL*(L-0.5))
                 PSISI= -0.3
                 VSANDI=BD(J)*SAND(J)/2.65
                 VCLAYI=BD(J)*CLAY(J)/2.65
                 FC(L)=FCI(J)
C
               DO 1031 K=1,NKH
                 BDC(L,K)=BD(J)
                 VSAND(L,K)=VSANDI
                 VCLAY(L,K)=VCLAYI
                 VH2OC(L,K)=THETAS(J)
                 IF(MSW3.EQ.1) VH2OC(L,K) = VH2OI(L)
                 TS(L,K)=TSI
                 PSIS(L,K)=PSISI
                 CXT(L,K)=0.21
                 COND(L,K)=SCOND(J)
1.04             RVR(L,K)=RVRL*SQRT(((D*(L-0.5))**2)+((W*(K-0.5))**2))
                 YRL(L,K)=RTWT(L,K)/RTWL
        1031 CONTINUE
        1010 CONTINUE
             IF(SITE.NE.2.OR.DATA.NE.1)GO TO 6667
             DO 6666 K= 1,NKH
             VH2OC(NL,K) = THETAI
        6666 CONTINUE
        6667 CONTINUE
C
```

```
     C      *** DISTRIBUTE FERTILISER AND ORGANIC MATTER THROUGH THE PLOW
     C          LAYER AND CALCULATE TOTAL SOIL POROSITY. ***
     C
            X201=1.122E-2/NLP/D
            X202=FERN*FNO3
            X203=FERN*FNH4
     C
            DO 1011 L=1,NLP
              DO 1021 K=1,NKH
                VMUCK(L,K)=OMA*X201/1000.0/1.30
                VNC(L,K)=OMA*X201*0.01
                VNO3C(L,K)=(RNNO3+X202)*X201
                VNH4C(L,K)=(RNNH4+X203)*X201
            TOPORS(L,K)=1.0-VSAND(L,K)-VCLAY(L,K)-VMUCK(L,K)
      1021 CONTINUE
      1011 CONTINUE
     C
     C      ***INITIALIZE SOME SOIL VARIABLES IN THE REMAINING CELLS. ***
     C
            J=NLP+1
            DO 1012 L=J,NL
              DO 1013 K=1,NKH
              VMUCK(L,K)=0.0
            VNO3C(L,K)=0.0
            TOPORS(L,K)=1.0-VSAND(L,K)-VCLAY(L,K)
      1013 CONTINUE
      1012 CONTINUE
     C
     C      *** CALCULATE THE THERMAL CONDUCTIVITY OF AIR-DRY SURFACE SOIL
     C          USING EQUATIONS FROM THERMS ***
     C
1.05            TCAIR=0.058+(1.7E-4*TS(1,NKH))
1.06            XSAND=((2/(1+(((20.4/TCAIR)-1)*0.125)))+(1/(1+(((20.4/TCAIR) -
          &             1)*0.75))))/3
                XCLAY=((2/(1+(((7/TCAIR)-1)*0.125)))+(1/(1+(((7/TCAIR)-1) *
          &             0.75))))/3
                XMUCK=((2/(1+(((0.6/TCAIR)-1)*0.5)))+1)/3
1.07            TCADS=((XSAND*VSAND(1,1)*20.4) +
          &               (XCLAY*VCLAY(1,1)*7.0)+(XMUCK*VMUCK(1,1)*0.6) +
          &               (TOPORS(1,1)
          &               *TCAIR))/((XSAND*VSAND(1,1)) +
          &               (XCLAY*VCLAY(1,1))+(XMUCK*VMUCK(1,1))+
          &               (TOPORS(1,1)))/1000*1.25
     C
     C      *** CALCULATE A CLOUD COVER FACTOR FOR THIS LATITUDE. ***
     C
            IF(LATUDE.LE.25.0)THEN
1.08        CLDFAC = 0.45-(LATUDE*0.004)
            ELSE
            CLDFAC = 0.30 + (LATUDE *0.002)
            END IF
     C
     C      *** SET VALUES FOR SOME ENVIRONMENT VARIABLES. ***
     C
            IDAY=0
            WATWK=RI*9.2592E-6
     C    9.2592E-6 = 0.45/13.5/3600.0
     C
     C      *** SET VALUES FOR SOME PLANT VARIABLES. ***
     C
1.09        TRIFMG=0.05+(0.015*(MG+1))
1.10        DIFIMG=15.2-(MG*0.4)
1.11        RTMINW=RTWL*SQRT(D**2.0+W**2.0)
```

```
      IF(LPNDT.EQ.1) WRITE(2,47)
C     ------------------------------------------------------------------
C     *************** CALCULATIONS MADE DAILY **************************
C     ------------------------------------------------------------------
      DO 1001 JDAY=JDFRST,JDLAST
        IDAY=IDAY+1
        IF(LPNDT.EQ.1)WRITE(2,59)
        CALL CLYMAT
C
C     *** AVERAGE LIGHT FLUX DENSITY ON CROP FOR PAST WEEK ***
C
        WATWK=(WATWK*0.856)+(RI*1.7857E-5/PERIOD/IPERD)
C     1.7857E-5 = 0.45/3600/7
C
C     *** THICKNESS OF NEW LEAVES ***
C
1.12    TNL=5.76E-4*WATWK/(1.0+(0.021*WATWK))+(7.E-6*(CO2-300))
C
C     *** NUMBER OF DAYS SINCE SEED FILL STARTED ***
C
        IF( SWR4.EQ.1 ) FILDAY=FILDAY+1
C
C     *** TIME SINCE MAINSTEM TERMINAL LEAF REACHED FULL EXPANSION ***
C
       IF(NBRNCH.EQ.0) GO TO 7779
        IF( TB(NBRNCH).GE.5.0) TLAGE=TLAGE+1.0
       GO TO 7775
 7779  IF(VSTAGE.GE.VSTMAH)TLAGE=TLAGE+1.0
C
C  *** NUMBER OF TOP LEAF ON MAIN STEM ***
C
 7775   NTOPLF = IFIX(VSTAGE+0.5)
        IF(NTOPLF.GE.VSTMAH)NTOPLF=VSTMAH-0.5
C
C     *** CALCULATE A CROP SURFACE ROUGHNESS PARAMETER ***
C     -
1.13    ROUGH=1.0/(ABS((Z/ROWSP)-0.5)+0.5)
        IF( ROUGH.LT.1.0 ) ROUGH=1.0
C
C     *** ON STILL DAYS, CONVECTION CURRENTS AT CROP LEVEL ASSURE SOME
C         AIR MOVEMENT. ***
C
1.14    IF( WIND.LE.0.36 ) WIND=0.36
C
C     *** CALCULATIONS MADE (IPERD) TIMES DAILY ***
C
        DO 1002 ITIME=1,IPERD
        IF(MATURE.EQ.1) GO TO 1027
          CALL EVTR
          CALL TMPSOL
          CALL PNET
          CALL SHTGRO
          CALL RUTGRO
          CALL NITRO
          CALL TRADE
          CALL GROWTH
          CALL PLTMAP
          CALL NODULE
          IF( ITIME.EQ.IPERD ) GOTO 1004
          CALL UPTAKE
          IF( RAIN.GT.0.0 ) CALL GRAFLO
          CALL CAPFLO
 1002   CONTINUE
```

```
1004     CALL NIGHT
C      ----------------------------------------------------------------
C      *** PRINT OUT THE MODEL PREDICTIONS IF TODAY IS A MULTIPLE OF
C          THE REQUIRED FREQUENCY OR THE CROP HAS REACHED MATURITY ***
C      ----------------------------------------------------------------
       IF(((IDAY.LT.NEXT).AND.(MATURE.EQ.0)).AND.(JDAY.LT.JDLAST))
     &    GO TO 1001
1027    NTOPLF = IFIX(VSTAGE+0.5)
        IF(NTOPLF.GE.VSTMAH)NTOPLF=VSTMAH-0.5
C
         TOPWT = STEMWT + PETWT+(SCPOOL*2.5*(STEMWT+PETWT)/SHTWT)+
     &           (STPOOL*2.5)
         TBS = 0.0
         IF(NBRNCH.EQ.0) GO TO 6661
         DO 1040 J = 1,NBRNCH
         IF(ILOWB(J).GE.TBMAH(J)) GO TO 1040
         NBTPLF(J) = IFIX(TB(J)+0.5)
         IF(NBTPLF(J).GT.TBMAH(J)) NBTPLF(J)= IFIX(TBMAH(J))
         TBS =TBS+ NBTPLF(J)-ILOWB(J)+1
1040     CONTINUE
6661     FRTWT = PODWT + SEEDWT
         LEAVES = NTOPLF-ILOW + 1
         IF(ILOW.EQ.0) LEAVES = LEAVES-1
         LFWT=LEAFWT+(SCPOOL*2.5*LEAFWT)/SHTWT
         IF(LFWT.GT.0.) GO TO 1050
         SLA = 0.0
         GO TO 1060
1050     SLA = LAREAT/LFWT
1060     TOPLF = FLOAT(NTOPLF)
         EAVES = FLOAT(LEAVES)
         BRNCH = FLOAT(NBRNCH)
         DAY = FLOAT(IDAY+1)
         WRITE(7,60)DAY,LAREAT,LAREAM,LFWT,TOPWT,FRTWT,MSTEMH,TOPLF,
     &                 EAVES,TBS,BRNCH,SLA,ROOTWT
C
         IDAYX = IDAY + 1
         JDAYX = JDAY + 1
         JPERD = IPERD + 1
           WRITE(2,17) IDAYX,JDAYX
           WRITE(2,18) LEAFWT,PETWT,STEMWT,PODWT,SEEDWT,ROOTWT,NODWT
           WRITE(2,19) PODS,SEEDS,LAREAM,LAREAB,LAREAT
           WRITE(2,20) VSTAGE,NTOPLF,RSTAGE,VLOST
           WRITE(2,21) PDNODC,NRATIO
           WRITE(2,22) ILOW,Z,NBRNCH
           WRITE(2,23) SCPOOL,RCPOOL,XTRACS,FIXCS,USEDCS
           WRITE(2,40) (RADINT(ITIME),ITIME=1,IPERD)
           WRITE(2,41) (EO(ITIME),ITIME=1,JPERD)
           WRITE(2,42) (PSIL(ITIME),ITIME=1,IPERD),PSILD,PILD,PSISM
           WRITE(2,45) (ASGT(ITIME),ITIME=1,JPERD)
           WRITE(2,46) (ASCRAT(ITIME),ITIME=1,JPERD)
           WRITE(2,34) EORS,EORSCS,AWUPSS,AFWUPS,EWUS,DRAIN
C
           IF(LW.EQ.1)THEN
           WRITE(2,24)
           WRITE(2,44)
           IF(NTOPLF.LT.1) GO TO 1023
           DO 1014 M=1,NTOPLF
             I=NTOPLF+1-M
             IF(I.GT.25) I=25
             IF( I.LE.20 ) GOTO 1015
               WRITE(2,25) I,MPETW(I),MLEAFW(I)
               GOTO 1014
1015         WRITE(2,26)I,MPETW(I),MLEAFW(I),BPETW(I,1),
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```
     &                BLEAFW(I,1),BPETW(I,2),BLEAFW(I,2),BPETW(I,3),
     &                BLEAFW(I,3),BPETW(I,4),BLEAFW(I,4),BPETW(I,5),
     &                BLEAFW(I,5),BPETW(I,6),BLEAFW(I,6),BPETW(I,7),
     &                BLEAFW(I,7),BPETW(I,8),BLEAFW(I,8)
 1014     CONTINUE
 1023     CONTINUE
          WRITE(2,27) UPETW,ULEAFW
          WRITE(2,43) MSTEMW,(BRNCHW(KK),KK=1,8)
          END IF
C
          IF(LA.EQ.1)THEN
          WRITE(2,24)
          WRITE(2,28)
          IF(NTOPLF.LT.1) GO TO 1024
          DO 1017 M=1,NTOPLF
            I=NTOPLF+1-M
            IF(I.GT.25) I=25
            IF( I.LE.20 ) GOTO 1018
              WRITE(2,35) I,MPETL(I),MLAREA(I)
              GOTO 1017
 1018       WRITE(2,30)I,MPETL(I),MLAREA(I),BPETL(I,1),
     &                BLAREA(I,1),BPETL(I,2),BLAREA(I,2),BPETL(I,3),
     &                BLAREA(I,3),BPETL(I,4),BLAREA(I,4),BPETL(I,5),
     &                BLAREA(I,5),BPETL(I,6),BLAREA(I,6),BPETL(I,7),
     &                BLAREA(I,7),BPETL(I,8),BLAREA(I,8)
 1017     CONTINUE
 1024     CONTINUE
          WRITE(2,38) UPETL,ULAREA
          WRITE(2,29) MSTEMH,(BRNCHH(KK),KK=1,8)
          END IF
C
          IF(LRTS.EQ.1)THEN
 1019     WRITE(2,31)
          DO 1016 L=1,NL
 1016     WRITE(2,32)(RTWT(L,K),K=1,NKH),(RGCF(L,K),K=1,NKH)
          END IF
C
          IF(LVHC.EQ.1)THEN
          WRITE(2,33)
          DO 1020 L=1,NL
 1020     WRITE(2,37) (VH2OC(L,K),K=1,NKH),(TS(L,K),K=1,NKH)
          END IF
C
          IF(LOXT.EQ.1)THEN
          WRITE(2,39)
          DO 1035 L=1,NL
 1035     WRITE(2,32)(CXT(L,K),K=1,NKH),(NFIXWT(L,K),K=1,NKH)
          END IF
C
          IF(LTOPS.EQ.1) CALL SOYPLT
          IF(LPNDT.EQ.1)WRITE(2,47)
          NEXT=IDAY+NFRQ
          IF( MATURE.EQ.1 ) STOP
 1001   CONTINUE
      STOP
      END
```

Notes for MAIN

The data to initialize plant variable at growth stage VC come from observations on Forrest soybean and may be different for other cultivars. However, organ sizes and dry weights for this early stage are rarely available. The data are entered in BLK:D.

1.01 Initial root distribution is chosen based on a knowledge of impenetrable layers in the soil but generally assuming the amount of root in each soil cell decreases with distance from the stem base.

1.02 The soil parameters are moved from arrays classified by horizon into arrays classified by soil cell number.

1.03 Initial soil temperature by layers is calculated assuming that the logarithm of temperature varies linearly between the fixed deep-soil temperature and air temperature.

1.04 Root vascular resistance between the stem base and each soil cell is here made directly proportional to the distance between them.

1.05 See THERMS, note T.06.

1.06 See THERMS, note T.08.

1.07 See THERMS, note T.10

1.08 The latitude-dependent factor for calculating radiation attenuation from cloud cover, CLDFAC is used in CLYMAT line 190. See note 3.13. The values of CLDFAC have been taken from Budyko (1974, his page 48, Table 4) and are plotted here in Figure 11. The lines described by the equations in the code are shown in

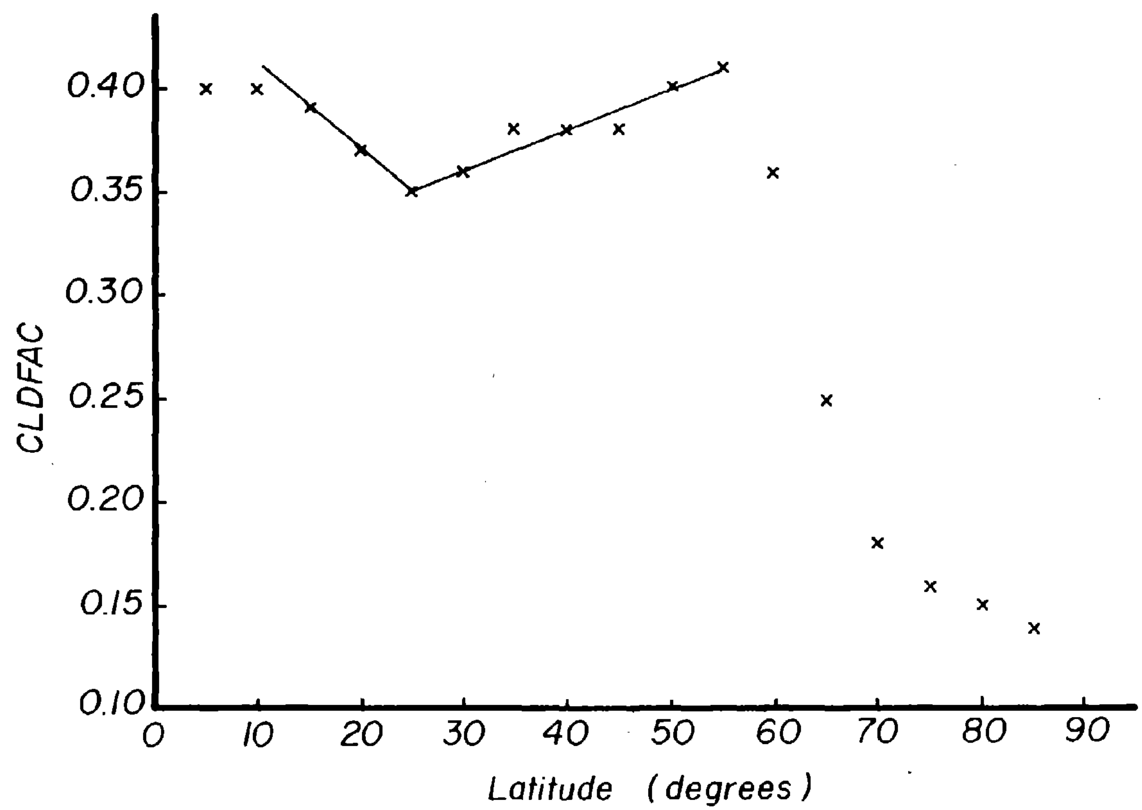


Fig. 11. Latitude-dependent factor for calculating radiation attenuation from cloud cover. Data from Budyko (1974).

the same figure. Clearly, the equations do not apply outside the range of latitude from 10 to 55°N.

1.09 TRIFMG is a variable that adjusts the relationship between temperature and the rate of trifoliolate expansion. It varies between cultivars and is assumed here to be proportional to maturity group number (See SHTGRO, note 7.01). It is used in SHTGRO on line 94 and in PLTMAP on line 94.

1.10 DIFIMG is a variable relating photoperiod to flowering. It varies between cultivars and correlates well with maturity group number (See PLTMAP, note 12.03). It is used in PLTMAP on lines 109 and 111.

1.11 RTMINW is dry weight per unit length of root times diagonal distance across the soil cell.

1.12 New leaf thickness has been made a function of mean light flux density over the previous week following the ideas of Charles-Edwards and Ludwig (1975).

1.13 The crop surface roughness parameter is used in calculating evapotranspiration. It arbitrarily increases from 1 to 2 as the crop grows to cover half the ground area and then decreases to 1 as the canopy closes (See EVTR, note 4.06).

1.14 The idea of a minimum affective windspeed caused by convection currents on hot still days is taken from Gates (1968).

NIGHT

The subroutine NIGHT is called once each 24 hours from MAIN. It performs a similar function to MAIN, controlling the sequence in which other subroutines are called during the overnight time step. However, there are some subroutines (like PNET) which are clearly not appropriate to an overnight step yet contain the equations for some processes (like respiration in PNET) which are appropriate. In cases like these, the necessary equations have been copied from the subroutine into NIGHT. Subroutines that are not called from NIGHT but have contributed equations to it are EVTR, PNET, and TRADE. Mostly the processes are simulated in NIGHT in exactly the same way as in the daytime subroutine but there are exceptions. Stomatal conductance is reduced to a minimum value which depends on the recent exposure of the plant to water stress. This accommodates the fact that stomata close more completely after the plant has experienced water stress. Water transpired overnight is extracted from the soil profile in the same pattern as it was extracted during the last period of the day. Leaf turgor pressure is assumed to increase logarithmically up to its dawn value and shoot growing time is the amount of time that the leaf turgor pressure is above 2 bars.

In NIGHT, the following tests are made: (a) A test to see if the lowest leaves on the canopy have been self-supporting for the past 24 hours and, if they have not, to set a flag to drop them in NITRO. (b) A test to see if branch development is pushing leaves off the mainstem and, if it is, to set a flag to drop them in NITRO. NIGHT also calculates dawn leaf water potential and an osmoregulation factor which depend on the amount of water stress experienced recently by the plants.

```
-----------------------( NIGHT )-----------------------------------
THIS ROUTINE FOLLOWS CROP ACTIVITY AND MOVEMENT OF MATERIALS
OVERNIGHT, MOSTLY USING THE SAME ROUTINES CALLED DURING THE DAY
BUT USING SIMPLIFIED VERSIONS OF EQUATIONS FROM OTHER ROUTINES.
-------------------------------------------------------------------
SET OVERNIGHT VALUES FOR SOME VARIABLES

CALCULATE TRANSPIRATION RATE AT MEAN NIGHT TEMPERATURE USING
   EQUATIONS FROM EVTR AND ALLOWING FOR STOMATAL CLOSURE.

CALCULATE WATER UPTAKE FROM A HALF SOIL SLAB BETWEEN DUSK AND
   DAWN.

EXTRACT HALF THIS WATER FROM SOIL CELLS IN PROPORTION TO THE
   EXTRACTION RATE AT ITIME=IPERD AND PREDICT CHANGE IN WATER
   CONTENT IN EACH SOIL CELL BETWEEN ITIME=IPERD AND TIME WHEN
   TAIR=TNYT.

CALL ROUTINES TO ADJUST SOIL WATER CONTENT, ETC. UP TO THE
   TIME WHEN TAIR=TNYT.

  CALL UPTAKE

   if there has been no rain or irrigation, jump down

  CALL GRAFLO
  CALL CAPFLO

IF CROP CANOPY HAS NOT YET CLOSED, CALCULATE RATE OF EVAPO-
   RATION OF WATER FROM THE SOIL USING EQUATIONS FROM EVTR.

TEST TO SEE IF LOWEST LEAVES IN THE CANOPY ARE SELF SUPPORTING.
   IF NOT DROP THEM.

TEST TO SEE IF BRANCH DEVELOPMENT IS PUSHING OFF LEAVES

CALCULATE OVERNIGHT CHANGE IN SOIL TEMPERATURE, POTENTIAL
   RATES OF SHOOT AND ROOT GROWTH AND PLANT NITROGEN STATUS.

  CALL TMPSOL
  CALL SHTGRO
  CALL RUTGRO
  CALL NITRO

CALCULATE DAWN LEAF WATER POTENTIALS

CALCULATE WHAT PROPORTION OF TIME BETWEEN ITIME=IPERD AND DAWN
   THE SHOOT TURGOR IS ABOVE 2.0 ( = TIME FOR WHICH SHOOT GROWS)

ALLOCATE TRANSLOCATED CARBON TO FRUIT, NODULES AND VEGETATIVE
   SHOOT PLUS ROOT

CALCULATE MEAN RATE AT WHICH CARBON IS SUPPLIED TO THE ROOTS
   USING EQUATIONS FROM TRADE.
```

```
CALCULATE A CARBON TO DRY WEIGHT CONVERSION FACTOR AND A
   RESPIRATION FACTOR FOR THE ROOTS AS IN TRADE.

FOR EACH SOIL CELL, CALCULATE OVERNIGHT ROOT GROWTH AND
   PREDICT WATER UPTAKE BETWEEN THE TIME WHEN TAIR=TNYT AND
   ITIME=1, USING EQUATIONS ADAPTED FROM TRADE.

REDISTRIBUTE ROOT GROWTH IN THE SOIL PROFILE

CALCULATE LENGTH OF YOUNG ROOTS IN CELLS

CALCULATE AMOUNT OF CARBON IN THE ROOT STORAGE POOL

CALCULATE SHOOT GROWTH, NODULE GROWTH AND NODULE FIXATION.

  CALL GROWTH
  CALL PLTMAP
  CALL NODULE

CALL ROUTINES TO ADJUST SOIL WATER CONTENT, ETC., UP TO
   ITIME=1.

  CALL UPTAKE

  if there has been no rain or irrigation, jump down

  CALL GRAFLO
  CALL CAPFLO
  CALL NITRIF
  CALL SOLOXY

RETURN
```

```
      SUBROUTINE NIGHT
C     --------------------( NIGHT )------------------------------------
C     THIS ROUTINE FOLLOWS CROP ACTIVITY AND MOVEMENT OF MATERIALS
C     OVERNIGHT, MOSTLY USING THE SAME ROUTINES CALLED DURING THE DAY
C     BUT USING SIMPLIFIED VERSIONS OF EQUATIONS FROM OTHER ROUTINES.
C     -----------------------------------------------------------------
      INCLUDE 'COMMON.FOR'
C     -----------------------------------------------------------------
C     *** SET OVERNIGHT VALUES FOR SOME VARIABLES ***
C
        NYTE = 1
        ITIME = IPERD+1
        EXTRA=PERIOD/2.0
        PERIOD=24.0-(PERIOD*IPERD)
        TAIR(ITIME)=TNYT
2.01    BMAINN = ((11.733*LEAFWT)+(366.67*LEAFWT*NRATIO*0.05))/
     &           24.0*0.27273/1000.0*0.25*EXP(0.0693*TNYT)
        USEDCS = USEDCS+(BMAINN*PERIOD)
2.02    X6 = SHTWT*0.06
        IF(TLAGE.GT.0.0)X6 = SHTWT*PRM(18)
        IF((VSTAGE.GE.6.0).AND.(SCPOOL.GT.X6))THEN
        TRANSC=X6*0.125
        X7=(SCPOOL-X6)/TRANSC
        IF(X7.LT.PERIOD)TRANSC=(SCPOOL-X6*(0.875**(PERIOD-X7)))/PERIOD
        ELSE
2.03    TRANSC=SCPOOL*(1.0-(0.875**PERIOD))/PERIOD
        END IF
        IF(BMAINN.GT.TRANSC)THEN
        SCPOOL=SCPOOL-((BMAINN-TRANSC)*PERIOD)
        TRANSC=0.0
        ELSE
        SCPOOL=SCPOOL-(TRANSC*PERIOD)
        TRANSC=TRANSC-BMAINN
        END IF
        IF(SCPOOL.LE.0.0) THEN
        USEDCS=USEDCS+SCPOOL
        SCPOOL = 0.0
        END IF
        TRNDAY=TRNDAY+(TRANSC*PERIOD)
        COVER=Z/ROWSP
        IF( COVER.GE.1.0 ) COVER=1.0
2.04    SCF=0.1
        IF(PILOSM.LT.-9.0)SCF = 0.01
C
C    *** CALCULATE TRANSPIRATION RATE AT MEAN NIGHT TEMPERATURE USING
C          EQUATIONS FROM EVTR AND ALLOWING FOR STOMATAL CLOSURE. ***
C
2.05    SVPA=0.61078*EXP((17.2693882*TAIR(ITIME))/(TAIR(ITIME)+237.3))
        DEL=(0.61078*EXP((17.2693882*(TAIR(ITIME)+1.0)) /
     &      (TAIR(ITIME)+1.0+237.3)))-SVPA
        VPD=SVPA-AVP
        IF(VPD.LE.0.0)VPD=0.0
          IF( MSW2.EQ.1 ) WIND=CLIMAT(5)
2.06    EPO=VPD*109.375*(ROUGH+(0.149*WIND))/((DEL/GAMMA)+1.0)*COVER
        ET=EPO*ROWSP/POPROW/100.0*SCF
        EO(ITIME) = ET/SCF
C
C    *** CALCULATE WATER UPTAKE FROM A HALF SOIL SLAB BETWEEN DUSK AND
C          DAWN. ***
C
        UPH2ON=ET*POPROW/100.0/2.0*PERIOD
        EORS=EORS+UPH2ON
        EORSCS = EORSCS+UPH2ON
```

```
               AWUPSS = AWUPSS+UPH2ON
C
C        *** EXTRACT HALF THIS WATER FROM SOIL CELLS IN PROPORTION TO THE
C            EXTRACTION RATE AT ITIME=IPERD AND PREDICT CHANGE IN WATER
C            CONTENT IN EACH SOIL CELL BETWEEN ITIME=IPERD AND TIME WHEN
C            TAIR=TNYT. ***
C
               DO 1006 L=1,NLR
                 DO 1005 K=1,NKR
                   IF(AWUPS.GT.0.0)AFWUP(L,K)=AFWUP(L,K)+(AWUP(L,K)*UPH2ON/
      &                       AWUPS/2.0)
  1005             CONTINUE
  1006             CONTINUE
C
C        *** CALL ROUTINES TO ADJUST SOIL WATER CONTENT, ETC. UP TO THE
C            TIME WHEN TAIR=TNYT. ***
C
               CALL UPTAKE
               IF( RAIN.GT.0.0 ) CALL GRAFLO
               CALL CAPFLO
C
C        *** IF CROP CANOPY HAS NOT YET CLOSED, CALCULATE RATE OF EVAPO-
C            RATION OF WATER FROM THE SOIL USING EQUATIONS FROM EVTR. ***
C
2.07           IF( COVER.LT.1.0 ) GOTO 7772
               DO 1026 K=1,NKH
  1026         EWU(K)=0.0
               GO TO 7771
  7772           KLJ=(IFIX(2/2.0/W)+1)
                 NEXP=0
                 VH2OCS=0.0
C
                 DO 1007 K=KLJ,NKH
                   NEXP=NEXP+1
  1007           VH2OCS=VH2OCS+VH2OC(1,K)
C
                 VH2OCM=VH2OCS/NEXP
2.08             ESO=(VPD*109.375*(1.0+(0.149*WIND*(1.0-COVER)))) /
      &               ((DEL/GAMMA)+1.0)*NKH/NEXP
C
                 DO 1008 K=KLJ,NKH
2.09         TNAIRD(K)=2.0*(1.0-(((VH2OC(1,K)-AIRDR(1))/(THETAS(1)-AIRDR(1)))
      &                ** 0.1))
2.10           DIFF = DIFF0(1)*EXP(BETA(1)*(VH2OC(1,K)-THETA0(1)))
2.11         DIFFM=DIFF*DIFF0(1)/((DIFF*TNAIRD(K))+((1.0-TNAIRD(K))*DIFF0(1)))
2.12         ESP = DIFFM*(VH2OC(1,K)-AIRDR(1))/0.5/D*10000.0/24.0
                   IF( ESO.GE.ESP ) EWU(K)=ESP*PERIOD/10000.0*W
                   IF( ESO.LT.ESP ) EWU(K)=ESO/10000.0*W*PERIOD
                   IF(EWU(K).LT.0.) EWU(K)=0.0
2.13               REW=ESP/3600.0*(2500.8-(2.3668*TAIR(ITIME)))
                   RHS(K)=-REW
                   IF( ESO.LT.ESP ) RHS(K)=0.0
  1008           CONTINUE
                 IF( KLJ.EQ.1 ) GOTO 7771
                   KLJM1=KLJ-1
                   DO 1033 K=1,KLJM1
  1033               EWU(K)=0.0
C
C      *** TEST TO SEE IF LOWEST LEAVES IN THE CANOPY ARE SELF SUPPORTING.
C          IF NOT DROP THEM. ***
C
  7771 DROP = 0
2.14         IF(PNLLFS.LT.0.) DROP = 1
```

```
      PNLLFS = 0.0
C
C     ***TEST TO SEE IF BRANCH DEVELOPMENT IS PUSHING OFF LEAVES ***
C
      DO 1041 J=1,NBRNCH
      DAYBR(J)=DAYBR(J)+1
 1041 IF(DAYBR(J).GT.PRM(21).AND.J.GE.ILOW)ABSCIS=1
C
C     *** CALCULATE OVERNIGHT CHANGE IN SOIL TEMPERATURE, POTENTIAL
C         RATES OF SHOOT AND ROOT GROWTH AND PLANT NITROGEN STATUS. ***
C
        CALL TMPSOL
        CALL SHTGRO
        CALL RUTGRO
        CALL NITRO
C
C     *** CALCULATE DAWN LEAF WATER POTENTIALS ***
C
        PSILD=-2.0
        IF( PSISM.LT.PSILD ) PSILD=PSISM
2.15    PILOSM=PILOSM-OSMREG
        IF(PILOSM.LT.-12.0)PILOSM=-12.0
        IF(PILOSM.GT.-9.0)PILOSM=-9.0
        OSMREG=-0.25
2.16    OSMFAC = (SCPOOL/SHTWT)+0.5
        PILD=((PSILD+2.0)*OSMFAC)+PILOSM
        PTPLD = TPLD
        TPLD=PSILD-PILD
C
C     *** CALCULATE WHAT PROPORTION OF TIME BETWEEN ITIME=IPERD AND DAWN
C         THE SHOOT TURGOR IS ABOVE 2.0 ( = TIME FOR WHICH SHOOT GROWS)
C
        SGT = 1.0
        SGTLI = 0.0
        IF(TPL.GT.2.0) GO TO 9999
2.17    SGT=(EXP(TPLD)-EXP(2.0))/(EXP(TPLD)-EXP(TPL))
        SGTLI = 1.0 - SGT
        GO TO 8888
 9999   IF(PTPLD.EQ.BTPL) GO TO 8888
        SGT = SGT + (PTPLD-TPL)/(PTPLD-BTPL) * SGTLT
 8888   ASGT(ITIME) = SGT
C
C     ***ALLOCATE TRANSLOCATED CARBON TO FRUIT, NODULES AND VEGETATIVE
C        SHOOT PLUS ROOT***
C
        NODC=PDNODC
        IF (NODC.GT.TRANSC)THEN
         NODC=TRANSC
         FRUITC=0.0
         VEGSRC=0.0
        ELSE
         FRUITC=PDPODC+PDSC
           IF(FRUITC.GE.(TRANSC-PDNODC))THEN
            FRUITC=TRANSC-PDNODC
            VEGSRC=0.0
           ELSE
            VEGSRC=TRANSC-FRUITC-PDNODC
              IF(VEGSRC.LE.0.0)THEN
               VEGSRC=0.0
               NODC=TRANSC-FRUITC
              ELSE
               NODC=PDNODC
               X30=RSTPC-PDNODC
```

```
                IF(X30.LE.0.0) GO TO 2222
                   IF(X30.LT.VEGSRC) THEN
                    VEGSRC=VEGSRC-X30
                    FRUITC=FRUITC+X30
                   ELSE
                    VEGSRC=0.0
                    FRUITC=FRUITC+VEGSRC
                   END IF
                END IF
             END IF
          END IF
      C
      C     *** CALCULATE MEAN RATE AT WHICH CARBON IS SUPPLIED TO THE ROOTS
      C           USING EQUATIONS FROM TRADE. ***
      C
       2222   X6 = ROOTWT*0.02
2.18          IF((VSTAGE.GE.6.0).AND.(RCPOOL.GT.X6))THEN
              RCPAV=X6*0.05
              X7=(RCPOOL-X6)/RCPAV
              IF(X7.LT.PERIOD)RCPAV=(RCPOOL-X6*(0.75**(PERIOD-X7)))/PERIOD
              ELSE
2.19          RCPAV=RCPOOL*(1.0-(0.75**PERIOD))/PERIOD
              END IF
              PCRL=RCPAV*POPROW/200.
2.20          PCRQ=(0.95*VEGSRC*POPROW/200.)+PCRL
              PCRS = PCRL
              IF(SGT.LT.1.0)PCRS=PCRL+((PCRQ-PCRL)*(1.0-SGT))
      C
      C     *** CALCULATE A CARBON TO DRY WEIGHT CONVERSION FACTOR AND A
      C           RESPIRATION FACTOR FOR THE ROOTS AS IN TRADE. ***
      C
              X20=NRATIO*0.1563
      C    0.1563 = 2.5*6.25/100.0
2.21          CONVR=(X20*0.6)+((1.0-X20)*0.47)
              CONRES=(X20*0.24)+((1.0-X20)*0.08)
      C
      C     *** FOR EACH SOIL CELL, CALCULATE OVERNIGHT ROOT GROWTH AND
      C           PREDICT WATER UPTAKE BETWEEN THE TIME WHEN TAIR=TNYT AND
      C           ITIME=1, USING EQUATIONS ADAPTED FROM TRADE. ***
      C
              PCRTS=0.0
              SW20=0
      C
2.22          DO 1009 J=1,LKRP1
                L=PCRPAR(J)/1000
                K=PCRPAR(J)-L*1000
                PCRTS=PCRTS+(PDWR(L,K)*CONVR)
                IF( PCRS.LE.PCRTS ) GOTO 75
                  ADWR(L,K)=PDWR(L,K)
                  GOTO 67
      C
         75     CONTINUE
                IF( SW20.EQ.1 ) GOTO 85
                  SW20=1
                  ADWR(L,K)=((PDWR(L,K)*CONVR)-PCRTS+PCRS)/CONVR
                  GOTO 67
      C
         85     ADWR(L,K)=0
2.23     67     AWR(L,K)=((ADWR(L,K)+PADWR(L,K))*EXTRA/2.0)+ADWR(L,K)*PERIOD
                PADWR(L,K)=ADWR(L,K)
                ROOTWT=ROOTWT+(AWR(L,K)*2.0*100.0/POPROW)
                X22 = 0.0
                IF(AWUPS.GT.0.0)X22=AWUP(L,K)*UPH2ON/AWUPS/PERIOD
```

```
              PWUP(L,K)=X22
2.23          AFWUP(L,K)=X22*((PERIOD/2.0)+EXTRA)
      1009  FWUP(L,K)=X22*EXTRA
            IF(SW20.EQ.1) PCRTS = PCRS
      C
      C     *** REDISTRIBUTE ROOT GROWTH IN THE SOIL PROFILE ***
      C
              DO 110 L=1,NLR
                DO 111 K=1,NKR
                  X28=RTWT(L,K)*0.2
                  IF(X28.GT.PERTWT(L,K)) PERTWT(L,K)=X28
              ADRL(L,K)=0.0
      111     CONTINUE
      110     CONTINUE
      C
              NLRP = NLR
              NKRP = NKR
              DO 121 L=1,NLR
                DO 120 K=1,NKR
                  IF( AWR(L,K).EQ.0.0 ) GOTO 120
2.24                IF( RTWT(L,K).LE.RTMINW ) GOTO 7774
                    LP1=L+1
                    IF( L.EQ.NL ) LP1=NL
                    IF( LP1.GT.NLR ) NLRP = LP1
                    KP1=K+1
                    IF( K.EQ.NKH ) KP1=NKH
                    IF( KP1.GT.NKR ) NKRP = KP1
                    LM1=L-1
                    IF(L.EQ.1)LM1=1
                    KM1=K-1
                    IF( K.EQ.1 ) KM1=1
2.25                X23=RGCF(L,K)+RGCF(L,KM1)+1.5*RGCF(L,KP1)
         &          +RGCF(LM1,K)+5.0*RGCF(LP1,K)
                    IF (X23.EQ.0.0) GO TO 7774
                     X24=AWR(L,K)*RGCF(L,K)/X23
                     RTWT(L,K)=RTWT(L,K)+X24
2.26                 RESPS(L,K) = RESPS(L,K) + (X24*CONRES)
                     ADRL(L,K)=ADRL(L,K)+X24/RTWL
                    X25=AWR(L,K)*RGCF(L,KM1)/X23
                    RTWT(L,KM1)=RTWT(L,KM1)+X25
                    RESPS(L,KM1) = RESPS(L,KM1) + (X25*CONRES)
                    ADRL(L,KM1)=ADRL(L,KM1)+X25/RTWL
                     X26=AWR(L,K)*1.5*RGCF(L,KP1)/X23
                     RTWT(L,KP1)=RTWT(L,KP1)+X26
                     RESPS(L,KP1) = RESPS(L,KP1) + (X26*CONRES)
                     ADRL(L,KP1)=ADRL(L,KP1)+X26/RTWL
                    X29=AWR(L,K)*RGCF(LM1,K)/X23
                    RTWT(LM1,K)=RTWT(LM1,K)+X29
                    RESPS(LM1,K) = RESPS(LM1,K) + (X29*CONRES)
                    ADRL(LM1,K)=ADRL(LM1,K)+X29/RTWL
                     X27=AWR(L,K)*5.0*RGCF(LP1,K)/X23
                     RTWT(LP1,K)=RTWT(LP1,K)+X27
                     RESPS(LP1,K) = RESPS(LP1,K) + (X27*CONRES)
                     ADRL(LP1,K)=ADRL(LP1,K)+X27/RTWL
                    GO TO 120
      7774          RTWT(L,K) = RTWT(L,K) + AWR(L,K)
                    RESPS(L,K) = RESPS(L,K) + (AWR(L,K)*CONRES)
                    ADRL(L,K) = ADRL(L,K) + AWR(L,K)/RTWL
        120   CONTINUE
        121 CONTINUE
              NLR = NLRP
              NKR = NKRP
      C
```

```
C      *** CALCULATE LENGTH OF YOUNG ROOTS IN CELLS ***
C
           DO 123 L=1,NLR
             DO 122 K=1,NKR
2.27             YRL(L,K) = (YRL(L,K)*(1.0-(0.021*PERIOD)))+ADRL(L,K)
C                0.021=1/2/24
                 RUTDEN(L,K)=RUTDEN(L,K)+(YRL(L,K)*0.021*PERIOD/D/W)
   122       CONTINUE
   123     CONTINUE
C
C      *** CALCULATE AMOUNT OF CARBON IN THE ROOT STORAGE POOL ***
C
         X15=(PCRS-PCRTS)*200./POPROW
         IF( X15.LE.0.0 ) X15=0.0
         X16 = SGT
         IF(X16.GT.1.0) X16 = 1.0
2.28     RCPOOL=RCPOOL+(((VEGSRC*0.05*(1.0-X16))-RCPAV+X15)*PERIOD)
         USEDCS = USEDCS + (PCRTS*200./POPROW*PERIOD)
C
C      *** CALCULATE SHOOT GROWTH, NODULE GROWTH AND NODULE FIXATION. ***
C
  7777   CALL GROWTH
         CALL PLTMAP
         CALL NODULE
C
C      *** CALL ROUTINES TO ADJUST SOIL WATER CONTENT, ETC., UP TO
C          ITIME=1 ***
C
         CALL UPTAKE
         IF(RAIN.GT.0.0) CALL GRAFLO
         CALL CAPFLO
         CALL NITRIF
         CALL SOLOXY
         NYTE=0
         SCF=1.0
         RETURN
         END
```

Notes for NIGHT

Most of the equations used to calculate plant activity and movement of materials overnight are simplified versions of equations from other subroutines. References to those equations and notes explaining them are made without further comment.

2.01 See PNET, note 6.16.

2.02 See PNET, note 6.19

2.03 See PNET, note 6.20.

2.04 Overnight the stomata close. Here it is assumed that the fluxes of carbon dioxide and water are reduced to 0.1 of the maximum rate when the stomata have not experienced water stress. This value is deduced from the data of Boyer (1970). If the plants have recently experienced stress, the stomata are assumed to close to the point where fluxes are reduced to 0.001 of the maximum rate.

2.05 See EVTR, note 4.05.

2.06 See EVTR, note 4.06.

2.07 See EVTR, note 4.08

2.08 See EVTR, note 4.14.

2.09 See EVTR, note 4.15 and 4.16.

2.10 See EVTR, note 4.17.

2.11 See EVTR, note 4.18.

2.12 See EVTR, note 4.19.

2.13 See EVTR, note 4.21.

2.14 PNLLFS is the net photosynthetic rate of the lowest leaves in the canopy averaged over 24 hours. It is calculated in PNET on line 261 and is used to set DROP and drop leaves that are no longer self-supporting. See PNET, note 6.23.

2.15 The leaf osmotic potential at dawn can change by 0.25 bars per day according to the water status of the leaves during the previous day. See TRADE, note 10.18.

2.16 OSMFAC is a factor which describes the ability of the plant to osmoregulate during the day. See TRADE, note 10.03.

2.17 Leaf turgor pressure is assumed to increase logarithmically to its new dawn value.

2.18 See TRADE, note 10.08.

2.19 See TRADE, note 10.09.

2.20 See TRADE, note 10.10.

2.21 See TRADE, note 10.11.

2.22 See TRADE, note 10.22.

2.23 The EXTRA term in these two equations is equal to the number of hours between dawn and the middle of the next calculation period. Its effect is to extend the period of root growth and water uptake for this number of hours.

2.24 See TRADE, note 10.24.

2.25 See TRADE, note 10.25.

2.26 See TRADE, note 10.26.

2.27 See TRADE, note 10.27.

2.28 See TRADE, note 10.28.

CLYMAT

The subroutine CLYMAT is called once each 24 hours from MAIN. It reads one line of data from the weather file each time it is called and adjusts the units if necessary. It calculates solar declination, daylength, and solar radiation incident on the top of the earth's atmosphere. Disposition of the solar radiation in the earth's atmosphere is modeled according to the diagram in Figure 12 using equations from the Smithsonian Meteorological Tables (1963). A fraction of it (determined by sun angle) is absorbed and a fraction of it (determined by atmospheric transmission coefficient) is scattered. Of the scattered radiation, half is scattered forward and reaches the earth as diffuse radiation. The sum of potential diffuse and direct radiation is compared with actual radiation receipt for the day and a cloud cover factor is calculated.

For each of ten equally-spaced points in time during the photoperiod, CLYMAT calculates: (a) the amount of diffuse and direct radiation incident on the crop, (b) the proportion of diffuse and direct radiation intercepted by the crop and (c) the proportion of light (photosynthetically active radiation) intercepted by the crop.

Diurnal temperature variation is modeled according to the diagram in Figure 13. It is assumed that the minimum temperatures occurs at dawn and that air temperature follows a half sine wave during the day. The time of maximum temperature is calculated from a semi-mechanistic equation. Overnight the temperature is assumed to fall, from its value at dusk, logarithmically to the minimum temperature for the next day at dawn.

Finally, CLYMAT calculates actual water vapor pressure for each period of the day and the effective photoperiod for soybean on that day. The

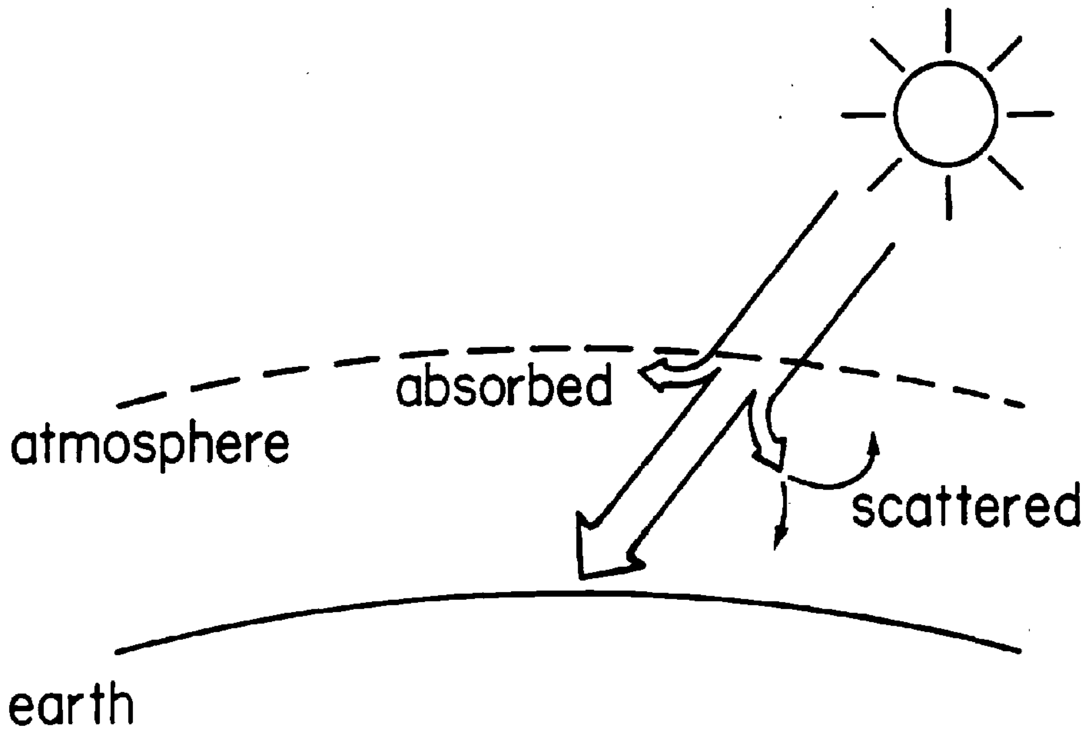


Fig. 12. Disposition of solar radiation in the earth's atmosphere.

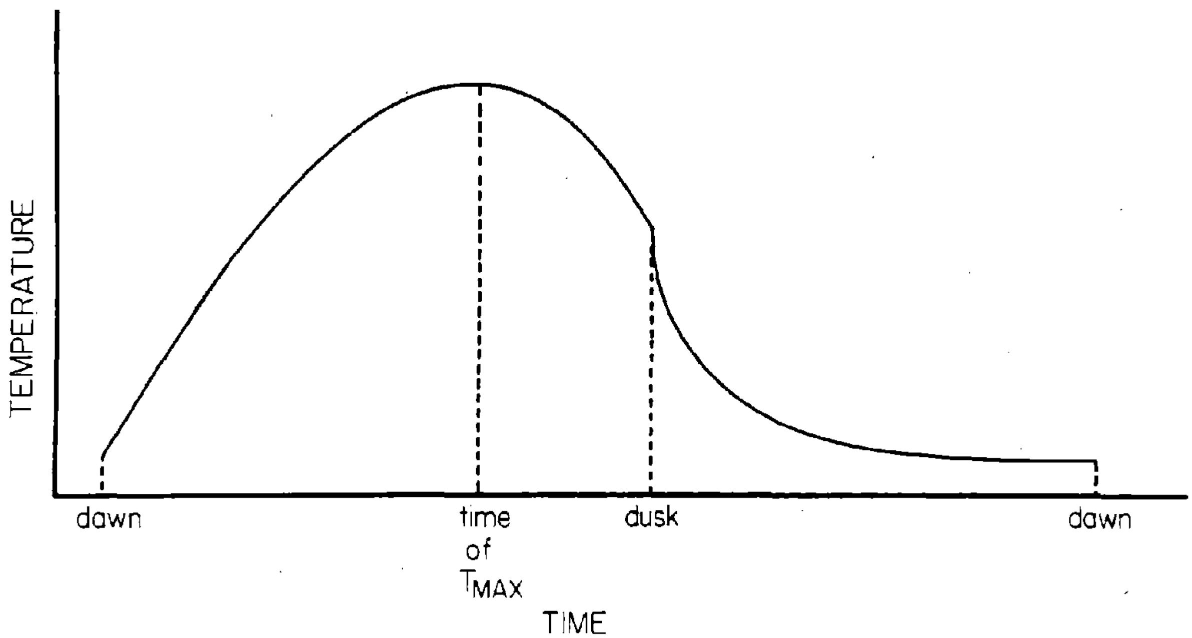


Fig. 13. Model of diurnal temperature variation used in GLYCIM.

effective photoperiod differs from daylength by an amount determined by cloud cover.

```
-----------------------( CLYMAT )---------------------------------
THIS ROUTINE CALCULATES DAYLENGTH, EFFECTIVE PHOTOPERIOD FOR
SOYBEAN, CLOUD COVER, DIFFUSE AND DIRECT LIGHT INTERCEPTION BY
"SOLID" CROP ROWS AND AIR TEMPERATURE AT TEN TIMES DURING THE
DAY. IT ALSO CALCULATES MEAN DAY AND NIGHT AIR TEMPERATURE.
------------------------------------------------------------------
ADJUST UNITS FOR WEATHER DATA

READ DATA FROM WEATHER FILE

CALCULATE SOLAR DECLINATION.

CALCULATE DAYLENGTH.

CALCULATE SOLAR RADIATION INCIDENT ON TOP OF EARTH'S ATMO-
   SPHERE AT SOLAR NOON.

CALCULATE AN ATMOSPHERIC TRANSMISSION COEFFICIENT FOR THIS
   LATUDE AND TIME OF YEAR.

CALCULATE POTENTIAL DIRECT + DIFFUSE RADIATION INCIDENT ON
   CROP AT SOLAR NOON.

CALCULATE ACTUAL RADIATION INCIDENT ON CROP AT SOLAR NOON
   GIVEN DAILY INTEGRAL AND ASSUMING RADIATION FLUX DENSITY
   VARIES SINUSOIDALLY OVER 24 HOURS.

CALCULATE CLOUD COVER.

DIVIDE PHOTOPERIOD INTO TEN EQUAL INCREMENTS AND CALCULATE
   HOUR ANGLES BETWEEN THE MIDPOINTS AND SOLAR NOON.

CALCULATE SOLAR ALTITUDE FOR EACH TIME.

for each tenth (or other fraction) of the photoperiod,
|  do the following:
|
| CALCULATE TOTAL RADIATION INCIDENT ON THE CROP AT EACH TIME.
|
| CALCULATE PROPORTION OF TOTAL RADIATION THAT IS DIFFUSE AT
|    EACH TIME.
|
| CALCULATE PHOTOSYNTHETICALLY ACTIVE RADIATION (PAR)INCIDENT ON
|    CROP ASSUMING PAR = 0.43 DIRECT + 0.57 DIFFUSE RADIATION.
|______________

IF CROP CANOPY HAS CLOSED, OMIT ALL CALCULATIONS PERTAINING
   TO DIRECT AND DIFFUSE RADIATION INTERCEPTION.

if the crop canopy has closed, jump down
|
| IF CLOUD COVER IS COMPLETE, OMIT ALL CALCULATIONS PERTAINING
|    TO DIRECT RADIATION INTERCEPTION.
V
A
```

A

if clouds cover the sky, jump down

CALCULATE SOLAR AZIMUTH FOR EACH TIME.

CALCULATE ANGLE BETWEEN ROW AND SOLAR AZIMUTH FOR EACH TIME.

CALCULATE INTERCEPTION OF DIRECT RADIATION BY "SOLID" ROWS.

CALCULATE INTERCEPTION OF DIFFUSE RADIATION BY "SOLID" ROWS USING NUMERICAL INTEGRATION.

FIRST DIVIDE ROWSPACING INTO 20 EQUAL PARTS AND FIND THE MIDPOINTS FROM ROW TO MID ROW.

for each of the 20 points, do the following:

CALCULATE PROPORTION OF SKY OBSCURED BY "SOLID" ROWS AT EACH ROWINC.

INTEGRATE INTERCEPTION FROM THE ROW TO MID ROW.

CALCULATE FRACTION OF PAR INTERCEPTED BY "SOLID" ROWS OF CROP ASSUMING THAT PAR = 0.43 DIRECT + 0.57 DIFFUSE RADIATION.

for each tenth of the photoperiod, do the following:

CALCULATE FRACTION OF SHORT WAVE RADIATION INTERCEPTED BY "SOLID ROWS" OF CROP

CALCULATE THE TIME AFTER DAWN IN HOURS WHEN MAXIMUM TEMPERATURE IS REACHED.

CALCULATE AVERAGE DAYTIME AIR TEMPERATURE IN DEGREES C.

CALCULATE AIR TEMPERATURE AT DUSK.

CALCUATE AIR TEMPERATURE AT EACH TIME.

CALCULATE AVERAGE NIGHT-TIME AIR TEMPERATURE.

if daily wet bulb temperatures are available, jump down

IF DAILY WET BULB TEMPERATURES ARE NOT AVAILABLE, ASSUME MINIMUM TEMPERATURE = DEW POINT TEMPERATURE AND CALCULATE WATER VAPOR PRESSURE. ASSUME A VALUE FOR THE PSYCHROMETRIC "CONSTANT".

jump down

SINCE DAILY WET BULB TEMPERATURES ARE AVAILABLE CALCULATE SATURATION VAPOR PRESSURE AT THE WET BULB TEMPERATURES.

B

B

CALCULATE THE HUMIDITY RATIO, X9, LATENT HEAT OF EVAPORATION,
X10 AND THE PSYCHROMETRIC "CONSTANT".

CALCULATE ACTUAL WATER VAPOR PRESSURE.

CALCULATE SOLAR ELEVATION WHEN LIGHT INTENSITY IS 10 LUX
AND 100 LUX.

CALCULATE EFFECTIVE PHOTOPERIOD FOR SOYBEAN.

RETURN

```
      SUBROUTINE CLYMAT
C     -------------------------( C L Y M A T )------------------------
C      THIS ROUTINE CALCULATES DAYLENGTH, EFFECTIVE PHOTOPERIOD FOR
C     SOYBEAN, CLOUD COVER, DIFFUSE AND DIRECT LIGHT INTERCEPTION BY
C     "SOLID" CROP ROWS AND AIR TEMPERATURE AT TEN TIMES DURING THE
C     DAY. IT ALSO CALCULATES MEAN DAY AND NIGHT AIR TEMPERATURE.
C     ------------------------------------------------------------------
      DOUBLE PRECISION C1,DEC,PHI,DEGRAD,XLAT,R,X1,X2,X3,TENLUX,HUNLUX,
     &                   SOLALV,SOLALW
      DIMENSION C1(9),D1(48)
      DIMENSION SINALT(10),SOLALT(10),SINAZI(10),SARANG(10),DIFINT(10),
     &          ROWINC(10),DIFWAT(10),DIRINT(10),HRANG(10),SOLAZI(10)
      INCLUDE 'COMMON.FOR'
3.01  DATA C1/0.3964D-0,0.3631D 1,0.3838D-1,0.7659D-1,0.0000D 0,
     &         -0.2297D 2,-0.3885D 0,-0.1587D-0,-0.1021D-1/
3.02  DATA D1/ 1.9,11.6,36.5,84.5,162.9,279.2,440.5,653.9,926.2,1263.8,
     &         1673.2,2160.1,2730.3,3389.0,4141.1,4991.1,5943.0,7000.6,
     &         8166.8,9444.6,10836.1,12343.0,13966.7,15707.8,17566.7,
     &         19543.0,21636.1,23844.6,26166.8,28600.6,31143.0,33791.1,
     &         36541.1,39389.0,42330.3,45360.1,48473.2,51663.8,54926.2,
     &         58253.9,61640.5,65079.2,68562.9,72084.5,75636.5,79211.6,
     &         82801.9,86400.0/
      PI=3.1416
      TPI=6.2832
C     ------------------------------------------------------------------
C
C     *** ADJUST UNITS FOR WEATHER DATA ***
C
3.03  RI=CLIMAT(1)*A
      TMAX=(CLIMAT(2)-C)/B
      TMIN=(CLIMAT(3)-C)/B
      RAIN=CLIMAT(4)*E
3.04  IF( MSW2.EQ.0 ) GOTO 3333
        WIND=CLIMAT(5)*F
C
 3333 IF( MSW1.EQ.0 ) GOTO 2222
      TWET=(CLIMAT(6)-C)/B
      TDRY=(CLIMAT(7)-C)/B
C
C     *** READ DATA FROM WEATHER FILE ***
C
 2222 IF( JDAY.EQ.JDLAST ) GOTO 1111
3.05    READ(1,*) X0,(CLIMAT(I),I=1,NCD)
      TMINT=(CLIMAT(3)-C)/B
        GOTO 9988
C
 1111 TMINT=TMIN
C
C     *** CALCULATE SOLAR DECLINATION. ***
C
 9988 DEGRAD=0.017453
C   0.017453 = 3.1416/180.0
3.06  XLAT=LATUDE*DEGRAD
      DEC=C1(1)
      DO 2 I=2,5
        N=I-1
        J=I+4
        PHI=N*0.01721*JDAY
    2 DEC=DEC+C1(I)*DSIN(PHI)+C1(J)*DCOS(PHI)
      DEC=DEC*DEGRAD
C
C     *** CALCULATE DAYLENGTH. ***
C
```

```
3.07       DAYLNG=DACOS((-0.014544-DSIN(XLAT)*DSIN(DEC))/(DCOS(XLAT) *
          &        DCOS(DEC)))*7.6394
      C    7.6394 = 180/3.1416/360*24*2
      C
      C    *** CALCULATE SOLAR RADIATION INCIDENT ON TOP OF EARTH'S ATMO-
      C        SPHERE AT SOLAR NOON. ***
      C
3.08       R=1+(0.01674*DSIN((JDAY-93.5)*0.9863*DEGRAD))
           X1=DSIN(XLAT)*DSIN(DEC)
           X2=DCOS(XLAT)*DCOS(DEC)
           X3=X1+X2
3.09       WATATM=1325.4*X3/R**2
      C
      C    *** CALCULATE AN ATMOSPHERIC TRANSMISSION COEFFICIENT FOR THIS
      C        LATITUDE AND TIME OF YEAR. ***
      C
3.10       IF(JDAY.LE.220) THEN
            IF(LATUDE.LE.40.0) THEN
            X11 = 0.7E-4+(LATUDE*0.31E-4)
            ELSE
            X11 = 0.5E-3 + (LATUDE*0.2E-4)
            END IF
            Q = 0.66 + X11*(200-JDAY)
           ELSE
            IF(LATUDE.LE.30.0) THEN
            X11 = LATUDE*0.49E-4
            ELSE
            X11 = 0.63E-3 + (LATUDE*0.285E-4)
            END IF
            Q = 0.68 + X11 * (JDAY-240)
           END IF
           IF(Q.LE.0.7) Q =0.7
      C
      C    *** CALCULATE POTENTIAL DIRECT + DIFFUSE RADIATION INCIDENT ON
      C        CROP AT SOLAR NOON. ***
      C
3.11       WATPOT=WATATM*0.5*(0.93-(0.02/X3)*Q**(1/X3))
      C
      C    *** CALCULATE ACTUAL RADIATION INCIDENT ON CROP AT SOLAR NOON
      C        GIVEN DAILY INTEGRAL AND ASSUMING RADIATION FLUX DENSITY
      C        VARIES SINUSOIDALLY OVER 24 HOURS. ***
      C
3.12       X12 = (24.0-DAYLNG) * 1800.0
      C    1800=3600/2
           WATADJ = 1.0 + SIN (X12*7.272E-5+4.7124)
      C    7.272E-5=3.1416*2/24/3600
      C    4.7124=3.1416*3/2
           I = INT(X12/900.0)
           X13 = 1+SIN(I*900.0*7.272E-5+4.7124)
           X14 = D1(I) + ((WATADJ + X13)*(X12-(900.0*I)))
           RIADJ = (X14+(WATADJ*DAYLNG*3600.0))/86400.0
      C    86400 = 24*3600
           WATACT = (RI/(1.0-RIADJ))/86400.0*(2.0-WATADJ)
           IF(SITE.EQ.3)TMAX=TMAX+(WATACT*0.00616667)
      C
      C    *** CALCULATE CLOUD COVER. ***
      C
           WATRAT = WATACT/WATPOT
           IF(WATRAT.GE.1.0) THEN
           CLOUD = 0.0
           ELSE
3.13       CLOUD = (CLDFAC-SQRT((CLDFAC*CLDFAC)+(1.52*(1-WATRAT))))/-0.76
      C    1.52 = 4*0.38
```

```
C      0.76 = 2*0.38
       IF(CLOUD.GE.1.0)CLOUD =1.0
       END IF
C
C      *** DIVIDE PHOTOPERIOD INTO TEN EQUAL INCREMENTS AND CALCULATE
C          HOUR ANGLES BETWEEN THE MIDPOINTS AND SOLAR NOON. ***
C
       IHPERD = IPERD/2
       DAYFRC=IHPERD-0.5
       DO 26 J=1,IHPERD
         HRANG(J)=3.1416*DAYLNG/12.0/IPERD*DAYFRC
         DAYFRC=DAYFRC-1.0
   26    HRANG(IPERD-(J-1))=HRANG(J)
       PERIOD=DAYLNG/IPERD
C
C      *** CALCULATE SOLAR ALTITUDE FOR EACH TIME. ***
C
       DO 30 ITIME=1,IPERD
3.14     SINALT(ITIME)=X1+(X2*COS(HRANG(ITIME)))
         SOLALT(ITIME)=ASIN(SINALT(ITIME))
C
C      ** CALCULATE TOTAL RADIATION INCIDENT ON THE CROP AT EACH TIME. **
C
3.15     WATTSM(ITIME)=(RI/(1.0-RIADJ))/86400.0*(1.0-WATADJ +
     &         SIN(PI + HRANG(ITIME) + 4.7124))
C      4.7124=3.1416*3/2
C
C      *** CALCULATE PROPORTION OF TOTAL RADIATION THAT IS DIFFUSE AT
C          EACH TIME. ***
         X4 = SINALT(ITIME)
3.16     X15 = (0.93-(0.02/X4)-Q**(1/X4))/(0.93-(0.02/X4)+Q**(1/X4))
         IF(CLOUD.GT.0.0) THEN
3.17     DIFWAT(ITIME)= 1.0-((1.0-X15)*(1.0-CLOUD)/(1.0-((CLDFAC+(0.38
     &   *CLOUD))*CLOUD)))
         ELSE
         DIFWAT(ITIME) = X15
         END IF
         IF( DIFWAT(ITIME).LT.0.0 ) DIFWAT(ITIME)=0.0
         IF( DIFWAT(ITIME).GT.1.0 ) DIFWAT(ITIME)=1.0
C
C      *** CALCULATE LIGHT INCIDENT ON CROP ASSUMING THAT
C          LIGHT = 0.43 DIRECT + 0.57 DIFFUSE RADIATION. ***
C
3.18     PAR(ITIME)=WATTSM(ITIME)*((DIFWAT(ITIME)*0.57)+((1.0 -
     &                 DIFWAT(ITIME))*0.43))
   30  CONTINUE
C
C      *** IF CROP CANOPY HAS CLOSED, OMIT ALL CALCULATIONS PERTAINING
C          TO DIRECT AND DIFFUSE RADIATION INTERCEPTION. ***
C
       IF( Z/ROWSP.LT.1 ) GOTO 8888
       DO 32 K=1,IPERD
         RADINT(K)=1.0
   32    PARINT(K)=1.0
       GO TO 7777
C
C      *** IF CLOUD COVER IS COMPLETE, OMIT ALL CALCULATIONS PERTAINING
C          TO DIRECT RADIATION INTERCEPTION. ***
C
 8888  IF( CLOUD.LT.1.0 ) GOTO 4444
         DO 25 K=1,IPERD
   25      DIRINT(K)=0.0
         GOTO 9999
```

```
C
C      *** CALCULATE SOLAR AZIMUTH FOR EACH TIME. ***
C
 4444 DO 35 ITIME=1,IPERD
3.19      SINAZI(ITIME)=-DCOS(DEC)*SIN(HRANG(ITIME))/COS(SOLALT(ITIME))
   35 CONTINUE
      DO 40 ITIME=1,IHPERD
        SOLAZI(ITIME)=3.1416+ASIN(SINAZI(ITIME))
   40 CONTINUE
        JHPERD = IHPERD + 1
      DO 50 ITIME=JHPERD,IPERD
        SOLAZI(ITIME)=3.1416-ASIN(SINAZI(ITIME))
   50 CONTINUE
C
C      *** CALCULATE ANGLE BETWEEN ROW AND SOLAR AZIMUTH FOR EACH TIME. *
C
      DO 55 ITIME=1,IPERD
        SARANG(ITIME)=DABS(SOLAZI(ITIME)-(ROWANG*DEGRAD))
       IF(SARANG(ITIME).GT.4.7124) SARANG(ITIME)=ABS(SARANG(ITIME)-TPI)
        IF(SARANG(ITIME).GT.1.5708) SARANG(ITIME)=ABS(SARANG(ITIME)-PI)
C
C      *** CALCULATE INTERCEPTION OF DIRECT RADIATION BY "SOLID" ROWS. **
C
3.20    DIRINT(ITIME)=Z/ROWSP/SIN(ATAN(TAN(SOLALT(ITIME)) /
     &                  SIN(SARANG(ITIME))))
        IF( DIRINT(ITIME).GT.1.0 ) DIRINT(ITIME)=1.0
   55 CONTINUE
C
C      *** CALCULATE INTERCEPTION OF DIFFUSE RADIATION BY "SOLID" ROWS
C          USING NUMERICAL INTEGRATION. ***
C      *** FIRST DIVIDE ROWSPACING INTO 20 EQUAL PARTS AND FIND THE
C          MIDPOINTS FROM ROW TO MID ROW. ***
C
 9999 DIFFIN=0.0
      DO 60 I=1,10
        ROWINC(I)=ROWSP/20.0*(I-0.5)
C
C      *** CALCULATE PROPORTION OF SKY OBSCURED BY "SOLID" ROWS AT EACH
C          ROWINC. ***
C
3.21    DIFINT(I)=(ATAN(Z/2.0/(ROWSP-ROWINC(I)))+ATAN(Z/2.0 /
     &                ROWINC(I)))/1.5708
        IF( DIFINT(I).GT.1.0 ) DIFINT(I)=1.0
C
C      *** INTEGRATE INTERCEPTION FROM THE ROW TO MID ROW. ***
C
   60   DIFFIN=DIFFIN+DIFINT(I)
      DIFFIN=DIFFIN/10
C
C      *** CALCULATE FRACTION OF LIGHT INTERCEPTED BY "SOLID" ROWS OF CROP
C          ASSUMING THAT LIGHT = 0.43 DIRECT + 0.57 DIFFUSE RADIATION. **
C
      DO 70 ITIME=1,IPERD
3.22    PARINT(ITIME)=WATTSM(ITIME)*((DIFFIN*0.57*DIFWAT(ITIME)) +
     &                DIRINT(ITIME)*0.43*(1.0-DIFWAT(ITIME)))/PAR(ITIME)
C
C      *** CALCULATE FRACTION OF SHORT WAVE RADIATION INTERCEPTED BY
C      ***  "SOLID ROWS" OF CROP    ***
C
        RADINT(ITIME)=(DIFFIN*DIFWAT(ITIME))+(DIRINT(ITIME)*
     &                (1.0-DIFWAT(ITIME)))
   70 CONTINUE
C
```

```
C      *** CALCULATE THE TIME AFTER DAWN IN HOURS WHEN MAXIMUM TEMPER-
C          ATURE IS REACHED. ***
C
  7777 IF(SITE.EQ.3)WATACT=WATACT/0.8
       X5=0.0945-(WATACT*8.06*10.0**(-5.0))+(TMAX*6.77*10**(-4.0))
       TMX5WA=TMAX/X5/WATACT
       IF( TMX5WA.GE.1.0 ) TMX5WA=1.0
3.23   TMAXHR=DAYLNG/PI*(PI-(ASIN(TMX5WA)))
C
C      *** CALCULATE AVERAGE DAYTIME AIR TEMPERATURE IN DEGREES C. ***
C
       X6=((TMAX-TMIN)/2.0)
       X7=(PI/TMAXHR)
       X8=1.5*PI
3.24   TDAY=X6*(1-(TMAXHR/PI/DAYLNG*COS((X7*DAYLNG)+X8)))+TMIN
C
C      *** CALCULATE AIR TEMPERATURE AT DUSK. ***
C
       TAIRSS=(X6*(1.0+SIN((X7*DAYLNG+X8))))+TMIN
C
C      *** CALCUATE AIR TEMPERATURE AT EACH TIME. ***
C
       DO 90 ITIME=1,IPERD
         TAIR(ITIME)=(X6*(1.0+SIN((X7*DAYLNG*(ITIME-0.5)/IPERD)+
     &                 X8)))+TMIN
    90 CONTINUE
C
C      *** CALCULATE AVERAGE NIGHT-TIME AIR TEMPERATURE. ***
C
        IF(TMINT.GT.0.0) GO TO 9977
        MATURE = 1
        TNYT = 1.0
  11    FORMAT(' PLANTS ARE FROZEN TO DEATH')
        WRITE(2,11)
        GO TO 9966
  9977  IF( TAIRSS.LE.TMINT ) GO TO 9955
        TNYT=(TAIRSS-TMINT)/(-ALOG(TMINT/TAIRSS))
        GO TO 9966
  9955  TNYT=(TAIRSS+TMINT)/2.0
C
C      *** IF DAILY WET BULB TEMPERATURES ARE NOT AVAILABLE, ASSUME
C          MINIMUM TEMPERATURE = DEW POINT TEMPERATURE AND CALCULATE
C          WATER VAPOUR PRESSURE.  ASSUME A VALUE FOR THE PSYCHROMETRIC
C          "CONSTANT". ***
C
  9966 IF( MSW1.EQ.1 ) GOTO 6666
3.25     AVP=0.61*EXP((17.27*TMIN)/(TMIN+237.3))
         GAMMA=0.0645
         GOTO 5555
C
C      *** SINCE DAILY WET BULB TEMPERATURES ARE AVAILABLE CALCULATE
C          SATURATION VAPOUR PRESSURE AT THE WET BULB TEMPERATURES. ***
C
  6666 SVPW=0.61*EXP((17.27*TWET)/(TWET+237.3))
C
C      *** CALCULATE THE HUMIDITY RATIO, X9, LATENT HEAT OF EVAPORATION,
C          X10 AND THE PSYCHROMETRIC "CONSTANT". ***
C
       X9=0.622*(SVPW/(101.3-SVPW))
       X10=2500.8-(2.37*TWET)
       GAMMA=0.62*(1.006+(1.846*X9))/((0.622+X9)**2.0*X10) *
     &      101.3
C
```

```
C      *** CALCULATE ACTUAL WATER VAPOR PRESSURE. ***
C
       AVP=SVPW-(GAMMA*(TDRY-TWET))
C
C      *** CALCULATE SOLAR ELEVATION WHEN LIGHT INTENSITY IS 10 LUX
C          AND 100 LUX. ***
C
 5555  TENLUX=10.0/750.0/WATRAT
       HUNLUX=100.0/750.0/WATRAT
       SOLALV=DEGRAD*DLOG(TENLUX)
       SOLALW=DEGRAD*DLOG(HUNLUX)
C
C      *** CALCULATE EFFECTIVE PHOTOPERIOD FOR SOYBEAN. ***
C
       PHOTO=((DACOS((DSIN(SOLALV)-DSIN(XLAT)*DSIN(DEC))/(DCOS(XLAT) *
     &        DCOS(DEC)))+(DACOS((DSIN(SOLALW)-DSIN(XLAT)*DSIN(DEC)) /
     &       (DCOS(XLAT)*DCOS(DEC))))))*3.8197
       RETURN
       END
```

3.26

Notes for CLYMAT

3.01 Data in the C1 array are used in the calculation of solar declination on line 119.

3.02 Data in the D1 array are factors used to calculate the area under both tails of a full sine curve cut off at various distances from the peak. See note 3.12 for a more complete explanation.

3.03 Today's weather data have already been read in to the CLIMAT file either in MAIN line 215 or in CLYMAT line 108.

The factors A, B, C, E and F, allow easy conversion of units if this is necessary.

3.04 MSW1 and MSW2 are flags to indicate the presence of wet and dry bulb temperature data and wind run respectively. If the flag is set to zero, the data are not present in the weather file. WIND is either read here daily or an average value for the site has already been read in MAIN line 205.

3.05 Tomorrow's weather data are read in to the CLIMAT file in order to make available tomorrow's minimum temperature. This is used in the calculation of mean night air temperature.

3.06 Solar declination is calculated using D. W. Stewart's subroutine which incorporates the algorithm of Robertson and Russelo (1968). The code was supplied by Dr. L. H. Allen, University of Florida.

3.07 The equation relating solar altitude and declination to latitude and hour angle is commonly found in books on celestial mechanics and is also given in the Smithsonian Meteorological

Tables, 6th edition, revised 1963 (S.M.T) on their page 497 equation (1). Daylength is defined in S.M.T. their page 506 as the interval between sunrise and sunset and at both those times the center of the sun's disc is 50' below the horizon. Setting solar altitude equal to -50' we solve for hour angle and then convert this to hours from solar noon. Doubling the answer gives daylength.

3.08 The radius vector of the earth is here calculated from an empirical equation fitted to data in S.M.T. their page 495, Table 169.

3.09 The equations for solar radiation incident on top of the earth's atmosphere are given in S.M.T. their page 417. The conditional solar constant, which is the radiation flux reaching the upper layers of the troposphere is assumed to the 1.90 cal cm^{-2} min^{-1} = 1325.4 W m^{-2} (Budyko, 1974).

3.10 Budyko (1974 in his Table 3, pages 46 and 47) gives the total radiation incident on the surface of the earth under a cloudless sky, for various latitudes and times of year. These data were used, along with many of the equations in this subroutine, to calculate atmospheric transmission coefficients for various latitudes and times of year. The program FINDQ was used to make these calculations. Some of the results of the calculations are shown in Figure 14. The estimates of atmospheric transmission coefficient provided by the equations in this section are shown on the same figure. There are clear limitations in both the data and the equations used to fit them.

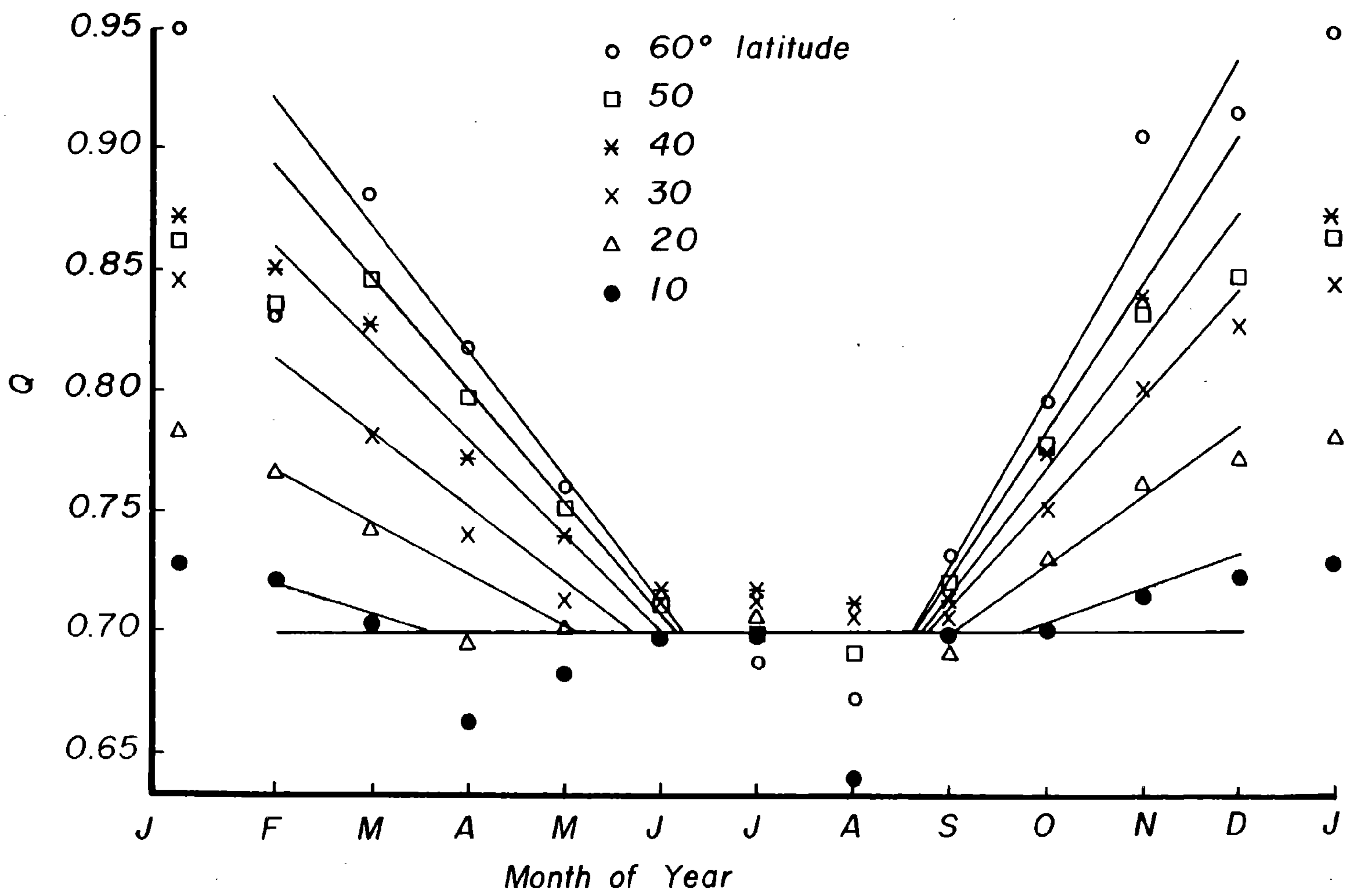


Fig. 14. Atmospheric transmission coefficients for various latitudes and times of year calculated from data in Budyko (1974) and the equations in CLYMAT.

3.11 This equation essentially reconstitutes the data given in Budkyo (1974, his Table 3). It is derived from the equation and calculation procedure given in S.M.T. their page 420. Instead of assuming that a constant 9% of the extra-terrestrial radiation is absorbed by water vapor and ozone, the percent absorption is allowed to vary with atmospheric path length, i.e. solar altitude, according to data given by Miller (1981, his page 42, Table IV). For further information about the derivation of the equation see note 3.16.

3.12 The diurnal variation in radiation flux density can be described by a sine wave with a 24 hour wave length. Daylength defines the part of the sine wave that is appropriate in a given situation. Thus, when daylength is 12 hours, the diurnal variation in radiation is described by a half sine wave. The procedure for calculating radiation flux density at noon in a day of any daylength will be illustrated using Figure 15.

In Figure 15, A = 24 hours, D = daylength and E = radiation integral. These are all known.

$$C = (A - D)/2$$

The height of the wave, B at a distance, C along it is given by

$$B = (E + F)/A*(1 + \sin(((2C/A) + 1.5)*3.1416))$$

Area F is not known but we can evaluate

$$B*A/(E + F) = (1 + \sin(((2C/A) + 1.5)*3.1416)$$

and this is equivalent to WATADJ in the equations below.

The relative area of the curve under the two tails defined by the dotted lines is calculated next. The data in

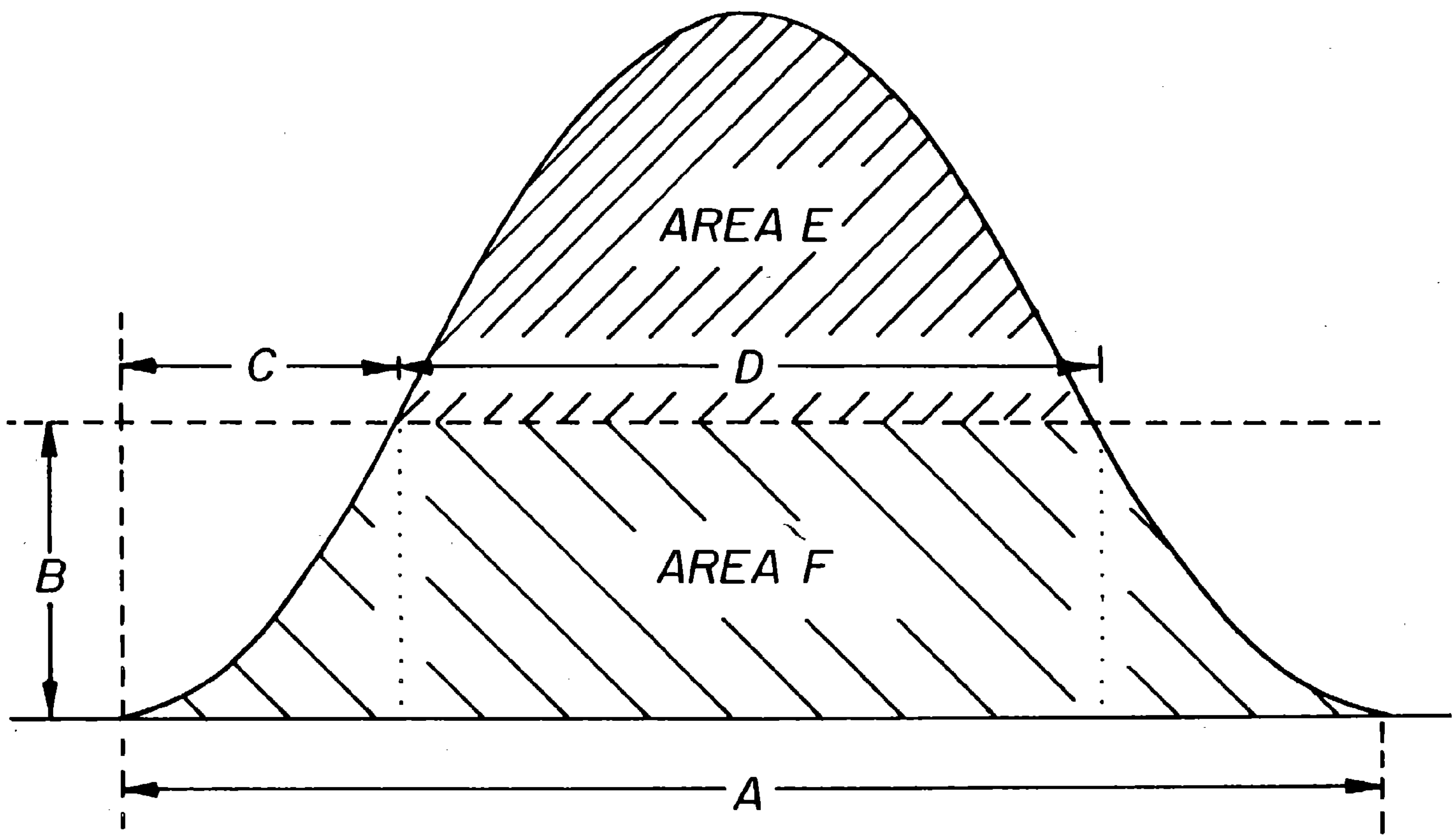


Fig. 15. Diagram to explain how diurnal variation in radiation flux density is related to daylength.

array D1 give the proportion of A/(E + F) under the two tails as C increases in 15 minute increments from 15 mins to 12 hours.

In the code below, the nearest value is taken from array D1 and an increment added to bring it up to the exact value of C. This is then added to D*B*A/(E + F) to give

F*A/(E + F)

and divided by A to give

F/(E + F)

This is equivalent to RIADJ in the equations below.

Then E/(1 - (F/(E + F))) = E + F

This value is substituted into the equation formerly used to evaluate B, but this time it is solved for C = 12 hours. B is subtracted from the answer to give the actual radiation incident on the crop at noon.

3.13 Cloud cover is calculated from the ratio of actual to potential radiation at noon using the equation of Berliand which is cited by Budyko (1974). The equation gives a good fit to radiation data for the north-central region of the USA (Baker, 1975). The equation is

WATACT = WATPOT*(1 - ((CLDFAC + 0.38*CLOUD)*CLOUD))

where CLDFAC is a latitude-dependent factor calculated in MAIN on line 365. Here the equation is solved for CLOUD.

3.14 See note 3.07 for the source of these equations.

3.15 The total radiation incident on the crop at various times of day is calculated from the same equation as WATACT, the total radiation at noon. See note 3.12 for further details.

3.16 According to the equation and procedures given in S.M.T. page 420, direct solar radiation flux at the earth's surface,

I = WATATM*Q**(1/SINALT)

Diffuse radiation flux at the earth's surface

= ((U*WATATM) - I)/2

= WATATM/2*(U - Q**(1/SINALT))

where U is the proportion of radiation not absorbed by ozone or water vapor.

Hence direct + diffuse radiation flux at the earth's surface

= (WATATM/2*(U - Q**(1/SINALT))) + (WATATM*Q**(1/SINALT))

= WATATM/2*(U + Q**(1/SINALT))

Therefore the ratio of diffuse to total radiation,

DIFWAT = (U - Q**(1/SINALT))/(U + Q**(1/SINALT))

In the equation used there, U is a function of solar altitude as explained in note 3.11. This equation gives the proportion of diffuse radiation on cloudless days.

3.17 On cloudy days, the direct radiation will be inversely proportional to the percentage of sky covered by cloud. This is illustrated in Figure 16.

DIFWAT = F/(F + R) is know already.

R/(F + R) = 1 - DIFWAT

Therefore R = (1 - DIFWAT)*(F + R)

For a given cloud cover, the flux density of direct radiation,

r = (1 - DIFWAT)*(F + R)*(1 - CLOUD)

and the total radiation given by Berliand's equation (See note

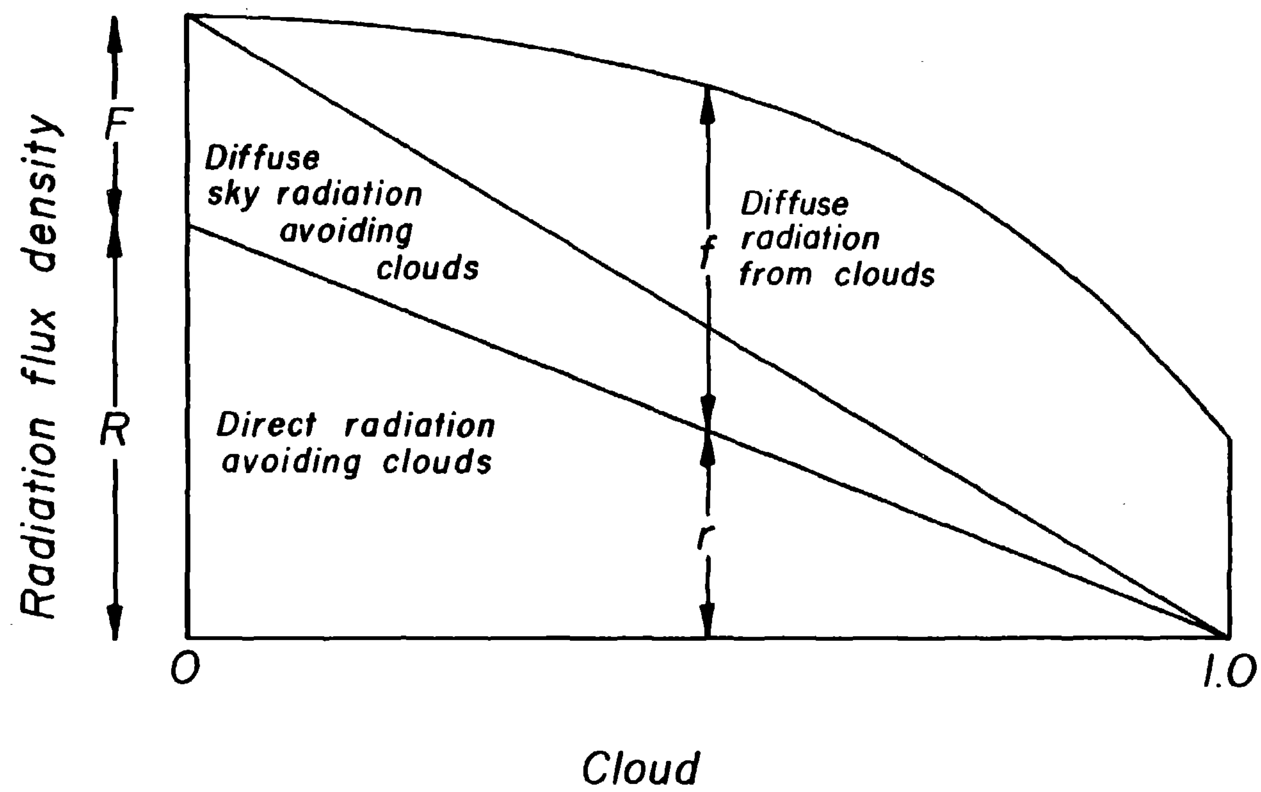


Fig. 16. Diagram to show the relationship between cloud cover and the proportion of diffuse and direct radiation.

3.13),

$$f + r = (F + R)*(1 - ((CLDFAC + 0.38*CLOUD)*CLOUD))$$

The equation in the code gives

$$1 - (r/(f + r) = f/(f + r)$$

which is the proportion of diffuse radiation on a cloudy day.

3.18 The proportions of light (i.e. PAR) in diffuse and direct radiation are taken from "Plant Photosynthetic Production: Manual of Methods," by Sestak, Catsky and Jarvis on their p. 423.

3.19 This equation appears in many books including S.M.T., their page 497, equation (2).

3.20 Prior to canopy closure, the crop rows are approximated as opaque cylinders. The interception of diffuse and direct radiation by these cylinders is calculated for each period of the day. Later, in EVTR and PNET, allowance is made for light filtering through the leaves in these cylinders of crop.

The calculation of direct radiation interception is an exercise in solid geometry (see Figure 17). First, solar position and row orientation are used to calculate an apparent solar elevation at right angles to the row that would give the same shadow length.

$$\tan a' = Q/Y = (Q/X)/(Y/X) = \tan a/\sin\sigma$$

Then the proportion of direct radiation intercepted is shadow width divided by row width (R_W),

$$= Z/R_W/\sin(\tan^{-1}(\tan a/\sin\sigma))$$

3.21 The logic for calculating diffuse radiation interception

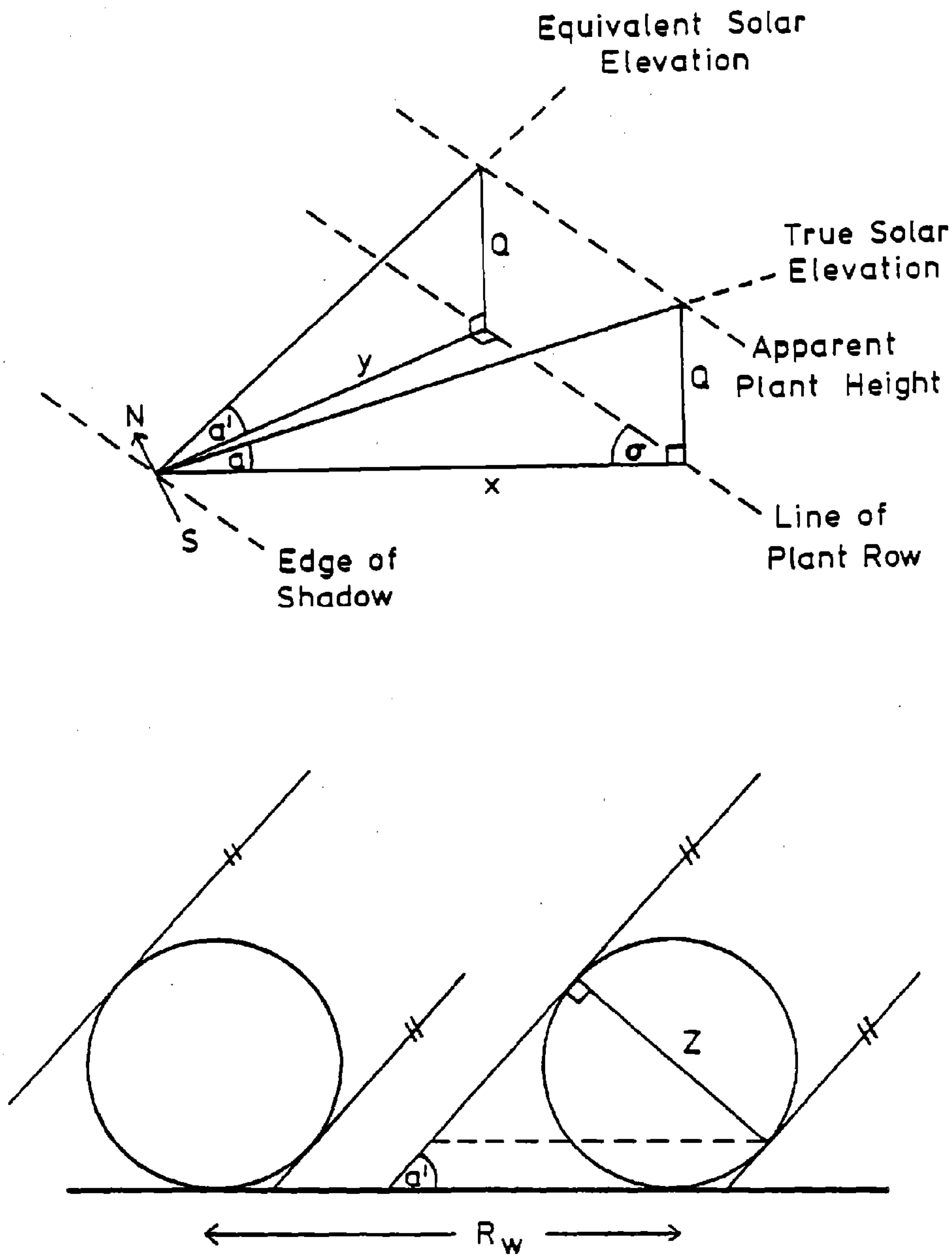


Fig. 17. Diagrams to explain the logic used in calculating direct radiation interception.

is given in Figure 18. For any given position on the soil surface, the portion of the hemisphere of sky obscured by a given row of plants is proportional to angle P or Q.

$$\tan Q/2 = Z/2d$$

$$\tan P/2 = Z/(2R_W - 2d)$$

These angles delineate segments of the sky hemisphere shaped like segments of an orange. Assuming that this hemisphere is a uniform source of diffuse light, the proportion of diffuse radiation intercepted as far as that point on the soil is concerned

$$= (\tan^{-1}(Z/(R_W - 2d)) + \tan^{-1}(Z/2d))/90$$

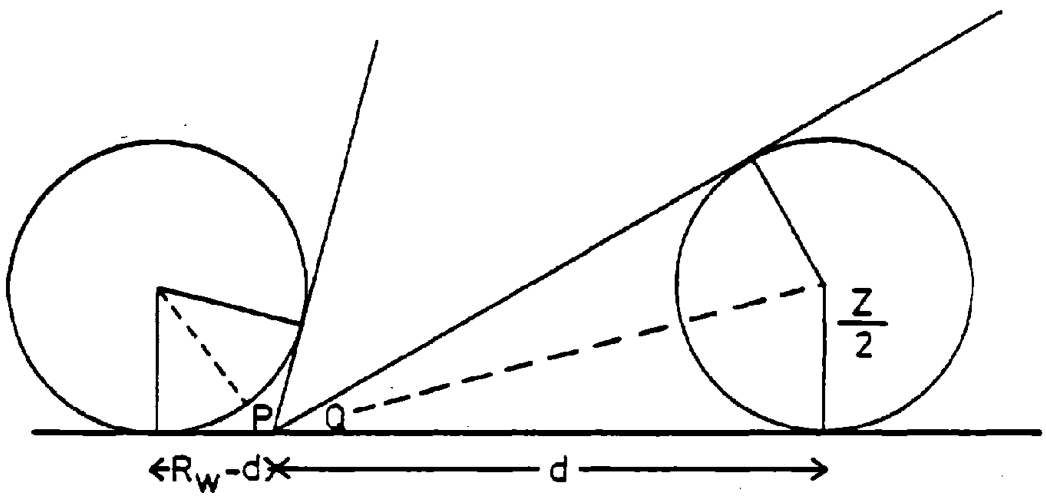
where 90 is in angular degrees.

This proportion is calculated for a number of points on the soil equally-spaced between the rows and a mean value is calculated. Of course, the sky hemisphere is not a uniform source of diffuse light but it approximates that on days of full cloud cover and the assumption of a "uniform overcast sky" is often used to good effect by meteorologists.

3.22 See note 3.18

3.23 The equation to calculate the time of day when maximum temperature is reached was derived from an empirical fitting of a Mississippi data set.

3.24 Minimum temperature is assumed to occur at dawn and air temperature variation during daylight is approximated by a half sine wave. At night the temperature is assumed to fall logarithmically.. These equations have given a good fit to



$$\tan \frac{Q}{2} = \frac{Z}{2d} \qquad \tan \frac{P}{2} = \frac{Z}{2R_W - 2d}$$

Fig. 18. Diagram to explain the logic used in calculating diffuse radiation interception.

other data sets from Mississippi and Arizona.

3.25 The algorithms used to calculate moist air properties leading to the water vapor pressure come from Weiss (1977)

3.26 Solar elevation at light intensities of 10 and 100 lux is given by an equation fitted to data cited by Francis (1970). These are the limiting light intensities for soybean photoperiod as found by Takimoto and Ikeda (1961). In both these papers the unit used for light flux density is lux. The unit is retained here because all we are doing is predicting the solar elevations that define the photoperiod for soybeans. Photoperiod is calculated from these solar elevations with the same equation as is used for daylength. See note 3.07.

EVTR

The subroutine EVTR is called once for each period of the day. It calculates potential transpiration rate for the crop using the Penman equation. The only innovation is that the roughness parameter changes gradually from 1 at emergence to 2 at half canopy and back to 1 at full canopy. The rationale for this is illustrated in Figure 19. Penman used a roughness parameter of 1 for short grass: a smooth surface. When soybean plants are very small (stage 1) and again after the canopy has closed (stage 3), the crop presents a smooth surface to the wind and causes little turbulence. Turbulence is maximal when the canopy is half closed (stage 2) and a roughness parameter of 2 fits the available data.

EVTR calculates potential evaporation rate for water at the soil surface using the Penman equation. It compares this with the rate at which water can move to the soil surface and actual evaporation rate is set equal to whichever of these is lowest. When evaporation is limited by soil water availability, EVTR calculates how much radiation is used to evaporate water and how much is available to heat the soil.

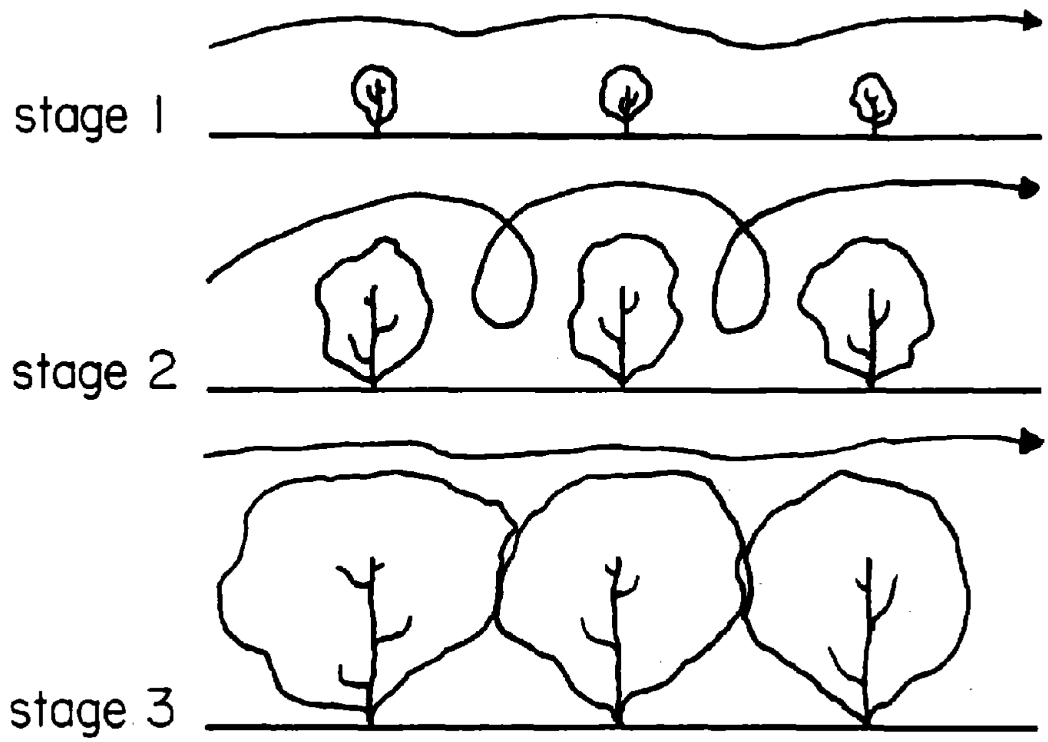


Fig. 19. Diagram to show how wind turbulence and the crop roughness parameter change as the crop develops.

```
----------------------------( EVTR )--------------------------------
THIS ROUTINE CALCULATES POTENTIAL WATER EVAPORATION RATE FROM THE
CROP AND THE SOIL AND ACTUAL WATER LOSS FROM THE EXPOSED SURFACE
SOIL CELLS.  IT ALSO GIVES THE AMOUNT OF RADIATION AVAILABLE FOR
SOIL HEATING.
--------------------------------------------------------------------
CALCULATE POTENTIAL TRANSPIRATION RATE FOR THE CROP
--------------------------------------------------------------------
CALCULATE LEAF AREA PER UNIT AREA COVERED BY CROP CANOPY

CALCULATE NET UPWARD LONG-WAVE RADIATION

CALCULATE NET RADIATION ON THE CROP

CALCULATE WATER SATURATION VAPOR PRESSUE AT AIR TEMPERATURE
   AND THE SLOPE OF THE RELATIONSHIP AT THAT POINT.

CALCULATE WATER VAPOR PRESSURE DEFICIT

CALCULATE POTENTIAL TRANSPIRATION RATE FOR CROP ALLOWING FOR
   INCOMPLETE RADIATION INTERCEPTION

CALCULATE POTENTIAL TRANSPIRATION RATE FOR AN AVERAGE PLANT
--------------------------------------------------------------------
CALCULATE POTENTIAL WATER EVAPORATION RATE FROM THE SOIL
   SURFACE
--------------------------------------------------------------------
IF CROP CANOPY HAS NOT YET CLOSED, CALCULATE EVAPORATION OF
   WATER FROM THE SOIL. OTHERWISE RETURN.

if the crop canopy has closed, jump down
|
|  CALCULATE A MEAN ALBEDO FOR THE EXPOSED SOIL CELLS
|
|  CALCULATE NET RADIATION ON THE EXPOSED SOIL CELLS
|     CORRECTING FOR PROPORTION OF RADIATION INTERCEPTED AND NUMBER
|     OF CELLS EXPOSED.
|
|  CALCULATE POTENTIAL EVAPORATION RATE FROM EXPOSED SOIL CELLS
V
A
```

A

CALCULATE RATE OF EVAPORATION POSSIBLE WITH SOIL WATER AVAILABLE IN EXPOSED SOIL CELLS

for each exposed surface cell, do the following:

CALCULATE RATE AT WHICH WATER CAN MOVE TO THE SOIL SURFACE.

CALCULATE WATER LOST FROM EACH EXPOSED SURFACE SOIL CELL OVER THE GIVEN TIME INTERVAL AS LIMITED BY EITHER ATMOSPHERIC DEMAND OR THE ABILITY OF THE SOIL TO SUPPLY WATER.

WHERE EVAPORATION IS LIMITED BY SOIL MOISTURE, CALCULATE HOW MUCH RADIATION IS USED TO EVAPORATE WATER AND HOW MUCH IS AVAILABLE TO HEAT THE SOIL.

NO WATER IS LOST FROM SOIL COVERED BY THE CROP.

RETURN

```
      SUBROUTINE EVTR
C     ----------------------------( E V T R )---------------------------
4.01 C     THIS ROUTINE CALCULATES POTENTIAL WATER EVAPORATION RATE FROM THE
C     CROP AND THE SOIL AND ACTUAL WATER LOSS FROM THE EXPOSED SURFACE
C     SOIL CELLS.  IT ALSO GIVES THE AMOUNT OF RADIATION AVAILABLE FOR
C     SOIL HEATING.
C     ------------------------------------------------------------------
      INCLUDE 'COMMON.FOR'
C     ------------------------------------------------------------------
C     *** CALCULATE POTENTIAL TRANSPIRATION RATE FOR THE CROP ***
C     ------------------------------------------------------------------
C     *** CALCULATE LEAF AREA PER UNIT AREA COVERED BY CROP CANOPY ***
C
      ROWIDE = ROWSP
      IF ( ROWSP.GT.2 ) ROWIDE = 2
      COVER = ROWIDE/ROWSP
      LCAI=LAREAT*POPROW/100.0/ROWIDE
C
C     *** CALCULATE NET UPWARD LONG-WAVE RADIATION ***
C
4.02  RNLU=(1.6E-3*WATRAT*(100.0-TAIR(ITIME)))*697.6
C
C     *** CALCULATE NET RADIATION ON THE CROP ***
C
4.03  LAMDAC=0.23
4.04  RNC=(((1.0-LAMDAC)*WATTSM(ITIME))-RNLU)*RADINT(ITIME)/COVER
      IF(RNC.LE.0.0) RNC=0.0
C
C     *** CALCULATE WATER SATURATION VAPOUR PRESSUE AT AIR TEMPERATURE
C         AND THE SLOPE OF THE RELATIONSHIP AT THAT POINT. ***
C
4.05  SVPA=0.61*EXP((17.27*TAIR(ITIME))/(TAIR(ITIME) +
     &      237.3))
      DEL=(0.61*EXP((17.27*(TAIR(ITIME)+1.)) /
     &     (TAIR(ITIME)+1.0+237.3)))-SVPA
C
C     *** CALCULATE WATER VAPOUR PRESSURE DEFICIT ***
C
      VPD=SVPA-AVP
        IF(VPD.LE.0.0)VPD=0.0
C
C     *** CALCULATE POTENTIAL TRANSPIRATION RATE FOR CROP ALLOWING FOR
C         INCOMPLETE RADIATION INTERCEPTION ***
C
4.06  EPO=((DEL/GAMMA*RNC*3600.0/(2500.8-(2.3668*TAIR(ITIME)))) +
     &      (VPD*109.375*(ROUGH+(0.149*WIND))))/((DEL/GAMMA)+1.0) *
4.07 &       COVER*(1.-EXP(-LCAI*CEC))
C
C     *** CALCULATE POTENTIAL TRANSPIRATION RATE FOR AN AVERAGE PLANT **
C
      EO(ITIME)=EPO*ROWSP/POPROW/100.0
C
C     ------------------------------------------------------------------
C     *** CALCULATE POTENTIAL WATER EVAPORATION RATE FROM THE SOIL
C         SURFACE ***
C     ------------------------------------------------------------------
C
C     *** IF CROP CANOPY HAS NOT YET CLOSED, CALCULATE EVAPORATION OF
C         WATER FROM THE SOIL. OTHERWISE RETURN. ***
C
4.08  IF( COVER.LT.1.0 ) GOTO 6666
      DO 15 K=1,NKH
   15 EWU(K)=0.0
```

```
           GO TO 7777
      C
      C    *** CALCULATE A MEAN ALBEDO FOR THE EXPOSED SOIL CELLS ***
      C
4.09   6666 LAMDAS=0.3-(0.5*AIRDR(1))
           IF(RAIN.GT.0.0) LAMDAS=0.3-(0.5*THETAS(1))
      C
      C    *** CALCULATE NET RADIATION ON THE EXPOSED SOIL CELLS
      C        CORRECTING FOR PROPORTION OF RADIATION INTERCEPTED AND NUMBER
      C        OF CELLS EXPOSED. ***
      C
4.10       KLJ=(IFIX(Z/2.0/W)+1.0)
           RNS=(((1.0-LAMDAS)*(WATTSM(ITIME)))-RNLU)
4.11      &       *(1.0-(RADINT(ITIME)*(1-EXP(-LCAI*CEC))))
4.12      &       *NKH/(NKH+1-KLJ)
           IF(RNS.LE.0.0) RNS=0.0
      C
      C *** CALCULATE POTENTIAL EVAPORATION RATE FROM EXPOSED SOIL CELLS ***
      C
4.13        X1 = 2.0-(2.0*COVER)
            IF(X1.GE.1.0) X1=1.0
4.14       ESO=((DEL/GAMMA*RNS*3600.0/(2500.8-(2.3668*TAIR(ITIME)))) +
          &      (VPD*109.375*(1.0+(0.149*WIND*X1))))/((DEL/GAMMA) +
          &       1.0)
      C    -------------------------------------------------------------
      C    *** CALCULATE RATE OF EVAPORATION POSSIBLE WITH SOIL WATER AVAIL-
      C        ABLE IN EXPOSED SOIL CELLS ***
      C    -------------------------------------------------------------
      C   *** CALCULATE RATE AT WHICH WATER CAN MOVE TO THE SOIL SURFACE.***
      C
4.15       DO 25 K=KLJ,NKH
4.16       TNAIRD(K)=2.0*(1.0-(((VH2OC(1,K)-AIRDR(1))/(THETAS(1)-AIRDR(1)))
          &          ** 0.1))
4.17         DIFFT = DIFF0(1)*EXP(BETA(1)*(VH2OC(1,K)-THETA0(1)))
4.18       DIFFM=DIFFT*DIFF0(1)/((DIFFT*TNAIRD(K))+((1.0-TNAIRD(K))
          &*DIFF0(1)))
4.19       ESP = DIFFM*(VH2OC(1,K)-AIRDR(1))/0.5/D*10000.0/24.0
      C
      C    *** CALCULATE WATER LOST FROM EACH EXPOSED SURFACE SOIL CELL OVER
      C        THE GIVEN TIME INTERVAL AS LIMITED BY EITHER ATMOSPHERIC
      C        DEMAND OR THE ABILITY OF THE SOIL TO SUPPLY WATER. ***
      C
4.20         IF( ESO.GE.ESP ) EWU(K)=ESP/10000.0*W*PERIOD
             IF( ESO.LT.ESP ) EWU(K)=ESO/10000.0*W*PERIOD
      C
      C    *** WHERE EVAPORATION IS LIMITED BY SOIL MOISTURE, CALCULATE HOW
      C        MUCH RADIATION IS USED TO EVAPORATE WATER AND HOW MUCH IS
      C        AVAILABLE TO HEAT THE SOIL. ***
      C
4.21         REW=ESP/3600*(2500.8-(2.3668*TAIR(ITIME)))
             RHS(K)=RNS-REW
             IF( ESO.LT.ESP ) RHS(K)=0.0
        25 CONTINUE
      C
      C    *** NO WATER IS LOST FROM SOIL COVERED BY THE CROP. ***
      C
           IF( KLJ.EQ.1 ) GOTO 7777
             KLJM1=KLJ-1
             DO 30 K=1,KLJM1
        30 EWU(K)=0.0
      7777 RETURN
           END
```

Notes for EVTR

4.01 All practical equations for evapotranspiration are empirical but the Penman (1963) equation is at least semi-mechanistic and has been widely tested. It is used here for both soil and leaf water loss.

4.02 Net upward long-wave radiation is calculated using an approximation derived by Linacre (1968, his appendix 1). The original equation has a term defining the ratio of actual to potential radiation using the duration of sunshine as in the Angstrom equation (Budyko, 1974, his page 44 equation 2.1). Here that term is replaced with the ratio which was calculated in CLYMAT on line 186.

4.03 The choice of value for crop albedo is based on data cited by Fritschen (1967), Linacre (1968 and 1969) and Ritchie (1971). Their values range from 0.2 to 0.25 for several species with most values clustered around 0.23.

4.04 The equation for net radiation on unit area of crop canopy (enclosed in the outer parentheses) is used by many authors, e.g. Linacre (1968). Here it is multiplied by the proportion of radiation intercepted by the crop and divided by the proportion of the field covered by the crop. This allows for the fact that spaced plants or rows of plants receive diffuse radiation on all sides.

4.05 The algorithms used to calculate moist air properties leading to water vapor pressure deficit come from Weiss (1977).

4.06 The Penman equation used to calculate potential transpiration rate contains the additional term ROUGH. ROUGH replaces a parameter which varied with crop surface roughness in the original equation. Penman used a value of 1.0 for short grass, i.e. a smooth surface. Here ROUGH increases from 1 to 2 as the crop grows to cover half the ground area and then decreases to 1 as the canopy closes. This realistic but arbitrary choice of values gives parameter values similar to those used by Ritchie for field crops (1972). ROUGH is calculated in MAIN on line 417.

4.07 The potential transpiration rate is multiplied by the proportion of field covered by crop and then by a term that makes a Beer's law correction for radiation that passes between the leaves in the canopy.

According to Beer's law of radiation interception, which is also variously attributed to Bouguer and Lambert, the logarithm of light intensity at any point in the canopy is proportional to the number of leaf layers passed. Thus

I = Io*exp(-CEC*LCAI)

where I = radiation flux density below the canopy and Io = radiation flux density above the canopy. Hence the proportion of radiation passing through the canopy, I/Io = exp(-CEC*LCAI).

4.08 It is assumed that after the crop canopy closes, evaporation of water from the soil will be negligible. Even after leaf fall, the leaf litter on the soil will prevent most of the evaporation.

4.09 The albedo of exposed soil is based on data for a range of soils with various water contents cited by Bowers and Hanks (1965).

The surface soil is assumed to be air dry unless it is raining in which case it is assumed to be saturated.

4.10 The number of the first soil cell beyond the edge of the crop canopy is calculated.

4.11 The equation for net radiation on unit soil area is multiplied by the proportion of that radiation that impinges on the soil. See notes 4.4 and 4.7 for further information and to check that all radiation has been accounted for.

4.12 The radiation impinging on the soil is assumed to be all spread uniformly over the soil cells exposed between the crop rows.

4.13 X1 is used to gradually alter windspeed at the soil surface as the crop grows and the canopy closes. Until the crop covers half the soil area, increasing turbulence brings wind to the soil surface. Subsequently, wind is gradually excluded.

4.14 The Penman equation again.

4.15 The idea of limiting soil water evaporation rate to the rate at which water can move to the soil surface is taken from Rowse (1975) and Rowse and Stone (1978). Their original equation limits evaporation too much on days immediately following rain so a modified version is used here. This mechanism replaces the empirical approach used by Ritchie (1972) with the aim of making it independent of soil types and responsive to water uptake by roots in the surface layers of soil.

4.16 The proportion of the depth of each surface cell that is air dry is here made an arbitrary function of mean cell water content. As mean cell water content decreases, the thickness of the air dry layer increases slowly at first and then more rapidly as the whole cell approaches air dry. The factor 2 makes TNAIRD the proportion of half the cell depth that is air dry.

4.17 The hydraulic diffusivity of the soil at the mean water content in each surface cell is calculated. See CAPFLO note 16.3 for source of equation.

4.18 The mean hydraulic diffusivity between the top and the center of each surface cell is calculated. The equation gives the geometric mean assuming that the air dry surface layer is underlain by soil with the mean cell water content.

4.19 The potential rate of flow of water between the cell surface and cell center is calculated using Darcy's law. See CAPFLO note 16.04 for source of equation.

4.20 The rate of evaporation of water from the soil surface as driven by climatic factors is compared to the rate of movement of water to the surface as limited by hydraulic diffusivity. The lowest of these is the actual soil water evaporation rate.

4.21 The Penman equation assumes that soil surface temperature is similar to air temperature and it incorporates terms dealing with latent heat loss. Thus, when soil water evaporation is limited by climatic conditions it is assumed here that all radiation on the soil is dissipated as latent heat and there is none left to heat the soil. However, when the soil water evaporation is limited by soil conditions, radiation not used to evaporate water is assumed to heat the soil.

TMPSOL

The subroutine TMPSOL is called once for each period of the day and night. It calculates a heat balance at the soil surface accounting for the factors shown in Figure 20. Solar radiation available for heating the soil, i.e., solar radiation minus the latent heat loss, has already been calculated in EVTR. In TMPSOL, the equations for reradiation and convection are expressed in terms of a temperature gradient between the soil surface and the air. Soil heat flux is expressed in terms of a temperature gradient between the soil surface and the center of the top soil layer (usually 5 cm.). The equations are solved simultaneously to eliminate the soil surface temperature and estimate soil heat flux.

TMPSOL calculates the thermal diffusivities for each cell in the soil profile, the heat fluxes between cells and the temperature in each cell.

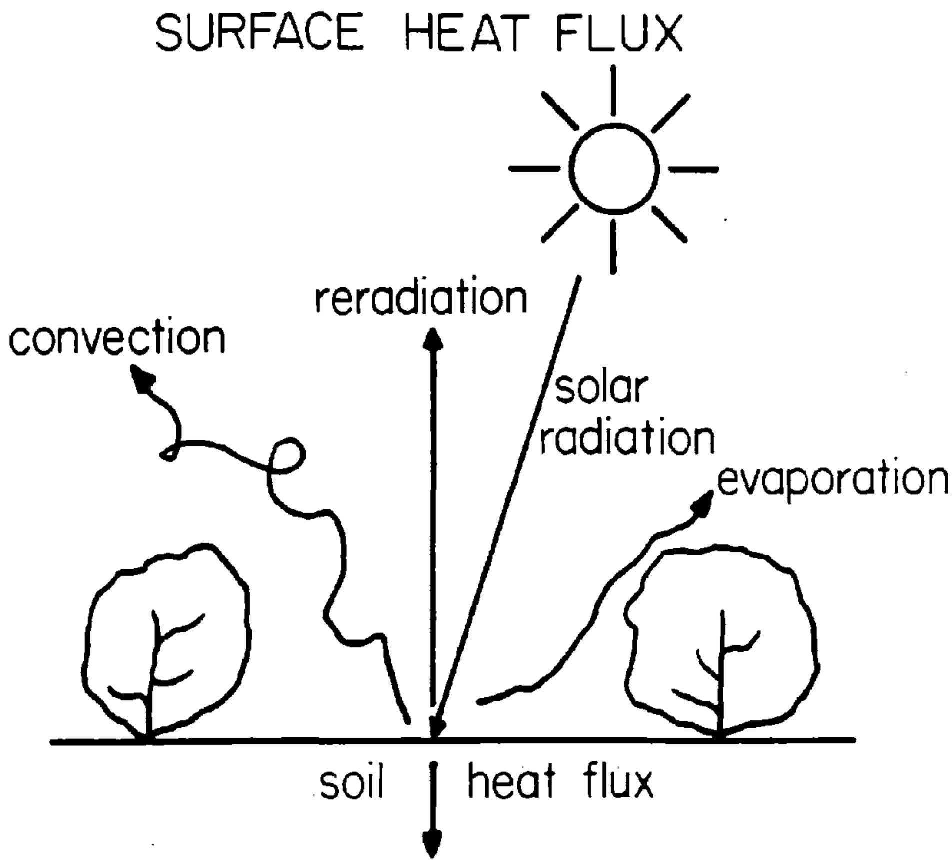


Fig. 20. Pictorial representation of the heat flux at the soil surface.

```
--------------------------( TMPSOL )------------------------------
THIS ROUTINE CALCULATES A RADIATION BALANCE FOR THE SOIL SURFACE
EXPOSED BETWEEN PLANT ROWS, HEAT FLUX INTO THE SOIL AND
TEMPERATURE OF EACH SOIL CELL.
------------------------------------------------------------------
for each soil cell, do the following:
|
| CALCULATE THE THERMAL DIFFUSIVITIES.
|
| CALCULATE MEAN THERMAL DIFFUSIVITY OF EACH PAIR OF
|    ADJACENT CELLS
|______________

IF THE CROP CANOPY HAS ALREADY CLOSED, MOVE DOWN THE ROUTINE

if the crop canopy has closed, jump down
|
| ----------------------------------------------------------------
| SOLVE SIMULTANEOUS EQUATIONS FOR HEAT FLUX AT THE SOIL
|    SURFACE AND DETERMINE HEAT FLUX DOWN INTO SOIL
| ----------------------------------------------------------------
| CALCULATE A FACTOR FOR EXTRA LONG-WAVE RADIATION FROM THE SOIL
|    SURFACE WHEN IT IS HOTTER THAN THE AIR
|
| CALCULATE A FACTOR FOR SENSIBLE HEAT CONVECTED FROM THE SOIL
|    SURFACE WHEN IT IS HOTTER THAN THE AIR
|
| CALCULATE FLUX OF HEAT DOWNWARDS INTO EACH EXPOSED SOIL CELL
V
CALCULATE FLUX OF HEAT DOWNWARD INTO EACH SOIL CELL COVERED
   BY THE CROP
------------------------------------------------------------------
CALCULATE SOIL HEAT FLUXES AND NEW SOIL CELL TEMPERATURES
------------------------------------------------------------------
for each soil cell, do the following:
|
| CALCULATE VERTICAL FLOW OF HEAT UPWARD OUT OF EACH CELL AND
|    CHANGE IN CELL TEMPERATURE.
|
| CALCULATE HORIZONTAL FLOW OF HEAT TO THE LEFT OUT OF EACH CELL
|    AND CHANGE IN CELL TEMPERATURE.
|______________

RETURN
```

```
           SUBROUTINE TMPSOL
     C     -------------------------( T M P S O L )-------------------------
     C     THIS ROUTINE CALCULATES A RADIATION BALANCE FOR THE SOIL SURFACE
     C     EXPOSED BETWEEN PLANT ROWS, HEAT FLUX INTO THE SOIL AND TEMPER-
     C     ATURE OF EACH SOIL CELL.
     C     ----------------------------------------------------------------
           DIMENSION HCS(20,10),TCS(20,10),FHL(20,11),FHU(21,10),
          &TDL(20,10),TDU(20,10),DSHF(20),TDS(20,10)
           INCLUDE 'COMMON.FOR'
     C     ----------------------------------------------------------------
     C     *** CALCULATE THE THERMAL DIFFUSIVITIES.***
     C     ----------------------------------------------------------------
     C
           DO 11 L=1,NL
           J = NJ(L)
             LM1 = L-1
             DO 10 K=1,NKH
             KM1 = K-1
5.01       TDS(L,K) = TDSX(1,J)+(TDSX(2,J)*TS(L,K))+(TDSX(3,J)*TS(L,K)
          &             *VH2OC(L,K))+(TDSX(4,J)*VH2OC(L,K))+(TDSX(5,J)
          &             *(VH2OC(L,K)**2))
     C
     C     *** CALCULATE MEAN THERMAL DIFFUSIVITY OF EACH PAIR OF
     C         ADJACENT CELLS ***
     C
            IF(K.EQ.1) GO TO 8888
5.02        TDL(L,K)= 2.* TDS(L,K)*TDS(L,KM1)/(TDS(L,K)+TDS(L,KM1))
      8888  IF(L.EQ.1) GO TO 10
            TDU(L,K) = 2.*TDS(L,K)*TDS(LM1,K)/(TDS(L,K)+TDS(LM1,K))
        10 CONTINUE
        11 CONTINUE
     C
     C     ** IF THE CROP CANOPY HAS ALREADY CLOSED, MOVE DOWN THE ROUTINE **
     C
5.03       IF( Z/ROWSP.GE.1.0 ) GOTO 9999
     C
     C     ----------------------------------------------------------------
     C     *** SOLVE SIMULTANEOUS EQUATIONS FOR HEAT FLUX AT THE SOIL
     C         SURFACE AND DETERMINE HEAT FLUX DOWN INTO SOIL ***
     C     ----------------------------------------------------------------
     C
     C     *** CALCULATE A FACTOR FOR EXTRA LONG-WAVE RADIATION FROM THE SOIL
     C         SURFACE WHEN IT IS HOTTER THAN THE AIR ***
     C
5.04       FELWR=(6.8E-3+(9.0E-5*TAIR(ITIME)))
     C
     C     *** CALCULATE A FACTOR FOR SENSIBLE HEAT CONVECTED FROM THE SOIL
     C         SURFACE WHEN IT IS HOTTER THAN THE AIR ***
     C
5.05       X6 = 2.0-(2.0*Z/ROWSP)
           IF(X6.GE.1.0) X6 = 1.0
5.06       FCSH=2.0E-1+(1.39E-3*WIND*X6)
     C  PARAMETER 2E-1 WAS 4E-3
     C
     C     ** CALCULATE FLUX OF HEAT DOWNWARDS INTO EACH EXPOSED SOIL CELL **
     C
           X2=FELWR+FCSH
           X3=D/2.0
           DO 30 K=KLJ,NKH
5.07       TCS(1,K) = TDS(1,K) *((0.46*(VSAND(1,K)+VCLAY(1,K)))+(0.6
          &             *VMUCK(1,K))+VH2OC(1,K))
5.08       TCSM=TCADS*TCS(1,K)/((TCS(1,K)*TNAIRD(K))+((1.0-TNAIRD(K))*TCADS))
             X1=RHS(K)/697.6
```

```
           X4=(X1/X2)+TAIR(ITIME)
           X5=TCSM*60.0/X3
5.09    30 DSHF(K)=X5*(X4-TS(1,K))/(1.0+(X5/X2))/60
C
C      *** CALCULATE FLUX OF HEAT DOWNWARD INTO EACH SOIL CELL COVERED
C          BY THE CROP ***
C
  9999 KLJM1=NKH
       IF( 2/ROWSP.LT.1.0 ) KLJM1=KLJ-1
       IF(KLJM1.EQ.0) GO TO 7777
       DO 40 K=1,KLJM1
5.10    40   DSHF(K)=(TAIR(ITIME)-TS(1,K))*(0.058+(1.7E-4*TAIR(ITIME)) +
     &           (0.052*EXP(0.058*TAIR(ITIME))))/1000
C
C      ------------------------------------------------------------------
C      *** CALCULATE SOIL HEAT FLUXES AND NEW SOIL CELL TEMPERATURES ****
C      ------------------------------------------------------------------
C
  7777   WPN=2.0*PERIOD*3600/W/W
         DPN=2.0*PERIOD*3600/D/D
C
C      *** CALCULATE VERTICAL FLOW OF HEAT UPWARD OUT OF EACH CELL AND
C          CHANGE IN CELL TEMPERATURE.***
C
         DO 55 K = 1,NKH
5.11   HCS(1,K) = (0.46*(VSAND(1,K)+VCLAY(1,K)))+(0.6 *VMUCK(1,K))
     &            +VH2OC(1,K)
5.12     FHU(1,K) =-DSHF(K)*3600*PERIOD/HCS(1,K)/D
    55   TS(1,K) = TS(1,K) - FHU(1,K)
         DO 60 L=2,NL
           LM1=L-1
           DO 61 K=1,NKH
5.13   FHU(L,K)=(TS(L,K)-TS(LM1,K))*0.5*(1-EXP(-TDU(L,K)*DPN))
       TS(L,K) = TS(L,K)-FHU(L,K)
       TS(LM1,K) = TS(LM1,K)+FHU(L,K)
       IF(L.EQ.NL) THEN
5.14   FHU(NL+1,K) = (TC-TS(L,K))*0.5*(1-EXP(-TDU(L,K)*DPN))
       TS(L,K)= TS(L,K)+FHU(NL+1,K)
       END IF
    61 CONTINUE
    60 CONTINUE
C
C      *** CALCULATE HORIZONTAL FLOW OF HEAT TO THE LEFT OUT OF EACH CELL
C          AND CHANGE IN CELL TEMPERATURE.***
C
         DO 50 L=1,NL
           DO 51 K=2,NKH
             KM1=K-1
       FHL(L,K)=(TS(L,K)-TS(L,KM1))*0.5*(1-EXP(-TDL(L,K)*WPN))
       TS(L,K) = TS(L,K)-FHL(L,K)
       TS(L,KM1)= TS(L,KM1)+FHL(L,K)
    51     CONTINUE
    50     CONTINUE
       RETURN
       END
```

Notes for TMPSOL

5.01 The equations used to predict thermal diffusivity of the soil are taken from "Physics of Plant Environment" (Van Wijk, 1966). These equations are solved for each soil horizon in a separate program called THERMS. (See the notes on that program for the derivation of the equations.) For each horizon of a soil, thermal diffusivity is a function of soil temperature and water content. Data for a range of temperatures and water contents generated in THERMS have been fitted to the equation shown here to determine the values of the TDSX parameters. Now estimates of thermal diffusivity are regenerated. This procedure saves computation time.

5.02 The equation for mean thermal diffusivity between adjacent pairs of cells uses geometric averaging.

5.03 Completely mechanistic equations for convective, conductive and radiative heat fluxes at the soil surface require a knowledge of the soil surface temperature (see for example Eshel and Curry, 1980). This is difficult to calculate and it is also difficult to measure. The assumption that soil surface temperature equals the mean temperature of the top layer in the soil profile results in large errors. Therefore, in this subroutine, simplified, approximate equations for heat flux have been used. They all have the form

Heat flux = $\int$(temperature difference)

where the temperature difference is that between soil surface and air or soil surface and a deeper soil layer. These equations are then solved simultaneously to eliminate soil surface temperature and estimate downward heat flux into the soil.

5.04 First a factor is calculated, which, when multiplied by the difference between air and soil surface temperature, gives the extra upward long-wave radiation. The equation is based on an idea by Linacre (1968, his appendix 2).

5.05 X 6 is used to gradually alter windspeed at the soil surface. cf EVTR note 4.13.

5.06 Next a factor is calculated for sensible heat convected away from the soil surface. The equation parameter values are based on Linacre (1968, his appendix 2) but the equation has been expanded to include explicitly the effects of windspeed above the canopy and of crop cover.

5.07 The thermal conductivity of soil in each exposed cell is calculated from its thermal diffusivity using equations from Van Wijk (1966).

5.08 Mean thermal conductivity between the center and top of each exposed soil cell is calculated using the geometric mean. The value of TCADS was calculated in MAIN on line 355.

5.09 The equations to be solved simultaneously are

Reradiative heat flux = FELWR*(Tss - TAIR)

where Tss = soil surface temperature

Convective heat flux = FCSH*(Tss - TAIR)

Radiation flux available for heating the soil,

RHS = Reradiative flux + Convective flux + Downward heat flux into the soil, DSHF.

The equations ignore heat conduction between the soil surface and air because this is assumed to be a negligible part of the heat flux on exposed soil cells.

Combining the equations above

RHS - DSHF = (FELWR + FCSH)*(Tss - TAIR)

and rearranging

Tss = (RHS - DSHF + ((FELWR + FCSH)*TAIR)/(FELWR + FCSH)

We can also write

DSHF = TCSM/D/2*(Tss - TS)

Substituting for Tss and rearranging we get

DSHF = TCSM/D/2*((RHS/(FELWR + FCSH)) + TAIR + TS)

/(1 + (TCSM/D/2(FELWR + FCSH)))

This is the equation solved here with appropriate adjustments for units.

5.10 It is assumed that, for soil cells covered by the crop canopy, convection and reradiation are minimal and that conduction is the principal model of heat transfer. It is further assumed that the air boundary layer at the soil surface is 1 cm thick and that the air is saturated with water vapor. The part of the equation following the first * is the calculation of the thermal conductivity of saturated air. It is a regression equation fitted to a graph of conductivity v. temperature prepared by Van Wijk (1966).

5.11 Soil heat capacity is calculated from an equation by Van Wijk (1966). See also note 5.07.

5.12. The downward flux of heat into the surface soil cells over the period of interest is used to calculate temperature changes in the cells.

5.13 The equations for the flow of heat between adjacent cells employ Fourier's law integrated over time and allowing for the fact that as flow occurs the gradient driving the flow decreases.

For vertical heat flow, the instantaneous flow rate,

dF/dT = TCS*D/W*(TS(1,1) - TS(2,1))

To get a dimensionless flow, F' divide by HCS and by cell volume, D*W

Then since TCS/HCS = TDS

dF'/dt = TDS/W/W*(TS(1,1) - TS(2,1))

As flow occurs, the gradient driving it decreases and the temperature of both soil cells approaches the mean temperature asymptotically. The equation used here has the required characteristics. It was derived by analogy with equations describing leakage of gas from an enclosure. Derivation from first principles is doubtless possible. The equation has been tested by comparing its predictions with those of the instantaneous flow rate equation when the latter is incremented with very short time steps.

5.14 It is assumed that the soil at the bottom of the profile has a constant temperature, TC.

PNET

The subroutine PNET is called once for each period of the day. It calculates canopy photosynthetic parameters from single leaf parameters weighted and integrated over the canopy. Leaf conductance to CO_2 is varied according to the carbon supply/demand ratio in the plant, the stage of growth and the light flux density in which the leaves developed. Canopy gross photosynthetic rate is calculated from an equation which gives it a hyperbolic dependence on light interception and CO_2 concentration as shown in Figure 21. On that figure, the intial slope is the light utilization efficiency and the asymptote equals CO_2 concentration times canopy conductance to CO_2

Once gross photosynthetic rate has been calculated it is adjusted downward for any limitations caused by temperature, leaf nitrogen content or leaf senescence. Next, light respiration and maintenance respiration rates are calculated and subtracted from gross photosynthetic rate to give net photosynthethic rate. This is further reduced, if the plant is in water stress, by a stomatal closure factor calculated in TRADE. The fixed carbon is either stored in the shoot carbon pool or is translocated to growing organs. A proportion of the carbon in the shoot carbon pool is also translocated.

Finally, PNET calculates the net photosynthetic rate for the lowest leaves on the plant and accumulates this over 24 hours to see if the leaves are self-supporting.

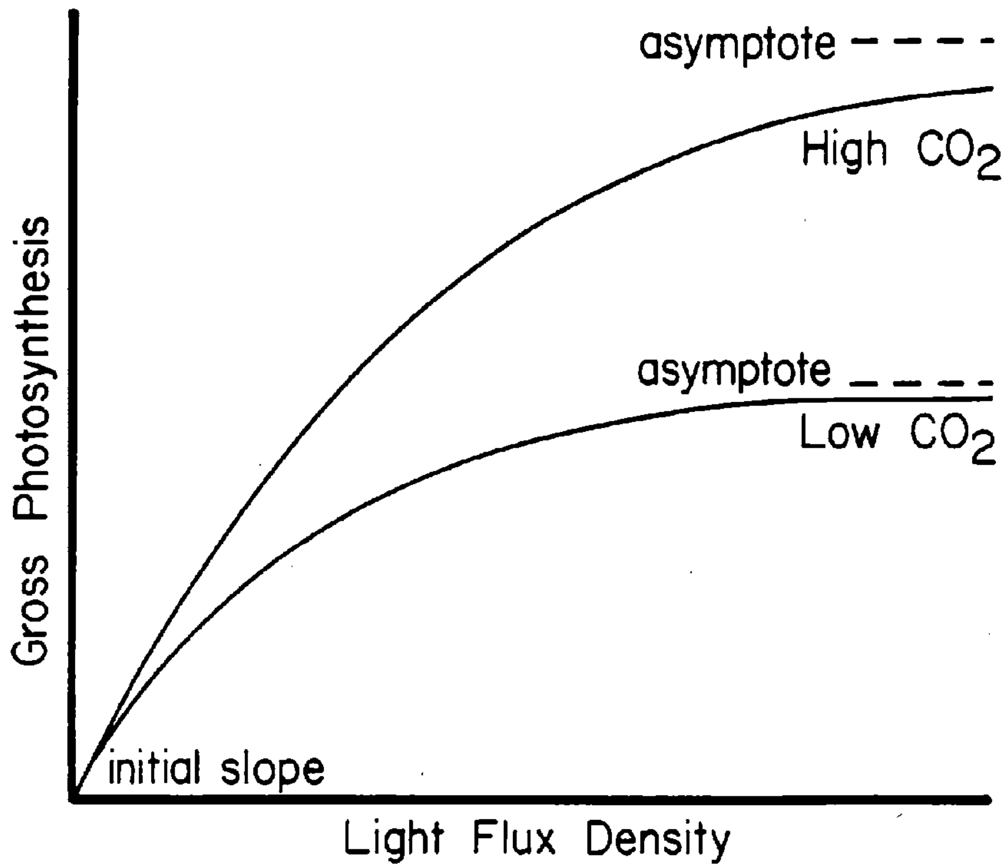


Fig. 21. Form of the relationship between gross photosynthesis, light flux density, and CO_2 concentration used in GLYCIM.

---------------------------(PNET)-----------------------------

THIS ROUTINE USES SINGLE LEAF PHOTOSYNTHETIC PARAMETERS TO CALCULATE CROP PARAMETERS AND GROSS PHOTOSYNTHESIS. RESPIRATION RATES FROM VARIOUS CAUSES ARE CALCULATED AND SUBTRACTED TO GIVE NET PHOTOSYNTHESIS RATE WHICH IS CORRECTED FOR STOMATAL CLOSURE. FROM THIS ARE CALCULATED THE NET CARBON FIXATION RATE AND THE RATE OF CARBON TRANSLOCATION OUT OF THE LEAF.

CANOPY PHOTOSYNTHETIC PARAMETERS ARE CALCULATED ONCE A DAY FROM LEAF PARAMETERS WEIGHTED AND INTEGRATED OVER THE CANOPY.

---

at any time other than dawn, jump down

CALCULATE CANOPY LIGHT UTILIZATION EFFICIENCY

CALCULATE CANOPY CONDUCTANCE TO CO2 TRANSFER

CANOPY GROSS PHOTOSYNTHETIC RATE AND ITS REDUCTION BY LOW TEMPERATUREC,LOW LEAF NITROGEN CONTENT AND LEAF SENESCENCE ARE CALCULATED.

CALCULATE CANOPY GROSS PHOTOSYNTHETIC RATE ASSUMING TEMPERATURE NOT LIMITING.

CALCULATE UPPER LIMIT OF LEAF GROSS PHOTOSYNTHESIS AT AIR TEMPERATURE

CALCULATE TEMPERATURE-LIMITED GROSS PHOTOSYNTHESIS RATE FOR EACH LEAF LAYER OF THE CANOPY WHERE PHOTOSYNTHESIS IS SO LIMITED

CALCULATE TEMPERATURE-LIMITED GROSS PHOTOSYNTHESIS, ENTIRE CANOPY

CALCULATE A PHOTOSYNTHESIS MULTIPLICATION FACTOR BASED ON LEAF N CONTENT.

CALCULATE A PHOTOSYNTHESIS MULTIPLICATION FACTOR BASED ON AGE OF TERMINAL MAINSTEM LEAF.

LIGHT RESPIRATION RATES AND MAINTENANCE RESPIRATION RATES ARE SUBTRACTED FROM GROSS PHOTOSYNTHETIC RATE. THE CARBON FIXED IS ALLOCATED TO STORAGE IN THE SHOOT OR IS TRANSLOCATED OUT OF THE LEAVES TO GROWING ORGANS.

CALCULATE LIGHT RESPIRATION RATE

CALCULATE MAINTENANCE RESPIRATION RATE

CALCULATE NET PHOTOSYNTHETIC RATE UNDER WELL WATERED CONDITIONS

CALCULATE NET PHOTOSYNTHETIC RATE ALLOWING FOR STOMATAL CLOSURE

CALCULATE NET CARBON FIXATION RATE

CALCULATE POTENTIAL RATE OF CARBON TRANSLOCATION FROM THE
SHOOT TO GROWING ORGANS AND AMOUNT STORED IN SHOOT IN THIS
PERIOD

NET PHOTOSYNTHETIC RATE FOR LEAVES AT THE BOTTOM OF THE
CANOPY IS INTEGRATED OVER 24 HOURS TO SEE IF THEY ARE STILL
SELF-SUFFICIENT.

CALCULATE GROSS PHOTOSYNTHETIC RATE FOR LEAVES AT THE BOTTOM
OF THE CANOPY

CALCULATE RESPIRATION RATES OF LEAVES AT THE BOTTOM OF THE CANOPY

CALCULATE ACCUMULATED NET PHOTOSYNTHETIC RATE FOR LEAVES AT
THE BOTTOM OF THE CANOPY

RETURN

```
          SUBROUTINE PNET
      C   ----------------------------( P N E T )--------------------------
      C   *** THIS ROUTINE USES SINGLE LEAF PHOTOSYNTHETIC PARAMETERS TO
      C   CALCULATE CROP PARAMETERS AND GROSS PHOTOSYNTHESIS.  RESPIRATION
      C   RATES FROM VARIOUS CAUSES ARE CALCULATED AND SUBTRACTED TO GIVE
      C   NET PHOTOSYNTHESIS RATE WHICH IS CORRECTED FOR STOMATAL CLOSURE.
      C   FROM THIS ARE CALCULATED THE NET CARBON FIXATION RATE AND THE RATE
      C   OF CARBON TRANSLOCATION OUT OF THE LEAF. ***
      C   -----------------------------------------------------------------
          INCLUDE 'COMMON.FOR'
      C   -----------------------------------------------------------------
6.01  C   *** CANOPY PHOTOSYNTHETIC PARAMETERS ARE CALCULATED ONCE A DAY
      C     FROM LEAF PARAMETERS WEIGHTED AND INTEGRATED OVER THE CANOPY.***
      C   -----------------------------------------------------------------
      C
      C   *** CALCULATE CANOPY LIGHT UTILIZATION EFFICIENCY ***
      C
          IF(ITIME.GT.1) GO TO 5555
          X31=EXP(-(LCAI*CEC))
          CLUE=LLUE*(1.0-X31)/(1.0-LTC)
      C
      C   *** CALCULATE CANOPY CONDUCTANCE TO CO2 TRANSFER ***
      C
6.02      TAUM =8.5E-5
          TAUNIN = PRM(9)
6.03      IF(RSTAGE.GE.3.0)TAUNIN = PRM(16)
6.04      TAUN = TAUN+(SINK*PRM(7))
          IF(TAUN.LT.TAUNIN) TAUN = TAUNIN
          IF(TAUN.GT.PRM(17))TAUN = PRM(17)
6.05      X32=0.021*LLUE*CEC*WATWK
          X33=LLUE*(1.0-LTC)
6.06      CCCT=ALOG((X32+X33)/(X32*X31+X33))*TAUM/TAUN/CEC
      C   -----------------------------------------------------------------
      C   *** CANOPY GROSS PHOTOSYNTHETIC RATE AND ITS REDUCTION BY LOW
      C       TEMPERATUREC,LOW LEAF NITROGEN CONTENT AND LEAF SENESCENCE ARE
      C       CALCULATED.***
      C   -----------------------------------------------------------------
      C   *** CALCULATE CANOPY GROSS PHOTOSYNTHETIC RATE ASSUMING
      C       TEMPERATURE NOT LIMITING. ***
      C
6.07 5555 WATPL = PAR(ITIME)*PARINT(ITIME)/COVER
          PG=CLUE*WATPL*CCCT*CO2*539.66/(273.15+TAIR(ITIME)) /
         &   ((CLUE*WATPL)+(CCCT*CO2*539.66/(273.15+TAIR(ITIME))))
         &   *COVER
      C
      C   *** CALCULATE UPPER LIMIT OF LEAF GROSS PHOTOSYNTHESIS AT AIR
      C   TEMPERATURE ***
      C
          IF( TAIR(ITIME).LE.37.0 ) GOTO 10
6.08      PGLIM=(50.0-TAIR(ITIME))*0.385
          GOTO 20
       10 PGLIM=TAIR(ITIME)*0.135
      C
      C   *** CALCULATE TEMPERATURE-LIMITED GROSS PHOTOSYNTHESIS RATE FOR
      C   EACH LEAF LAYER OF THE CANOPY WHERE PHOTOSYNTHESIS IS SO LIMITED *
      C
6.09   20 PGCOR=0.0
          X34=1.0
          M=IFIX(LCAI)+1
          DO 30 J=1,M
          IF(J.EQ.M) X34=LCAI+1-M
            WATLF=WATPL*CEC/(1.0-LTC)*EXP(-CEC*(J-0.5))
            WWKLF=WATLF*WATWK/PAR(ITIME)
```

```
           CCTLF=TAUM*WWKLF/(1.0+TAUN*WWKLF)
6.10       PGLF=LLUE*WATLF*CCTLF*CO2*539.66/(273.15+TAIR(ITIME)) /
     &         (LLUE*WATLF+CCTLF*CO2*539.66/(273.15+TAIR(ITIME)))
     &         *X34*COVER
         PGLIM=PGLIM*X34*COVER
           IF(PGLF.LE.PGLIM) GOTO 40
6.11    30 PGCOR=PGCOR+PGLF-PGLIM
      C
      C       *** CALCULATE TEMPERATURE-LIMITED GROSS PHOTOSYNTHESIS, ENTIRE
      C       CANOPY ***
      C
        40 PGROSS=PG-PGCOR
      C
      C       *** CALCULATE A PHOTOSYNTHESIS MULTIPLICATION FACTOR BASED ON LEAF
      C           N CONTENT. ***
      C
          IF(NRATIO.GT.0.4) GO TO 9999
          REDN=0.
          GO TO 8888
6.12  9999 REDN=1./((1./(1.65*((NRATIO*5.0)-2.)))+0.8)
      C
      C       *** CALCULATE A PHOTOSYNTHESIS MULTIPLICATION FACTOR BASED ON
      C           AGE OF TERMINAL MAINSTEM LEAF. ***
      C
      8888 IF( TLAGE.LT.10) GOTO 55
6.13      SEN=SIN(3.1416/2.0*(1.0+(TLAGE-10.)/(TLIFE-10.)))
          GOTO 50
        55 SEN=1.0
6.14    50 REDSEN = REDN
          IF(SEN.LT.REDN) REDSEN = SEN
      C       ---------------------------------------------------------------
      C       *** LIGHT RESPIRATION RATES AND MAINTENANCE RESPIRATION RATES ARE
      C           SUBTRACTED FROM GROSS PHOTOSYNTHETIC RATE. THE CARBON FIXED IS
      C           ALLOCATED TO STORAGE IN THE SHOOT OR IS TRANSLOCATED OUT OF
      C           THE LEAVES TO GROWING ORGANS.***
      C       ---------------------------------------------------------------
      C
      C       *** CALCULATE LIGHT RESPIRATION RATE ***
      C
6.15      LYTRES=PGROSS*REDSEN*151890.0/CO2*0.00012*EXP(0.0295 *
     &         TAIR(ITIME))
      C    151890 = 1.429*0.21*273/(273+TAIR)/1.9767E-6/273*(273+TAIR)
      C
      C       *** CALCULATE MAINTENANCE RESPIRATION RATE ***
      C
6.16      BMAIN=((11.733*(LEAFWT))+(366.67*LEAFWT*NRATIO*0.05))/86400.0
     &         /ROWSP*POPROW*100.0*0.25*EXP(0.0693*TAIR(ITIME))
      C
      C       *** CALCULATE NET PHOTOSYNTHETIC RATE UNDER WELL WATERED CONDI-
      C       TIONS ***
      C
          PINDEX=(PGROSS*REDSEN)-LYTRES-BMAIN
      C
      C       *** CALCULATE NET PHOTOSYNTHETIC RATE ALLOWING FOR STOMATAL
      C       CLOSURE ***
      C
6.17      PN=PINDEX*SCF
          IF( PN.LE.0.0 ) PN=0.0
      C
      C       *** CALCULATE NET CARBON FIXATION RATE ***
      C
6.18      FIXC=PN*0.27273*3600/1000/POPROW/100*ROWSP
          FIXCS = FIXCS+FIXC*PERIOD
```

```
C
C      *** CALCULATE POTENTIAL RATE OF CARBON TRANSLOCATION FROM THE
C          SHOOT TO GROWING ORGANS AND AMOUNT STORED IN SHOOT IN THIS
C          PERIOD ***
C
6.19   X36 = SCPOOL
       X6 = SHTWT*0.06
       IF(TLAGE.GT.0.0)X6 = SHTWT*PRM(18)
       IF((VSTAGE.GE.6.0).AND.(SCPOOL.GT.X6)) X36 = X6
6.20   TRANSC=(FIXC*0.35)+(X36*(1.0-(0.875**PERIOD))/PERIOD)
       TRNDAY=TRNDAY+(TRANSC*PERIOD)
6.21   SCPOOL=SCPOOL+((FIXC-TRANSC)*PERIOD)
       SCPMAX=0.4*SHTWT
       XTRAC=SCPOOL-SCPMAX
       IF( XTRAC.LE.0.0 ) XTRAC=0.0
       IF( SCPOOL.GT.SCPMAX ) SCPOOL=SCPMAX
       XTRACS=XTRACS+XTRAC
       QZ = 0.0124+FIXCS-USEDCS-XTRACS-SCPOOL-RCPOOL-(TRANSC*PERIOD)
       LINE=1449
 300   FORMAT(1X,I4,3(E10.3),2(F6.3),F5.2,E10.3,3(F6.3),3(E10.3)
      &,E10.3,1X,F6.2)
        IF(LPNDT.EQ.1) WRITE(2,300) LINE,TRANSC,FIXC,SCPOOL,SEN,REDN,
      &SCF,RCPOOL,QZ,SCRATO,SGTR,ROOTWT,USEDCS,FIXCS,TAUN,SINK
C      ----------------------------------------------------------------
C      *** NET PHOTOSYNTHETIC RATE FOR LEAVES AT THE BOTTOM OF THE
C          CANOPY IS INTEGRATED OVER 24 HOURS TO SEE IF THEY ARE STILL
C          SELF-SUFFICIENT.***
C      ----------------------------------------------------------------
C      ***CALCULATE GROSS PHOTOSYNTHETIC RATE FOR LEAVES AT THE BOTTOM
C         OF THE CANOPY ***
C
6.22   IF(VSTAGE.LT.4.0) RETURN
       WATLF = PAR(ITIME)*CEC/(1.0-LTC)*EXP(-CEC*LCAI)
       WWKLF = WATLF * WATWK/PAR(ITIME)
         CCTLF=TAUM*WWKLF/(1.0+TAUN*WWKLF)
       PGLF  = LLUE * WATLF *CCTLF *CO2 *539.66/(273.15+TAIR(ITIME))/
      &(LLUE * WATLF+ CCTLF *CO2 *539.66/(273.15+TAIR(ITIME)))
C
C      *** CALCULATE RESPIRATION RATES OF LEAVES AT BOTTOM OF CANOPY ***
C
       LYTRES = PGLF * REDSEN *151890.0/CO2*0.00012*EXP(0.0295*
      &          TAIR(ITIME))
       IF(ILOW.EQ.0) THEN
       X35 = ULEAFW/ULAREA*10000.
       BMAIN = ((11.733 *X35) +(366.67 * X35 *NRATIO *0.05))/86400.0
      &        *0.25*EXP(0.0693*TAIR(ITIME))
       LLFWT=ULEAFW
       LLAREA=ULAREA
       GO TO 7777
       END IF
       LLFWT = MLEAFW(ILOW)
       LLAREA = MLAREA(ILOW)
       IF(ILOW.EQ.1) GO TO 70
       IF(NBRNCH.LE.0) GO TO 70
       DO 60 J = 1,NBRNCH
       IF(INITBR(J).GE.(ILOW-2)) GO TO 70
       I = ILOWB(J)
       LLFWT = LLFWT + BLEAFW(I,J)
       LLAREA = LLAREA + BLAREA(I,J)
 60    CONTINUE
 70    CONTINUE
       IF(LLAREA.EQ.0.0)GO TO 6666
       X35 = LLFWT/LLAREA * 10000.
```

```
           BMAIN = ((11.733 *X35) +(366.67 * X35 *NRATIO *0.04))/86400.0
     &             *0.25*EXP(0.0693*TAIR(ITIME))
C
C          *** CALCULATE ACCUMULATED NET PHOTOSYNTHETIC RATE FOR LEAVES AT
C              THE BOTTOM OF THE CANOPY ***
C
6.23  7777 PNLLFS = PNLLFS + ((PRM(4)*PGLF*REDSEN)-LYTRES-BMAIN)*SCF
           IF(ITIME.EQ.IPERD)PNLLFS = PNLLFS-(BMAIN*EXP(0.0693*TNYT)/
     &               EXP(0.0693*TAIR(ITIME))*((24.0/PERIOD)-IPERD))
C    &               +(SCPOOL*LLFWT/SHTWT/LLAREA*9166.7/(24.0-(IPERD*PERIOD)))
C          9166.7=1000*10000*3.3/3600
      6666 RETURN
           END
```

Notes for PNET

6.01 The canopy photosynthetic parameters are calculated from single leaf parameters weighted and integrated over the canopy. The equations to perform these integrations are taken from Acock et al. (1978). Canopy photosynthetic parameters derived in this way can be checked against measured canopy photosynthesis during model validation. The equation for canopy light utilization efficiency, CLUE is similar to equation (3.11), page 59 in Charles-Edwards (1981). The derivation is as given, except that, for calculating gross photosynthesis, the leaf photorespiration constant, $\beta = 0.0$ and therefore it disappears from the equation.

6.02 Leaf conductance to CO_2 (TAUM/TAUN) was initially estimated from data of Brun and Cooper (1976) but has since been adjusted to fit the authors' observations of canopy photosynthesis (Acock et al., 1982).

6.03 The upper limit on leaf conductance appears to increase when the plants enter the reproductive phase of growth. This increase is not related to an immediate increase in sink strength or demand for carbon. It is probably controlled by hormones. Since we have no mechanism for control we use the artificial device here to fit observed plant behavior.

6.04 Soybean plants regulate their photosynthetic rate according to the sink, or demand, for carbon (e.g. Thorne and Koller, 1974 and Clough, Peet and Kramer, 1981). In this line of code, leaf conductance to CO_2 is varied by the value of SINK which is an indicator of the source/sink balance for carbon. SINK is positive when the sink is smaller than the source. SINK is calculated in NODULE on line 370.

6.05 Canopy conductance to CO_2 is a function of the light flux density in which the leaves developed. Here we make it a function of average light flux density over the past week, WATWK. The equation for the relationship is taken from Acock et al. (1978) and the parameter values are based on data for tomato collected by Charles-Edwards and Ludwig (1975). WATWK is calculated in MAIN on line 392.

6.06 This equation is derived from equation 8, page 818 of Acock et al. (1978) by taking the limit as I_o approaches infinity.

6.07 Gross canopy photosynthetic rate is here given a hyperbolic dependence on both light intercepted by the crop and CO_2 concentration. Carbon dioxide concentration is handled explicitly throughout this model. Equations of this form are very common in the literature. For example, see the citations in Acock et al. (1971).

6.08 The shape of the relationship between temperature and leaf gross photosynthesis in O_2-free air is taken from Hofstra and Hesketh (1969). See Figure 22 for their original data points and the lines fitted by the equations given here. The absolute values of gross photosynthesis in their data are much too low to accommodate observed rates of photosynthesis. Possibly, this is because the plants were grown in a low light

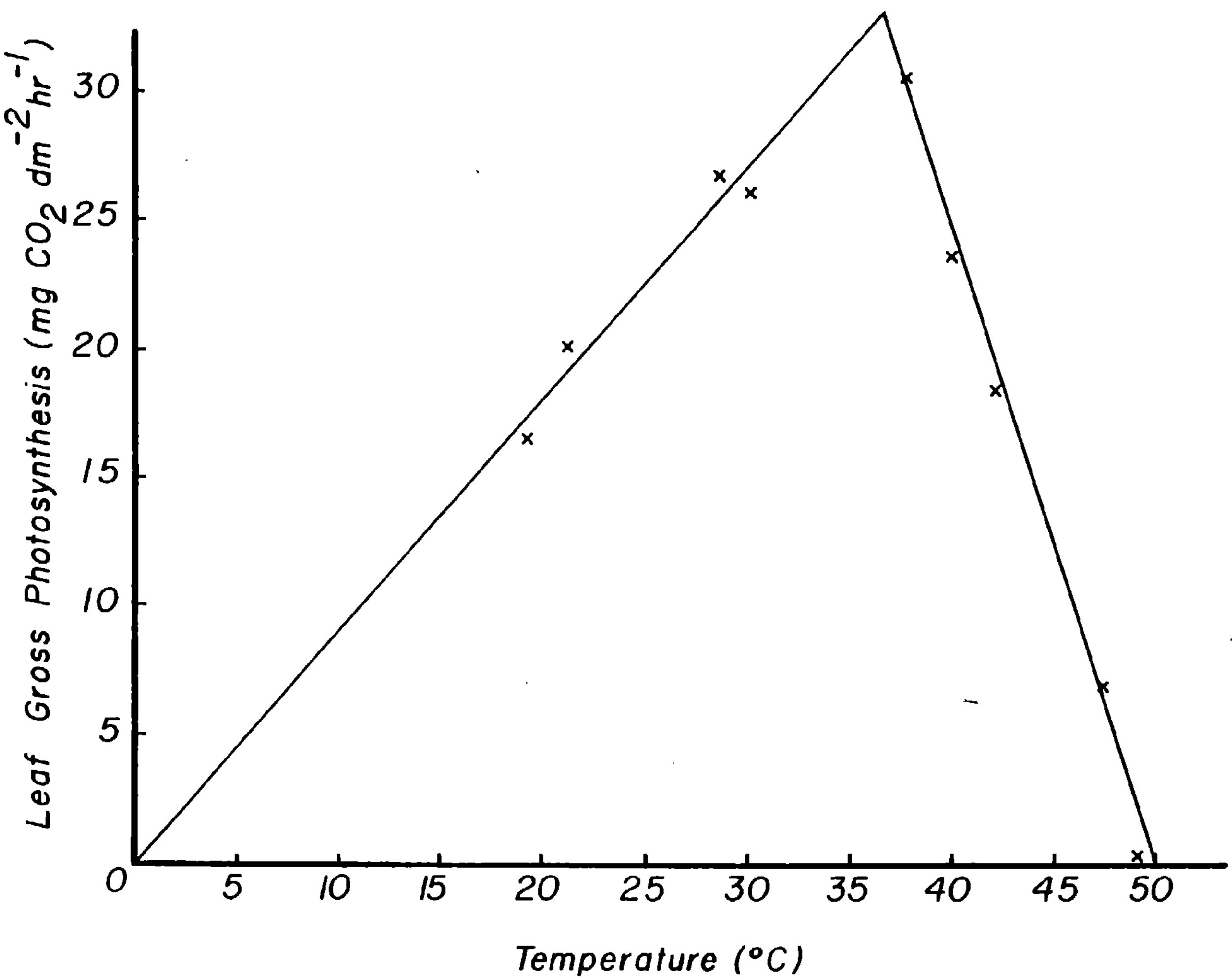


Fig. 22. Relationship between temperature and soybean leaf gross photosynthetic rate in O_2-free air (Hofstra and Hesketh, 1969).

flux density. Ludwig and Withers (1978) have reported a pronounced effect of light level during growth on the temperature response of tomato leaves. Therefore a peak value of gross photosynthetic rate was chosen that would allow observed photosynthetic rates to occur in the model.

6.09 The calculation of temperature limitations on gross photosynthesis for each leaf layer uses equations by Acock (1980) based on observations by Ludwig and Withers (1978).

6.10 Starting with the top leaf layer, the photosynthetic characteristics and gross photosynthesis for each leaf layer are calculated using the same equations that were used for canopy photosynthesis. Light flux density on the leaf layer is calculated using Beer's law (See EVTR note 4.07). The photosynthetic characteristics of leaves in the layer are derived using equations from Acock et al. (1978).

6.11 If gross photosynthesis for the leaf layer is greater than the temperature-limited leaf gross photosynthesis, the difference is subtracted from gross canopy photosynthesis.

6.12 The equation relating leaf photosynthetic rate to leaf nitrogen content was derived by fitting a curve to data of Boote et al. (1978). See Figure 23.

6.13 SEN is a senescence factor to reduce crop photosynthesis after leaf expansion stops. The equation assumes that the fall in canopy photosynthesis will be similar to the fall in leaf photosynthesis for a fully-exposed leaf as reported by Woodward (1976). See Figure 24.

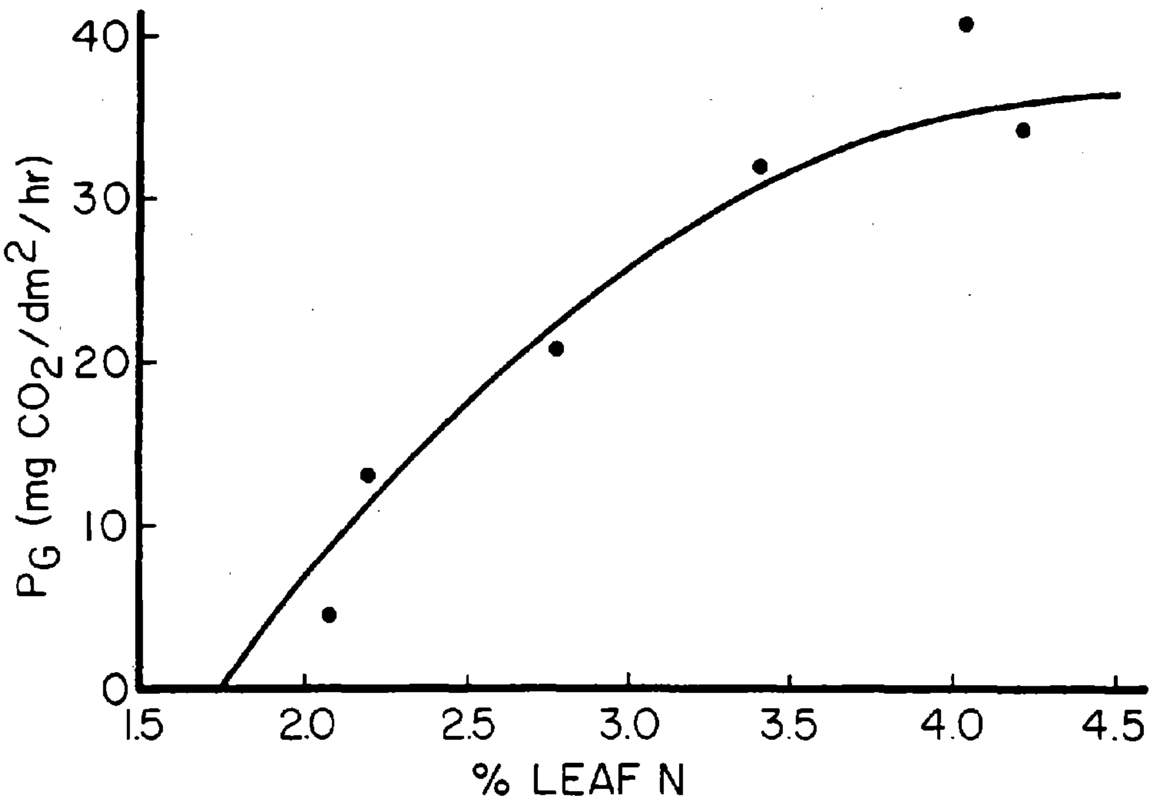


Fig. 23. Relationship between gross photosynthesis rate and leaf nitrogen for soybean. Data of Boote et al., 1978.

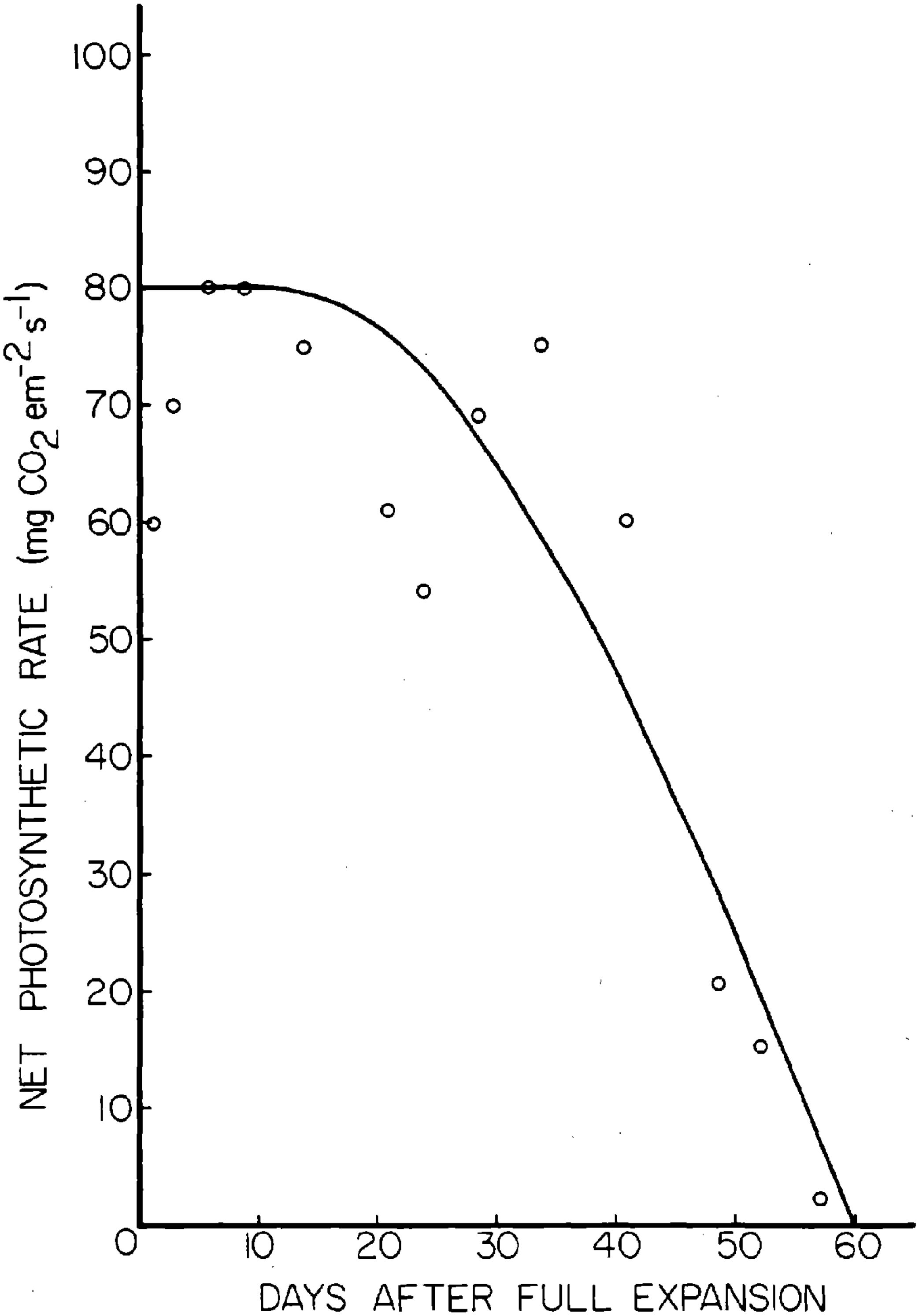


Fig. 24. Change in net photosynthetic rate for a fully exposed soybean leaf with age. Data of Woodward (1976).

6.14 Gross photosynthesis is reduced either by low leaf nitrogen or by leaf senescence, whichever factor is the most limiting.

6.15 Light respiration is calculated from temperature and the ratio of O_2 and CO_2 concentrations. For the derivations of the equation see Charles-Edwards (1978). Parameter values for soybean are based on data of Laing et al. (1974).

6.16 The calculation of the maintenance respiration rate required to turn over leaf protein and combat ion leakage uses the ideas and data of Penning de Vries (1975). It is assumed that only leaf tissue requires appreciable maintenance and also that each gram of leaf N represents 6.25 grams of leaf protein. Maintenance respiration rate is further assumed to have a Q_{10} of 2.

6.17 The stomatal conductance factor, SCF is proportional to stomatal conductance. It reduces photosynthetic rate when the plant is under water stress. Its value is calculated in TRADE on lines 298, 308, 323, or 342. The value of SCF calculated in one calculation period is used to reduce photosynthesis in the <u>next</u> calculation period.

6.18 Net canopy photosynthetic rate is converted to a net carbon fixation rate per plant.

6.19 The effect of this and the next 3 lines of code is to limit the rate of remobilization of carbon from the shoot carbon pool. Up to V6 there is no limitation, so that the plant can freely utilize carbon from the cotyledons. After V6, the effective size of the shoot carbon pool, i.e. the amount available for remobilization, is made a function of the size of the storage site, specifically shoot weight. The logic behind this is that carbon accumulates as grains of starch and that the rate of remobilization will be a function of grain surface area rather than grain volume.

When the leaves stop expanding, they start behaving as storage organs (Wittenbach, 1982). This is simulated here by further restricting the rate of carbon remobilization.

6.20 Carbon translocated out of the leaf is a function of both current photosynthetic rate and stored photosynthate. The parameter values are for tomato and are taken from Ho (1978). No comparable data are available for soybean.

6.21 Any current photosynthate not translocated is put in shoot storage up to a maximum of 40% of the shoot structural dry weight. Any carbon that cannot be stored is lost to the plant and is accumulated in XTRACS. This is a simple method for incorporating the fact that starch accumulating in the leaves is often associated with the suppression of photosynthesis.

6.22 The calculation of net photosynthetic rate for the lowest leaves in the canopy employs the same equations as used above for the entire canopy.

6.23 Net photosynthetic rate for the lowest leaves is accumulated during the day. Just before dusk, the overnight maintenance respiration rate is subtracted and an adjustment is made for the number of calculation periods it represents. The resulting 24-hour net photosynthetic rate for the lowest leaves is later used in NIGHT line 191 to see if the leaves are self-supporting for carbon. It they are not self-supporting they are dropped from the plant.

SHTGRO

The subroutine SHTGRO is called once each period of the day and night. It calculates the rates of appearance and expansion of all organs on the shoot as determined by temperature. These are, therefore, potential rates. From the expansion rates are calculated potential rates of dry weight gain for the upper and lower limits of density found in each organ. From these potential rates of dry weight gain and a knowledge of the biochemical cost of producing the compounds in that dry matter, we can calculate the potential rates of carbon use by the growing organs. These rates are summed over all the organs to calculate the potential rates of carbon use by the shoot when growing organs with either the maximum or the minimum density.

--------------------------(SHTGRO)------------------------------
THIS ROUTINE CALCULATES POTENTIAL SHOOT GROWTH ASSUMING THAT CARBOHYDRATE, NITROGEN, AND WATER ARE NOT LIMITING BUT ALLOWING FOR TEMPERATURE LIMITATION.

CALCULATE POTENTIAL RATE OF CHANGE IN VSTAGE GIVEN ADEQUATE SHOOT TURGOR.

CALCULATE A CARBON/DRY WEIGHT CONVERSION FACTOR FOR EACH ORGAN DEPENDING ON ITS PROTEIN, (I.E., N), CONTENT.

CALCULATE TEMPERATURE LIMITED RATES OF LEAF AREA AND PETIOLE EXTENSION.

LIMIT THE VEGETATIVE GROWTH OF DETERMINATE PLANTS.

CALCULATE POTENTIAL RATES OF CHANGE IN MAINSTEM LEAF AND PETIOLE DIMENSIONS AND DRY WEIGHTS. CALCULATE MINIMUM RATE OF CARBON SUPPLY NEEDED TO SYNTHESIZE THESE.

---

for the top three leaves on the mainstem, do the following:

POTENTIAL RATE OF CHANGE IN AREA OF EACH GROWING MAINSTEM LEAF.

TEMPERATURE LIMITED RATE OF MAINSTEM LEAF AREA GROWTH.

RATE OF CHANGE IN MAINSTEM LEAF AREA.

POTENTIAL RATE OF CHANGE IN LENGTH OF EACH GROWING MAIN-STEM PETIOLE.

TEMPERATURE LIMITED RATE OF MAINSTEM PETIOLE EXTENSION.

RATE OF CHANGE IN MAINSTEM PETIOLE DRY WEIGHT AND MINIMUM RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS.

RATE OF CHANGE IN MAINSTEM LEAF VOLUME AND DRY WEIGHT, AND MINIMUM RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS.

CALCULATE POTENTIAL RATES OF CHANGE IN BRANCH LEAF AND PETIOLE DIMENSIONS AND DRY WEIGHT. CALCULATE MINIMUM RATE OF CARBON SUPPLY NEEDED TO SYNTHESIZE THESE.

---

for the top three leaves on each branch, do the following:

POTENTIAL RATE OF CHANGE IN AREA OF EACH GROWING BRANCH LEAF

TEMPERATURE LIMITED RATE OF BRANCH LEAF AREA GROWTH.

RATE OF CHANGE IN BRANCH LEAF AREA.

POTENTIAL RATE OF CHANGE IN LENGTH OF EACH GROWING BRANCH PETIOLE.

A

A

TEMPERATURE LIMITED RATE OF BRANCH PETIOLE EXTENSION.

RATE OF CHANGE IN BRANCH PETIOLE DRY WEIGHT AND MINIMUM RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS.

RATE OF CHANGE IN BRANCH LEAF VOLUME AND DRY WEIGHT, AND MINIMUM RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS.

CALCULATE POTENTIAL RATE OF CHANGE IN STEM AND BRANCH LENGTH AND DRY WEIGHT AND MINIMUM RATE OF CARBON SUPPLY NEEDED FOR THEIR SYNTHESIS.

CALCULATE POTENTIAL RATE OF CHANGE IN DRY WEIGHT OF FLOWERS PODS AND SEEDS AND THE CARBON SUPPLY NEEDED FOR THEIR SYNTHESIS.

CALCULATE A CARBON/DRY WEIGHT CONVERSION FACTOR FOR PODS

MOVE TO APPROPRIATE PART OF ROUTINE TO CALCULATE GROWTH OF PODS AND BEANS.

if reproductive stage < 1, jump down

if reproductive stage > or = 5, jump down

if reproductive stage > or = 3, jump down

CALCULATE POTENTIAL RATE OF CHANGE IN RSTAGE BETWEEN R1 AND R3.

CALCULATE POTENTIAL RATE OF CHANGE IN FLOWER DRY WEIGHT AND RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS.

jump down

CALCULATE POTENTIAL RATE OF CHANGE IN RSTAGE BETWEEN R3 AND R5.

CALCULATE POTENTIAL RATE OF CHANGE IN POD DRY WEIGHT AND RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS.

CALCULATE POTENTIAL RATE OF CHANGE IN SEED DRY WEIGHT AND RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS.

if reproductive stage > 4, jump down

BETWEEN R3 AND R4 CALCULATE THE CARBON THAT WOULD BE NEEDED BY SEEDS IF THEY WERE GROWING.

MAXIMUM AND MINIMUM RATES OF CARBON USE BY SHOOT EXPANDING AT POTENTIAL RATE.

RETURN

```
      SUBROUTINE SHTGRO
C     ----------------------------( SHTGRO )----------------------------
C     THIS ROUTINE CALCULATES POTENTIAL SHOOT GROWTH ASSUMING
C     THAT CARBOHYDRATE, NITROGEN, AND WATER ARE NOT LIMITING BUT ALLOW-
C     ING FOR TEMPERATURE LIMITATION.
C     ------------------------------------------------------------------
      INCLUDE 'COMMON.FOR'
C     ------------------------------------------------------------------
C     *** CALCULATE POTENTIAL RATE OF CHANGE IN VSTAGE GIVEN ADEQUATE
C         SHOOT TURGOR. ***
C
      PDMPC=0.0
      PDMLC=0.0
      PDMC=0.0
      PDBPC=0.0
      PDBLC=0.0
      PDBC=0.0
      PDFC=0.0
      PDFW=0.0
      PDSC=0.0
      PDSW=0.0
      PDPODC=0.0
      SLOW = 1.0
      IF(RSTAGE.GE.1.0) SLOW = PRM(10)
      TAIRL=TAIR(ITIME)
      IF( TAIR(ITIME).GT.30.0 ) TAIRL=30.0
7.01  IF( VSTAGE.GE.0.0 ) PDV=(0.0157*TAIRL+0.0024)/24.0
      IF( VSTAGE.GE.1.0 ) PDV=(0.012*TAIRL-0.08)/24.0
      IF(VSTAGE.GE.2.0) PDV=(PRM(23)*TAIRL-TRIFMG)/24.
C     PARAMETER PRM(23) WAS 0.018
      IF( PDV.LE.0.0 ) PDV=0.0
C
C     *** CALCULATE A CARBON/DRY WEIGHT CONVERSION FACTOR FOR EACH
C         ORGAN DEPENDING ON ITS PROTEIN, (I.E., N), CONTENT. ***
C
7.02  X41=6.25*NRATIO/100
      X42=5.0*X41
      CONVL=(X42*0.6)+((1.0-X42)*0.47)
      X43=2.0*X41
      CONVPE=(X43*0.6)+((1.0-X43)*0.47)
      CONVST=CONVPE
C
C     *** CALCULATE TEMPERATURE LIMITED RATES OF LEAF AREA AND PETIOLE
C         EXTENSION. ***
C
7.03  PDLAT=8.5*(TAIR(ITIME)-9.0)*PDV/3.0
C     PARAMETER 8.5 WAS 10.
      IF(PDLAT.LT.0.) PDLAT=0.0
      PDPLT=1.5*(TAIR(ITIME)-9.0)*PDV/3.0
      IF(PDPLT.LT.0.) PDPLT=0.0
C
C     *** LIMIT THE VEGETATIVE GROWTH OF DETERMINATE PLANTS. ***
C
      IF( VSTAGE.GE.VSTMAH ) GOTO 9988
C     ------------------------------------------------------------------
C     *** CALCULATE POTENTIAL RATES OF CHANGE IN MAINSTEM LEAF AND
C         PETIOLE DIMENSIONS AND DRY WEIGHTS.  CALCULATE MINIMUM
C         RATE OF CARBON SUPPLY NEEDED TO SYNTHESIZE THESE.
C     ------------------------------------------------------------------
C  ** POTENTIAL RATE OF CHANGE IN AREA OF EACH GROWING MAINSTEM LEAF. **
C
      PDMLA=0.
      PDMPW=0.
```

```
      DO 10 I=1,3
        X45=VSTAGE+1.5-I
        IF(X45.LT.0.) GO TO 7777
        ITRIF=IFIX(X45)
        IF( ITRIF.GE.1 ) PDLAM(I)=((ITRIF*35.0)+15.0)*PDV/3.0
C   PARAMETER 35.0 WAS 18.6 AND 15.0 WAS 31.4
        IF( ITRIF.EQ.0 ) PDLAM(I)=20.0*PDV/3.0
C
C       *** TEMPERATURE LIMITED RATE OF MAINSTEM LEAF AREA GROWTH. ***
C
        IF( PDLAT.LT.PDLAM(I) ) PDLAM(I)=PDLAT
        PDLAM(I) = PDLAM(I) * SLOW
C
C       *** RATE OF CHANGE IN MAINSTEM LEAF AREA. ***
C
        PDMLA=PDMLA+PDLAM(I)
C
C       *** POTENTIAL RATE OF CHANGE IN LENGTH OF EACH GROWING MAIN-
C           STEM PETIOLE. ***
C
        PDPLM(I)=((ITRIF*4.0)+1.0)*PDV/3.0
        IF( ITRIF.LE.0.0 ) PDPLM(I)=0.
C
C       *** TEMPERATURE LIMITED RATE OF MAINSTEM PETIOLE EXTENSION. ***
C
        IF( PDPLT.LT.PDPLM(I) ) PDPLM(I)=PDPLT
        PDPLM(I) = PDPLM(I) * SLOW
C
C       *** RATE OF CHANGE IN MAINSTEM PETIOLE DRY WEIGHT AND MINIMUM
C           RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS. ***
C
       PDPWM(I)=0.0
        IF( ITRIF.EQ.0 ) GOTO 10
7.04    PDPWM(I)=(PDPLM(I)*PRM(11))+(8.7E-5*(PDPLM(I)**2.0+
     &            (2.0*PDPLM(I)*MPETL(ITRIF))))
C       PARAMETER PRM(11) WAS 0.0024
   10   PDMPW=PDMPW+PDPWM(I)
 7777 PDMPC=PDMPW*CONVPE
C
C     *** RATE OF CHANGE IN MAINSTEM LEAF VOLUME AND DRY WEIGHT, AND
C         MINIMUM RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS. ***
C
      PDMLV=PDMLA*TNL
      PDMLW=PDMLV/9.0*PRM(26)
C  PARAMETER 9.0 WAS 9.0
      PDMLC=PDMLW*CONVL
C     ---------------------------------------------------------------
C     *** CALCULATE POTENTIAL RATES OF CHANGE IN BRANCH LEAF AND PETIOLE
C         DIMENSIONS AND DRY WEIGHT. CALCULATE MINIMUM RATE OF CARBON
C         SUPPLY NEEDED TO SYNTHESIZE THESE. ***
C     ---------------------------------------------------------------
C     *** POTENTIAL RATE OF CHANGE IN AREA OF EACH GROWING BRANCH LEAF *
C
 9988 PDBLA=0.0
      PDBPW=0.0
      IF(NBRNCH.LT.1) GO TO 3333
      DO 21 J=1,NBRNCH
        IF( TB(J).GE.TBMAH(J) ) GOTO 21
          DO 20 II=1,3
            I=IFIX(TB(J)+1.5-II)
            IF( I.LE.0 ) GOTO 20
            PDLAB(I,J)=((I*PRM(24))+5.0)*PDV/3.0
C   PARAMETER PRM(24) WAS 37.2 AND 5.0 WAS 12.8
```

```
C
C     *** TEMPERATURE LIMITED RATE OF BRANCH LEAF AREA GROWTH. ***
C
            IF((PDLAT*PRM(22)).LT.PDLAB(I,J) ) PDLAB(I,J)=PDLAT*PRM(22)
            PDLAB(I,J) = PDLAB(I,J) * SLOW
C
C     *** RATE OF CHANGE IN BRANCH LEAF AREA. ***
C
            PDBLA=PDBLA+PDLAB(I,J)
C
C     *** POTENTIAL RATE OF CHANGE IN LENGTH OF EACH GROWING BRANCH
C         PETIOLE. ***
C
            PDPLB(I,J)=I*12.*PDV/3.0
C
C     *** TEMPERATURE LIMITED RATE OF BRANCH PETIOLE EXTENSION. ***
C
            IF( PDPLT.LT.PDPLB(I,J) ) PDPLB(I,J)=PDPLT
        PDPLB(I,J) = PDPLB(I,J) * SLOW
C
C     *** RATE OF CHANGE IN BRANCH PETIOLE DRY WEIGHT AND MINIMUM RATE
C         OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS. ***
C
            PDPWB(I,J)=(PDPLB(I,J)*PRM(11))+(8.7E-5*(PDPLB(I,J)**2.0+
     &                 (2.0*PDPLB(I,J)*BPETL(I,J))))
C       PARAMETER PRM(11) WAS 0.0024
            PDBPW=PDBPW+PDPWB(I,J)
   20     CONTINUE
   21     CONTINUE
 3333 PDBPC=PDBPW*CONVPE
C
C     *** RATE OF CHANGE IN BRANCH LEAF VOLUME AND DRY WEIGHT, AND
C         MINIMUM RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS. ***
C
      PDBLV=PDBLA*TNL
      PDBLW=PDBLV/12.0*PRM(26)
C  PARAMETER 12.0 WAS 12.0
      PDBLC=PDBLW*CONVL
C     ----------------------------------------------------------------
C     *** CALCULATE POTENTIAL RATE OF CHANGE IN STEM AND BRANCH LENGTH
C         AND DRY WEIGHT AND MINIMUM RATE OF CARBON SUPPLY NEEDED FOR
C         THEIR SYNTHESIS. ***
C     ----------------------------------------------------------------
7.05  PDMH=(PRM(19)+(PRM(5)*(VSTAGE+PDV/2.0)))*PDV*SLOW
C  PARAMETER PRM(5) WAS 0.36 AND PRM(19) WAS 1.5
      IF(VSTAGE.GE.(VSTMAH-PRM(12))) GO TO 8888
      PDMW=(PDMH*PRM(15))+(8.7E-5*(PDMH**2.0+(2.0*PDMH*MSTEMH)))
C PARAMETER PRM(15) WAS 0.011
      PDMC=PDMW*CONVST
 8888 PDBW=0.
      IF(NBRNCH.LT.1) GO TO 9977
      DO 30 J=1,NBRNCH
      IF(TB(J).GE.(TBMAH(J)-PRM(12))) GO TO 30
        PDBRW(J)=(PDMH*0.007)+(8.7E-5*(PDMH**2.0+(2.0*PDMH*
     &             BRNCHH(J))))
C PARAMETER 0.007 WAS 0.011
        PDBW=PDBW+PDBRW(J)
 30     CONTINUE
 9977 PDBC=PDBW*CONVST
C     ----------------------------------------------------------------
C     *** CALCULATE POTENTIAL RATE OF CHANGE IN DRY WEIGHT OF FLOWERS
C         PODS AND SEEDS AND THE CARBON SUPPLY NEEDED FOR THEIR
C         SYNTHESIS.***
```

```
      C     ----------------------------------------------------------------
      C     *** CALCULATE A CARBON/DRY WEIGHT CONVERSION FACTOR FOR PODS ***
      C
       1111 X44=4.0*NRATIO*6.25/100
            CONVPO=(X44*0.53)+((1.0-X44)*0.45)
      C
      C     *** MOVE TO APPROPRIATE PART OF ROUTINE TO CALCULATE GROWTH OF
      C         PODS AND BEANS. ***
      C
            PDR=0.0
            IF(RSTAGE.GE.5.0) GO TO 4444
            IF( RSTAGE.GE.3.0 ) GOTO 2222
            IF( RSTAGE.LT.1.0 ) GOTO 9999
      C
      C     *** CALCULATE POTENTIAL RATE OF CHANGE IN RSTAGE BETWEEN R1 AND
      C         R3. ***
      C
            IF( JDAY.GE.173 ) GOTO 6666
7.06        PDR=2.0/(4.0+(45.0*(PHOTO-DIFIMG)**1.5))/24.0
            GOTO 5555
       6666 PDR=2.0/(4.0+(10.0*(PHOTO-DIFIMG)**1.5))/24.0
       5555 IF( PHOTO.LE.DIFIMG ) PDR=2.0/4.0/24.0
      C
      C     *** CALCULATE POTENTIAL RATE OF CHANGE IN FLOWER DRY WEIGHT AND
      C         RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS. ***
      C
            PDFW=NFLWRS*0.002/2.*PDR
            PDFC=PDFW*CONVPO
            GO TO 9999
      C
      C     *** CALCULATE POTENTIAL RATE OF CHANGE IN RSTAGE BETWEEN R3
      C         AND R5. ***
      C
       2222 PDR=1.0/((2.0*PHOTO)-13.0)/24.0
      C
      C     *** CALCULATE POTENTIAL RATE OF CHANGE IN POD DRY WEIGHT AND
      C         RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS. ***
      C
            PDPODW=(PODS*0.07)*PDR
            IF(RSTAGE.GE.4.0) PDPODW = PDPODW*(5.0-RSTAGE)
            PDPODC=PDPODW*CONVPO
      C
      C     *** CALCULATE POTENTIAL RATE OF CHANGE IN SEED DRY WEIGHT AND
      C         RATE OF CARBON SUPPLY NEEDED FOR ITS SYNTHESIS. ***
      C
7.07   4444 OIL=(FILDAY+4.)/100.
            IF(RSTAGE.LT.4.0)OIL=0.2
            IF(OIL.GT.0.2) OIL=0.2
            PROTIN=0.4063
      C    0.4063 = 6.5*6.25/100
            CONVSE=((OIL*0.70)+(PROTIN*0.53)+((1-OIL-PROTIN)*0.45))
            IF(SEEDS.EQ.0.0)GO TO 9999
            SDWLIM = 1.0
            X46 = SEEDWT/SEEDS/SDWMAX
7.08        IF(X46.GT.0.8)SDWLIM = (1.0 - X46)*5.0
7.09        PDSW=SEEDS*(3.0+((TAIR(ITIME)-15.5)*(FILL-3.0)/8.5))/24./1000.
          &        *SDWLIM
            IF(PDSW.LE.0.) PDSW=0.
            PDSC=PDSW*CONVSE
      C
      C     *** BETWEEN R3 AND R4 CALCULATE THE CARBON THAT WOULD BE NEEDED BY
      C         SEEDS IF THEY WERE GROWING. ***
      C
```

```
           IF(RSTAGE.LT.4.0)THEN
7.10       EQSC=PDSC+(PDSW*0.13)
           RSTPC=EQSC-PDPODC
           PDSC=0.0
           PDSW=0.0
           ELSE
           RSTPC=0.0
           END IF
      C
      C    *** MAXIMUM AND MINIMUM RATES OF CARBON USE BY SHOOT EXPANDING AT
      C        POTENTIAL RATE. ***
      C
7.11  9999 PCST=((PDMPC+PDMLC+PDBPC+PDBLC)*1.6)+((PDMC+PDBC)*2.5)+PDFC+
          &        PDPODC+PDSC+RSTPC
           QCST=PDMPC+PDMLC+PDBPC+PDBLC+PDMC+PDBC+PDFC+PDPODC+PDSC+RSTPC
           RETURN
           END
```

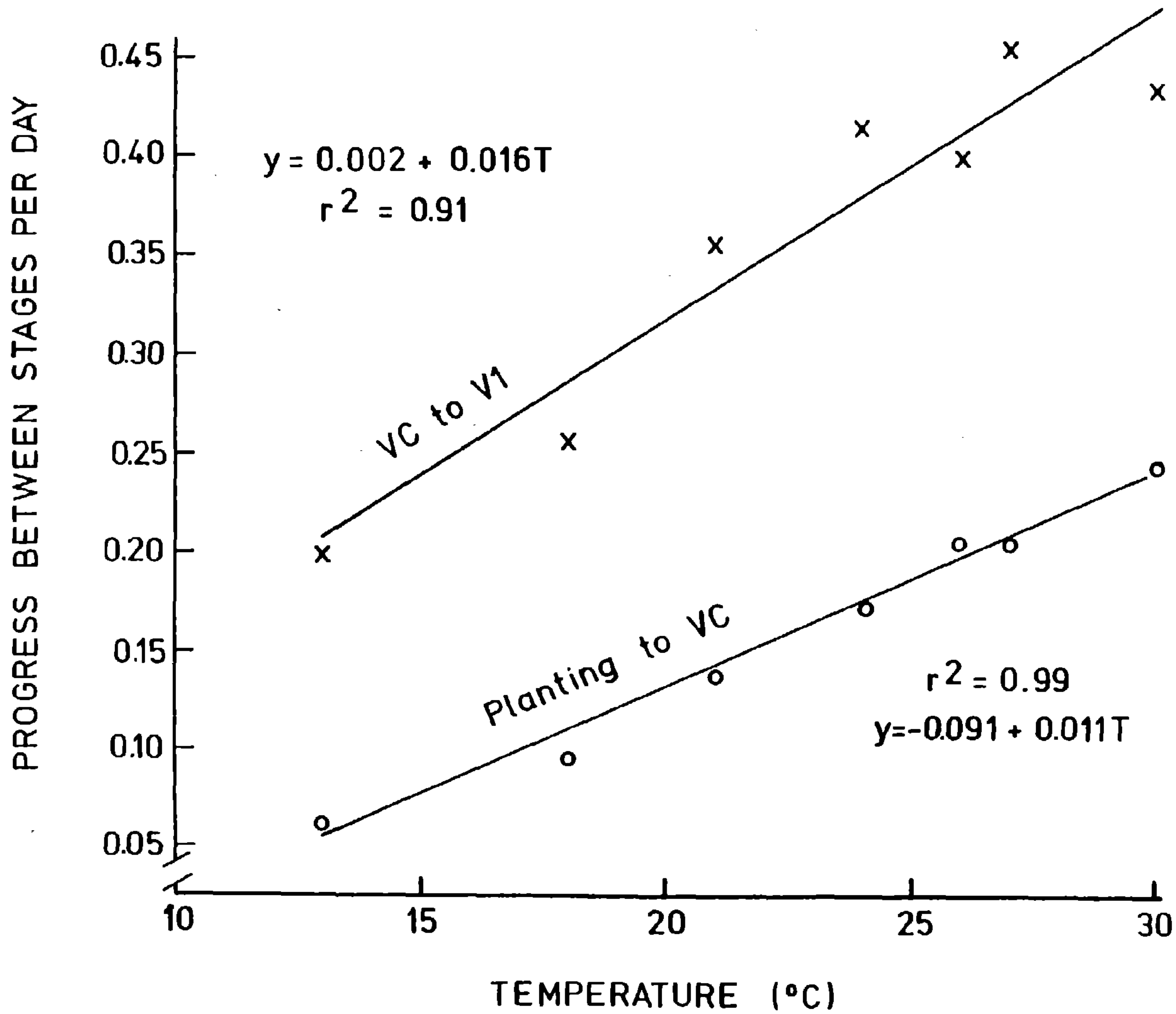
Notes for SHTGRO

All organ weights calculated in the model are structural dry weights and do not include stored carbon.

7.01 Potential rates of change of vegetative growth stage are calculated using relationships like those shown in Figures 25 and 26, which are based on the data of Hesketh et al. (1973). These data are for Wayne, a maturity group 3 cultivar. Thomas and Raper (1976) using Ransom (group 7) found similar rates except that the rate of appearance of trifoliolates was lower. Their data lie on a line parallel to the upper line on Figure 26 and with a different intercept. In the absence of better data it has been assumed that the intercept of the trifoliolate line, TRIFMG is proportional to maturity group number as in Figure 27. There is, however, some evidence that maturity groups 00 through 4 may have similar trifoliolate expansion rates. TRIFMG is calculated in MAIN on line 378.

7.02 Carbon to dry weight conversion factors are estimated using parameter values calculated by Penning de Vries (1975), and assuming that organ protein content = 6.25 x organ N content.

7.03 Organ growth rates as functions of temperature are taken from our own data for Forrest soybeans grown in SPAR units at 600 ppm CO_2 to ensure that photosynthate was not limiting. The final size of each organ appeared to be genetically-limited on the lower nodes and environmentally-limited on the upper nodes. See Figure 28 for example. At any time, it was assumed that



(Data of Hesketh et al. 1973)

Fig. 25. Daily progress between growth stages as a function of temperature for events prior to V1.

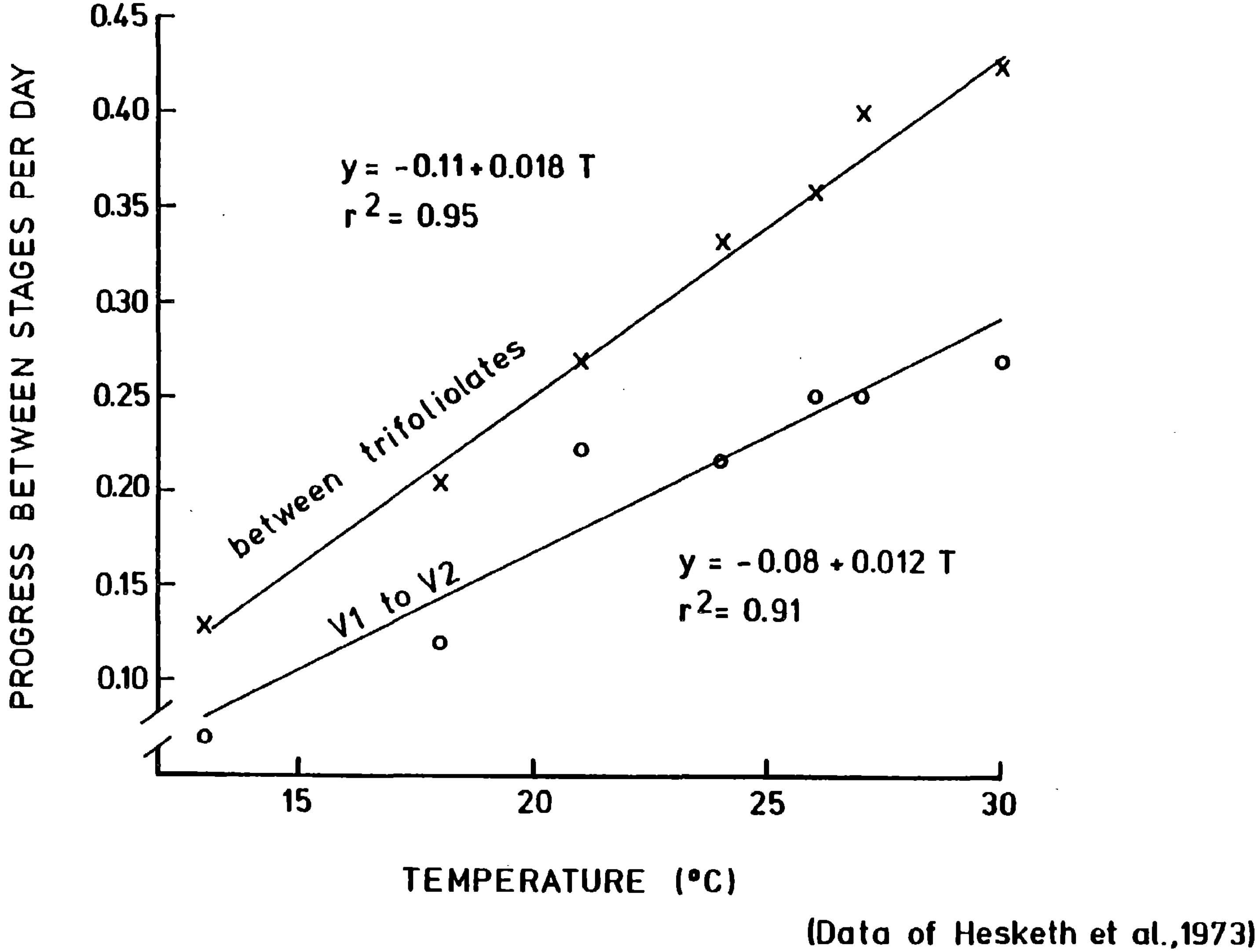


Fig. 26. Daily progress between growth stages as a function of temperature for events after V1.

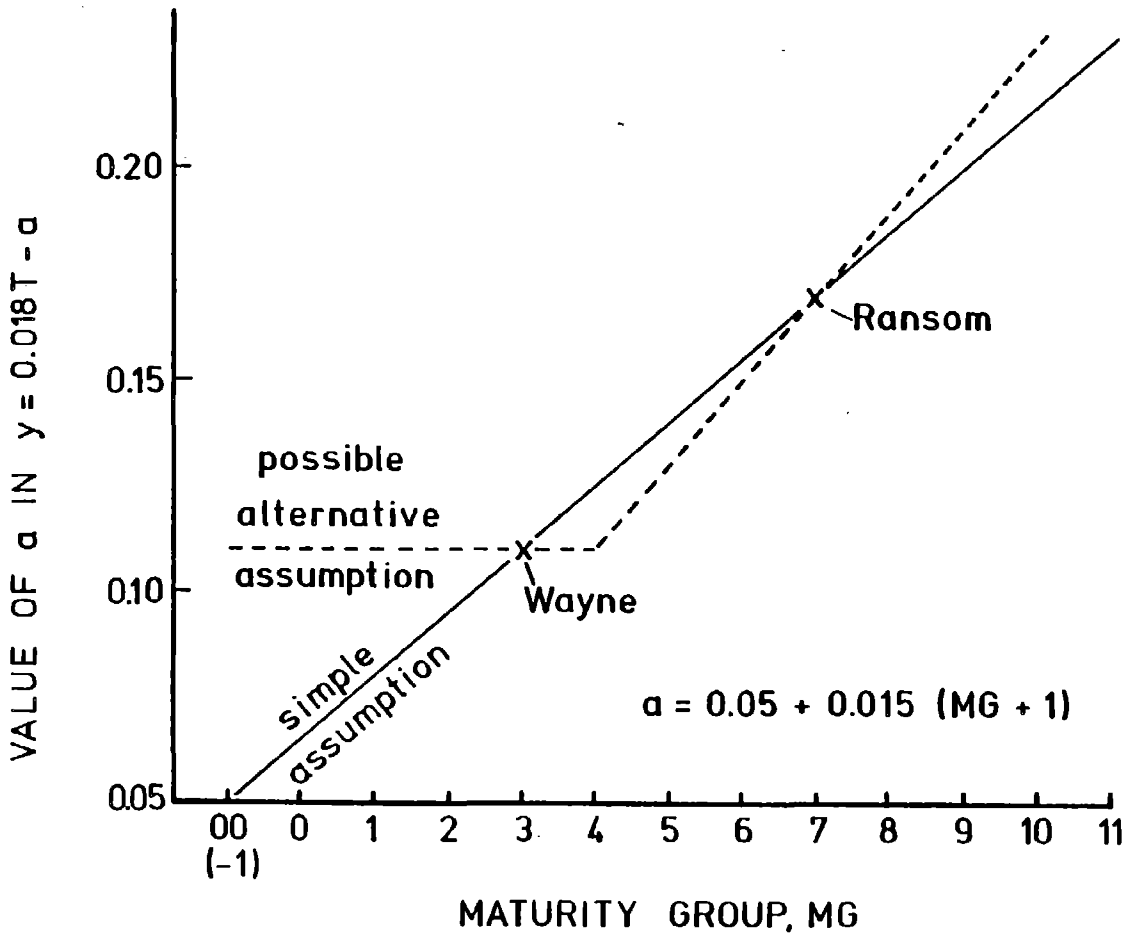


Fig. 27. Relationship between a parameter controlling trifoliolate expansion rate and maturity group number.

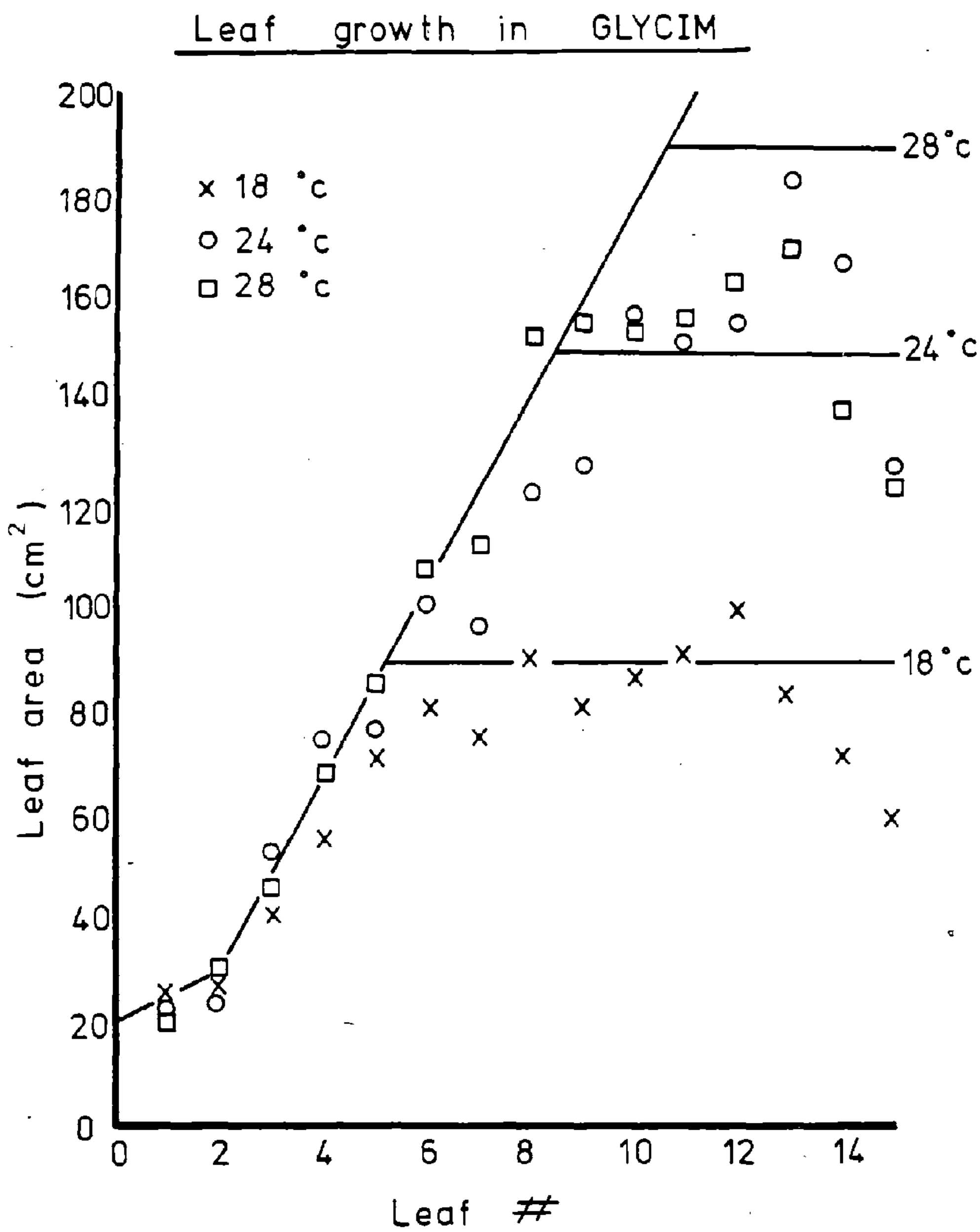


Fig. 28. Relationship between final leaf area and leaf position on the mainstem for soybean grown in different temperatures. Lines indicate model used in GLYCIM.

only 3 leaves and petioles were growing on each axis and the sigmoid growth curves were approximated with straight lines as shown in Figure 29.

7.04 The amount of dry matter required to add unit length to a petiole increases as the petiole length increases. A growing petiole must increase in diameter as well as length. Change in petiole dry weight is therefore a function of both existing petiole length and the change in length.

7.05 Stem and branch growth are related here to vegetative stage of growth (Figure 30) because our data did not reveal any model for the growth of individual internodes. Change in stem dry weight is, like petiole dry weight, a function of both existing length and the change in length.

7.06 Potential rates of change in reproductive stage are calculated from equations discussed in PLTMAP, note 12.03.

7.07 The carbon to dry weight conversion factor for seeds takes into account the oil content which rises to about 20% over the first 20 days of seed fill (Yazdi-Samadi, et al., 1977).

7.08 After the seeds have reached 80% of their final dry weight the seed fill rate is progressively reduced to zero as the seeds become completely full.

7.09 Seed fill rate is a function of temperature and the genetic potential of the variety. The equation used here is based on a small amount of data by Egli et al. (1978b). The genetic potential of varieties can be estimated from their relative seed sizes at harvest. It tends to decrease as

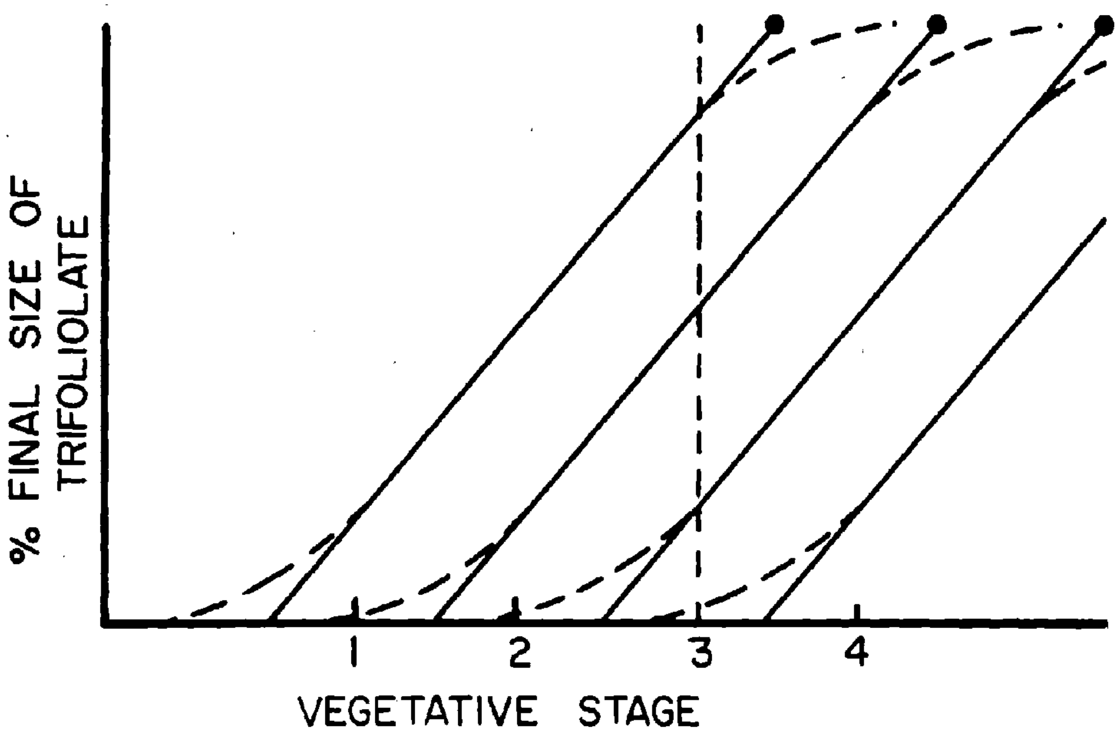


Fig. 29. Diagram showing how the sigmoid growth curve of soybean leaves is approximated by a straight line in GLYCIM and how, at any time, 3 leaves are growing.

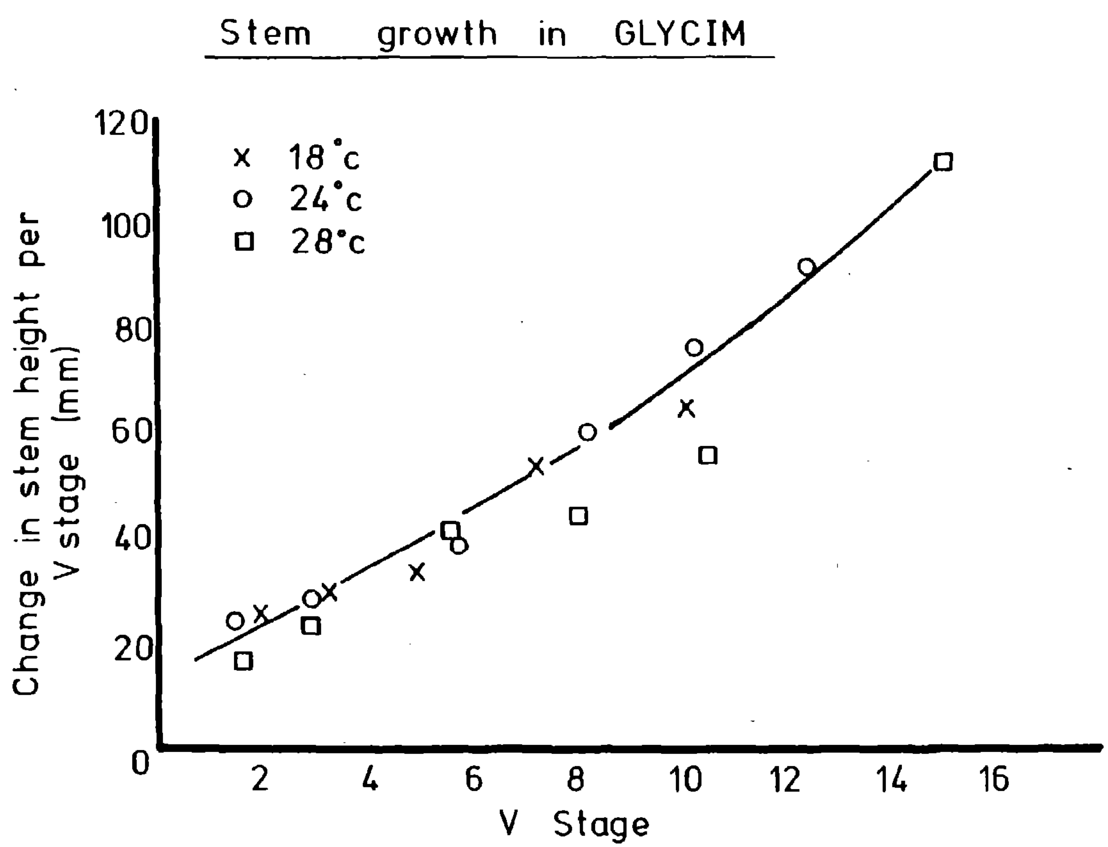


Fig. 30. Stem growth as a function of vegetative stage of growth in soybean.

maturity group number increases.

7.10 From R3 to R4, pods are developing on the plant but seed growth is minimal. In determinate cultivars, the growth of vegetative organs is much reduced or zero. The photosynthetic efficiency of the plant is near its peak and a lot of the carbon fixed is being stored. Despite this excess of carbon available, the plant is aborting pods that it could be supporting. However, the plant could not support these pods to full maturity because later on seed-fill places a much greater demand for carbon on the plant. Whatever the true mechanism involved, the plant appears to be matching its ability to support seeds to the number of pods, i.e. seed production sites, that it is developing. To do this, it seems to throw photosynthesis into high gear, develop as many pods as it could fill with seeds at that rate of carbon fixation, and, after feeding the pods, put the rest of the carbon into storage. The main storage sites appear to be stems and petioles.

7.11 Maximum and minimum rates of carbon supply needed to synthesize organ tissues have been estimated from maximum and minimum structural weights observed in our experiments.

RUTGRO

The subroutine RUTGRO is called once each period of the day and night. It calculates the potential root growth rate as limited by temperature, oxygen concentration, soil mechanical resistance and root water potential. These are assumed to act as independent limiting factors. The potential root growth rate is calculated for each soil cell in the profile where roots are already present. These rates are then sorted into descending order and the coordinates of the cells in this order are stored in an array. The array is later used in TRADE or NIGHT to grow roots preferentially in the cells that have the highest potential root growth rate.

```
---------------------------( RUTGRO )-----------------------------
THIS ROUTINE CALCULATES POTENTIAL ROOT GROWTH IN EACH SOIL CELL
ASSUMING THAT CARBOHDRATE AND NITROGEN ARE NOT LIMITING. IT ALLOWS
FOR TEMPERATURE LIMITATION AND APPROXIMATES THE SOIL MOISTURE
LIMITATION. IT "KILLS" SOME ROOTS IN SOIL CELLS WITH UNFAVORABLE
CONDITIONS.
------------------------------------------------------------------
at any time other than dawn, jump down
|
|  for each soil cell with roots, do the following:
|  |
|  |  CALCULATE SOIL MECHANICAL RESISTANCE TO ROOT GROWTH IN EACH
|  |      CELL AT DAWN.
|  |
|  |  CALCULATE ROOT TURGOR PRESSURE IN EACH CELL AT DAWN MINUS
|  |     THRESHOLD TURGOR FOR GROWTH.
V  |___________
for each soil cell with roots, do the following:
|
|  IN SOIL CELLS WITH LOW OXYGEN OR LOW WATER POTENTIAL THE
|     ROOTS DIE.
|
|  CALCULATE SOIL MECHANICAL RESISTANCE TO ROOT GROWTH IN EACH
|     CELL.
|
|  CALCULATE PROPORTIONIAL REDUCTION OF ROOT GROWTH CAUSED BY
|     MECHANICAL RESISTANCE AND SOIL WATER POTENTIAL ASSUMING ROOT
|     IS AT SAME POTENTIAL AS SOIL.
|
|  CALCULATE PROPORTIONAL REDUCTION OF ROOT GROWTH CAUSED BY
|     SOIL TEMPERATURE.
|
|  CALCULATE PROPORTIONIAL REDUCTION OF ROOT GROWTH CAUSED BY
|     PARTIAL PRESSURE OF SOIL OXYGEN.
|
|  CALCULATE PROPORTIONAL REDUCTION OF ROOT GROWTH FROM THE
|     PHYSICAL CAUSE THAT IS MOST LIMITING.
|
|  CALCULATE POTENTIAL RATE OF CHANGE IN ROOT DRY WEIGHT IN EACH
|     SOIL CELL AND HALF SOIL SLAB.
|___________

PLACE VALUES OF RGCF(L,K) IN ORDER HIGHEST TO LOWEST AND
   PUT L,K IN AN ARRAY PCRPAR(J) IN THE SAME ORDER.

RETURN
```

```
      SUBROUTINE RUTGRO
C     ------------------------( R U T G R O )------------------------
C     THIS ROUTINE CALCULATES POTENTIAL ROOT GROWTH IN EACH SOIL CELL
C     ASSUMING THAT CARBOHDRATE AND NITROGEN ARE NOT LIMITING. IT ALLOWS
C     FOR TEMPERATURE LIMITATION AND APPROXIMATES THE SOIL MOISTURE
C     LIMITATION. IT "KILLS" SOME ROOTS IN SOIL CELLS WITH UNFAVORABLE
C     CONDITIONS.
C     ---------------------------------------------------------------
       DIMENSION TPRD(20,10),PSIRD(20,10)
      INCLUDE 'COMMON.FOR'
C     ---------------------------------------------------------------
C     *** CALCULATE SOIL MECHANICAL RESISTANCE TO ROOT GROWTH IN EACH
C         CELL AT DAWN. ***
C
          NLRP1 = NLR + 1
          IF(NLRP1.GE.NL) NLRP1 = NL
          NKRP1 = NKR + 1
          IF(NKRP1.GE.NKH) NKRP1 = NKH
          LKRP1 = NLRP1 * NKRP1
      IF( ITIME.GT.1 ) GOTO 7777
        DO 10 L=1,NL
          DO 11 K=1,NKH
8.01        MECHR=0.5+(14.0-VH2OC(L,K)*100.0)*EXP(-8.08*(1.66-BDC(L,K)))
            IF(SITE.EQ.2)MECHR=(BDC(L,K)-1.48)*94.0
            IF( MECHR.LE.0.0 ) MECHR=0.0
C
8.02 C     *** CALCULATE ROOT TURGOR PRESSURE IN EACH CELL AT DAWN MINUS
C         THRESHOLD TURGOR FOR GROWTH. ***
C
8.03        TPRD(L,K)=2.0+(0.1*PSIS(L,K)+(0.7*MECHR))
            PSIRD(L,K)=PSIS(L,K)
   11   CONTINUE
   10   CONTINUE
C
C     *** IN SOIL CELLS WITH LOW OXYGEN OR LOW WATER POTENTIAL THE
C         ROOTS DIE. ***
C
 7777 PDWRS=0.0
      DO 5 L=1,NLRP1
        DO 6 K=1,NKRP1
8.04      IF( CXT(L,K).LE.0.01 ) GOTO 6666
          GOTO 5555
C
 6666     X1=RTWT(L,K)-PERTWT(L,K)
          DEADRN=DEADRN+(X1*2.5/100.*NRATIO)
          ROOTWT=ROOTWT-(X1*2.0*100.0/POPROW)
          VNC(L,K)=VNC(L,K)+(X1*25.0*NRATIO/D/W)
C         25=2.5/100*1000
          RTWT(L,K)=PERTWT(L,K)
C
C     *** CALCULATE SOIL MECHANICAL RESISTANCE TO ROOT GROWTH IN EACH
C         CELL. ***
C
 5555 MECHR=0.5+(14.0-VH2OC(L,K)*100.0)*EXP(-8.08*(1.66-BDC(L,K)))
      IF( MECHR.LE.0.0 ) MECHR=0.0
C
C     *** CALCULATE PROPORTIONIAL REDUCTION OF ROOT GROWTH CAUSED BY
C         MECHANICAL RESISTANCE AND SOIL WATER POTENTIAL ASSUMING ROOT
C         IS AT SAME POTENTIAL AS SOIL. ***
C
8.05  RGCF1=((TPRD(L,K)-MECHR)/2.0)-((PSIRD(L,K)-PSIS(L,K))/4.0)
      IF( RGCF1.LT.0.0 ) RGCF1=0.0
      IF( RGCF1.GT.1.0 ) RGCF1=1.0
```

```
      C
      C     *** CALCULATE PROPORTIONAL REDUCTION OF ROOT GROWTH CAUSED BY
      C         SOIL TEMPERATURE. ***
      C
8.06        IF( TS(L,K).GE.30.0 ) GOTO 9999
            IF(TS(L,K).LE.0.0) TS(L,K)=1.0
              RGCF2=EXP((ALOG(TS(L,K))-2.31)*3.658)/50
              GOTO 8888
       9999 RGCF2=EXP(15.385*(3.66-ALOG(TS(L,K))))/50
       8888 IF( RGCF2.LT.0.0 ) RGCF2=0.0
            IF( RGCF2.GT.1.0 ) RGCF2=1.0
      C
      C     *** CALCULATE PROPORTIONIAL REDUCTION OF ROOT GROWTH CAUSED BY
      C         PARTIAL PRESSURE OF SOIL OXYGEN. ***
      C
8.07            RGCF3=(CXT(L,K)-0.02)*7.14
                IF( RGCF3.LE.0.0 ) RGCF3=0.0
                IF( RGCF3.GE.1.0 ) RGCF3=1.0
      C
      C     *** CALCULATE PROPORTIONAL REDUCTION OF ROOT GROWTH FROM THE
      C         PHYSICAL CAUSE THAT IS MOST LIMITING. ***
      C
                RGCF(L,K)=MIN(RGCF1,RGCF2,RGCF3)
      C
      C     *** CALCULATE POTENTIAL RATE OF CHANGE IN ROOT DRY WEIGHT IN EACH
      C         SOIL CELL AND HALF SOIL SLAB. ***
      C
8.08            PDWR(L,K)=RTWT(L,K)*0.2*RGCF(L,K)/24.0
                PDWRS=PDWRS+PDWR(L,K)
          6   CONTINUE
            5 CONTINUE
      C
      C     *** PLACE VALUES OF RGCF(L,K) IN ORDER HIGHEST TO LOWEST AND
      C         PUT L,K IN AN ARRAY PCRPAR(J) IN THE SAME ORDER. ***
      C
8.09        DO 30 I=1,NLRP1
              DO 31 J=1,NKRP1
                KOUNT=1
                DO 41 II=1,NLRP1
                  DO 40 JJ=1,NKRP1
                    IF( I.EQ.II.AND.J.EQ.JJ ) GOTO 40
                    IF( RGCF(I,J).LE.RGCF(II,JJ) ) KOUNT=KOUNT+1
                    IF( RGCF(I,J).NE.RGCF(II,JJ) ) GOTO 40
                    IF( (I*100+J).LT.(II*100+JJ) ) KOUNT=KOUNT-1
         40       CONTINUE
         41       CONTINUE
                PCRPAR(KOUNT)=I*1000+J
         31   CONTINUE
         30 CONTINUE
            RETURN
            END
```

Notes for RUTGRO

8.01 The calculation of soil mechanical resistance to root growth is based on curves fitted to the data of Greacen and Oh (1972) for "Parafield" loam. When time is available this will be replaced with the RIMPED subroutine by Frank Whisler, which contains data on several different soil types.

8.02 The ideas about how roots osmoregulate against changes in soil water potential and soil mechanical resistance, and much of the data for modeling them, come from Greacen and Oh (1972). They showed that the roots of germinating pea seeds osmoregulated with about 90% efficiency against soil water potential changes and with about 70% efficiency against changes in soil mechanical resistance (Figure 31). Sharp and Davies (1979) also claim that roots osmoregulate almost completely against soil water potential changes. However, it seems unlikely that roots can make these adjustments as rapidly as soil water potential can change during the day. Therefore, it is assumed in the model that root osmoregulation is complete at dawn but operates at half that efficiency during the day. It is further assumed that the expanding cells in the root tip, being in intimate contact with the soil, will have a water potential equal to that of the soil. Other hypotheses can be accommodated if data become available.

8.03 Based on the Greacen and Oh data, the root osmotic potential as dawn,

$$\begin{aligned} ORD &= 0.9*PSISD - 6.5 - 0.7*(MECHRD - 1.5) \\ &= 0.9*PSISD - 5.5 - 0.7*MECHRD \end{aligned}$$

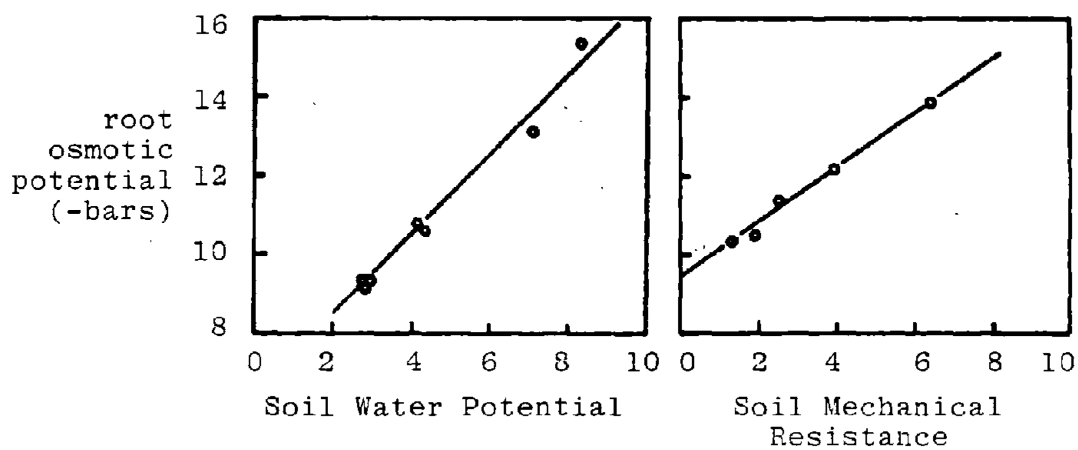


Fig. 31. Osmoregulation by germinating pea roots against changes in soil water potential and soil mechanical resistance (Greacen and Oh, 1972).

where PSISD = soil water potential at dawn and MECHRD = soil mechanical resistance at dawn. Now root turgor pressure at dawn,

TPRD = PSIRD - ORD,

where PSIRD = root water potential at dawn. Since root and soil water potentials are equal

TPRD = 0.1*PSISD + 5.5 + 0.7*MECHRD

Subtracting the threshold turgor pressure at which root growth starts (= 3.5 bars), we get

TPRD = 0.1*PSISD + 2.0 + 0.7*MECHRD

8.04 Roots in less than 1 % oxygen die and in this model a soil water potential of -15 bars (arbitrarily chosen) also kills them. Of the original root weight, 20% remains as indestructible pipeline roots from which regeneration can occur.

8.05 The turgor pressure available to expand the root after overcoming soil mechanical resistance,

EXTPRD = TPRD - MECHRD at dawn and, at any other time of day (dropping the last D in each variable name),

EXTPR = TPR - MECHR

The threshold pressure for root growth to occur is 3.5 bars and the maximum value that EXTPR can attain in these equations is 5.5 bars. Assuming root growth is proportional to EXTPR over this range, the proportion of maximum root growth possible at the prevailing soil water potential and soil mechanical resistance,

RGCF1 = (EXTPR - 3.5)/(5.5 - 3.5)

= (TPR - MECHR - 3.5)/2

If the root osmoregulates with half efficiency during the day, change in root turgor pressure will equal half of any change in root water potential. That is,

TPR = TPRD - ((PSIRD - PSIR)/2)

where PSIR = root water potential during the day.

Substituting in the earlier equation and remembering that PSIR = PSIS where PSIS = soil water potential during the day.

RGCIF1 = ((TPRD - 3.5)/2) - (MECHR/2) - ((PSIRD - PSIS)/4)

= ((TPRD - MECHR)/2) - ((PSIRD - PSIS)/4)

In the code, TPRD, and PSIRD are calculated at dawn and then used for the rest of the day to calculate RGCF1.

8.06 Root response to temperature is modeled on data for excised tomato roots by White (1937). See Fig. 32. This is the only data we know where we can be sure that the shoot was not controlling root growth.

8.07 Root response to oxygen concentration is based on the data of Eavis et al. (1971, their Fig. 2, curve A). See Figure 33. This assumes that oxygen transfer from shoot to root is negligible in mature plants.

8.08 Change in root weight in a given soil cell is a function of existing root weight in that cell. The function is arbitrarily chosen to obtain a realistic root distribution. Root growth is controlled ultimately by the shoot and not by this function.

8.09 The root growth correction factors are sorted into an array so that roots can be grown (in later subroutines)

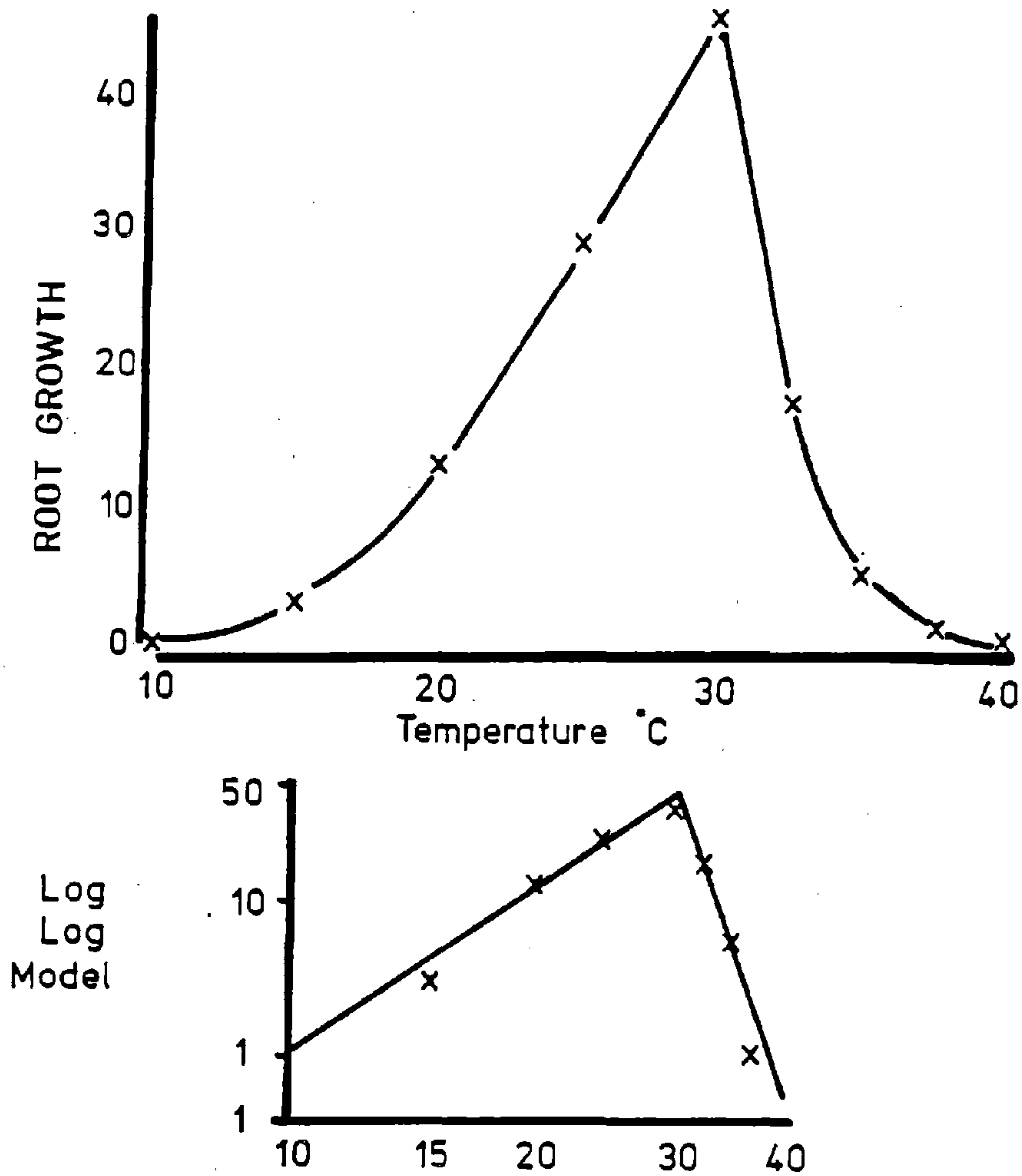


Fig. 32. Relationship between the growth of excised tomato roots and temperature. Data of White, 1937.

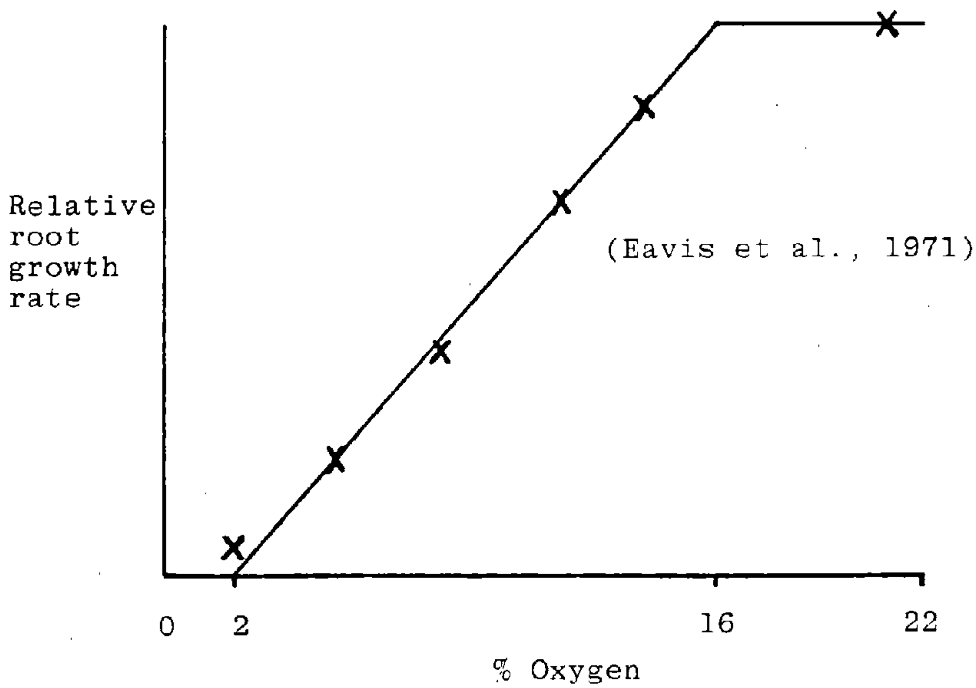


Fig. 33. Relationship between root growth rate and oxygen concentration used in GLYCIM. Data of Eavis et al., (1971).

preferentially in the most favorable cells.

NITRO

The subroutine NITRO is called once for each period of the day and night. It calculates the total amount of nitrate nitrogen in the plant and the nitrate reductase activity in the plant leaves. Whichever of these is limiting determines the amount of nitrate converted to ammonium nitrogen and thus made available to the plant. The energy cost of this reduction is subtracted from the translocated carbon. Nitrogen from this source plus nitrogen fixed in the nodules is added to the nitrogen already in the plant and the nitrogen lost in fallen plant parts is subtracted. The resulting figure is the supply of nitrogen in the plant. This is compared with the demand for nitrogen which is the maximum amount of nitrogen the plant could contain. The amount of carbon partitioned to the nodules is proportional to the difference between supply and demand. The nitrogen supply/demand ratio is used as a measure of the relative nitrogen content of all organs on the plant.

Leaves and petioles may be abscised in NITRO for any of the following reasons: (a) A branch developing in the axil of the leaf has reached a certain age. (b) During early vegetative growth, the nitrogen supply/demand ratio falls below a set level for five days. (c) Leaves at the bottom of the canopy are no longer self-supporting. (d) After seed growth has started (R5) the nitrogen supply/demand ratio falls below a set level. The oldest leaves abscise first and leaves of the same age abscise together whether they are on the mainstem or branches.

--------------------------(NITRO)------------------------------

THIS ROUTINE CALCULATES THE SUPPLY OF AND THE DEMAND FOR NITROGEN IN THE WHOLE PLANT AND DETERMINES AN "N" STRESS FACTOR WHICH IS USED TO PARTITION CARBON TO THE NODULES.

NITRATE REDUCTASE ACTIVITY AND THE ACTUAL AMOUNT OF NITRATE REDUCED BY IT ARE CALCULATED.

CALCULATE TOTAL NO3 CONTENT OF PLANT

if there is no NO3 in the plant or if reproductive stage > or = 4, jump down

IF THERE IS NO3 IN THE PLANT, CALCULATE THE NITRATE REDUCTASE ACTIVITY FOR THE WHOLE PLANT PRIOR TO R4.

NO3 REDUCTION IS LIMITED EITHER BY NO3 UPTAKE OR BY REDUCTASE ACTIVITY.

CALCULATE THE COST OF THIS NO3 REDUCTION AS A PRIMARY CHARGE ON PHOTOSYNTHESIS.

THE RATIO OF THE ACTUAL NITROGEN CONTENT OF THE PLANT TO ITS MAXIMUM POSSIBLE NITROGEN CONTENT IS USED TO CALCULATE A NITROGEN STRESS FACTOR.

CALCULATE TOTAL N IN REDUCED FORM IN THE PLANT.

RESET ACCUMULATORS

CALCULATE N REQUIRED BY PODS AND SEEDS

CALCULATE TOTAL N REQUIRED BY PLANT AT MAXIMUM N CONTENT

CALCULATE N SUPPLY/DEMAND FOR VEGETATIVE PARTS

CALCULATE AMOUNT OF CARBON PARTITIONED TO NODULES

IF THE LOWER LEAVES ARE PARASITIC OR PLANT IS SHORT OF N OR A BRANCH IS GROWING IN THE LEAF AXIL OR THERE IS TOO LITTLE N TO GROW SEEDS AND MAINTAIN HIGH LEAF N, ABSCISE LEAVES AND PETIOLES STARTING FROM BASE OF PLANT.

---

at any time other than dawn or if vegetative stage < 5, jump down

IS A BRANCH GROWING IN THE LEAF AXIL ?

if a branch growing in the leaf axil has reached a critical stage, jump down

A B

A B

IS THE PLANT SHORT OF NITROGEN ?

if the plant is short of nitrogen and there are more than 5 leaves on the mainstem, jump down

ARE THE LOWER LEAVES PARASITIC ?

if the lower leaves are parasitic and reproductive stage > 5, jump down

IS THERE TOO LITTLE N TO GROW SEEDS AND MAINTAIN LEAF N ?

if the plant is short of nitrogen and reproductive stage > 5, jump down

jump down

if the unifoliolates have already abscised, jump down

ABSCISE THE UNIFOLIOLATES

jump down

ABSCISE TRIFOLIOLATES ON THE MAINSTEM

if the branch trifoliolates are too young to die, jump down

ABSCISE TRIFOLIOLATES ON THE BRANCHES

if the plant is still short of nitrogen, jump back up

if there are still some leaves on the plant, jump down

WHEN ALL LEAVES HAVE ABSCISED, GROWTH STOPS.

RETURN

```
      SUBROUTINE NITRO
C     --------------------------( N I T R O )--------------------------
C     THIS ROUTINE CALCULATES THE SUPPLY OF AND THE DEMAND FOR NITROGEN
C     IN THE WHOLE PLANT AND DETERMINES AN "N" STRESS FACTOR WHICH IS
C     USED TO PARTITION CARBON TO THE NODULES.
C     -----------------------------------------------------------------
      INCLUDE 'COMMON.FOR'
      DATA LONDAY/0/
C     -----------------------------------------------------------------
C     *** NITRATE REDUCTASE ACTIVITY AND THE ACTUAL AMOUNT OF NITRATE
C         REDUCED BY IT ARE CALCULATED. ***
C     -----------------------------------------------------------------
C     *** CALCULATE TOTAL NO3 CONTENT OF PLANT ***
C
9.01  TOTNO3=REMNO3+SUPNO3
         X11 = 0.0
C
C     *** IF THERE IS NO3 IN THE PLANT, CALCULATE THE NITRATE REDUCTASE
C         ACTIVITY FOR THE WHOLE PLANT PRIOR TO R4. ***
C
9.02  IF((TOTNO3.LE.0.0).OR.(RSTAGE.GE.4.0)) GOTO 5555
        RASEN=0.0
        SUMLW=0.0
        BRASEN=4.0-RSTAGE
        IF( BRASEN.LE.0.0 ) BRASEN=0.0
        IF( RSTAGE.LE.1.0 ) BRASEN=3.0
        ITRIF=IFIX(VSTAGE+0.5)
C
        IF(ITRIF.LE.0) GO TO 5555
 8888   DRASEN=4.0*(3.0-(VSTAGE+0.5-ITRIF))
        IF( BRASEN.GT.DRASEN ) GOTO 9999
          RASEN=RASEN+(DRASEN*MLEAFW(ITRIF)*1.4E-4)
          SUMLW=SUMLW+MLEAFW(ITRIF)
          ITRIF=ITRIF-1
          IF( ITRIF.LE.0 ) GOTO 9999
        GOTO 8888
C
 9999   IF(NBRNCH.LE.0) GO TO 9955
        DO 10 J=1,NBRNCH
          I=IFIX(TB(J)+0.5)
          IF( I.LE.0 ) GOTO 10
 6666       DRASEN=4.0*(3.0-(TB(J)+0.5-I))
            IF( BRASEN.GT.DRASEN ) GOTO 10
              RASEN=RASEN+(DRASEN*BLEAFW(I,J)*1.4E-4)
              SUMLW=SUMLW+BLEAFW(I,J)
              I=I-1
              IF( I.LE.0 ) GOTO 10
            GOTO 6666
   10   CONTINUE
 9955 CONTINUE
C
        RASEN=RASEN+(BRASEN*1.4E-4*(LEAFWT-SUMLW))
        GOTO 9922
 7777 RASEN = 0.
C
C     *** NO3 REDUCTION IS LIMITED EITHER BY NO3 UPTAKE OR BY REDUCTASE
C         ACTIVITY. ***
C
 9922 X10=RASEN*PERIOD
      IF( TOTNO3.GE.X10 ) GOTO 4444
        FIXNO3=TOTNO3
        REMNO3=0.0
      GOTO 3333
```

```
C
 4444 FIXNO3=X10
      REMNO3=TOTNO3-X10
C
C     *** CALCULATE THE COST OF THIS NO3 REDUCTION AS A PRIMARY CHARGE
C         ON PHOTOSYNTHESIS. ***
C
9.03
 3333 X11=FIXNO3*2.0*SCF/PERIOD
      IF(NYTE.EQ.1)X11=X11/SCF
      TRANSC=TRANSC-X11
      IF(TRANSC.GE.0.) GO TO 9911
      FIXNO3=FIXNO3*(X11+TRANSC)/X11
      REMNO3=REMNO3+(FIXNO3*ABS(TRANSC)/X11)
      X11 = X11+TRANSC
      TRANSC=0.
      GO TO 9911
C
C     ----------------------------------------------------------------
C     *** THE RATIO OF THE ACTUAL NITROGEN CONTENT OF THE PLANT TO ITS
C         MAXIMUM POSSIBLE NITROGEN CONTENT IS USED TO CALCULATE A
C         NITROGEN STRESS FACTOR. ***
C     ----------------------------------------------------------------
C     *** CALCULATE TOTAL N IN REDUCED FORM IN THE PLANT. ***
C
 5555 FIXNO3=0.0
 9911 PLANTN=PLANTN+NODN+FIXNO3-ABSLPN-DEADRN-ABSPON
        USEDCS = USEDCS+(X11*PERIOD)
C
C     *** RESET ACCUMULATORS ***
C
      NODN=0.0
      FIXNO3=0.0
      ABSLPN=0.0
      DEADRN=0.0
      ABSPON=0.0
C
C     *** CALCULATE N REQUIRED BY PODS AND SEEDS ***
C
      PODN = PODWT*0.04
      SEEDN=SEEDWT*0.065
      X12 = PODN + SEEDN
C
C     *** CALCULATE TOTAL N REQUIRED BY PLANT AT MAXIMUM N CONTENT ***
C
9.04
      REQN=((((FLWRWT+LEAFWT)*5.0)+((STEMWT+PETWT)*2.0) +
     &          (ROOTWT*2.5))/100.)+PODN+SEEDN
C
C     *** CALCULATE N SUPPLY/DEMAND FOR VEGETATIVE PARTS ***
C
9.05
      NRATIO=(PLANTN-X12)/(REQN-X12)
      IF(NRATIO.LE.0.1) NRATIO=0.1
      NRATS = NRATS + (NRATIO*PERIOD)
C
C     *** CALCULATE HOW MUCH CARBON IS PARTITIONED TO NODULES ***
C
9.06
       PDNODC=(REQN-PLANTN)*PRM(25)
       IF(PDNODC.LE.0.0)PDNODC=0.0
C
C     ----------------------------------------------------------------
C     *** IF THE LOWER LEAVES ARE PARASITIC OR PLANT IS SHORT OF N
C         OR A BRANCH IS GROWING IN THE LEAF AXIL OR THERE IS TOO
C         LITTLE N TO GROW SEEDS AND MAINTAIN HIGH LEAF N, ABSCISE
C         LEAVES AND PETIOLES STARTING FROM BASE OF PLANT. ***
```

```
      C       -----------------------------------------------------------------
9.07          IF(ITIME.GT.1.OR.VSTAGE.LT.5.) GO TO 9944
              DEFICN = 0.0
      C
      C       *** IS A BRANCH GROWING IN THE LEAF AXIL ? ***
      C
9.08          IF(ABSCIS.EQ.1)THEN
              ABSCIS=0
        4     FORMAT(1X,3HDAY,I3,5H LEAF,I3,12H BRANCH PUSH)
              WRITE(2,4)IDAY,ILOW
              GO TO 8866
              END IF
      C
      C       *** IS THE PLANT SHORT OF NITROGEN ? ***
      C
              AVNRAT = NRATS/24.0
              NRATS = 0.0
              IF(RSTAGE.GT.1.0) GO TO 8855
9.09          IF(AVNRAT.LT.PRM(1).AND.(VSTAGE-5.0).GT.ILOW)
             &            LONDAY = LONDAY + 1
              IF(LONDAY.LT.5) GO TO 8855
              LONDAY = 0
        1     FORMAT(1X,3HDAY,I3,5H LEAF,I3,14H MILD N STRESS)
              WRITE(2,1) IDAY,ILOW
              GO TO 8866
      C
      C       *** ARE THE LOWER LEAVES PARASITIC ? ***
      C
9.10   8855 IF(DROP.EQ.0.OR.RSTAGE.LT.5.0) GO TO 8844
        2     FORMAT(1X,3HDAY,I3,5H LEAF,I3,10H PARASITIC)
              WRITE(2,2) IDAY,ILOW
              GO TO 8866
      C
      C       *** IS THERE TOO LITTLE N TO GROW SEEDS AND MAINTAIN LEAF N ? ***
      C
9.11   8844 IF( AVNRAT.GE.PRM(6).OR.RSTAGE.LT.5.0) GOTO 9944
        3     FORMAT(1X,3HDAY,I3,5H LEAF,I3,21H PLUS OTHERS N STRESS)
              WRITE(2,3) IDAY,ILOW
              DEFICN=(NRATIO*(REQN-PODN-SEEDN))-(PLANTN-PODN-SEEDN)
       8866     NEXLP=0.0
      C
      C     *** ABSCISE THE UNIFOLIOLATES ***
      C
                IF( ILOW.GE.1 ) GOTO 9977
9.12              NEXLP=(((ULEAFW+0.02)*5.0)+(UPETW*2.0))*(NRATIO-0.4)/100
                  ABSLPN=(((ULEAFW+0.02)*2.0)+(UPETW*0.8))/100
                  LEAFWT=LEAFWT-ULEAFW-0.02
                  PETWT=PETWT-UPETW
                  LAREAM=LAREAM-ULAREA-4.00
                  ULEAFW=0.0
                  UPETW=0.0
                  ULAREA=0.0
                  UPETL=0
                  ILOW=ILOW+1
                  IF(DROP.EQ.1) GO TO 1111
                  IF( NEXLP.GE.DEFICN ) GOTO 1111
      C
      C      *** ABSCISE TRIFOLIOLATES ON THE MAINSTEM ***
      C
       9977   IF( ILOW.GT.NTOPLF ) GOTO 9966
                  I=ILOW
                  NEXLP=NEXLP+(((MLEAFW(I)*5.0)+(MPETW(I)*2.0))*(NRATIO-0.4)
             &           /100)
```

```
            ABSLPN=ABSLPN+(((MLEAFW(I)*2.0)+(MPETW(I)*0.8))/100)
            LEAFWT=LEAFWT-MLEAFW(I)
            PETWT=PETWT-MPETW(I)
            LAREAM=LAREAM-MLAREA(I)
            MLEAFW(I)=0.0
            MPETW(I)=0.0
            MLAREA(I)=0.0
            MPETL(I)=0.0
            ILOW=ILOW+1
C
C      *** ABSCISE TRIFOLIOLATES ON THE BRANCHES ***
C
 9966        IF(NBRNCH.LE.0) GO TO 9988
               DO 40 J = 1,NBRNCH
9.13             IF(INITBR(J).GE.(ILOW-2)) GO TO 9988
                 IF(ILOWB(J).GT.(IFIX(TBMAH(J)-0.5))) GO TO 40
                 JJ=ILOWB(J)
                 NEXLP=NEXLP+(((BLEAFW(JJ,J)*5.0)+(BPETW(JJ,J)*2.0))
     &                *(NRATIO-0.4)/100)
                 ABSLPN=ABSLPN+(((BLEAFW(JJ,J)*2.0)+(BPETW(JJ,J)*0.8)) /
     &                  100)
                 LEAFWT=LEAFWT-BLEAFW(JJ,J)
                 PETWT=PETWT-BPETW(JJ,J)
                 LAREAB=LAREAB-BLAREA(JJ,J)
                 BLEAFW(JJ,J)=0.0
                 BPETW(JJ,J)=0.0
                 BLAREA(JJ,J)=0.0
                 BPETL(JJ,J)=0.0
                 ILOWB(J) = ILOWB(J)+1
   40          CONTINUE
                 X13 = ILOWB(NBRNCH)
                 IF(BLAREA(X13,NBRNCH).LE.0.0) GO TO 2222
C
 9988    IF(DROP.EQ.1) GO TO 1111
            IF( NEXLP.LT.DEFICN ) GO TO 9977
 1111       LAREAT = LAREAM+LAREAB
            DROP  =  0
            GOTO 9944
C
C      *** WHEN ALL LEAVES HAVE ABSCISED, GROWTH STOPS. ***
C
 2222     MATURE=1
          LAREAM = 0.
          LAREAB = 0.
          LAREAT=0.0
          LEAFWT=0.0
          PETWT=0.0
 9944 RETURN
      END
```

Notes for NITRO

9.01 In this model, plants are allowed to take up as much nitrate as they can but only reduced nitrogen is considered to be available to them.

9.02 The equations to calculate nitrate reductase activity are based on data for soil-grown plants by Harper and Hageman (1972). Young leaves have the greatest activity per unit dry weight but their dry weight is small. Activity in mature leaves gradually decreases after R1 until it is negligible by R4. Numerous studies show that soybeans fail to respond to nitrogen salts applied to either the leaves or the soil in the reproductive phase of growth.

9.03 The cost of reducing nitrate to ammonium is estimated from Penning de Vries (1975) to be about 2 gms of carbon per gm of nitrogen. The cost of converting the ammonium to protein has already been charged in the carbon to dry matter conversion factors in SHTGRO.

9.04 The maximum nitrogen contents of various organs are based on data by Streeter (1972 and 1978), Sinclair and de Wit (1976), Brevadan et al. (1977 and 1978), Bhangoo and Albritton (1976), Pal and Saxena (1976), Latimore et al. (1977), Egli et al. (1978a), Hanway and Weber (1971) and Hardy and Havelka (1976).

9.05 Nitrogen is very mobile in plants and seems to be shared among plant parts according to their needs (Boote et al., 1978). Therefore the nitrogen supply/demand ratio is also the proportion of the maximum nitrogen content present in all organs.

9.06 The rate at which carbon is supplied to the nodules is a function of the absolute nitrogen deficit in the plant. It is sufficient to remedy the deficit in less than an hour if the plant already has enough active nodules.

9.07 Leaves and petioles are abscissed in this model for one of four reasons given below.

9.08 When a branch developing in the axil of a leaf reaches a certain age, the leaf is abscissed. The test of branch age is made in NIGHT, line 198 and the flag ABSCIS is set there.

9.09 If NRATIO falls below a certain level for five days during early vegetative growth, leaves on the mainstem are abscissed providing at least 5 remain.

9.10 Leaves at the bottom of the canopy that are so shaded that they are no longer self-supporting may be abscised. The test for this condition is made and the flag DROP is set in NIGHT, line 191.

9.11 After R5, if the seeds extract nitrogen from the leaves and NRATIO falls below a certain level some lower leaves are abscised. The nitrogen extracted from these leaves before they fall increases NRATIO again.

9.12 Data on the nitrogen content of fallen leaves and petioles comes from Egli et al. (1979a).

9.13 Leaves started abscising from the branches when leaves of a similar age on the mainstem abscise.

TRADE

The subroutine TRADE is called once for each period of the day. It maintains a functional balance between root and shoot growth in the following way. The leaf water potential which would just prevent all shoot growth and the water potentials corresponding to turgor pressures of 0 and 2 bars are calculated. Possible water uptake rates at these key leaf water potentials are calculated and compared with the potential transpiration rate calculated in EVTR. If the potential transpiration rate is below the water uptake rate at a leaf turgor pressure of 2 bars, it is assumed that uptake equals transpiration. Using the relationship between leaf water potential and water uptake rate previously determined, it is possible to estimate leaf water potential and the relative amounts of root and shoot growth that will occur. If the potential transpiration rate is above the water uptake rate at a leaf turgor pressure of 2 bars, stomatal closure occurs. The stomata are fully closed at a turgor pressure of 0 bars in the model. This stomatal closure reduces actual transpiration rate below the potential rate until the uptake rate equals actual transpiration rate. It also reduces net photosynthesis rate in PNET.

Starting with the soil cell with the highest potential root growth rate (calculated in RUTGRO), TRADE grows roots in the cells until all the carbon allocated to root growth is used up. Then the root growth in each cell is redistributed between that cell and the four adjacent cells according to their suitability for root growth plus a weighting factor for geotropism. Water uptake from each of the soil cells is also calculated and, assuming that uptake rate describes a half sine wave during the day, water uptake from each cell for the next period of the day is predicted.

----------------------------(TRADE)----------------------------

THIS ROUTINE MAINTAINS A FUNCTIONAL BALANCE BETWEEN ROOT AND SHOOT BY GROWING ROOT AS NECESSARY TO MEET TRANSPIRATION DEMAND. NEW ROOT HAS LESS RESISTANCE TO WATER UPTAKE. SINCE TRANSPIRATION RATE, PHOTOSYNTHESIS RATE AND HENCE POTENTIAL ROOT GROWTH RATE ARE ROUGHLY PROPORTIONAL, LEAF WATER POTENTIAL IS NOT AFFECTED MUCH BY TRANSPIRATION RATE. HOWEVER, LEAF WATER POTENTIAL MUST FALL AS TRANSPIRATION INCREASES BECAUSE THE THRESHOLD TURGOR PRESSURE FOR STOPPING SHOOT GROWTH AND DIVERTING CARBOHYDRATES TO ROOTS DECREASES WITH TIME. HENCE THE ROUTINE ALSO PREDICTS CHANGES IN TOTAL POTENTIAL, OSMOTIC POTENTIAL AND TURGOR PRESSURE IN LEAVES DURING THE DAY.

THE THRESHOLD LEAF WATER POTENTIAL AND TURGOR PRESSURE THAT WILL JUST PREVENT ALL SHOOT EXPANSION ARE CALCULATED.

---

at any time other than dawn, jump down

CALCULATE THE THRESHOLD LEAF WATER POTENTIAL AND TURGOR PRESSURE THAT WILL JUST PREVENT ALL SHOOT GROWTH AT ITIME=1

CALCULATE THE SAME VARIABLES FOR ANY OTHER ITIME

WATER UPTAKE BY YOUNG AND OLD ROOTS UNDER VARIOUS CIRCUMSTANCES IS CALCULATED.

---

for each soil cell with roots, do the following:

CALCULATE RADIUS OF SOIL CYLINDER THROUGH WHICH WATER MUST TRAVEL TO REACH ROOTS. THIS IS APPROXIMATED BY A SIMPLE FUNCTION OF SOIL WATER POTENTIAL TO AVOID ITERATION

CALCULATE RESISTANCE TO WATER FLOW IN SOIL CELL

CALCULATE RATE OF WATER UPTAKE BY EXISTING ROOTS IF LEAF WATER POTENTIAL IS AT THE THRESHOLD

ALLOCATE TRANSLOCATED CARBON TO FRUIT, NODULES AND VEGETATIVE SHOOT PLUS ROOT

CALCULATE RATE AT WHICH CARBON IS SUPPLIED TO THE GROWING ROOTS IN A HALF SOIL SLAB WHEN:
(1) NO CARBON IS TRANSLOCATED FROM THE SHOOT
(2) SHOOTS DO NOT GROW AND ROOTS GET ALL TRANSLOCATED CARBON

CALCULATE A CARBON TO DRY WEIGHT CONVERSION FACTOR FOR THE ROOTS.

for each soil cell with roots, do the following:

CALCULATE GROWTH BY NEW ROOTS IF ROOTS GREW AT POTENTIAL RATE DURING PAST PERIOD. START WITH SOIL CELL WHERE CONDITIONS ARE MOST FAVORABLE FOR GROWTH.

A

A

DETERMINE TOTAL WATER UPTAKE BY ALL NEW ROOTS AND WEIGHTED AVERAGE SOIL WATER POTENTIAL WHEN:
I. ROOTS GET ONLY THE CARBON LEFT OVER FROM SHOOT GROWTH OR
II. ROOTS GET ALL THE CARBON TRANSLOCATED.

CALCULATE WATER UPTAKE RATE IF LEAF WATER POTENTIAL HAS NOT CHANGED IN PAST PERIOD.

CALCULATE WATER UPTAKE RATE IF LEAF TURGOR PRESSURE = 2 BARS OR = 0 BARS.

DETERMINE HOW POTENTIAL TRANSPIRATION RATE IS RELATED TO THE VARIOUS POTENTIAL WATER UPTAKE RATES AND GO TO THE APPROPRIATE SECTION OF THE CODE. CALCULATE LEAF WATER POTENTIAL,STOMATAL CONDUCTANCE,THE PROPORTION OF TIME FOR WHICH THE SHOOT GROWS, THE AMOUNT OF CARBON ACTUALLY NEEDED FOR ROOT GROWTH AND THE AMOUNT OF OSMOREGULATION THAT WILL OCCUR.

DETERMINE HOW POTENTIAL TRANSPIRATION RATE IS RELATED TO THE VARIOUS POTENTIAL WATER UPTAKE RATES.

if potential transpiration rate < water uptake rate when leaf water potential has not changed in past period, jump down

if potential transpiration rate < water uptake rate when leaf water potential just prevents all shoot growth, jump down

if potential transpiration rate < water uptake rate when leaf turgor pressure = 2 bars, jump down

SINCE POTENTIAL TRANSPIRATION RATE IS GREATER THAN THE POSSIBLE WATER UPTAKE RATE WHEN LEAF TURGOR PRESSURE IS 2 BARS STOMATAL CLOSURE WILL DECREASE TRANSPIRATION RATE.

CALCULATE LEAF WATER POTENTIAL AND STOMATAL CLOSURE FACTOR.

jump down

SINCE POTENTIAL TRANSPIRATION RATE IS BETWEEN POSSIBLE WATER UPTAKE RATES WHEN LEAF TURGOR PRESSURE IS AT THRESHOLD FOR GROWTH AND AT 2 BARS, CALCULATE LEAF WATER POTENTIAL.

FOR ALL LEAF WATER POTENTIALS ABOVE THE THRESHOLD, THE SHOOT DOES NOT GROW.

jump down

SINCE POTENTIAL TRANSPIRATION RATE IS SUCH THAT SOME GROWTH OF BOTH ROOTS AND SHOOTS IS PERMITTED, CALCULATE APPROXIMATE LEAF WATER POTENTIAL AND PERCENTAGE OF TIME FOR WHICH SHOOT GROWS.

B C

B C

jump down

SINCE POTENTIAL TRANSPIRATION RATE IS SUCH THAT NO ROOT
GROWTH IS NECESSARY, CALCULATE LEAF WATER POTENTIAL.

LEAF WATER POTENTIAL CANNOT RISE ABOVE ITS DAWN VALUE

CALCULATE LEAF TURGOR PRESSURE

IF TURGOR PRESSURE IS LESS THAN 5 BARS SHOOT GROWTH
POTENTIAL IS REDUCED.

IF TURGOR PRESSURE HAS INCREASED AND IS ABOVE 2.0 BARS,
SHOOT GROWTH POTENTIAL IS INCREASED.

CALCULATE AN INTEGRATION FACTOR FOR PREDICTING WATER UPTAKE
FROM EACH SOIL CELL FOR THE COMING PERIOD OF THE DAY ASSUMING
UPTAKE VARIES AS A HALF SINE WAVE.

CALCULATE CHANGE IN ROOT DRY WEIGHT AND LENGTH, AND WATER
UPTAKE FROM EACH SOIL CELL FOR PAST PERIOD, AND PREDICT
WATER UPTAKE FOR THE NEXT PERIOD OF THE DAY.

---

for each soil cell with roots, do the following:

GROW ROOTS IN THE MOST FAVORABLE CELLS UNTIL CARBON AVAILABLE
IS USED,AND CALCULATE CURRENT WATER UPTAKE RATE FOR EACH CELL.

CALCULATE MEAN ROOT GROWTH RATE AND MEAN WATER UPTAKE RATE
FOR THE PAST PERIOD AND PREDICT WATER UPTAKE FOR THE NEXT
PERIOD OF THE DAY.

REDISTRIBUTE ROOT GROWTH IN THE SOIL PROFILE AND CALCULATE
THE ROOT WEIGHT,NEW ROOT LENGTH AND CO2 RESPIRED IN EACH CELL.

CALCULATE LENGTH OF YOUNG ROOTS IN CELLS

CALCULATE AMOUNT OF CARBON IN THE ROOT STORAGE POOL

RETURN

```
      SUBROUTINE TRADE
C     -----------------------------( TRADE )----------------------------
10.01 C      THIS ROUTINE MAINTAINS A FUNCTIONAL BALANCE BETWEEN ROOT AND
C     SHOOT BY GROWING ROOT AS NECESSARY TO MEET TRANSPIRATION DEMAND.
C     NEW ROOT HAS LESS RESISTANCE TO WATER UPTAKE. SINCE TRANSPIRATION
C     RATE, PHOTOSYNTHESIS RATE AND HENCE POTENTIAL ROOT GROWTH RATE
C     ARE ROUGHLY PROPORTIONAL, LEAF WATER POTENTIAL IS NOT AFFECTED
C     MUCH BY TRANSPIRATION RATE. HOWEVER, LEAF WATER POTENTIAL MUST
C     FALL AS TRANSPIRATION INCREASES BECAUSE THE THRESHOLD TURGOR
C     PRESSURE FOR STOPPING SHOOT GROWTH AND DIVERTING CARBOHYDRATES
C     TO ROOTS DECREASES WITH TIME. HENCE THE ROUTINE ALSO PREDICTS
C     CHANGES IN TOTAL POTENTIAL, OSMOTIC POTENTIAL AND TURGOR PRESSURE
C     IN LEAVES DURING THE DAY.
C     ------------------------------------------------------------------
      DIMENSION SR(20,10),WUPM(20,10),WUPN(20,10),DRL(20,10),
     &          WUPT(20,10),UPTH2O(20,10)
      INCLUDE 'COMMON.FOR'
C     ------------------------------------------------------------------
      DO 13 L=1,NLR
        DO 14 K=1,NKR
          SR(L,K)=0.0
          WUPM(L,K)=0.0
          WUPN(L,K)=0.0
          WUPT(L,K)=0.0
          DRL(L,K)=0.0
   14     CONTINUE
   13     CONTINUE
          SGTLT = 0.0
C
C     ------------------------------------------------------------------
C     *** THE THRESHOLD LEAF WATER POTENTIAL AND TURGOR PRESSURE THAT
C         WILL JUST PREVENT ALL SHOOT EXPANSION ARE CALCULATED. ***
C     ------------------------------------------------------------------
C     *** CALCULATE THE THRESHOLD LEAF WATER POTENTIAL AND TURGOR
C         PRESSURE THAT WILL JUST PREVENT ALL SHOOT GROWTH AT ITIME=1 **
C
      IF( ITIME.GT.1 ) GOTO 9999
10.02   X11=(TPLD-2.0)*(1.0-EXP(-0.7*PERIOD/2.0))
        IF(X11.LE.0.0) X11 = 0.0
10.03   PSILT=PSILD-(X11/(1.0-OSMFAC))
        TPLT=TPLD-X11
        PSILO=PSILT-(TPLT/(1.0-OSMFAC))
        PPSIL=PSILD
        TPL = TPLD
        BTPL = TPLD
        SGTLT = 0.0
        GOTO 8888
C
C     *** CALCULATE THE SAME VARIABLES FOR ANY OTHER ITIME ***
C
 9999 PPSIL=PSIL(ITIME-1)
      X11=(TPL-2.0)*(1.0-EXP(-0.7*PERIOD))
        IF(X11.LE.0.0) X11 = 0.0
      PSILT=PPSIL-(X11/(1.0-OSMFAC))
      TPLT=TPL-X11
      PSILO=PSILT-(TPLT/(1.0-OSMFAC))
C
C     ------------------------------------------------------------------
C     *** WATER UPTAKE BY YOUNG AND OLD ROOTS UNDER VARIOUS
C         CIRCUMSTANCES IS CALCULATED.***
C     ------------------------------------------------------------------
C     *** CALCULATE RADIUS OF SOIL CYLINDER THROUGH WHICH WATER MUST
C         TRAVEL TO REACH ROOTS. THIS IS APPROXIMATED BY A SIMPLE
```

```
C          FUNCTION OF SOIL WATER POTENTIAL TO AVOID ITERATION ***
C
 8888 WUPMS=0.
      WUPSI=0.
      DO 11 L=1,NLRP1
        DO 10 K=1,NKRP1
            X22=PSIS(L,K)-PSILT
            IF( X22.LE.0.0 ) GOTO 10
10.04         SC=0.017-(PSIS(L,K)*0.05)
C
C     *** CALCULATE RESISTANCE TO WATER FLOW IN SOIL CELL ***
C
          IF(COND(L,K).LE.0.0)GO TO 10
10.05     SR(L,K)=ALOG(SC*SC/3.0E-4)/(4.0*3.1416*COND(L,K)*1019.7/24.0)
C
C     *** CALCULATE RATE OF WATER UPTAKE BY EXISTING ROOTS IF LEAF WATER
C         POTENTIAL IS AT THE THRESHOLD ***
C
10.06     WUPM(L,K)=(((PSIS(L,K)-PSILT)*D*W*RUTDEN(L,K))/
     &               (SR(L,K)+RRRM+RVR(L,K)))*0.02275*(24.5+TS(L,K))
          WUPN(L,K)=(((PSIS(L,K)-PSILT)*YRL(L,K))/
     &               (SR(L,K)+RRRY+RVR(L,K)))*0.02275*(24.5+TS(L,K))
10.07     WUPMS=WUPMS+WUPM(L,K)+WUPN(L,K)
          WUPSI=WUPSI+(PSIS(L,K)*(WUPM(L,K)+WUPN(L,K)))
   10     CONTINUE
   11     CONTINUE
C
C     ***ALLOCATE TRANSLOCATED CARBON TO FRUIT, NODULES AND VEGETATIVE
C        SHOOT PLUS ROOT***
C
        NODC=PDNODC
        IF (NODC.GT.TRANSC)THEN
         NODC=TRANSC
         FRUITC=0.0
         VEGSRC=0.0
        ELSE
         FRUITC=PDPODC+PDSC
           IF(FRUITC.GE.(TRANSC-PDNODC))THEN
            FRUITC=TRANSC-PDNODC
            VEGSRC=0.0
           ELSE
            VEGSRC=TRANSC-FRUITC-PDNODC
              IF(VEGSRC.LE.0.0)THEN
               VEGSRC=0.0
               NODC=TRANSC-FRUITC
              ELSE
               NODC=PDNODC
               X30=RSTPC-PDNODC
              IF(X30.LE.0.0) GO TO 2222
                 IF(X30.LT.VEGSRC) THEN
                  VEGSRC=VEGSRC-X30
                  FRUITC=FRUITC+X30
                 ELSE
                  VEGSRC=0.0
                  FRUITC=FRUITC+VEGSRC
                 END IF
              END IF
           END IF
        END IF
C
C     *** CALCULATE RATE AT WHICH CARBON IS SUPPLIED TO THE GROWING
C         ROOTS IN A HALF SOIL SLAB WHEN:
C         (1) NO CARBON IS TRANSLOCATED FROM THE SHOOT
```

```
      C        (2) SHOOTS DO NOT GROW AND ROOTS GET ALL TRANSLOCATED CARBON
      C
       2222    X36 = RCPOOL
10.08          IF((VSTAGE.GE.6.0).AND.(RCPOOL.GT.(ROOTWT*0.02)))X36=ROOTWT*0.02
10.09          RCPAV=X36*(1.0-(0.75**PERIOD))/PERIOD
               PCRL=RCPAV*POPROW/200.
10.10          PCRQ=(0.95*VEGSRC*POPROW/200.)+PCRL
      C
      C     *** CALCULATE A CARBON TO DRY WEIGHT CONVERSION FACTOR FOR THE
      C          ROOTS. ***
      C
               X20=NRATIO*0.1563
      C    0.1563 = 2.5*6.25/100
10.11          CONVR=(X20*0.6)+((1.0-X20)*0.47)
               CONRES=(X20*0.24)+((1.0-X20)*0.08)
      C
      C     *** CALCULATE GROWTH BY NEW ROOTS IF ROOTS GREW AT POTENTIAL
      C          RATE DURING PAST PERIOD. START WITH SOIL CELL WHERE CONDITIONS
      C          ARE MOST FAVORABLE FOR GROWTH. ***
      C
               PCRTS=0.
               WUPTS=0.
               SW10=0
10.12          DO 30 J=1,LKRP1
                 L=PCRPAR(J)/1000
                 K=PCRPAR(J)-L*1000
                   PDRL=PDWR(L,K)/RTWL
                   DRL(L,K)=(PDRL+PPDRL(L,K))*PERIOD/2.0
                   PPDRL(L,K)=PDRL
                   WUPT(L,K)=(((PSIS(L,K)-PSILT)*DRL(L,K))/(SR(L,K)+RRRY+
           &                 RVR(L,K)))*0.02275*(24.5+TS(L,K))
                   IF( WUPT(L,K).LE.0.0 ) WUPT(L,K)=0.0
                   WUPTS=WUPTS+WUPT(L,K)
                   WUPSI=WUPSI+(PSIS(L,K)*WUPT(L,K))
      C
      C     *** DETERMINE TOTAL WATER UPTAKE BY ALL NEW ROOTS AND WEIGHTED
      C          AVERAGE SOIL WATER POTENTIAL WHEN:
      C          I. ROOTS GET ONLY THE CARBON LEFT OVER FROM SHOOT GROWTH OR
      C          II. ROOTS GET ALL THE CARBON TRANSLOCATED. ***
      C
                   PCRTS=PCRTS+(PDWR(L,K)*CONVR)
                   IF( PCRL.LE.PCRTS ) GOTO 40
                   GOTO 60
      C
          40       CONTINUE
                   IF( SW10.EQ.1 ) GOTO 60
                     SW10=1
                     X17 = 0.0
            IF(PDWR(L,K).GT.0.)X17=WUPT(L,K)*(1.0-((PDWR(L,K)*CONVR)
           &          -PCRTS+PCRL)/(PDWR(L,K)*CONVR))
                     WUPRS=WUPTS-X17
10.13       IF((WUPMS+WUPRS).GT.0.)PSISM=(WUPSI-(PSIS(L,K)*X17))/(WUPMS+WUPRS)
          60       CONTINUE
      C
                   IF( PCRQ.LE.PCRTS ) GOTO 70
          30     CONTINUE
      C
       6666 IF((WUPMS+WUPTS).GT.0.)PSIST=WUPSI/(WUPMS+WUPTS)
                 IF( SW10.EQ.0 ) WUPRS=WUPTS
                 IF( SW10.EQ.0 ) PSISM=PSIST
                 GOTO 20
      C
          70     X18 = 0.0
```

```
              IF(PDWR(L,K).GT.0.0) X18 = WUPT(L,K)*(1.0-((PDWR(L,K)*CONVR)
     &               -PCRTS+PCRQ) / (PDWR(L,K)*CONVR))
              WUPTS=WUPTS-X18
10.13     IF((WUPMS+WUPTS).GT.0.)PSIST=(WUPSI-(PSIS(L,K)*X18))/(WUPMS+WUPTS)
C
C       *** CALCULATE WATER UPTAKE RATE IF LEAF WATER POTENTIAL HAS NOT
C           CHANGED IN PAST PERIOD. ***
C
   20         X1=(PSISM-PPSIL)/(PSISM-PSILT)
10.14         WUPDS=(WUPMS+WUPRS)*X1
C
C       *** CALCULATE WATER UPTAKE RATE IF LEAF TURGOR PRESSURE = 2 BARS
C           OR = 0 BARS. ***
C
              DPSI02 = 2.0/(1.0-OSMFAC)
10.15         WUP2S=(WUPMS+WUPTS)*(PSIST-PSILO-DPSI02)/(PSIST-PSILT)
10.16         WUP0S=(WUPMS+WUPTS)*(PSIST-PSILO)/(PSIST-PSILT)
C
C       -----------------------------------------------------------------
C       *** DETERMINE HOW POTENTIAL TRANSPIRATION RATE IS RELATED TO THE
C           VARIOUS POTENTIAL WATER UPTAKE RATES AND GO TO THE APPROPRIATE
C           SECTION OF THE CODE. CALCULATE LEAF WATER POTENTIAL,STOMATAL
C           CONDUCTANCE,THE PROPORTION OF TIME FOR WHICH THE SHOOT GROWS,
C           THE AMOUNT OF CARBON ACTUALLY NEEDED FOR ROOT GROWTH AND THE
C           AMOUNT OF OSMOREGULATION THAT WILL OCCUR. ***
C       -----------------------------------------------------------------
C       *** DETERMINE HOW POTENTIAL TRANSPIRATION RATE IS RELATED TO THE
C           VARIOUS POTENTIAL WATER UPTAKE RATES. ***
C
              EOR=EO(ITIME)*POPROW/100./2.
              EORS=EORS+EOR*PERIOD
              IF( EOR.LE.WUPDS ) GOTO 80
              X13=WUPMS+WUPTS
              IF( EOR.LE.X13 ) GOTO 90
              IF( EOR.LE.WUP2S ) GOTO 100
C
C       *** SINCE POTENTIAL TRANSPIRATION RATE IS GREATER THAN THE
C           POSSIBLE WATER UPTAKE RATE WHEN LEAF TURGOR PRESSURE IS 2 BARS
C           STOMATAL CLOSURE WILL DECREASE TRANSPIRATION RATE.
C           CALCULATE LEAF WATER POTENTIAL AND STOMATAL CLOSURE FACTOR. **
C
   15         PSIL(ITIME)=PSILO+DPSI02-((EOR-WUP2S)
     &               /(EOR/DPSI02)+((WUP0S-WUP2S)/DPSI02))
10.17         SCF=1.0-((PSILO+DPSI02)-PSIL(ITIME))/DPSI02
10.18         OSMREG=0.25
              GOTO 25
C
C       *** SINCE POTENTIAL TRANSPIRATION RATE IS BETWEEN POSSIBLE WATER
C           UPTAKE RATES WHEN LEAF TURGOR PRESSURE IS AT THRESHOLD FOR
C           GROWTH AND AT 2 BARS, CALCULATE LEAF WATER POTENTIAL. ***
C
  100         IF(OSMREG.LE.0.) OSMREG=0.
              PSIL(ITIME)=PSIST-((PSIST-PSILT)*EOR/(WUPMS+WUPTS))
              SCF=1.0
C
C       *** FOR ALL LEAF WATER POTENTIALS ABOVE THE THRESHOLD, THE SHOOT
C           DOES NOT GROW. ***
C
   25         SGT=0.0
              GOTO 35
C
C       *** SINCE POTENTIAL TRANSPIRATION RATE IS SUCH THAT SOME GROWTH
C           OF BOTH ROOTS AND SHOOTS IS PERMITTED, CALCULATE APPROXIMATE
```

```
C          LEAF WATER POTENTIAL AND PERCENTAGE OF TIME FOR WHICH SHOOT
C          GROWS. ***
C
    90     X16=(PPSIL-PSILT)/2.0
           PSIL(ITIME)=(PPSIL+PSILT)/2.0
           SCF=1.0
           IF(X16.LE.0.0)THEN
           SGT=1.0
           GO TO 35
        END IF
10.19      DO 45 J=1,3
             WUPGS=(PSISM-PSIL(ITIME))/(PSISM-PSILT)*(WUPMS+WUPRS+
     &              ((WUPTS-WUPRS)*(PPSIL-PSIL(ITIME))/(X16*2)))
             IF( EOR.GT.WUPGS ) PSIL(ITIME)=PSIL(ITIME)-(X16/2**J)
             IF( EOR.LT.WUPGS ) PSIL(ITIME)=PSIL(ITIME)+(X16/2**J)
    45     CONTINUE
C
    55     SGT=(PSIL(ITIME)-PSILT)/(X16*2.0)
           GOTO 35
C
C      *** SINCE POTENTIAL TRANSPIRATION RATE IS SUCH THAT NO ROOT
C          GROWTH IS NECESSARY, CALCULATE LEAF WATER POTENTIAL. ***
C
    80     PSIL(ITIME)=PSISM-((PSISM-PSILT)*EOR/(WUPMS+WUPRS))
           SCF=1.0
           SGT=1.0
C
C      *** LEAF WATER POTENTIAL CANNOT RISE ABOVE ITS DAWN VALUE ***
C
   35      IF(PSIL(ITIME).GT.PSILD) PSIL(ITIME)=PSILD
C
C      *** CALCULATE LEAF TURGOR PRESSURE ***
C
           PTPL=TPL
           TPL=((PSIL(ITIME)-PPSIL)/2.)+TPL
           IF(BTPL.GT.TPL)BTPL = TPL
C
C      *** IF TURGOR PRESSURE IS LESS THAN 5 BARS SHOOT GROWTH
C          POTENTIAL IS REDUCED. ***
C
10.20      IF(TPL.GT.2.0) GO TO 96
           SGTLI = 1.0
           SGT = 0.0
           PCRS = PCRQ
           GO TO 95
   96      IF(TPL.GT.5.0.OR.TPL.GT.PTPL) GO TO 97
           SGT = SGT * (TPL-2.0)/3.0
           SGTLT =SGTLT+1.0-SGT
           GO TO 1111
C
C      *** IF TURGOR PRESSURE HAS INCREASED AND IS ABOVE 2.0 BARS,
C          SHOOT GROWTH POTENTIAL IS INCREASED. ***
C
10.21  97  IF(BTPL.LT.2.0) BTPL = 2.0
           IF(TPLD.EQ.BTPL) GO TO 1111
           SGT=SGT+(TPL-PTPL)/(TPLD-BTPL)*SGTLT
  1111     PCRS = PCRL
           IF(SGT.LT.1.0)PCRS=PCRL+((PCRQ-PCRL)*(1.0-SGT))
   95      ASGT(ITIME) = SGT
C
C      *** CALCULATE AN INTEGRATION FACTOR FOR PREDICTING WATER UPTAKE
C          FROM EACH SOIL CELL FOR THE COMING PERIOD OF THE DAY ASSUMING
C          UPTAKE VARIES AS A HALF SINE WAVE. ***
```

```
C
          X12=ITIME+0.5
          RPERD = FLOAT(IPERD)
          IF( X12.GT.RPERD) X12=RPERD
          FINT=PERIOD*RPERD/3.1416/SIN(3.1416*(ITIME-0.5)/RPERD) *
     &         (COS(3.1416*(ITIME-0.5)/RPERD)-COS(3.1416*X12/RPERD))
C    -----------------------------------------------------------------
C    *** CALCULATE CHANGE IN ROOT DRY WEIGHT AND LENGTH, AND WATER
C        UPTAKE FROM EACH SOIL CELL FOR PAST PERIOD, AND PREDICT
C        WATER UPTAKE  FOR THE NEXT PERIOD OF THE DAY. ***
C    -----------------------------------------------------------------
C    *** GROW ROOTS IN THE MOST FAVORABLE CELLS UNTIL CARBON AVAILABLE
C     IS USED,AND CALCULATE CURRENT WATER UPTAKE RATE FOR EACH CELL.***
C
          AWUPS=0.0
          PCRTS=0.0
          UPH2OS=0.0
          SW20=0
          SW30=0
          EORSCF=EOR*SCF
10,22     DO 65 J=1,LKRP1
            L=PCRPAR(J)/1000
            K=PCRPAR(J)-L*1000
            X28=(PSIS(L,K)-PSIL(ITIME))/(PSIS(L,K)-PSILT)
            IF(PSIS(L,K).LE.PSIL(ITIME)) X28=0.
            IF(PDWR(L,K).LE.0.0) GO TO 85
            PCRTS=PCRTS+(PDWR(L,K)*CONVR)
            IF( PCRS.LE.PCRTS ) GOTO 75
              ADWR(L,K)=PDWR(L,K)
              AWUP(L,K)=(WUPM(L,K)+WUPN(L,K)+WUPT(L,K))*X28
              GOTO 67
C
   75       CONTINUE
            IF( SW20.EQ.1 ) GOTO 85
              SW20=1
              ADWR(L,K)=((PDWR(L,K)*CONVR)-PCRTS+PCRS)/CONVR
              AWUP(L,K)=(WUPM(L,K)+WUPN(L,K)+(WUPT(L,K)*ADWR(L,K)/
     &                  PDWR(L,K)))*X28
              GOTO 67
C
   85       ADWR(L,K)=0
            AWUP(L,K)=(WUPM(L,K)+WUPN(L,K))*X28
C
   67       IF(SW30.EQ.1) GO TO 69
            AWUPS=AWUPS+AWUP(L,K)
            IF(AWUPS.LT.EORSCF) GO TO 66
            SW30=1
            AWUP(L,K)=AWUP(L,K)-AWUPS+EORSCF
            AWUPS=EORSCF
            GO TO 66
   69       AWUP(L,K)=0.
C
C    *** CALCULATE MEAN ROOT GROWTH RATE AND MEAN WATER UPTAKE RATE
C        FOR THE PAST PERIOD AND PREDICT WATER UPTAKE FOR THE NEXT
C        PERIOD OF THE DAY.***
C
   66       AWR(L,K)=(ADWR(L,K)+PADWR(L,K))*PERIOD/2.0
            UPTH2O(L,K)=(AWUP(L,K)+PWUP(L,K))*PERIOD/2.0
            IF( (ITIME.GT.1).OR.(IDAY.EQ.1) ) GOTO 68
              AWR(L,K)=AWR(L,K)/2.0
              UPTH2O(L,K)=UPTH2O(L,K)/2.0
   68         PADWR(L,K)=ADWR(L,K)
            ROOTWT=ROOTWT+(AWR(L,K)*2.0*100.0/POPROW)
```

```
                  PWUP(L,K)=AWUP(L,K)
10.23             AFWUP(L,K)=UPTH2O(L,K)-FWUP(L,K)+AWUP(L,K)*FINT
                  FWUP(L,K)=AWUP(L,K)*FINT
                  UPH2OS=UPH2OS+UPTH2O(L,K)
      65        CONTINUE
                IF(SW20.EQ.1) PCRTS = PCRS
            AWUPSS=AWUPSS+AWUPS*PERIOD
            EORSCS=EORSCS+EORSCF*PERIOD
C
C           *** REDISTRIBUTE ROOT GROWTH IN THE SOIL PROFILE AND CALCULATE
C            THE ROOT WEIGHT,NEW ROOT LENGTH AND CO2 RESPIRED IN EACH CELL.***
C
                  NLRP = NLR
                  NKRP = NKR
                DO 121 L=1,NLR
                  DO 120 K=1,NKR
                    ADRL(L,K) = 0.0
                    IF(AWR(L,K).EQ.0.0) GO TO 120
10.24               IF( RTWT(L,K).LE.RTMINW ) GOTO 5555
                    LP1=L+1
                    IF( L.EQ.NL ) LP1=NL
                    IF( LP1.GT.NLR) NLRP = LP1
                    KP1=K+1
                    IF( K.EQ.NKH ) KP1=NKH
                    IF( KP1.GT.NKR) NKRP = KP1
                    LM1=L-1
                    IF(L.EQ.1)LM1=1
                    KM1=K-1
                    IF( K.EQ.1 ) KM1=1
10.25               X23=RGCF(L,K)+RGCF(L,KM1)+1.5*RGCF(L,KP1)
     &               +RGCF(LM1,K)+5.0*RGCF(LP1,K)
                    IF(X23.EQ.0.) GO TO 5555
                     X24=AWR(L,K)*RGCF(L,K)/X23
                     RTWT(L,K)=RTWT(L,K)+X24
10.26                RESPS(L,K) = RESPS(L,K) + (X24*CONRES)
                     ADRL(L,K)=ADRL(L,K)+X24/RTWL
                    X25=AWR(L,K)*RGCF(L,KM1)/X23
                    RTWT(L,KM1)=RTWT(L,KM1)+X25
                    RESPS(L,KM1) = RESPS(L,KM1) + (X25*CONRES)
                    ADRL(L,KM1)=ADRL(L,KM1)+X25/RTWL
                     X26=AWR(L,K)*1.5*RGCF(L,KP1)/X23
                     RTWT(L,KP1)=RTWT(L,KP1)+X26
                     RESPS(L,KP1) = RESPS(L,KP1) + (X26*CONRES)
                     ADRL(L,KP1)=ADRL(L,KP1)+X26/RTWL
                    X29=AWR(L,K)*RGCF(LM1,K)/X23
                    RTWT(LM1,K)=RTWT(LM1,K)+X29
                    RESPS(LM1,K) = RESPS(LM1,K) + (X29*CONRES)
                    ADRL(LM1,K)=ADRL(LM1,K)+X29/RTWL
                     X27=AWR(L,K)*5.0*RGCF(LP1,K)/X23
                     RTWT(LP1,K)=RTWT(LP1,K)+X27
                     RESPS(LP1,K) = RESPS(LP1,K) + (X27*CONRES)
                     ADRL(LP1,K)=ADRL(LP1,K)+X27/RTWL
                    GO TO 120
    5555            RTWT(L,K)=RTWT(L,K)+AWR(L,K)
                    RESPS(L,K) = RESPS(L,K) + (AWR(L,K)*CONRES)
                    ADRL(L,K)=ADRL(L,K)+AWR(L,K)/RTWL
     120   CONTINUE
     121 CONTINUE
                    NLR = NLRP
                    NKR = NKRP
C
C          *** CALCULATE LENGTH OF YOUNG ROOTS IN CELLS ***
C
```

```
            DO 123 L=1,NLR
              DO 122 K=1,NKR
10.27               YRL(L,K) = (YRL(L,K)*(1.0-(0.021*PERIOD)))+ADRL(L,K)
C                   0.021=1/2/24
                    RUTDEN(L,K)=RUTDEN(L,K)+(YRL(L,K)*0.021*PERIOD/D/W)
  122         CONTINUE
  123       CONTINUE
C
C     *** CALCULATE AMOUNT OF CARBON IN THE ROOT STORAGE POOL ***
C
        X15=(PCRS-PCRTS)*200./POPROW
        IF(X15.LE.0.0)X15 =0.0
        X16 = SGT
        IF(X16.GT.1.0) X16 = 1.0
10.28   RCPOOL=RCPOOL+(((VEGSRC*0.05*(1.0-X16))-RCPAV+X15)*PERIOD)
        USEDCS = USEDCS+(PCRTS*200/POPROW*PERIOD)
      RETURN
      END
```

Notes for TRADE

10.01 This subroutines contains a new conceptual model of the mechanism used by plants to partition carbon between the shoot and the root. The physical and physiological processes involved are all well known but they are assumed to interact in a novel way. Because of the novelty of these ideas, our rationale is presented below before we discuss the subroutine in detail.

In the subroutine for each period of the day we generate a relationship between leaf water potential and potential transpiration rate like that shown in Fig. 34. The shape of the relationship is similar to some of those reported by Stoker and Weatherley (1971), Hailey et al. (1973) and Bunce (1978). Our interpretation of the relationship is that, between A and B., potential transpiration rate is such that the plant can extract sufficient water with existing old roots that its leaf water potential need not decrease. Between B and C., leaf water potential must decrease and this causes a loss of shoot turgor, loss of potential shoot growth, diversion of carbon to the root and hence root growth. This root growth creates new root which has a higher conductance to water and also increases the total amount of root available for extracting water. For both these reasons, the plant can meet a higher transpiration demand with less of a change in leaf water potential. At C, the flow rate of carbon to the roots is the maximum permitted so between C and D the relationship becomes linear again. At D the leaf water potential has decreased to the point where the stomata begin to close. Above D, stomatal closure means that the plant only has

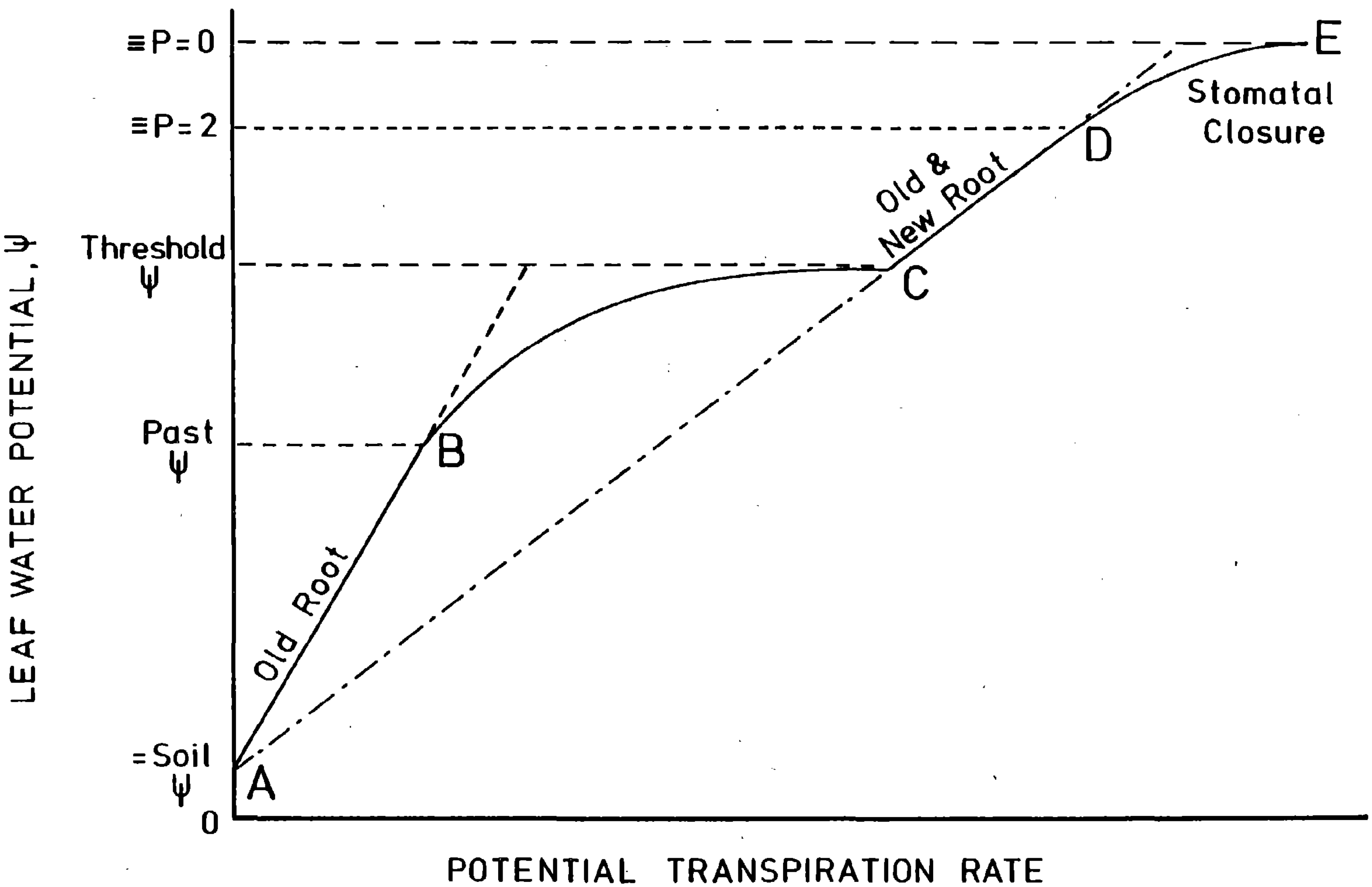


Fig. 34. Typical relationship between leaf water potential and potential transpiration rate.

to meet a fraction of the potential transpiration demand so the relationship become almost horizontal.

To generate the relationship we calculate the average soil water potential weighted by the rate at which water is being extracted from each soil cell. This is point A. Next the water uptake for existing roots is calculated at the leaf water potential found in the previous period of the day (point B). The change in leaf water potential necessary to prevent all shoot growth fixes point C on one axis and water uptake for new plus old roots at that leaf water potential fixes it on the other axis. Point D is marked at the leaf water potential corresponding with a turgor pressure of 2 bars and the stomata are assumed to be fully closed at 0 bars (point E).

Having generated the relationship and knowing the potential transpiration rate we can calculate leaf water potential, growth rates of root and shoot and stomatal conductance.

10.02 Green et al. (1971) have shown for Nitella that a decrease in cell turgor pressure is followed by a decrease in the threshold turgor pressure required for cell growth. Thus to prevent growth over a period of time, turgor pressure must fall more than threshold turgor pressure can adjust in that time. The rate of threshold adjustment is higher at high turgor pressure (Figure 35) and this is reflected in the equation to calculate X11, the amount by which the threshold will fall in a given period. The equation was derived by fitting the data of Green et al.

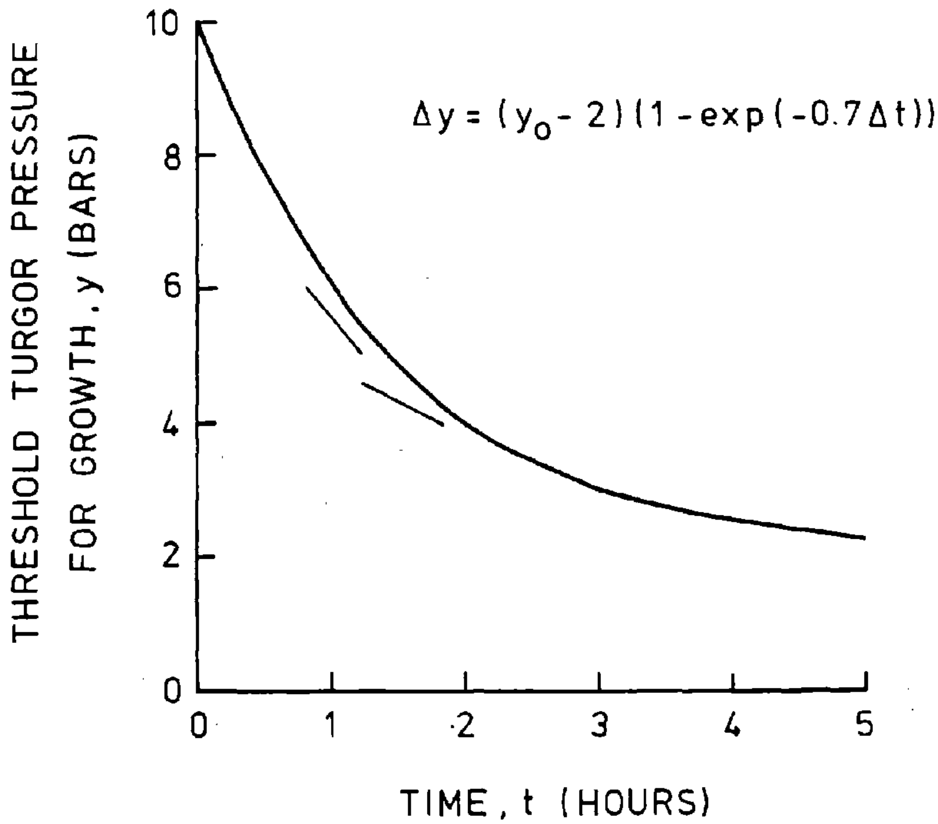


Fig. 35. Diagram showing how the rate at which the threshold turgor pressure can change is a function of turgor pressure. The short bars are data from Green et al. (1971).

10.03 Part of any change in leaf water potential during a day is balanced by a change in leaf osmotic potential. The leaf, in fact, osmoregulates on a diurnal basis. Wenkert et a. (1978a) in experiments with soybean found a change in leaf osmotic potential equal to half the change in leaf water potential. Cutler et al. (1977) observed similar changes in cotton. The senior author's unpublished observations on this topic suggest that one way plants become conditioned to water stress is that they increase their ability to osmoregulate diurnally. OSMFAC is the factor in the model which simulates this conditioning to water stress. It is calculated in NIGHT on line 216. OSMFAC varies from 0.5 to a seldom-obtained maximum of 0.9 as the sugar plus starch content of the shoot increases over its permitted range.

When OSMFAC is 0.5, a change in turgor pressure of one bar will cause a change in leaf water potential of 2 bars. Since we know how much turgor pressure must change to prevent leaf expansion we can calculate how much leaf water potential must change and hence, for a given time of day, the leaf water potential that just prevents all shoot expansion.

The relationship between shoot growth and leaf water potential can be seen in the data of Wenkert et al (1978b, their Fig. 5 for August 9). On this day there was no leaf elongation in the morning and leaf water potential changed at the rate predicted by this model.

PSILT, the leaf water potential that just prevents all shoot expansion fixes point C of Figure 34 on the water potential axis. See line 151 and note 10.07 for the other axis.

10.04 The radius of the soil cylinder through which water must travel to reach the roots is normally considered to be a function of root density. However, it is actually a function of soil water content, hydraulic diffusivity and root water uptake rate. In this model, to avoid interaction it is made a simple function of soil water potential. The equation is based on data of Cowan (1965).

10.05 Resistance to water flow to the root is calculated by assuming that we are dealing with steady-state diffusion of water across the soil cylinder to the root surface. This equation was originally proposed by Gardner (1960). The root radius is assumed to be 0.017 cm (Taylor and Klepper, 1975).

10.06 Root water uptake is made a function of water potential gradient, root length, resistances to water flow and water viscosity. Calculation of water viscosity follows the method of Dalton and Gardner (1978). The resistances include soil resistance, root vascular resistance and root radial resistance.

Plant resistances for soybean were derived as follows. Overall plant resistance was calculated from Bunce (1978) because it could be obtained in suitable units. This overall resistance was then partitioned between leaf, vascular and root radial resistances in the proportions reported by Boyer (1971). Since the roots in both the Bunce and Boyer data were probably not growing very rapidly (transpiration demand was low) the root radial resistance obtained by this method was assumed to be the one for "old" roots. "Young roots have been shown to have a lower resistance than "old" ones by Graham et al. (cited in "Plant Root Systems", Scott Russel, 1977), and Brouwer, in "Water in Living Organisms" (1965). The ratio of "old" to "young" resistances found by Brouwer for Vicia faba was used to estimate a radial resistance for "young" soybean roots. According to Brouwer, differentiation, suberisation and lignification of root tissues all occur at a distance from the root tip which is a function of root growth rate. Salim (1966) has presented data showing that root tissues lignify about 1.7 hours after they are formed. However this time seems much too short when one considers the life of root hairs and it proved much too short during validation runs. Therefore in this model the roots remain "young" for 2 days after they are grown.

10.07 WUPMS, water uptake rate by existing roots fixes point C of Figure 34 on the transpiration (and water uptake) axis. See lines 105 and 119 and note 10.03 for the other axis.

10.08 Carbon in the root carbon pool may not all be available for remobilization. See PNET note 6.19 for the logic of this.

10.09 Of the carbon available for remobilization, 25% per hour is remobilized.

10.10 All the carbon translocated out of the leaves, comes to the roots when the shoot is not expanding. Of this, 95% is immediately available for growth and 5% goes in to the root carbon storage pool.

10.11 See SHTRGRO note 7.02.

10.12 Roots grow first in soil cells where conditions are most favorable for growth and then in less favorable cells until all the carbon available to them is used.

10.13 The average soil water potential weighted by the rate at which water is being extracted from each cell is, of course, a function of transpiration rate. Therefore, two values are calculated. One of these, PSISM, is the value when the roots do not receive any carbon translocated from the shoot. This is used when transpiration rate is low. The other, PSIST, is the value when the roots are growing at their maximum rate and is used when transpiration rate is high.

Depending on transpiration rate, one of these values fixes point A on Figure 34.

10.14 WUPDS, water uptake rate if leaf water potential has not changed in the past period fixes point B on Figure 34.

10.15 WUP2S, water uptake rate if leaf turgor pressure equals 2 bars, fixes point D on Figure 34.

10.16 WUP0S, water uptake rate if leaf turgor pressure equals 0 bars, fixes point E on Figure 34.

10.17 No direct effect of humidity on stomata is inlcuded in this model because it has never been reported for soybean and it seems to be most common in desert species. See Johnson and Caldwell (1976).

10.18 In addition to diurnal osmoregulation (See note 10.03), the model allows changes in the baseline or dawn osmotic potential. If the stomata close at any time during the day, leaf osmotic potential decreases (becomes more negative). If the shoot keeps growing all day, osmotic potential increases.

10.19 Under conditions where some growth of both roots and shoots occurs (portion BC of the curve in Fig. 34), leaf water potential has to be calculated iteratively.

10.20 At a leaf turgor pressure below 2 bars, the shoot does not grow and this growing time is lost irretrievably. At leaf turgor pressures between 2 and 5 bars, shoot growth is reduced. However, this lost growth is retrievable.

10.21 As leaf turgor pressure increases at the end of the day, the growth that was lost temporarily is recovered.

10.22 The amount of carbon actually needed for root growth is used, as before, to grow roots preferentially in soil cells where conditions are most favorable.

10.23 For each soil cell the current rate of water uptake by roots is calculated and multiplied by an integration factor to predict the water uptake up to the middle of the next calculation period. Any under or over-estimate for the previous calculation period is added algebraically. The integration factor assumes that diurnal water uptake rate follows a half sine wave.

10.24 If root weight in a cell is less than the weight needed to cross the cell diagonally, the root cannot grow into adjacent cells. RTWNIM is calculated in MAIN on line 380.

10.25 Any increase in root weight in a cell is distributed between that cell and its neighbors in proportion to their desirability for root growth. The cells below the cell in question and on the side furthest from the plant have extra weighting factors.

10.26 Carbon dioxide respired by the growing root in each cell is recorded for further use in SOLOXY. The calculation employs the ideas and data of Penning de Vries (1975).

10.27 See note 10.06 for the definition of "young" roots.

10.28 The root carbon pool, RCPOOL, is debited for the amount of carbon remobilized (see note 10.09) and credited with 5% of the carbon translocated down from the shoot (see note 10.10) plus any carbon not used for growth because of limited root growth potential.

GROWTH

The subroutine GROWTH is called once for each period of the day and night. It calculates the actual amount of growth made by each organ of the shoot. First it grows existing pods and seeds and initiates some new pods if carbon is available. The vegetative organs on the shoot are grown with any carbon left over.

Potential organ growth rates have been calculated in SHTGRO and these are often limited by shoot growing time (a function of leaf turgor pressure calculated in TRADE). If there is insufficient carbon to satisfy these growth rates for organs with the minimum density, then actual growth rates are further reduced. If, however, there is sufficient carbon, then these growth rates are the actual growth rates and a tissue density between the maximum and minimum is calculated. The sizes, dry weight, etc. of the organs are incremented.

Depending on the stage of growth of the plant and the availability of carbon and sites, branches and seeds may be initiated.

```
---------------------------( GROWTH )-----------------------------
THIS ROUTINE CALCULATES THE GROWTH IN SIZE AND DRY WEIGHT OF
ORGANS ON THE SHOOT AND INITIATES BRANCHES,PODS AND SEEDS.
-----------------------------------------------------------------
GROW EXISTING PODS AND SEEDS AND INITIATE PODS
-----------------------------------------------------------------
GROW EXISTING PODS AND SEEDS

INITIATE PODS IF C IS AVAILABLE
-----------------------------------------------------------------
CALCULATE CARBON AVAILABILITY AND LIMITATIONS TO THE CHANGE IN
   SIZE AND WEIGHTS OF VEGETATIVE ORGANS ON THE SHOOT
-----------------------------------------------------------------
if there is no carbon left after growing pods and seeds, jump down
|
|  IF THERE IS CARBON LEFT AFTER GROWING THE PODS AND SEEDS,
|     CALCULATE HOW MUCH IS AVAILABLE TO GROW THE REST OF THE SHOOT.
|
|  CALCULATE MAXIMUM AND MINIMUM AMOUNTS OF CARBON NEEDED TO
|     GROW SHOOTS AT POTENTIAL RATE LIMITED ONLY BY TEMPERATURE
|     AND SHOOT TURGOR.
|
|  DETERMINE IF CARBON AVAILABILITY FURTHER LIMITS SHOOT GROWTH
|     AND CALCULATE A SHOOT CARBON SUPPLY/DEMAND RATIO.
|  --------------------------------------------------------------
|  CALCULATE ACTUAL CHANGES IN SIZE AND DRY WEIGHT OF ORGANS
|     ON THE SHOOT DURING THIS PERIOD.
|  --------------------------------------------------------------
|  CALCULATE CHANGES IN MAINSTEM LEAF AND PETIOLE SIZES AND
|     DRY WEIGHTS.
|
|  CALCULATE CHANGES IN BRANCH LEAF AND PETIOLE DIMENSIONS AND
|     DRY WEIGHTS.
|
|  CALCULATE CHANGES IN MAINSTEM AND BRANCH LENGTHS AND DRY
|     WEIGHTS.
|
|  CALCULATE HEIGHT OF TOP LEAVES ABOVE SOIL.
|
|  CALCULATE CHANGE IN DRY WEIGHT OF FLOWERS.
v
A
```

```
A
|
|  ------------------------------------------------------------------
|  DEPENDING ON THE STAGE OF GROWTH AND AVAILABILITY OF
|     SUBSTRATES, INITIATE BRANCHES AND SEEDS
|  ------------------------------------------------------------------
|  DEPENDING ON STAGE OF GROWTH GO TO APPROPRIATE PART OF
|     ROUTINE
|
|-at any time other than night, or if it is more than 20 days
|     since seedfill started, or reproductive stage > or = 1
|
|    if reproductive stage > or = 4
|    |
|    |  INITIATE BRANCHES IF C AND SITES ARE AVAILABLE.
|    |
|----+---jump down
|    v
|    INITIATE SEEDS IF C IS AVAILABLE
v
BALANCE CARBON ACCOUNTS

RETURN
```

```
      SUBROUTINE GROWTH
C     ---------------------------( GROWTH )---------------------------
C      THIS ROUTINE CALCULATES THE GROWTH IN SIZE AND DRY WEIGHT OF
C     ORGANS ON THE SHOOT AND INITIATES BRANCHES,PODS AND SEEDS.
C     ----------------------------------------------------------------
      INCLUDE 'COMMON.FOR'
C     ----------------------------------------------------------------
C     *** GROW EXISTING PODS AND SEEDS AND INITIATE PODS ***
C     ----------------------------------------------------------------
C     *** GROW EXISTING PODS AND SEEDS ***
C
       X72=FRUITC
       USEDCS=USEDCS+(FRUITC*PERIOD)
       IF(RSTAGE.LT.3.0) GO TO 5555
       IF(RSTAGE.LT.4.0) GO TO 6666
       X73 = PDSC
       IF(X72.LE.X73)X73 = X72
       SEEDWT = SEEDWT+(X73*PERIOD/CONVSE)
       X72=FRUITC-X73
       IF(RSTAGE.GT.5.0) GO TO 5555
 6666  X74 = PDPODC
11.01  IF(X72.LE.X74) X74 = X72
       PODWT = PODWT+(X74*PERIOD/CONVPO)
11.02  X75=X72-X74
       IF(RSTAGE.LT.4.0)THEN
       STPOOL=STPOOL+(X75*PERIOD)
C
C   *** INITIATE PODS IF C IS AVAILABLE ***
C
       IF(ITIME.NE.(IPERD+1)) GO TO 5555
       X76=PODS+(NODES*0.5)
11.03  SEQSC=EQSC/SEEDS*X76*2.5*PRM(27)
        IF(SEQSC.GT.(TRNDAY/24.0))THEN
        X77=TRNDAY/24.0/EQSC*SEEDS/2.5/PRM(27)
         IF(X77.GT.PODS) PODS = X77
        ELSE
        PODS = X76
        END IF
       SEEDS=PODS*2.5
       END IF
C     ----------------------------------------------------------------
C     *** CALCULATE CARBON AVAILABILITY AND LIMITATIONS TO THE CHANGE IN
C         SIZE AND WEIGHTS OF VEGETATIVE ORGANS ON THE SHOOT ***
C     ----------------------------------------------------------------
C     *** IF THERE IS CARBON LEFT AFTER GROWING THE PODS AND SEEDS,
C     CALCULATE HOW MUCH IS AVAILABLE TO GROW THE REST OF THE SHOOT. ***
C
 5555 SGTR=0.
      SCRATO=0.
      ASCRAT(ITIME) = 0.0
11.04 CASG=VEGSRC*SGT*PERIOD
      IF(SGT.GT.1.0)CASG = VEGSRC*PERIOD
      IF(CASG.LE.0.0) GO TO 9977
      REMC = CASG
      X74 = PCST-PDPODC-PDSC
      IF(X74.EQ.0.) GO TO 9955
      REMC = 0.0
      USEDCS = USEDCS + CASG
C
C     *** CALCULATE MAXIMUM AND MINIMUM AMOUNTS OF CARBON NEEDED TO
C         GROW SHOOTS AT POTENTIAL RATE LIMITED ONLY BY TEMPERATURE
C         AND SHOOT TURGOR. ***
C
```

```
11.05       CMAXSG=(PCST-PDPODC-PDSC)*SGT*PERIOD
            CMINSG=(QCST-PDPODC-PDSC)*SGT*PERIOD
      C
      C     *** DETERMINE IF CARBON AVAILABILITY FURTHER LIMITS SHOOT GROWTH
      C         AND CALCULATE A SHOOT CARBON SUPPLY/DEMAND RATIO. ***
      C
            IF( CASG.LT.CMINSG ) THEN
11.06       SGTR=SGT*CASG/CMINSG
            SCRATO=0.0
            GO TO 2222
            ELSE
            SGTR=SGT
            SCRATO=1.0
            END IF
            IF( CASG.GT.CMAXSG ) THEN
            USEDCS = USEDCS-CASG+CMAXSG
            REMC=CASG-CMAXSG
            GOTO 2222
            ELSE
            XDUM=CMAXSG-CMINSG
11.07       IF(XDUM.NE.0.) SCRATO=(CASG-CMINSG)/XDUM
            END IF
       2222 ASCRAT(ITIME) = SCRATO
            SCRATS=SCRATS+(SCRATO*PERIOD)
            IF( VSTAGE.GE.VSTMAH ) GOTO 1111
      C     ---------------------------------------------------------------
      C     *** CALCULATE ACTUAL CHANGES IN SIZE AND DRY WEIGHT OF ORGANS
      C         ON THE SHOOT DURING THIS PERIOD. ***
      C     ---------------------------------------------------------------
      C     *** CALCULATE CHANGES IN MAINSTEM LEAF AND PETIOLE SIZES AND
      C         DRY WEIGHTS. ***
      C
11.08       DO 20 I=1,3
            X71 = VSTAGE+ 1.5 - I
            IF(X71.LT.0.0) GO TO 1111
              ITRIF=IFIX(X71)
                ADLA=PDLAM(I)*SGTR*PERIOD
                ADLW=ADLA*TNL/9.0*(1.0+(0.6*SCRATO))
                ADPL=PDPLM(I)*SGTR*PERIOD
                ADPW=PDPWM(I)*SGTR*PERIOD*(1.0+(0.6*SCRATO))
                IF( ITRIF.GE.1 ) GOTO 9988
                  ULAREA=ULAREA+ADLA
                  ULEAFW=ULEAFW+ADLW
                  UPETL=UPETL+ADPL
                  UPETW=UPETW+ADPW
                  LEAFWT=LEAFWT+ADLW
                  PETWT=PETWT+ADPW
      C           WRITE(2,*) LAREAT,ADLA
                  LAREAM=LAREAM+ADLA
                  GOTO 1111
      C
       9988     MLAREA(ITRIF)=MLAREA(ITRIF)+ADLA
                MLEAFW(ITRIF)=MLEAFW(ITRIF)+ADLW
                MPETL(ITRIF)=MPETL(ITRIF)+ADPL
                MPETW(ITRIF)=MPETW(ITRIF)+ADPW
      C           WRITE(2,*) LAREAT,ADLA
                LEAFWT=LEAFWT+ADLW
                PETWT=PETWT+ADPW
      C           WRITE(2,*) LAREAT,ADLA
                LAREAM=LAREAM+ADLA
         20 CONTINUE
      C
      C     *** CALCULATE CHANGES IN BRANCH LEAF AND PETIOLE DIMENSIONS AND
```

```
C           DRY WEIGHTS. ***
C
 1111 IF( NBRNCH.EQ.0 ) GOTO 9948
      DO 31 J=1,NBRNCH
        IF(TB(J).GE.TBMAH(J)) GO TO 31
        DO 30 II=1,3
          I=IFIX(TB(J)+1.5-II)
          IF( I.LE.0 ) GOTO 30
            ADLA=PDLAB(I,J)*SGTR*PERIOD
            ADLW=ADLA*TNL/12.0*(1.0+(0.6*SCRATO))
            ADPW=PDPWB(I,J)*SGTR*PERIOD*(1.0+(0.6*SCRATO))
            BLAREA(I,J)=BLAREA(I,J)+ADLA
            BLEAFW(I,J)=BLEAFW(I,J)+ADLW
            BPETL(I,J)=BPETL(I,J)+(PDPLB(I,J)*SGTR*PERIOD)
            BPETW(I,J)=BPETW(I,J)+ADPW
            LEAFWT=LEAFWT+ADLW
            PETWT=PETWT+ADPW
            LAREAB=LAREAB+ADLA
   30 CONTINUE
   31 CONTINUE
 9948  LAREAT = LAREAM+LAREAB
C
C     *** CALCULATE CHANGES IN MAINSTEM AND BRANCH LENGTHS AND DRY
C         WEIGHTS. ***
C
      ADMH=PDMH*SGTR*PERIOD
      IF(VSTAGE.GE.(VSTMAH-PRM(12))) GO TO 4444
      ADMW=PDMW*SGTR*PERIOD*(1.0+(1.5*SCRATO))
      MSTEMH=MSTEMH+ADMH
      MSTEMW=MSTEMW+ADMW
      STEMWT=STEMWT+ADMW
C
 4444 IF( NBRNCH.LE.0 ) GOTO 9933
      DO 40 J=1,NBRNCH
      IF(TB(J).GE.(TBMAH(J)-PRM(12))) GO TO 40
        ADBW=PDBRW(J)*SGTR*PERIOD*(1.0+(1.5*SCRATO))
        BRNCHH(J)=BRNCHH(J)+ADMH
        BRNCHW(J)=BRNCHW(J)+ADBW
      STEMWT=STEMWT+ADBW
 40   CONTINUE
C
C     *** CALCULATE HEIGHT OF TOP LEAVES ABOVE SOIL. ***
C
 9933 IF(NTOPLF.LE.0) NTOPLF = 1
      ZNEW=MSTEMH+MPETL(NTOPLF)+5.0
      IF(ZNEW.GE.Z) Z=ZNEW
C
C     *** CALCULATE CHANGE IN DRY WEIGHT OF FLOWERS. ***
C
      FLWRWT=FLWRWT+PDFW*SGTR*PERIOD
C
C     -----------------------------------------------------------------
C     *** DEPENDING ON THE STAGE OF GROWTH AND AVAILABILITY OF
C         SUBSTRATES, INITIATE BRANCHES AND SEEDS ***
C     -----------------------------------------------------------------
C     *** DEPENDING ON STAGE OF GROWTH GO TO APPROPRIATE PART OF
C         ROUTINE ***
C
 9955  IF(ITIME.NE.(IPERD+1)) GO TO 9999
       IF(FILDAY.GT.20) GO TO 9999
      IF( RSTAGE.GE.4.0 ) GOTO 9944
      IF( RSTAGE.GE.1.0 ) GOTO 9999
C
```

```
      C      *** INITIATE BRANCHES IF C AND SITES ARE AVAILABLE. ***
      C
11.09           BSITES=IFIX(VSTAGE-PRM(14))
      C     PARAMETER PRM(14) WAS 1.5
                X75=SCRATS/24.0
                IF( NBRNCH.GE.BSITES ) GOTO 9999
                IF(X75.LT.PRM(3)) GO TO 9999
                NBRNCH = NBRNCH + 1
11.10           INITBR(NBRNCH) = NTOPLF
                  GOTO 9999
      C
      C    *** INITIATE SEEDS IF C IS AVAILABLE ***
      C
       9944   IF(REMC.LE.0.0) GO TO 9999
11.11         FSEEDS = SEEDS*REMC/PERIOD/PDSC
              IF(FSEEDS.GT.(PODS*0.3))FSEEDS=PODS*0.3
              SEEDS = SEEDS + FSEEDS
      C
      C    *** BALANCE CARBON ACCOUNTS ***
      C
       9999   SCPOOL = SCPOOL+REMC
       9977   SHTWT = LEAFWT + PETWT + STEMWT
              IF(ITIME.EQ.(IPERD+1))TRNDAY=0.0
              RETURN
              END
```

Notes for GROWTH

11.01 Between R4 and R5 when pods and seeds are growing together, the seeds have the highest priority.

11.02 Any carbon left over after growing pods and seeds is added to STPOOL, the irretrievable pool of carbon stored in stems and petioles. Almost all the additions to STPOOL occur between R3 and R4 and come from RSTPC which is calculated in SHTGRO on line 320 and defined in note 7.09. RSTPC is added to FRUITC in NIGHT on line 258 and in TRADE on line 180.

11.03 The pod load on the plant is increased by one pod for every two nodes each day. This represents the continuing flowering and podset between R3 and R4. Assuming that there will be on average 2.5 seeds per pod, the potential rate of carbon use in synthesizing seeds to fill these pods is calculated. This is compared with the mean rate of carbon translocation from the leaves for the day. Pods accumulate on the plant until it reaches the maximum number that it could fill with its current carbon supply. The average number of seeds per pod set initially is based on data from many sources, e.g., Brevedan et al. (1978), and Patterson et al. (1977).

11.04 The carbon available for vegetative shoot growth, CASG is a proportion of the carbon allocated to the vegetative shoot and root. That proportion is defined by SGT, the short growing time. SGT is actually the proportion of time that shoot expansion is not limited by low shoot turgor. It is calculated in TRADE on lines 313, 325, 335, 343, 360, 364 and 373 depending on the circumstances.

11.05 The maximum (PCST) and minimum (QCST) potential rates of carbon use by the shoot are calculated in SHTGRO on lines 330 and 332.

11.06 IF CASG is less than the minimum required for shoot growth, then shoot growing time is further reduced in the calculation of SGTR. This SGTR is used to limit the change in area or volume of organs that can actually take place.

11.07 The shoot carbon ratio, SCRATO, is a measure of carbon availability relative to maximum and minimum amounts of carbon that can be used in the volume of organs to be grown. If SCRATO = 0, the organs grown will have the minimum structural dry weight permissible. If SCRATO = 1, the organs will have the maximum structural dry weight permissible.

11.08 Having calculated the limitations on growth in volume and dry weight as a fraction of the potential growth, actual growth of each organ on the shoot can now be determined. The potential growth rates used in the calculations all come from SHTGRO.

11.09 To initiate a branch, there must be available a site on the mainstem and sufficient carbon. The threshold level of carbon availability used in the model is completely arbitrary; there being no data on which to base it. Plant spacing affects branching in the model only via light interception by the individual plant and the effect that has on carbon availability.

11.10 The age of each new branch is recorded in INITBR as the number of the leaf then at the top of the mainstem.

11.11 After R4, the number of seeds can be increased above 2.5 per pod if there is sufficient carbon to support them.

PLTMAP

The subroutine PLTMAP is called once each period of the day and night. It calculates the stage of vegetative or reproductive growth reached by the plants. At floral initiation, the number of mainstem and branch nodes is calculated for determinate plants.

```
---------------------------( PLTMAP )---------------------------
THIS ROUTINE CALCULATES THE STAGE OF GROWTH AS DEFINED BY FEHR
AND CAVINESS, 1977, LIMITS THE VEGETATIVE GROWTH OF DETERMINATE
PLANTS AND DEFINES THE NUMBER OF FLOWERS.
-----------------------------------------------------------------
CALCULATE PROGRESS BETWEEN VSTAGES. CONVERT SURPLUS PROGRESS
   IN ONE STAGE TO PROGRESS IN THE NEXT.

MOVE TO APPROPRIATE PART OF ROUTINE TO CALCULATE STAGE OF
   REPRODUCTIVE GROWTH.

if reproductive stage > or = 5, jump down

  if reproductive stage > or = 1, jump down

    if reproductive stage > or = 0 (i.e. floral initiation),
       jump down

      CALCULATE PROGRESS TOWARDS R0 ( = FLORAL INDUCTION).

        if reproductive stage < 0 (i.e. floral initiation),
           jump down

      WHEN R0 IS REACHED, INITIATION OF FLOWERS IN ALL NODES OF
         DETERMINATE PLANTS FIXES THE MAXIMUM NUMBER OF NODES.

        jump down

    CALCULATE PROGRESS BETWEEN R0 AND R1.

      if reproductive stage < 1, jump down

    OVER THE PERIOD R1 TO R3 PLANTS OPEN AN AVERAGE OF 4 FLOWERS
       PER NODE.

      jump down

  CALCULATE PROGRESS BETWEEN R1 AND R5.

RETURN
```

```
          SUBROUTINE PLTMAP
C       ----------------------------( PLTMAP )---------------------------
C         THIS ROUTINE CALCULATES THE STAGE OF GROWTH AS DEFINED BY FEHR
C       AND CAVINESS, 1977, LIMITS THE VEGETATIVE GROWTH OF DETERMINATE
C       PLANTS AND DEFINES THE NUMBER OF FLOWERS.
C       ------------------------------------------------------------------
        DIMENSION TBMAX(20)
        INCLUDE 'COMMON.FOR'
C       ------------------------------------------------------------------
C       *** CALCULATE PROGRESS BETWEEN VSTAGES. CONVERT SURPLUS PROGRESS
C           IN ONE STAGE TO PROGRESS IN THE NEXT. ***
C
12.01   DV=PDV*PERIOD * (1.0-SGTLI)
        VLOST=VLOST+(PDV*PERIOD)-DV
        VSTAGE=VSTAGE+DV
        IF(VSTAGE.GE.VSTMAH) VSTAGE = VSTMAH
        IF(NBRNCH.EQ.0) GO TO 9999
        DO 20 J=1,NBRNCH
12.02 20 TB(J)=TB(J)+DV
 9999   IF(SWV2.GE.1 ) GOTO 8888
        IF( VSTAGE.LE.1.0 ) GOTO 7777
        TAIRL=TAIR(ITIME)
        IF( TAIR(ITIME).GE.30.0 ) TAIRL=30.0
        IF(SWV1.GE.1 ) GOTO 4444
        VSTAGE=1.0+((VSTAGE-1.)/(.0157*TAIRL+.0024)*(.012*TAIRL-.08))
        SWV1=1
        GOTO 8888
 4444   IF( VSTAGE.LE.2.0 ) GOTO 8888
        VSTAGE=2.0+((VSTAGE-2.)/(.012*TAIRL-.08)*(.018*TAIRL-TRIFMG))
        SWV2=1
C
C       *** MOVE TO APPROPRIATE PART OF ROUTINE TO CALCULATE STAGE OF
C           REPRODUCTIVE GROWTH. ***
C
12.03 8888 IF( RSTAGE.GE.5.0 ) GOTO 7777
        IF( RSTAGE.GE.1.0 ) GOTO 500
        IF( RSTAGE.GE.0.0 ) GOTO 400
C
C       *** CALCULATE PROGRESS TOWARDS R0 ( = FLORAL INDUCTION). ***
C
        IF( PHOTO.LE.DIFIMG ) RSTAGE=0.0
        IF( RSTAGE.EQ.0.0 ) GOTO 100
        IF( JDAY.GE.173 ) GOTO 6666
        DR=1/(45.0*(PHOTO-DIFIMG)**1.5)/24*PERIOD
        GOTO 5555
 6666   DR=1/(10.0*(PHOTO-DIFIMG)**1.5)/24*PERIOD
 5555   RSTAGE=RSTAGE+DR
        IF( RSTAGE.LT.0.0 ) GOTO 7777
C
C       *** WHEN R0 IS REACHED, INITIATION OF FLOWERS IN ALL NODES OF
C           DETERMINATE PLANTS FIXES THE MAXIMUM NUMBER OF NODES. ***
C
  100   IF( DET.EQ.1 ) GOTO 7777
12.04     VSTMAX=IFIX(2.5+PRM(13)*VSTAGE**0.73)
C     PARAMETER PRM(13) WAS 4.1
          VSTMAH=VSTMAX+0.5
          NODES=VSTMAX
C
          DO 10 J=1,NBRNCH
            TBMAX(J)=IFIX(TB(J)+VSTMAX-VSTAGE+0.5)
   10       TBMAH(J)=TBMAX(J)+0.5
            NBP1  =  NBRNCH + 1
            DO 30 J= NBP1,20
```

```
             TBMAX(J) = 5
 30          TBMAH(J) = 5.5
          GOTO 7777
C
C      *** CALCULATE PROGRESS BETWEEN R0 AND R1.
C
  400 TAIRL=TAIR(ITIME)
      IF( TAIR(ITIME).GT.24.0 ) TAIRL=24.0
      DR=(0.018*TAIRL-TRIFMG)/6./24.*PERIOD * (1.0-SGTLI)
      IF( DR.LE.0.0 ) DR=0.0
      RSTAGE=RSTAGE+DR
      IF( RSTAGE.LT.1.0 ) GOTO 7777
C
C      *** OVER THE PERIOD R1 TO R3 PLANTS OPEN AN AVERAGE OF 4 FLOWERS
C          PER NODE. ***
C
       DO 40 J = 1,NBRNCH
 40    NODES    = NODES + TBMAX(J)
12.05   NFLWRS=(4*NODES)
        GOTO 7777
C
C      *** CALCULATE PROGRESS BETWEEN R1 AND R5. ***
C
  500 DR=PDR*PERIOD * (1.0-SGTLI)
      RSTAGE=RSTAGE+DR
      IF((RSTAGE.GE.3.0).AND.(SWR3.EQ.0))THEN
      PODS=NODES*0.5
      SEEDS=PODS*2.5
      SWR3=1
      END IF
      IF( RSTAGE.LT.4.0 ) GOTO 7777
        SWR4=1
 7777 RETURN
      END
```

Notes for PLTMAP

12.01 The systems of Fehr and Caviness (1977) is used to enumerate the vegetative and reproductive growth stages. The prediction of vegetative stage has been discussed in SHTGRO, note 7.01. Here the potential change in vegetative stage, PDV is reduced by the proportion of shoot growing time lost irretrievably, SGTLI. PDV is calculated in SHTGRO on lines 92 through 96. SGTLI is calculated in TRADE on line 359.

12.02 TB(J) is equivalent to vegetative stage for branch J.

12.03 Predicting reproductive growth stage is a problem needing further research but the logic used in this model is as follows. Experiments in which plants were moved from a non-inductive to an inductive daylength indicate that the plants flowered after a period of time equivalent to that required to expand 6 trifoliolates, i.e. 6 plastochrons. This suggests that the time from floral initiation (which we call R0) to first flowering (R1) is a function of temperature and maturity group but not daylength. It seems reasonable when one realizes that there are several leaf primordia in the apex at all times and after R0 these must grow out before the flowers can be seen. Subtracting a period equivalent to 6 plastochrons from days between V1 and R1 yields days between V1 and R0.

Data derived in this way from many sources are plotted in Fig. 36. However, empirical curves based on this data failed to simulate the results of Garner and Allard (1930) which were obtained in naturally varying daylengths at Washington, DC. In fact their results can only be simulated by assuming that the

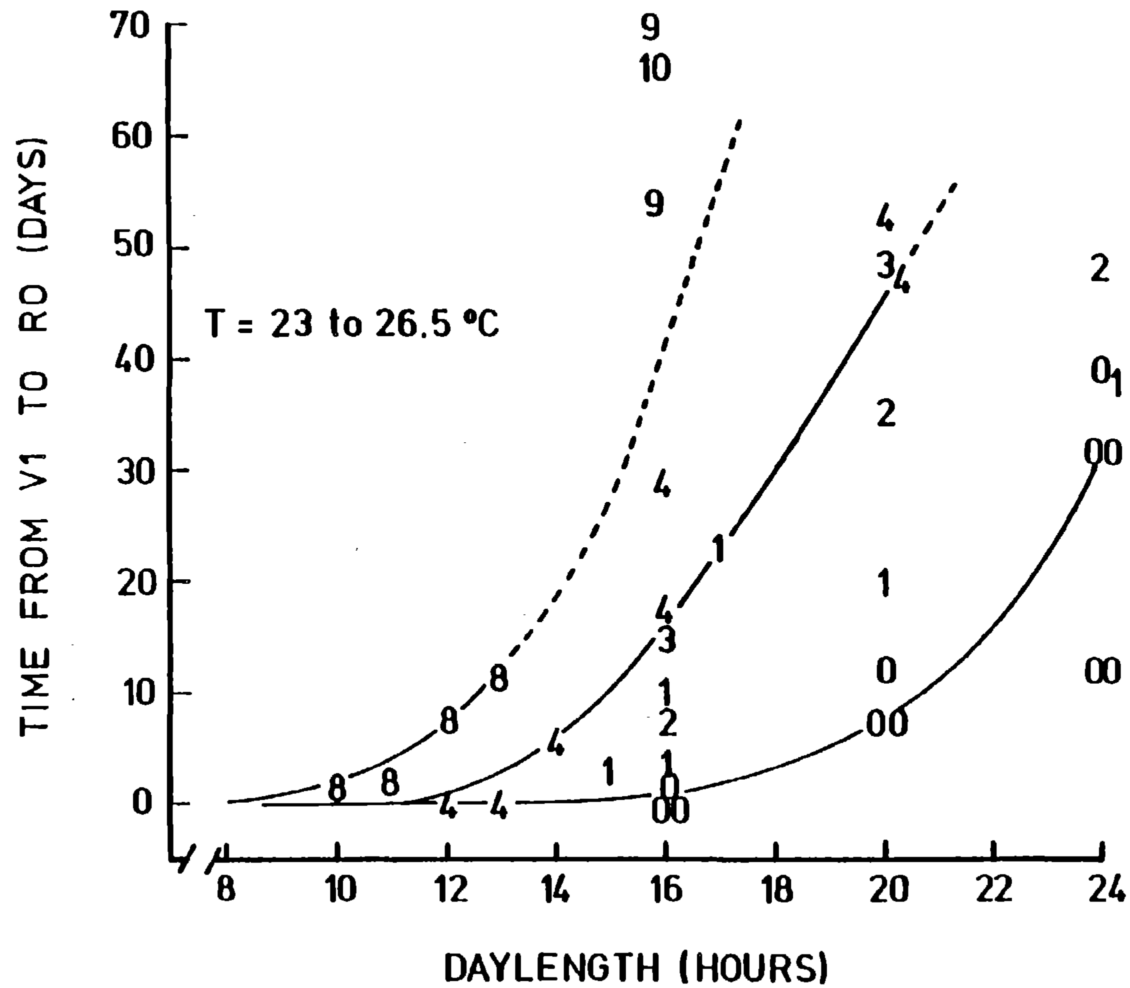


Fig. 36. Time between unifoliolate expansion and floral initiation for soybeans of various maturity groups grown in various daylengths and temperatures between 23 and 26.5^{o}C. Data from many literature sources.

response to daylength is different for rising and falling daylengths. These response curves are the dashed lines on Fig. 37, and the values they predict are shown in Fig. 38 as crosses superimposed on Garner and Allard's original data. The equations for the rate of change in reproductive stage between V1 and R0 are therefore based on the response curves in Fig. 37, while those for the rate between R0 and R1 are based on trifoliolate expansion rate.

Data on the length of time between R1 and R3 for various maturity groups and daylengths were obtained from Thomas and Raper (1976), Johnson et al. (1960), Criswell and Hume (1972) and Major et al. (1975). When these data were plotted, they looked similar to the data in Fig. 36, but the baseline was at 4 days instead of 0 days. Therefore the equations in the model to calculate progress between R1 and R3 are very similar to those used for progress between V1 and R0 (= floral initiation).

The sources quoted above also provided data on the length of time between R3 and R4 and this was a simple function of daylength independent of maturity group number.

12.04 The relationship between vegetative stage and the number of nodes on the main axis is taken from Johnson et al. (1960). See Figure 39.

12.05 The average number of flowers opened at each node is based on data of Brevedan et al. (1978) and Hansen and Shibles (1978).

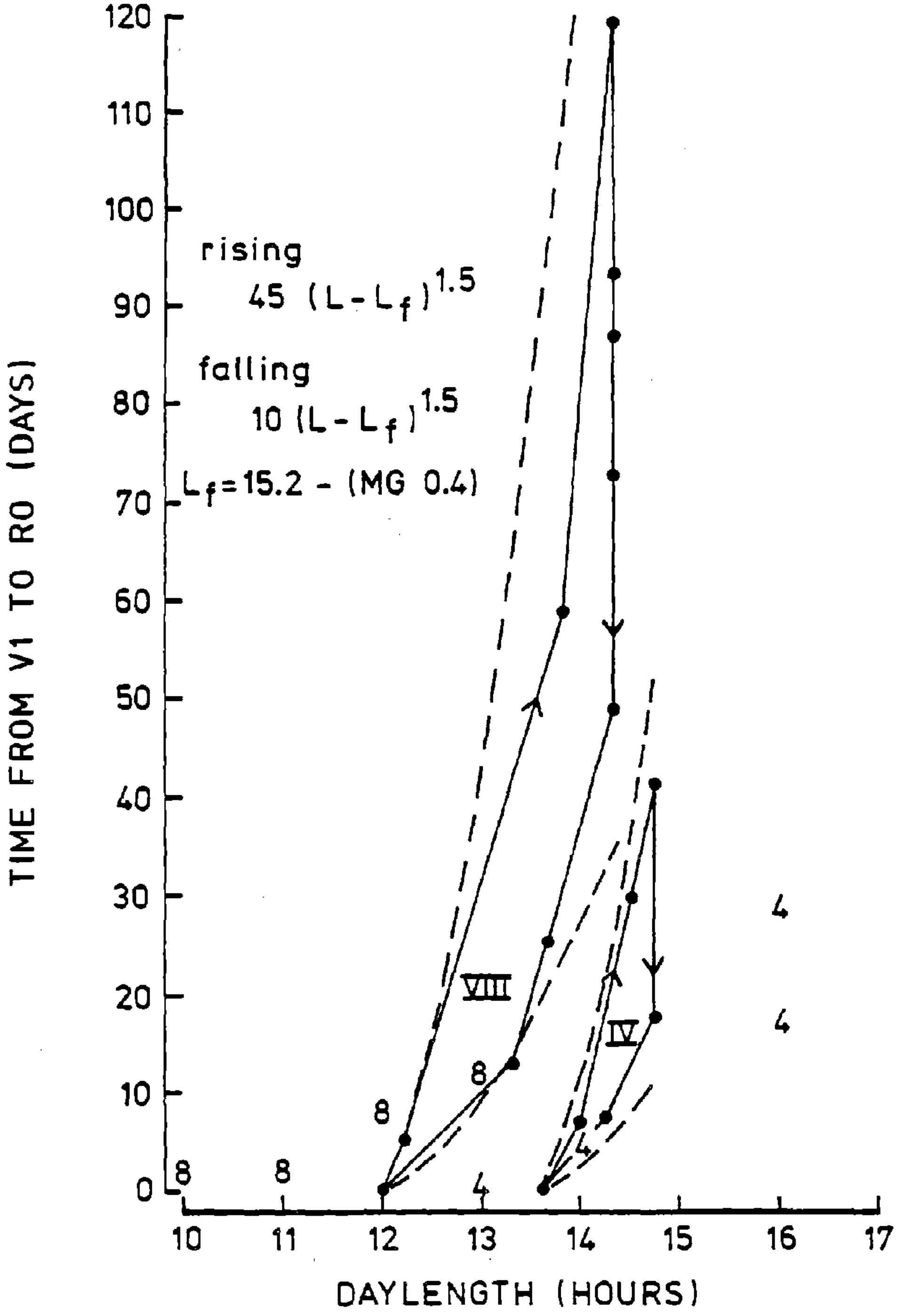


Fig. 37. Time between unifoliolate expansion and floral initiation for soybeans of maturity groups 4 and 8 in natural (changing) daylengths. The equations are for the dashed lines.

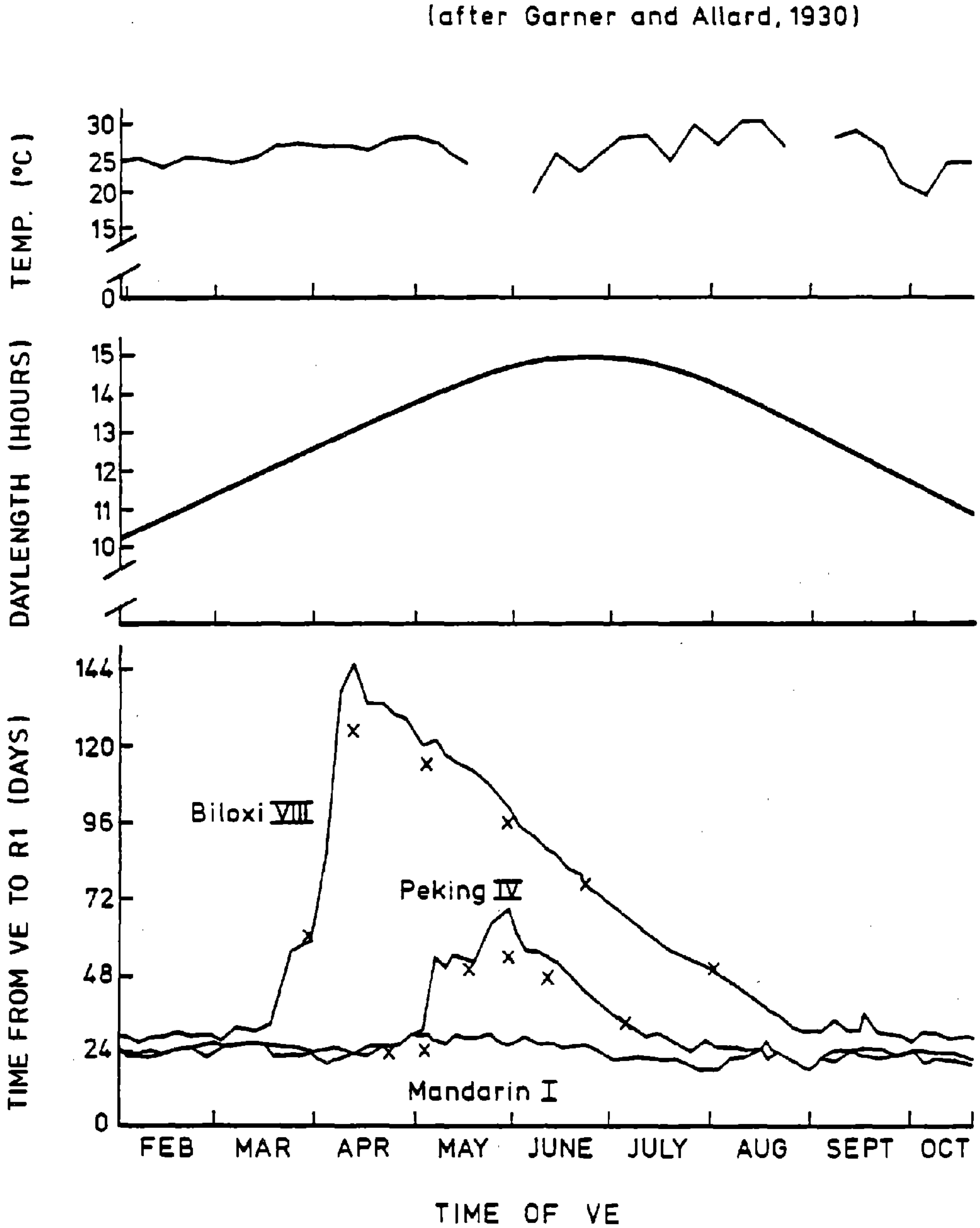


Fig. 38. Time between emergence (VE) and first flowering (R1) for three maturity groups planted at various times of year and grown in a glasshouse at Washington, D. C. The crosses are simulated points.

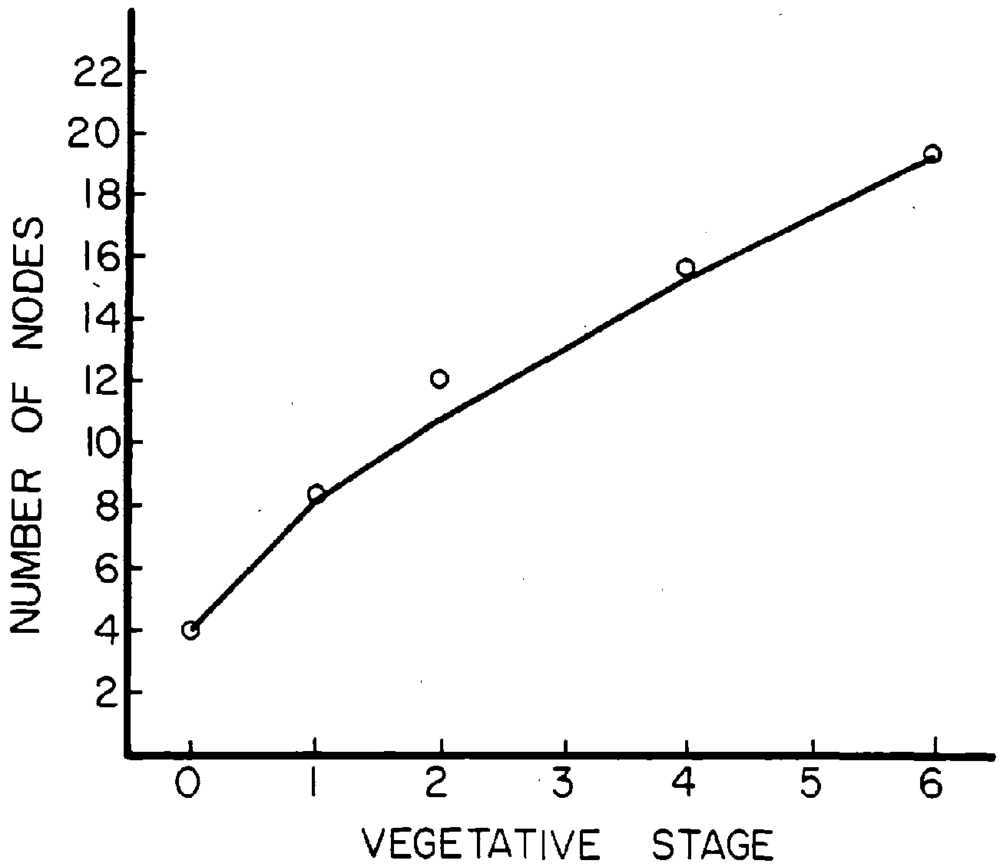


Fig. 39. The number of nodes on the mainstem including those in the apex as a function of vegetative stage (Johnson et al., 1960).

NODULE

The subroutine NODULE is called once for each period of the day and night. It calculates nodule growth and nitrogen fixation in the following way. The relative reduction of nitrogen fixation by soil temperature (from TMPSOL) and the reduction by soil oxygen concentration (from SOLOXY) are calculated for each soil cell that contains roots. Whichever produces the greatest reduction is the limiting factor for that cell. The cells are sorted into order starting with the one that has the most favorable conditions for nitrogen fixation. Then the coordinates of the cells in this order are stored in an array for later use.

The carbon that was partitioned to the nodules in NITRO is first used to fix nitrogen starting with the soil cells that have the most favorable conditions. Any remaining carbon is next used to fill the nodule carbon store and then grow nodules. Carbon still remaining after this is used to initiate new nodules. Nodules are assumed to start fixing nitrogen 3 days after they are initiated and they are assumed to die if they do not fix nitrogen for more than 10 days.

----------------------------(NODULE)----------------------------

THIS ROUTINE CALCULATES THE NITROGEN FIXING ACTIVITY, GROWTH AND DEATH OF NODULES IN EACH SOIL CELL. IT STARTS N FIXATION BY NODULES 3 DAYS AFTER THEY ARE INITIATED AND IT KILLS THOSE THAT HAVE NOT FIXED N FOR 10 DAYS.

AT THE START OF EACH DAY, AGE RECENTLY INITIATED NODULES BY ONE DAY.

CALCULATE RATE OF CARBON SUPPLY TO NODULES AND AMOUNT OF CARBON AVAILABLE TO NODULES IN A HALF SOIL SLAB DURING PRESENT PERIOD.

if there is no carbon for nodules, jump down

SINCE THERE IS NO CARBON FOR NODULES, ALL EXISTING NODULES ARE MARKED FOR DEATH IN 10 DAYS.

jump down

THE REDUCTION IN NODULE NITROGEN-FIXING ACTIVITY CAUSED BY TEMPERATURE AND OXYGEN CONCENTRATION IS CALCULATED FOR EACH SOIL CELL AND THE VALUES ARE SORTED INTO AN ARRAY TO INDICATE WHICH CELLS HAVE THE MOST FAVORABLE CONDITIONS.

----

for each soil cell with roots, do the following:

CALCULATE PROPORTIONAL REDUCTION OF NODULE N FIXATION BY SOIL TEMPERATURE.

CALCULATE PROPORTIONAL REDUCTION OF NODULE N FIXATION BY SOIL OXYGEN CONTENT.

CALCULATE PROPORTIONAL REDUCTION OF NODULE N FIXATION FROM ALL PHYSICAL CAUSES.

PLACE VALUES OF NFCF(L,K) IN ORDER HIGHEST TO LOWEST AND PUT L,K IN AN ARRAY NODWAR(J) IN THE SAME ORDER.

NITROGEN FIXING ACTIVITY,GROWTH AND INITIATION OF NODULES ARE CALCULATED FOR EACH SOIL CELL.

CALCULATE MAXIMUM POTENTIAL CHANGE IN DRY WEIGHT OF EACH NODULE DURING THIS PERIOD.

for each soil cell with roots, do the following:

CALCULATE POTENTIAL CHANGE IN DRY WEIGHT OF EXISTING NODULES AS LIMITED BY TEMPERATURE AND OXYGEN CONCENTRATION IN EACH SOIL CELL. START WITH SOIL CELL WHERE CONDITIONS ARE MOST FAVORABLE FOR N FIXATION.

A B

A B

if all carbon allocated to nodules has been used, jump down

CALCULATE C CONSUMPTION DURING N FIXATION IN EACH CELL UNTIL ALL C IS USED.

jump to end of loop

NODULES IN CELLS NOT RECEIVING C FOR N FIXATION ARE MARKED FOR DEATH IN 10 DAYS.

CALCULATE AMOUNT OF N FIXED PER PLANT DURING THIS PERIOD.

if all carbon allocated to nodules has been used, jump down

IF ANY C IS AVAILABLE FILL THE NODULES CARBON STORES.

CALCULATE C NEEDED TO GROW EXISTING NODULES AT POTENTIAL RATE IN A HALF SOIL SLAB.

CALCULATE HOW MUCH C IS AVAILABLE TO INITIATE NEW NODULES IN A HALF SOIL SLAB. IF THERE IS NONE, JUMP TO APPROPRIATE PART OF ROUTINE.

if there is no carbon available for initiating nodules, jump down

SINCE THERE IS C AVAILABLE FOR INITIATING NEW NODULES, GROW ALL EXISTING NODULES AT POTENTIAL RATE AND INITIATE NEW ONES UNTIL ALL C USED. START WITH SOIL CELL WHERE CONDITIONS ARE MOST FAVORABLE FOR N FIXATION.

jump down

SINCE THERE IS NO C AVAILABLE FOR INITIATING NEW NODULES, GROW EXISTING NODULES UNTIL ALL C USED. START WITH THE SOIL CELL WHERE CONDITIONS ARE MOST FAVORABLE FOR N FIXATION.

REDISTRIBUTE STORED CARBON BETWEEN ROOT AND SHOOT POOLS

EXAMINE SHOOT CARBON AVAILABILITY AND SET SINK TO ADJUST PHOTOSYNTHETIC RATE

RETURN

```
      SUBROUTINE NODULE
C     ----------------------------( NODULE )----------------------------
C      THIS ROUTINE CALCULATES THE NITROGEN FIXING ACTIVITY, GROWTH AND
C     DEATH OF NODULES IN EACH SOIL CELL.  IT STARTS N FIXATION BY NOD-
C     ULES 3 DAYS AFTER THEY ARE INITIATED AND IT KILLS THOSE THAT HAVE
C     NOT FIXED N FOR 10 DAYS.
C     ------------------------------------------------------------------
      INCLUDE 'COMMON.FOR'
C     ------------------------------------------------------------------
C     *** AT THE START OF EACH DAY, AGE RECENTLY INITIATED NODULES BY
C         ONE DAY. ***
C
      IF( ITIME.NE.1 ) GOTO 1111
C
        DO 120 L=1,NLN
          DO 121 K=1,NKN
            IF( NODNUM(L,K).EQ.0.0 ) GOTO 121
              NFIXNU(L,K)=NFIXNU(L,K)+NODNEW(3,L,K)
              NFIXWT(L,K)=NFIXWT(L,K)+NNEWWT(3,L,K)
C
              DO 80 I=1,2
                II=4-I
                JJ=3-I
13.01           NODNEW(II,L,K)=NODNEW(JJ,L,K)
   80         NNEWWT(II,L,K)=NNEWWT(JJ,L,K)
C
              NODNEW(1,L,K)=0.0
              NNEWWT(1,L,K)=0.0
  121   CONTINUE
  120   CONTINUE
C
C     *** CALCULATE RATE OF CARBON SUPPLY TO NODULES AND AMOUNT OF
C         CARBON AVAILABLE TO NODULES IN A HALF SOIL SLAB DURING
C         PRESENT PERIOD. ***
C
 1111 USEDCS = USEDCS + (NODC*PERIOD)
13.02 RNODC=(NODC*POPROW/200*PERIOD)+NCPOOL
      IF(RNODC.GT.0.0) GOTO 9999
C
C     *** SINCE THERE IS NO CARBON FOR NODULES, ALL EXISTING NODULES ARE
C         MARKED FOR DEATH IN 10 DAYS. ***
C
      NLNP=1
      NKNP=1
      DO 101 L=1,NLN
        DO 100 K=1,NKN
          IF( NODNUM(L,K).EQ.0.0 ) GOTO 100
13.03       NODIE(L,K)=NODIE(L,K)+(1.0/(IPERD+1.0))
            IF( NODIE(L,K).LT.10) GOTO 8888
                  NODWT=NODWT-NFIXWT(L,K)
              NFIXWT(L,K)=0.0
              NFIXNU(L,K)=0.0
              NODNUM(L,K)=0.
              GOTO 100
C
 8888       IF( L.GT.NLNP ) NLNP=L
            IF( K.GT.NKNP ) NKNP=K
  100 CONTINUE
  101 CONTINUE
      NLN=NLNP
      NKN=NKNP
      GOTO 5555
C
```

```
      C     -------------------------------------------------------------------
      C     *** THE REDUCTION IN NODULE NITROGEN-FIXING ACTIVITY CAUSED BY
      C         TEMPERATURE AND OXYGEN CONCENTRATION IS CALCULATED FOR EACH
      C         SOIL CELL AND THE VALUES ARE SORTED INTO AN ARRAY TO
      C         INDICATE WHICH CELLS HAVE THE MOST FAVORABLE CONDITIONS.***
      C     -------------------------------------------------------------------
      C     *** CALCULATE PROPORTIONAL REDUCTION OF NODULE N FIXATION BY SOIL
      C         TEMPERATURE. ***
      C
       9999 DO 21 L=1,NLR
13.04         DO 20 K=1,NKR
                IF( TS(L,K).LE.9.0 ) NFCF1=0.
                IF( (TS(L,K).GT.9.0).AND.(TS(L,K).LT.26.0) ) NFCF1=(TS(L,K) -
     &              9.0)*0.0588
                IF( (TS(L,K).GE.26.0).AND.(TS(L,K).LE.34.0) ) NFCF1=1.0
                IF( (TS(L,K).GT.34.0).AND.(TS(L,K).LT.44.0) ) NFCF1=(44.0 -
     &             TS(L,K))*0.1
                IF(TS(L,K).GE.44.0) NFCF1=0.
      C
      C     *** CALCULATE PROPORTIONAL REDUCTION OF NODULE N FIXATION BY SOIL
      C         OXYGEN CONTENT. ***
      C
13.05           NFCF2=(8.0*CXT(L,K))-(15.0*CXT(L,K)**2)
      C
      C     *** CALCULATE PROPORTIONAL REDUCTION OF NODULE N FIXATION FROM
      C         ALL PHYSICAL CAUSES. ***
      C
             NFCF(L,K)=MIN(NFCF1,NFCF2)
         20  CONTINUE
         21  CONTINUE
      C
      C     *** PLACE VALUES OF NFCF(L,K) IN ORDER HIGHEST TO LOWEST AND PUT
      C         L,K IN AN ARRAY NODWAR(J) IN THE SAME ORDER. ***
      C
            DO 51 I=1,NLR
              DO 50 J=1,NKR
                KOUNT=1
                DO 61 II=1,NLR
                  DO 60 JJ=1,NKR
                    IF( I.EQ.II.AND.J.EQ.JJ ) GOTO 60
                    IF( NFCF(I,J).LE.NFCF(II,JJ) ) KOUNT=KOUNT+1
                    IF( NFCF(I,J).NE.NFCF(II,JJ) ) GOTO 60
                    IF( (I*100+J).LT.(II*100+JJ) ) KOUNT=KOUNT-1
         60     CONTINUE
         61     CONTINUE
                NODWAR(KOUNT)=I*1000+J
         50 CONTINUE
         51 CONTINUE
      C
      C     -------------------------------------------------------------------
      C     *** NITROGEN FIXING ACTIVITY,GROWTH AND INITIATION OF NODULES ARE
      C         CALCULATED FOR EACH SOIL CELL.***
      C     -------------------------------------------------------------------
      C     *** CALCULATE MAXIMUM POTENTIAL CHANGE IN DRY WEIGHT OF EACH
      C         NODULE DURING THIS PERIOD. ***
      C
13.06       PDNWM=5.27E-5/24*PERIOD
      C
      C     *** CALCULATE POTENTIAL CHANGE IN DRY WEIGHT OF EXISTING NODULES
      C         AS LIMITED BY TEMPERATURE AND OXYGEN CONCENTRATION IN EACH
      C         SOIL CELL.  START WITH SOIL CELL WHERE CONDITIONS ARE MOST
      C         FAVORABLE FOR N FIXATION. ***
      C
```

```
              PDNWS=0.0
              CNFIXS=0.0
              NLNP=1
              NKNP=1
              LKR = NLR*NKR
13.07         DO 10 J=1,LKR
                L=NODWAR(J)/1000
                K=NODWAR(J)-L*1000
                IF( NODNUM(L,K).EQ.0.0 ) GOTO 10
                  PDNW(L,K)=NODNUM(L,K)*NFCF(L,K)*PDNWM
                  PDNWS=PDNWS+PDNW(L,K)
      C
      C       *** CALCULATE C CONSUMPTION DURING N FIXATION IN EACH CELL UNTIL
      C           ALL C IS USED. ***
      C
                  IF( NFIXWT(L,K).EQ.0.0 ) GOTO 9977
                    IF( RNODC.LE.0.0 ) GOTO 2222
13.08                 CNFIX=NFIXWT(L,K)*NFCF(L,K)*2.147E-3*PERIOD*2.0
                      NODIE(L,K)=0.
                      IF( RNODC.LE.CNFIX ) CNFIX=RNODC
                      RNODC=RNODC-CNFIX
                      CNFIXS=CNFIXS+CNFIX
                      RESPS(L,K) = RESPS(L,K) + (CNFIX*3.67)
                      GOTO 10
      C
      C       *** NODULES IN CELLS NOT RECEIVING C FOR N FIXATION ARE MARKED
      C           FOR DEATH IN 10 DAYS. ***
      C
13.09  2222         NODIE(L,K)=NODIE(L,K)+(1.0/(IPERD+1.0))
                    IF( NODIE(L,K).LT.10) GOTO 9977
                      NODWT=NODWT-NFIXWT(L,K)
                      NFIXWT(L,K)=0.0
                      NFIXNU(L,K)=0.0
                      NODNUM(L,K)=0.
                      GOTO 10
      C
       9977         IF( L.GT.NLNP ) NLNP=L
                    IF( K.GT.NKNP ) NKNP=K
          10  CONTINUE
              NLN=NLNP
              NKN=NKNP
      C
      C       *** CALCULATE AMOUNT OF N FIXED PER PLANT DURING THIS PERIOD. ***
      C
              NODN=CNFIXS/2.0*200/POPROW
      C
      C       *** IF ANY C IS AVAILABLE FILL THE NODULES CARBON STORES. ***
      C
              IF( RNODC.LE.0.0 ) GOTO 5555
                NCPOOL=RNODC
13.10           NCPM=NODWT*0.03
                IF( NCPOOL.GE.NCPM ) NCPOOL=NCPM
                IF( NCPOOL.LT.NCPM ) GOTO 7777
                  RNODC=RNODC-NCPOOL
      C
      C       *** CALCULATE C NEEDED TO GROW EXISTING NODULES AT POTENTIAL RATE
      C           IN A HALF SOIL SLAB. ***
      C
13.11             PDNCS=PDNWS*CONVN
      C
      C       *** CALCULATE HOW MUCH C IS AVAILABLE TO INITIATE NEW NODULES IN
      C           A HALF SOIL SLAB. IF THERE IS NONE, JUMP TO APPROPRIATE PART
      C           OF ROUTINE. ***
```

```
      C
13.12          CNINT=RNODC-PDNCS
               IF( CNINT.LT.0.0 ) GOTO 7777
      C
      C    *** SINCE THERE IS C AVAILABLE FOR INITIATING NEW NODULES, GROW
      C        ALL EXISTING NODULES AT POTENTIAL RATE AND INITIATE NEW ONES
      C        UNTIL ALL C USED. START WITH SOIL CELL WHERE CONDITIONS ARE
      C        MOST FAVOURABLE FOR N FIXATION. ***
      C
                 DO 30 J=1,LKR
                   L=NODWAR(J)/1000
                   K=NODWAR(J)-L*1000
                   IF(NFCF(L,K).EQ.0.) GO TO 30
                     X1=NFCF(L,K)*PDNWM
                   IF( NODNUM(L,K).EQ.0 ) GOTO 34
                     X2=X1*NFIXNU(L,K)
                     NFIXWT(L,K)=NFIXWT(L,K)+X2
                       NODWT=NODWT+X2
13.13              RESPS(L,K) = RESPS(L,K) + (X2*0.128)
                     DO 40 I=1,10
                       II=11-I
                       X3=X1*NODNEW(II,L,K)
                       NNEWWT(II,L,K)=NNEWWT(II,L,K)+X3
                       NODWT=NODWT+X3
        40         RESPS(L,K) = RESPS(L,K) + (X3*0.128)
         34          IF( CNINT.LE.0.0 ) GOTO 30
13.14                  NODSIT=ADRL(L,K)
                       PNONEW=CNINT/CONVN/PDNWM/NFCF(L,K)
                       IF( PNONEW.GE.NODSIT ) GOTO 6666
                         NODNEW(1,L,K)=NODNEW(1,L,K)+PNONEW
                         NODNUM(L,K)=NODNUM(L,K)+PNONEW
                         X4 = CNINT/CONVN
                         NNEWWT(1,L,K)=NNEWWT(1,L,K)+X4
                       NODWT=NODWT+X4
                         CNINT=0.0
                         GOTO 32
      C
       6666            NODNEW(1,L,K)=NODNEW(1,L,K)+NODSIT
                       NODNUM(L,K)=NODNUM(L,K)+NODSIT
                       X4=NODSIT*X1
                       NNEWWT(1,L,K)=NNEWWT(1,L,K)+X4
                       CNINT=CNINT-X4*CONVN
                       NODWT=NODWT+X4
      C
        32         RESPS(L,K) = RESPS(L,K) + (X4*0.128)
                       IF( L.GT.NLN ) NLN=L
         30      CONTINUE
            RCPOOL = RCPOOL+(CNINT*200./POPROW)
            USEDCS = USEDCS -(CNINT*200./POPROW)
                 GOTO 5555
      C
      C    *** SINCE THERE IS NO C AVAILABLE FOR INITIATING NEW NODULES, GROW
      C        EXISTING NODULES UNTIL ALL C USED. START WITH THE SOIL CELL
      C        WHERE CONDITIONS ARE MOST FAVOURABLE FOR N FIXATION. ***
      C
       7777    DO 70 J=1,LKR
                 L=NODWAR(J)/1000
                 K=NODWAR(J)-L*1000
                 IF( NODNUM(L,K).EQ.0.) GOTO 70
                 IF( RNODC.LE.0.0 ) GOTO 5555
                   PDNFW=NFIXNU(L,K)*NFCF(L,K)*PDNWM
                   X5=PDNFW*CONVN
                   IF( RNODC.GT.X5 ) GOTO 4444
```

```
                  X7 = RNODC/CONVN
                  NFIXWT(L,K)=NFIXWT(L,K)+X7
                  NODWT=NODWT+X7
                  RESPS(L,K) = RESPS(L,K) + (X7*0.128)
                  RNODC=0.0
                  GOTO 5555
C
 4444           NFIXWT(L,K)=NFIXWT(L,K)+PDNFW
                NODWT=NODWT+PDNFW
                RESPS(L,K) = RESPS(L,K) + (PDNFW*0.128)
                RNODC=RNODC-X5
                DO 90 I=1,5
                  II=6-I
                  PNNWWT=NODNEW(II,L,K)*NFCF(L,K)*PDNWM
                  X6=PNNWWT*CONVN
                  IF( RNODC.GT.X6 ) GOTO 3333
                    X8 = RNODC/CONVN
                    NNEWWT(II,L,K)=NNEWWT(II,L,K)+X8
                    NODWT=NODWT+X8
                    RESPS(L,K) = RESPS(L,K) + (X8*0.128)
                    RNODC=0.0
                    GO TO 5555
C
 3333             NNEWWT(II,L,K)=NNEWWT(II,L,K)+PNNWWT
                  NODWT=NODWT+PNNWWT
                  RESPS(L,K) = RESPS(L,K) + (PNNWWT*0.128)
   90           RNODC=RNODC-X6
   70       CONTINUE
C
C     *** REDISTRIBUTE STORED CARBON BETWEEN ROOT AND SHOOT POOLS ***
C
 5555       IF(ITIME.NE.(IPERD+1)) GO TO 9955
            IF((SCPOOL+RCPOOL).LE.0.0001)GO TO 9988
      RCPMAX=PRM(20)*ROOTWT
      SCPMAX=0.4*SHTWT
      XTRAC = RCPOOL-RCPMAX+SCPOOL-SCPMAX
      IF(XTRAC.LE.0.0) THEN
      X7=RCPOOL+SCPOOL
      X8=X7/(RCPMAX+SCPMAX)*RCPMAX
      IF(X8.GE.RCPOOL)GO TO 9988
      RCPOOL = X8
      SCPOOL=X7/(RCPMAX+SCPMAX)*SCPMAX
      GO TO 9988
      ELSE
      RCPOOL = RCPMAX
      SCPOOL = SCPMAX
      XTRACS = XTRACS + XTRAC
      END IF
C
C     *** EXAMINE SHOOT CARBON AVAILABILITY AND SET SINK TO ADJUST
C         PHOTOSYNTHETIC RATE ***
C
 9988 SINK = 0.0
      X75 = SCRATS/24.0
      IF(X75.GT.PRM(8))SINK = 1.0
      IF(X75.LT.PRM(8))SINK = -1.0
      SCRATS = 0.0
 9955 RETURN
      END
```

Notes for NODULE

13.01 Nodules are arbitrarily assumed to become active 3 days after initiation. A box-car train is used to carry recently-initiated nodules to activity.

13.02 The rate of carbon supply to the nodules is calculated either in NIGHT lines 237 through 265 or in TRADE lines 159 through 187. Here, that rate is converted from a rate per plant to an amount per half soil profile. The carbon in the nodule carbon pool is added to it.

13.03 Nodules are arbitrarily assumed to die if they have not fixed nitrogen for 10 days. A counter is used to determine when inactive nodules should die, but the counter can be reset to zero by a single period of nodule activity.

13.04 The effect of soil temperature on nodule N fixation is taken from the curve for plants grown at 27/22^{o}C by Gibson (1976, his Fig. 29.1). See Figure 40. A similar temperature curve has been published by Dart and Day (1971).

13.05 The effect of soil oxygen on nodule N fixation is modelled on the data of Criswell et al. (1977). See Figure 41.

Drought is assumed not to have any direct effect on N fixation but influences it via its effect on carbon supply. For evidence see Huang et al. (1975).

13.06 From the data of Weil and Ohlrogge (1975) we learn that the maximum rate of increase in volume of an average nodule is 1.054 mm^3 per day. Because nodules float half submerged in water we assume their density is 0.5 g of fresh weight per cc. Further assuming that 90% of the fresh weight is water gives us

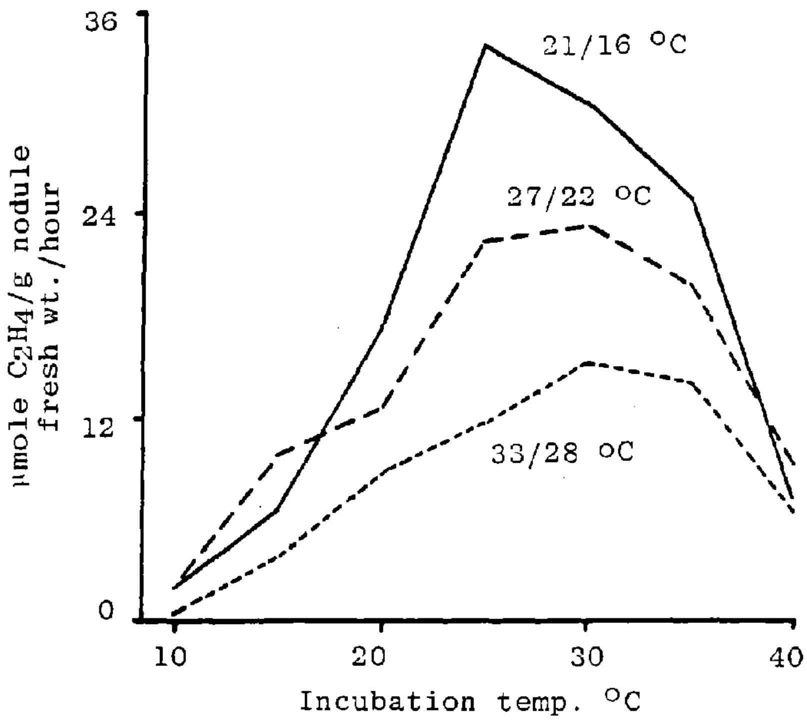


Fig. 40. Effect of temperature on acetylene reduction rate (= N_2 fixation). Figures on the curves indicate the temperature at which the plants were grown. Data of Gibson (1976).

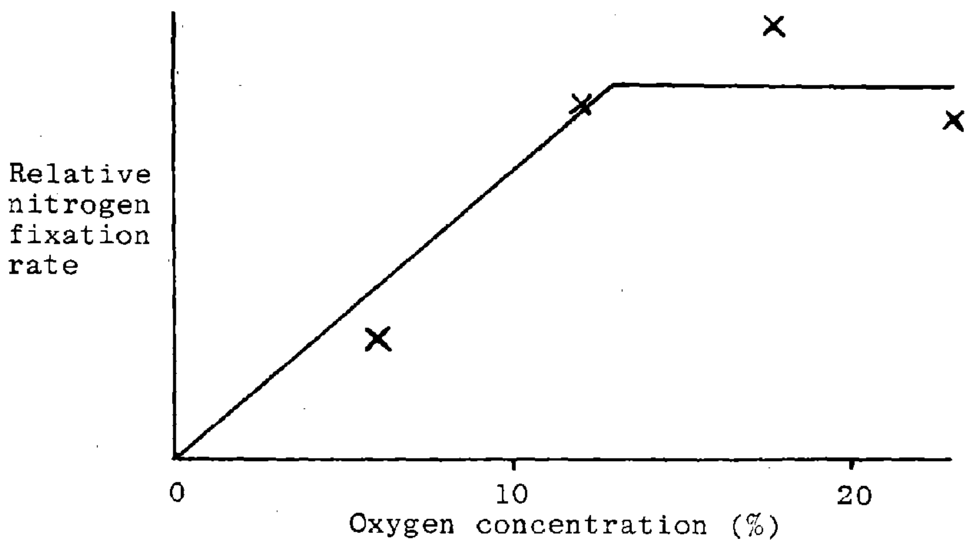


Fig. 41. Effect of oxygen concentration on nodule nitrogen fixation rate. Data of Criswell et al., (1977).

a figure for the maximum potential rate of change in nodule dry weight.

13.07 Starting with the soil cell that has the most favorable conditions, two calculations are made for each cell on the first pass. These are the potential change in dry weight of existing nodules and the amount of carbon used in nitrogen fixation.

13.08 The maximum potential rate of N fixation of 2.147×10^{-3} g N per g nodule dry weight per hour is taken from Gibson (1976, his Fig. 29.1). A similar figure has been found by Hardy et al. (1980).

Theoretically the amount of carbon needed to fix a gram of nitrogen from N_2 is the same as that needed to fix it from $^{-}NO_3$, i.e. about 2 g C per g N. However, measurements of root and nodule respiration rates and nodule N fixation rates have given estimates as high as 40 g C per g N. (See Hardy et al., 1980). This is unrealistic as the plant would need to use 74% of its carbon to fix N if it was to maintain a maximum N content. A plant using N_2 is quite obviously not at any disadvantage with respect to a plant using $^{-}NO_3$ because numerous studies have shown that they grow at the same rate. Either the cost of N fixation is essentially the same in both cases or carbon availability never limits growth. In this model the former is assumed to be true.

13.09 If all the available carbon is used up in nitrogen fixation, nodules in the remaining cells are marked for death.

13.10 Nodules are almost entirely dependent on current translocation for their carbon. In this model they have been endowed with the ability to store C up to 3% of their dry weight.

13.11 The carbon/dry weight conversion factor for nodules, CONVN, incorporates the assumption that the nodules contain 5% of structural N, i.e. nontranslocatable N. This is based on data by Lie et al. (1976) and by Gibson (1976).

13.12 If there is carbon remaining after all nodules have fixed nitrogen then a test is made to see whether or not there is enough to grow all existing nodules. If there is, the nodules all grow and some new ones are initiated. If not, only the nodules in the most favorable soil cells grow.

13.13 Carbon dioxide respired during nodule growth in each cell is added to the respiration term, RESPS, which already contains the carbon dioxide from root growth. See TRADE note 10.26.

13.14 The number of sites available for initiation of nodules is numerically equal to the length of "young" root in cm., since nodules are normally spaced at 1 cm intervals along roots and they start with the infection of root hairs which will only be present on "young" roots (Angulo et al. 1976).

UPTAKE

The subroutine UPTAKE is called once for each period of the day and night. The amount of water to be taken from each soil cell, which is calculated in NIGHT or TRADE, is actually removed in UPTAKE. An amount of nitrate appropriate to its concentration in solution in the cell is taken up with the water.

```
----------------------------( UPTAKE )------------------------------
THIS ROUTINE CALCULATES UPTAKE OF NO3 BY THE PLANTS AND CHANGES
IN WATER AND NO3 CONTENT OF EACH SOIL CELL.
--------------------------------------------------------------------
for each soil cell with roots, do the following:
|
|  CALCULATE PREDICTED NO3 UPTAKE FOR THE NEXT PERIOD OF THE DAY
|     AND NO3 REMAINING IN EACH SOIL CELL.
|
|  CALCULATE PREDICTED WATER CONTENT OF EACH SOIL CELL FOR THE
|     NEXT PERIOD OF THE DAY.
|_____________
RETURN
```

```
      SUBROUTINE UPTAKE
C     -------------------------( U P T A K E )-------------------------
C      THIS ROUTINE CALCULATES UPTAKE OF NO3 BY THE PLANTS AND CHANGES
C     IN WATER AND NO3 CONTENT OF EACH SOIL CELL.
C     -----------------------------------------------------------------
      INCLUDE 'COMMON.FOR'
C     -----------------------------------------------------------------
C     *** CALCULATE PREDICTED NO3 UPTAKE FOR THE NEXT PERIOD OF THE DAY
C         AND NO3 REMAINING IN EACH SOIL CELL. ***
C
      SUPNO3=0.0
      DO 11 L=1,NLR
        J=NJ(L)
        DO 10 K=1,NKR
          X1=AFWUP(L,K)*VNO3C(L,K)/VH2OC(L,K)
          SUPNO3=SUPNO3+(X1/1000*200/POPROW)
          VNO3C(L,K)=VNO3C(L,K)-(X1/D/W)
C
C     *** CALCULATE PREDICTED WATER CONTENT OF EACH SOIL CELL FOR THE
C         NEXT PERIOD OF THE DAY. ***
C
          VH2OC(L,K)=VH2OC(L,K)-(AFWUP(L,K)/D/W)
          AFWUPS=AFWUPS+AFWUP(L,K)
   10 CONTINUE
   11 CONTINUE
      DO 12 K = 1,NKH
          VH2OC(1,K)=VH2OC(1,K)-(EWU(K)/D/W)
          EWUS=EWUS+EWU(K)
          IF(VH2OC(1,K).LE.AIRDR(1))THEN
          LINE=3225
          WRITE(2,*)LINE,L,VH2OC(1,K),EWU(K)
          VH2OC(1,K)=AIRDR(1)
          END IF
   12 CONTINUE
      RETURN
      END
```

Notes for UPTAKE

This is a small part of the original RHIZOS subroutine (Lambert and Baker, 1984).

GRAFLO

The subroutine GRAFLO is called once for each period of the day and night but only for days when rain has fallen or irrigation has been applied. It calculates the movement of water down the profile layer by layer and out of the bottom. It also calculates the movement down and out of the profile of nitrate in solution.

```
-------------------------( GRAFLO )-------------------------------
THIS ROUTINE CALCULATES FLOW OF NO3 AND H2O DOWN PROFILE AFTER
RAIN OR IRRIGATION
------------------------------------------------------------------
for each column of soil cells, do the following:
|
|  RAIN (AND NITROGEN) SOAKING INTO TOP LAYER.
|
|  for each layer of soil cells, do the following:
|  |
|  |  TOTAL TEMPORARY AND RESIDUAL VOLUME OF H2O IN SOIL CELL.
|  |
|  |  H2O AND N SOAKING INTO LAYER BENEATH.
|  |
|  |  CALCULATE VOLUMETRIC H2O AND N CONTENTS OF ALL CELLS RECEIVING
|  |     GRAVITY FLOW.
|__|____________
RETURN
```

```
      SUBROUTINE GRAFLO
C     -----------------------( GRAFLO )---------------------------------
C     THIS ROUTINE CALCULATES FLOW OF NO3 AND H2O DOWN PROFILE AFTER
C     RAIN OR IRRIGATION
C     ------------------------------------------------------------------
      DIMENSION SOAKW(21),SOAKN(21)
      INCLUDE 'COMMON.FOR'
C     ------------------------------------------------------------------
C
C     *** RAIN (AND NITROGEN) SOAKING INTO TOP LAYER. ***
C
      DO 2 K=1,NKH
        SOAKW(NL+1)=0.0
        SOAKW(1)=RAIN*0.1/24.0*PERIOD
        IF(NYTE.EQ.1) SOAKW(1)=SOAKW(1)/2.0
        SOAKN(1)=0.0
C
C     TOTAL TEMPORARY AND RESIDUAL VOLUME OF H2O IN SOIL CELL.
C
        DO 3 L=1,NL
          H2O=SOAKW(L)+VH2OC(L,K)*D
C
C     H2O AND N SOAKING INTO LAYER BENEATH.
C
          SOAKW(L+1)=AMAX1(0.0,H2O-FC(L)*D)
          SOAKN(L+1)=AMAX1(0.0,SOAKW(L+1)*(VNO3C(L,K) /
     &                (VH2OC(L,K)+SOAKW(L)/D)))
    3   CONTINUE
        DRAIN=DRAIN+(SOAKW(NL+1)*10/NKH)
C
C     CALCULATE VOLUMETRIC H2O AND N CONTENTS OF ALL CELLS RECEIVING
C     GRAVITY FLOW.
C
        DO 4 L=1,NL
          IF( SOAKW(L).LE.0.0 ) GOTO 4
            VH2OC(L,K)=VH2OC(L,K)+(SOAKW(L)-SOAKW(L+1))/D
            VNO3C(L,K)=AMAX1(0.0,VNO3C(L,K)+(SOAKN(L)-SOAKN(L+1))/D)
    4   CONTINUE
    2 CONTINUE
      RETURN
      END
```

Notes for GRAFLO

This subroutine from RHIZOS (Lambert and Baker 1984) is modified to run in steps of less than a day.

Water only moves vertically down within each column of cells. There is no movement sideways. This can result in unrealistic loss of water out of the profile at the base of columns of cells between the plant rows if the subroutine only runs once a day (Figure 42). In this model it runs twelve times a day and CAPFLO gives realistic movement of water sideways.

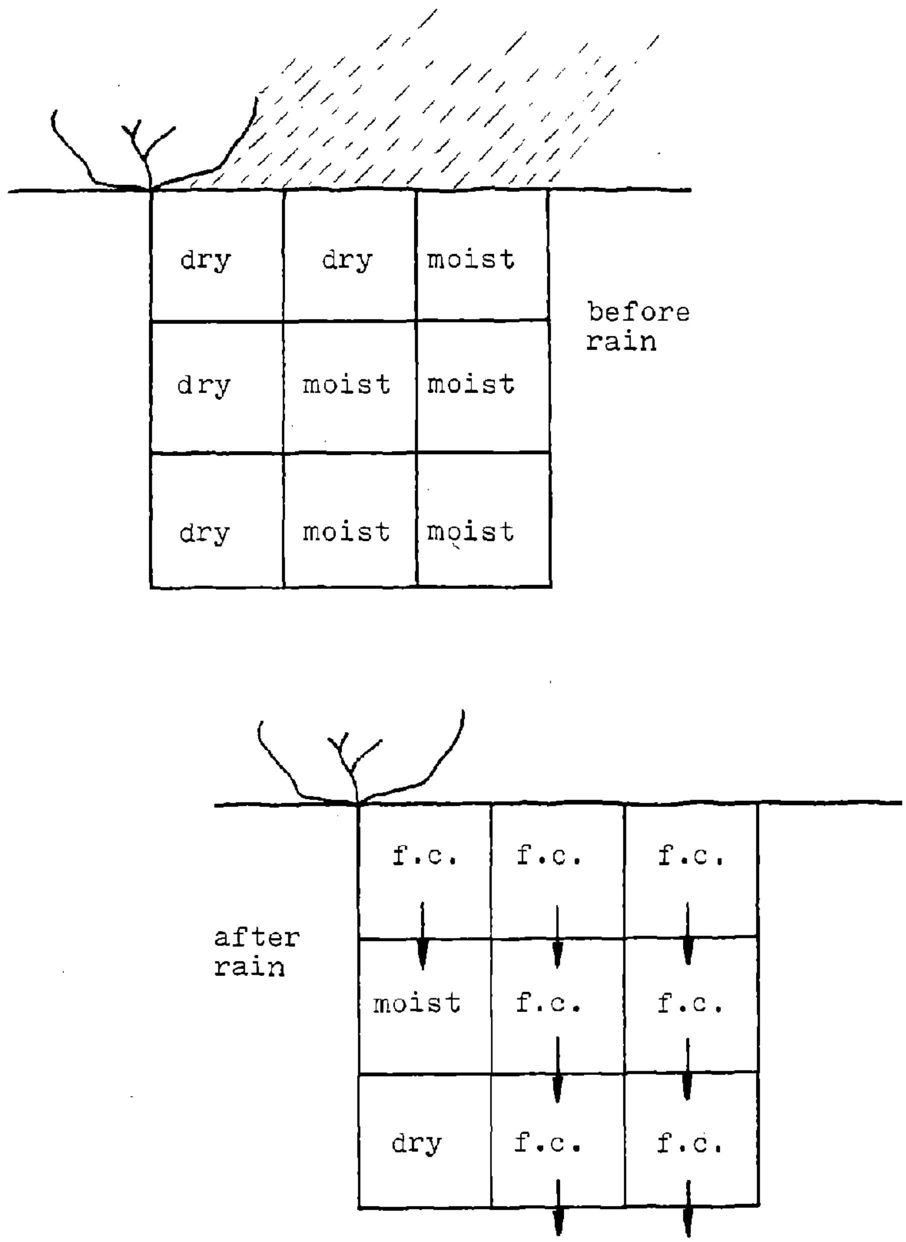


Fig. 42. Diagram illustrating known deficiencies in GRAFLO. Rain or irrigation water moves down columns of cells wetting each one to field capacity (f.c.). There is no horizontal movement of water, so unrealistic drainage out of the base of the profile can result. The soil is never saturated.

CAPFLO

The subroutine CAPFLO is called once for each period of the day and night. It calculates the hydraulic diffusivities for each cell in the soil profile, the fluxes of water between cells and the water content, hydraulic conductivity and soil water potential in each cell.

```
---------------------------( CAPFLO )-----------------------------
THIS ROUTINE CALCULATES THE CAPILLARY FLOW OF H2O AND NO3 IN ALL
DIRECTIONS WITHIN THE HALF SOIL SLAB.  IT ALSO CALCULATES HY-
DRAULIC CONDUCTIVITY AND SOIL H2O POTENTIAL IN EACH CELL.
------------------------------------------------------------------
for each layer of soil cells, do the following:
|
| CALCULATE A FACTOR FOR DETERMINING PSIS IN EACH HORIZON
|
| for each column of soil cells, do the following:
| |
| | CALCULATE HYDRAULIC DIFFUSIVITY IN EACH CELL
| |
| |
| | CALCULATE MEAN DIFFUSIVITY OF EACH PAIR OF ADJACENT CELLS
|_|__________
CALCULATE VERTICAL FLOWS OF H2O AND NO3

for each soil cell, do the following:
|
| FLOW OF H2O UPWARD OUT OF CELL
|
| CALCULATE CHANGE IN H2O CONTENT IN CELLS
|
| FLOW OF NITROGEN UPWARD OUT OF CELL.  WHEN FLOW IS NEGATIVE
|    USE VNO3C AND VH2OC VALUES FOR OTHER CELL.
|
| CALCULATE CHANGE IN NO3 CONTENT IN CELL
|__________
CALCULATE HORIZONTAL FLOW OF H2O AND NO3

for each soil cell, do the following:
|
| FLOW OF WATER LEFT OUT OF CELL
|
| CALCULATE CHANGE IN H2O CONTENT IN CELL
|
| FLOW OF NITROGEN TO THE LEFT OUT OF CELL.  WHEN FLOW IS NEGA-
|    TIVE USE VNO3C AND VH2OC VALUES FOR OTHER CELL.
|
| CALCULATE CHANGE IN NO3 CONTENT IN CELL
|__________
RESET BOTTOM BOUNDARY CONDITIONS FOR SOME SITES

for each soil cell, do the following:
|
| CALCULATE HYDRAULIC CONDUCTIVITY AND SOIL H2O POTENTIAL IN
|    EACH CELL
|__________
RETURN
```

```
      SUBROUTINE CAPFLO
C     --------------------------( CAPFLO )------------------------------
C     THIS ROUTINE CALCULATES THE CAPILLARY FLOW OF H2O AND NO3 IN ALL
C     DIRECTIONS WITHIN THE HALF SOIL SLAB.  IT ALSO CALCULATES HY-
C     DRAULIC CONDUCTIVITY AND SOIL H2O POTENTIAL IN EACH CELL.
C     ------------------------------------------------------------------
      DIMENSION DIFF(20,10),DIFL(20,10),DIFU(20,10),PSIFAC(5)
      INCLUDE 'COMMON.FOR'
C     ----------------------------------------------------------------
C     *** CALCULATE A FACTOR FOR DETERMINING PSIS IN EACH HORIZON ***
C
      DO 5 L=1,NL
        J = NJ(L)
        LM1 = L-1
      THETAR = THETA0(J)-0.001
      X2=(THETAR-AIRDR(J))/(THETAS(J)-AIRDR(J))
16.01 PSIFAC(J)=ALOG(15.0/ABS(PSISAT(J)))/ALOG(X2)
C
C     *** CALCULATE HYDRAULIC DIFFUSIVITY IN EACH CELL ***
C
        DO 4 K=1,NKH
        KM1 = K-1
          DIFF(L,K)= DIFF0(J)
          X10=BETA(J)*(VH2OC(L,K)-THETA0(J))
16.02     IF(VH2OC(L,K).GT.THETA0(J)) DIFF(L,K)=DIFF0(J)*EXP(BETA(J) *
     &                                 (VH2OC(L,K)-THETA0(J)))
C
C     *** CALCULATE MEAN DIFFUSIVITY OF EACH PAIR OF ADJACENT CELLS ***
C
      IF(K.EQ.1) GO TO 9999
16.03 DIFL(L,K)=2.*DIFF(L,K)* DIFF(L,KM1)/(DIFF(L,K)+DIFF(L,KM1))
 9999 IF(L.EQ.1) GO TO 4
      DIFU(L,K)=2.*DIFF(L,K)*DIFF(LM1,K)/(DIFF(L,K)+DIFF(LM1,K))
    4   CONTINUE
    5   CONTINUE
        DW=D*W
        WPN=2.0*PERIOD/24.0/W/W
        DPN=2.0*PERIOD/24.0/D/D
C
C     *** CALCULATE VERTICAL FLOWS OF H2O AND NO3 ***
C
        DO 7 L=2,NL
          J = NJ(L)
          LM1=L-1
          DO 6 K=1,NKH
C
C     *** FLOW OF H2O UPWARD OUT OF CELL ***
C
      IF(NJ(L).EQ.NJ(LM1))THEN
16.04 FWU=(VH2OC(L,K)-VH2OC(LM1,K))*(1-EXP(-DIFU(L,K)*DPN))*DW*0.5
      ELSE
      THETAR = THETA0(J)-0.001
16.05 VH2OEQ = THETAR+(((PSIS(LM1,K)/PSISAT(J))**(1.0/PSIFAC(J)))
     &           *(THETAS(J)-THETAR))
      FWU=(VH2OC(L,K)-VH2OEQ)*(1-EXP(-DIFU(L,K)*DPN))*DW*0.5
C
C     *** CALCULATE CHANGE IN H2O CONTENT IN CELLS ***
C
            VH2OC(L,K)=VH2OC(L,K)-(FWU/DW)
               IF(VH2OC(L,K).GT.THETAS(J))THEN
               VH2OC(LM1,K)=VH2OC(LM1,K)+VH2OC(L,K)-THETAS(J)
               VH2OC(L,K)=THETAS(J)
               END IF
```

```
               VH2OC(LM1,K)=VH2OC(LM1,K)+(FWU/DW)
                  M=NJ(LM1)
                  IF(VH2OC(LM1,K).GT.THETAS(M))THEN
                  VH2OC(L,K)=VH2OC(L,K)+VH2OC(LM1,K)-THETAS(M)
                  VH2OC(LM1,K)=THETAS(M)
                  END IF
               GO TO 8888
         END IF
               VH2OC(L,K)=VH2OC(L,K)-(FWU/DW)
               VH2OC(LM1,K)=VH2OC(LM1,K)+(FWU/DW)
C
C       *** FLOW OF NITROGEN UPWARD OUT OF CELL.  WHEN FLOW IS NEGATIVE
C           USE VNO3C AND VH2OC VALUES FOR OTHER CELL. ***
C
  8888         IF( FWU.LT.0.0 ) GOTO 300
                 FNU=FWU*VNO3C(L,K)/VH2OC(L,K)
                 GOTO 302
   300         FNU=FWU*VNO3C(LM1,K)/VH2OC(LM1,K)
C
C       *** CALCULATE CHANGE IN NO3 CONTENT IN CELL ***
C
   302         VNO3C(L,K)=VNO3C(L,K)-(FNU/DW)
               VNO3C(LM1,K)=VNO3C(LM1,K)+(FNU/DW)
     6    CONTINUE
     7    CONTINUE
C
C       *** CALCULATE HORIZONTAL FLOW OF H2O AND NO3 ***
C
          DO 52 L=1,NL
            DO 51 K=2,NKH
              KM1=K-1
C
C       *** FLOW OF WATER LEFT OUT OF CELL ***
C
        FWL=(VH2OC(L,K)-VH2OC(L,KM1))*(1-EXP(-DIFL(L,K)*WPN))*DW*0.5
C
C       *** CALCULATE CHANGE IN H2O CONTENT IN CELL ***
C
              VH2OC(L,K)=VH2OC(L,K)-(FWL/DW)
              VH2OC(L,KM1)=VH2OC(L,KM1)+(FWL/DW)
C
C       *** FLOW OF NITROGEN TO THE LEFT OUT OF CELL.  WHEN FLOW IS NEGA-
C           TIVE USE VNO3C AND VH2OC VALUES FOR OTHER CELL. ***
C
              IF( FWL.LT.0.0 ) GOTO 304
                FNL=FWL*VNO3C(L,K)/VH2OC(L,K)
                GOTO 306
   304        FNL=FWL*VNO3C(L,KM1)/VH2OC(L,KM1)
C
C       *** CALCULATE CHANGE IN NO3 CONTENT IN CELL ***
C
   306        VNO3C(L,K)=VNO3C(L,K)-(FNU/DW)
              VNO3C(L,KM1)=VNO3C(L,KM1)+(FNU/DW)
C
    51      CONTINUE
    52      CONTINUE
C
C       *** RESET BOTTOM BOUNDARY CONDITIONS FOR SOME SITES ***
C
        IF(SITE.NE.2.OR.DATA.NE.1) GO TO 18
        DO 18 K = 1,NKH
        IF(IDAY.GT.T0) GO TO 6667
        VH2OC(NL,K) = THETAI
```

```
      GO TO 18
C
6667  IF(VH2OC(NL,K).LE.(DUMC*THETAI)) GO TO 6668
      VH2OC(NL,K) = THETAI-DUMB*(IDAY-T0)
      GO TO 18
C
6668  VH2OC(NL,K) = DUMC*THETAI
18    CONTINUE
C
C     *** CALCULATE HYDRAULIC CONDUCTIVITY AND SOIL H2O POTENTIAL IN
C         EACH CELL ***
C
        DO 53 L=1,NL
          J=NJ(L)
          THETAR = THETA0(J)-0.001
          DO 50 K=1,NKH
          IF((VH2OC(L,K)-THETAR).LE.0.0)THEN
          COND(L,K) = 0.0
          ELSE
16.06     COND(L,K)=SCOND(J)*((VH2OC(L,K)-THETAR)/(THETAS(J)-THETAR))
     &                **(3.0*ETA(J)/(ETA(J)-2.0))
          END IF
          IF( COND(L,K).GT.SCOND(J)) COND(L,K)=SCOND(J)
      X4=(VH2OC(L,K)-AIRDR(J))/(THETAS(J)-AIRDR(J))
      PSIS(L,K) = -30.0
16.07 IF(X4.GT.0.0)PSIS(L,K)=PSISAT(J)*X4**PSIFAC(J)
   30   FORMAT(19H PROBLEM IN CAPFLO.)
      IF(VH2OC(L,K).GT.(THETAS(J)+.01)) WRITE(2,30)
   50   CONTINUE
   53   CONTINUE
      RETURN
      END
```

Notes for CAPFLO

This is a subroutine from RHIZOS (Lambert and Baker 1984) that has been greatly modified.

16.01 PSIFAC is a factor used in the calculation soil water potential on line 217. It is the part of the overall equation that is dependent only on soil type and not on soil water content.

16.02 Hydraulic diffusivity is calculated using the method and equations of Whisler (1976).

16.03 The calculation of mean diffusivity between adjacent pairs of cells uses geometric averaging.

16.04 The equations for the flow of water between adjacent cells employ Darcy's law integrated over time and allowing for the fact that as flow occurs the gradient driving the flow decreases. The derivation of the equations is the same as for those used to describe heat flow in TMPSOL (See note 5.13).

16.05 In the equations used to describe water flux, the hydraulic diffusivity form of the Darcy law is used, and a water content gradient drives the flux. This form of the law is easier to manipulate mathematically than the conductivity form, in which a water potential gradient drives the flux. However, in reality, water moves in response to water potential gradients. Use of the water content gradient only becomes a problem when we calculate movement of water across the boundary between soil horizons that have completely different pF curves.

Here, we overcome the problem of transfer across a horizon boundary by taking the soil water potential in horizon B and using it to calculate an equivalent water content from the pF curve for horizon A. This apparent gradient is used to drive the flux of water across the boundary.

16.06 Hydraulic conductivity is calculated using the equation given by Whisler (1976).

16.07 Soil water potential is calculated from an equation suggested by Dr. A. Marani, of the Hebrew University of Jerusalem, which gives a good fit to data for soils we have examined.

NITRIF

The subroutine NITRIF is called once each 24 hours from NIGHT. It calculates the amount of ammonium released by the breakdown of organic matter and the amount of ammonium converted to nitrate in each cell of the soil profile.

```
      SUBROUTINE NITRIF
C     ----------------------------( NITRIF )----------------------------
C     THIS SUBROUTINE CALCULATES THE NITRIFICATION OF NH4 AND ORGANIC
C     MATTER.
C     ------------------------------------------------------------------
      INCLUDE 'COMMON.FOR'
C     ------------------------------------------------------------------
      DO 6 L=1,NL
        DO 5 K=1,NKH
          IF (VNC(L,K).LE.0.0)GO TO 5
          FMIN=7300000.0*10.0**(-2758.0/(TS(L,K)+273.3))
          FNIT=0.05*10.0**(12.0-3573.0/(TS(L,K)+273.3))
          WFMIN=VH2OC(L,K)/FC(L)
          DNMIN=VNC(L,K)*FMIN*WFMIN
          VNC(L,K)=VNC(L,K)-DNMIN
          VMUCK(L,K)=VMUCK(L,K)-DNMIN*0.077
          VNH4C(L,K)=VNH4C(L,K)+DNMIN
          WFNIT=0.7-1.30*(FC(L)-VH2OC(L,K))/FC(L)
          IF( WFNIT.LT.0.0 ) WFNIT=0.0
          DNIT=VNH4C(L,K)*FNIT*WFNIT
          VNH4C(L,K)=VNH4C(L,K)-DNIT
          VNO3C(L,K)=VNO3C(L,K)+DNIT
17.01     RESPS(L,K) = RESPS(L,K) + (DNMIN*D*W*0.1452)
C         0.1452 = 99*0.4*3.67/1000
    5 CONTINUE
    6 CONTINUE
      RETURN
      END
```

Notes for NITRIF

This subroutine is based entirely on the data and model of Kafkafi et al. (1978). It was encoded by Dr. A. Marani of The Hebrew University of Jerusalem and has not been checked by the authors.

17.01 In calculating the concentration of organic nitrogen in the soil, VNC, it is assumed that soil organic matter is 1% nitrogen. If the amount of N nitrified to NH_4 is DNMIN, then the amount of organic matter respired will be DNMIN*99. There is 40% C in CH_2O and each gram of C respired releases 3.67 g CO_2. Thus respiration by soil micro-organisms in each soil cell can be calculated.

SOLOXY

The subroutine SOLOXY is called once each 24 hours from NIGHT. It calculates, for each soil cell, its aeration porosity and the apparent diffusivity of oxygen. Oxygen consumption rate for each soil cell is calculated from the respiration rates associated with root and nodule growth, nodule nitrogen fixation, and micro-organism activity. These respiration rates have already been determined in NIGHT, TRADE, NODULE and NITRIF. The movement of oxygen vertically down the profile is calculated first and then the movement horizontally. New oxygen concentrations are calculated for each cell.

```
------------------------( SOLOXY )--------------------------------
THIS SUBROUTINE CALCULATES THE OXYGEN CONCENTRATION IN THE
ROOT ZONE.
------------------------------------------------------------------
for each soil cell, do the following:
|
|  DETERMINATION OF SOIL POROSITY
|
|  DETERMINATION OF APPARENT DIFFUSION OF OXYGEN
|     USING ITS LINEAR RELATION WITH AERATION POROSITY
|____________
for each soil cell, do the following:
|
|  VERTICAL DIFFUSION OF OXYGEN
|____________
for each soil cell, do the following:
|
|  HORIZONTAL DIFFUSION OF OXYGEN
|____________
RETURN
```

```
      SUBROUTINE SOLOXY
C     ------------------------( SOLOXY )------------------------------
C     THIS SUBROUTINE CALCULATES THE OXYGEN CONCENTRATION IN THE
C     ROOT ZONE.
C     ----------------------------------------------------------------
      DIMENSION S(20,10),DPR(20,10),SORES(20,10),CB(10)
      INCLUDE 'COMMON.FOR'
C     ----------------------------------------------------------------
C      XI=1.
C      IF(XI.EQ.1.)RETURN
C
C      *** DETERMINATION OF SOIL POROSITY ***
C
      DO 6 L=1,NL
      DO 5 K=1,NKH
      S(L,K)=TOPORS(L,K)-VH2OC(L,K)
C
C      *** DETERMINATION OF APPARENT DIFFUSION OF OXYGEN
C          USING ITS LINEAR RELATION WITH AERATION POROSITY ***
C
      DPR(L,K)=0.0249+(.1387*S(L,K))
    5 CONTINUE
    6 CONTINUE
C
C     *** VERTICAL DIFFUSION OF OXYGEN ***
C
      T = 86400.
      DO 702 I=1,NKH
      CB(I)=0.21
  702 CONTINUE
      DO 25 L=1,NL
      AL=DIMPL-D*(L-0.5)
      IF(L.EQ.1)SX=D*.5
      IF(L.GT.1)SX=D
      DO 35 K=1,NKH
      SUM1=0.
      N=1
   26 BETA0=(2.*N-1.)/(2.*AL)
      IF((DPR(L,K)*T*(BETA0**2)).GE.70) GO TO 29
      DLSUM1=(SIN(BETA0*SX)*EXP(-DPR(L,K)*T*(BETA0**2)))/(2.*N-1.)
      SUM1=SUM1+DLSUM1
      TEST1=DLSUM1/SUM1
      IF(TEST1.LT.0.05)GO TO 29
      N=N+1
      IF(N.GE.20)GO TO 29
      GO TO 26
   29 SUM2=0.
      N=1
   27 BETA0=(2.*N-1.)/(2.*AL)
      IF((DPR(L,K)*T*(BETA0**2)).GE.70) GO TO 28
      DLSUM2=(SIN(BETA0*SX)*EXP(-DPR(L,K)*T*(BETA0**2)))/((2.
     &*N-1.)**3)
      SUM2=SUM2+DLSUM2
      TEST2=DLSUM2/SUM2
      IF(TEST2.LT.0.05)GO TO 28
      N=N+1
      IF(N.GE.20)GO TO 28
      GO TO 27
 28   IF(RESPS(L,K).LE.1.0E-30) RESPS(L,K) = 0.0
18.01 SORES(L,K) = RESPS(L,K)/D/W/T
      RESPS(L,K) = 0.0
      VSORES=-(SORES(L,K)*.0821*(273.+TS(L,K))*1000.)/273.
      TERM2=(VSORES*SX*(2.*AL-SX))/(2.*DPR(L,K))
```

```
        TERM3=(CXT(L,K)-CB(K))*4.*SUM1/3.1416
        TERM4=(16.*VSORES*(AL**2)*SUM2)/((3.1416**3
     &)*DPR(L,K))
        CXT(L,K)=CB(K)+TERM2+TERM3-TERM4
        IF(CXT(L,K).LT.1.0E-30) CXT(L,K)=0.0
        CB(K)=CXT(L,K)
   35 CONTINUE
   25 CONTINUE
C
C     *** HORIZONTAL DIFFUSION OF OXYGEN ***
C
18.02   DO 703 L = 1,NL
        CB(L)  = CXT(L,NKH)
  703   CONTINUE
        SX = W
        JJ = NKH-1
        DO 45 KK = 1,JJ
        K = NKH-KK
        AL = W*(K-0.5)
        DO 40 L = 1,NL
        SUM1 = 0.
        N = 1
 41     BETA0=((2.*N)-1.)/(2.*AL)
        IF((DPR(L,K)*T*(BETA0**2)).GE.70) GO TO 42
        DLSUM1=(SIN(BETA0*SX)*EXP(-DPR(L,K)*T*(BETA0**2)))
     &             /((2.*N)-1.)
        SUM1 = SUM1 + DLSUM1
        TEST1 = DLSUM1/SUM1
        IF(TEST1.LT.0.05) GO TO 42
        N = N + 1
        IF(N.GE.20) GO TO 42
        GO TO 41
C
 42     TERM3=(CXT(L,K)-CB(L))*4*SUM1/3.1416
        CXT(L,K) = CB(L)+TERM3
        IF(CXT(L,K).LT.1E-10) CXT(L,K)=0.0
        CB(L) = CXT(L,K)
  40    CONTINUE
  45    CONTINUE
        RETURN
        END
```

Notes for SOLOXY

This subroutine is based on the data and model of Melhuish et al. (1974) for oxygen diffusion as interpreted and encoded by J. B. Talib in his Ph.D dissertation (1980) under the direction of F. D. Whisler.

18.01 The respiration rate used here is calculated from root growth (NIGHT, lines 354 through 374 and TRADE, lines 478 through 498), from nodule growth and nitrogen fixation (NODULE) and from microbial activity (NITRIF, line 88). This replaces the empirical relationship between respiration rate, root density and soil depth used by earlier workers.

18.02 The equations used to calculate vertical diffusion of oxygen are now used to calculate horizontal diffusion. However, respiration has already been accounted for during vertical diffusion, so respiration rate is set equal to zero and that simplifies considerably the equations for horizontal diffusion.

SOYPLT

The subroutine SOYPLT is called, with the frequency requested by the user, from MAIN. It prepares a simple diagram of the soybean plant for output on the line printer.

```
-------------------------( SOYPLT )-------------------------------
THIS SUBROUTINE PREPARES A DIAGRAM OF THE SOYBEAN PLANT WITH THE
FOLLOWING SYMBOLS: NODE I, COTYLEDONS 0, UNIFOLIOLATES -0,
TRIFOLIOLATES ---000, FLOWERS *, GREENPODS D, DRYPODS $
------------------------------------------------------------------
SELECT APPROPRIATE SYMBOLS FOR REPRODUCTIVE STAGE

INITIALIZE THE ARRAY OF SYMBOLS FOR VEGETATIVE PLANT PARTS

THE ARRAY USED TO PLOT THE PLANT IS FILLED WITH BLANK
   CHARACTERS

INSERT THE FIRST TWO NODES INTO THE PLOT ARRAY AND IF THE
   COTYLEDONS AND UNIFOLIOLATES ARE STILL ON THE PLANT,
   ADD THESE TO THE ARRAY

if there are no trifoliolates on the mainstem, jump down
|
|  ADD TO THE PLOT ARRAY THE SYMBOLS FOR ORGANS AT EACH NODE ON
|     THE MAINSTEM
|
|---if there are no branches on the plant, jump down
|
|  SET UP THE PLOT ARRAY FOR THE BRANCHES IN THE SAME WAY AS
|     FOR THE MAINSTEM
|
v
WRITE OUT THE PLOT ARRAY TO SHOW A SYMBOLIC PLANT

RETURN
```

```
      SUBROUTINE SOYPLT
C     -------------------------( SOYPLT )-------------------------------
C     THIS SUBROUTINE PREPARES A DIAGRAM OF THE SOYBEAN PLANT WITH THE
C     FOLLOWING SYMBOLS: NODE I COTYLEDONS 0 UNIFOLIATES -0
C     TRIFOLIATES ---000 FLOWERS * GREENPODS D DRYPODS $
C     ------------------------------------------------------------------
      DIMENSION PMAP(21,11,9),ALP(5)
      INCLUDE 'COMMON.FOR'
C     ------------------------------------------------------------------
C     *** SELECT APPROPRIATE SYMBOLS FOR REPRODUCTIVE STAGE ***
C
      STG = '    '
      IF(RSTAGE.GE.1.0) STG = '****'
      IF(RSTAGE.GE.3.0) STG = 'DDDD'
      IF(RSTAGE.GE.5.0) STG = '$$$$'
C
C     *** INITIALIZE THE ARRAY OF SYMBOLS FOR VEGETATIVE PLANT PARTS ***
C
      ALP(1) = '----'
      ALP(2) = '----'
      ALP(3) = '0000'
      ALP(4) = '0000'
      ALP(5) = '0000'
C
C     *** THE ARRAY USED TO PLOT THE PLANT IS FILLED WITH BLANK
C         CHARACTERS ***
C
          DO 100 I = 1,21
          DO 100 J = 1,11
          DO 100 K = 1,9
19.01     PMAP(I,J,K) = '    '
  100 CONTINUE
C
C     *** INSERT THE FIRST TWO NODES INTO THE PLOT ARRAY AND IF THE
C         COTYLEDONS AND UNIFOLIOLATES ARE STILL ON THE PLANT,ADD
C         THESE TO THE ARRAY ***
C
      PMAP(21,6,1) = 'IIII'
      PMAP(20,6,1) = 'IIII'
      IF(MPETL(1).LE.0.) GO TO 120
      PMAP(21,5,1) = '0000'
      PMAP(21,7,1) = '0000'
      PMAP(20,4,1) = '0000'
      PMAP(20,5,1) = '----'
      PMAP(20,7,1) = '----'
      PMAP(20,8,1) = '0000'
  120 CONTINUE
      NODE = 21
      IF(NTOPLF.LT.1) GO TO 240
C
C     *** ADD TO THE PLOT ARRAY THE SYMBOLS FOR ORGANS AT EACH NODE ON
C         THE MAINSTEM ***
C
      DO 160 I = 1,NTOPLF
      NODE = 20-I
      NSTG = 5
      IF(I/2*2.NE.I) NSTG = 7
      PMAP(NODE,6,1) = 'IIII'
      PMAP(NODE,NSTG,1) = STG
      IF(MPETL(I).LE.0) GO TO 160
      DO 140 J = 1,5
      L = J
      IF(NSTG.GT.6) L = -J
```

```
              PMAP(NODE,6+L,1) = ALP(J)
        140   CONTINUE
        160   CONTINUE
              IF(NBRNCH.LE.0) GO TO 240
      C
      C       *** SET UP THE PLOT ARRAY FOR THE BRANCHES IN THE SAME WAY AS FOR
      C           THE MAINSTEM ***
      C
              DO 220 K = 1,NBRNCH
19.02         ITB = NBTPLF(K)
              IF(ITB.LE.0) GO TO 220
              DO 200 M = 1,ITB
              NODB = 20-M
              NSTG = 5
              IF(M/2*2.NE.M) NSTG = 7
              PMAP(NODB,6,K+1) = 'IIII'
              PMAP(NODB,NSTG,K+1) = STG
              IF(BPETL(M,K).LE.0.) GO TO 200
              DO 180 J = 1,5
              L = J
              IF(NSTG.GT.6) L = -J
              PMAP(NODB,6+L,K+1) = ALP(J)
        180   CONTINUE
        200   CONTINUE
        220   CONTINUE
        240   CONTINUE
      C
      C       *** WRITE OUT THE PLOT ARRAY TO SHOW A SYMBOLIC PLANT ***
      C
        290   FORMAT(//)
              WRITE(2,290)
19.03         DO 280 I = NODE,21
        260   FORMAT(9(3X,11A1))
              WRITE(2,260)((PMAP(I,J,K),J=1,11),K=1,9)
        280   CONTINUE
              RETURN
              END
```

Notes for SOYPLT

Subroutine written by Mr. S. B. Turner.

19.01 For PMAP (I,J,K), I is the number of the line on which a given node will be printed, J specifies the position on that line for each of the symbols representing organs at that node and K is the number of the axis (mainstem or branches) under consideration.

19.02 ITB is the number of nodes on the branch under consideration.

19.03 At this point in the calculation, NODE is the line number in the PMAP array occupied by the highest node on the mainstem.

THERMS

THERMS is a program which is separate from but ancillary to GLYCIM. It is written in standard FORTRAN 77. Its sole purpose is to save computing time.

In the first draft of the code for GLYCIM, a large block of equations was used to calculate soil thermal diffusivity for each soil cell, eleven times each simulated day. The variables used in these equations are the percentages of sand, silt, clay and organic matter, the volumetric water content and the soil temperature. Since sand, silt, clay and organic matter content very only with soil horizon, we realized that, for a given horizon, soil thermal diffusivity is a function of just water content and temperature. It was, therefore, possible to save computer time by replacing the block of equations with a single regression equation with only a small, known loss of accuracy. To do this, the block of equations was removed from TMPSOL and, with certain embellishments became THERMS.

THERMS is used to calculate thermal diffusivity for a range of soil temperatures and soil water contents given the characteristcs of the soil horizon under consideration. These data are then used for estimating the parameters in the regression equation. Most computer systems have suitable fitting routines to do this. The parameters are entered into GLYCIM along with other descriptions of the soil in each horizon and are used in TMPSOL to generate estimates of thermal diffusivity. This saves a lot of computing time. However, if computing time is not a problem, THERMS with a few deletions can be reinserted into the front of the TMPSOL subroutine.

```
              PROGRAM THERMS
C       -----------------------------------------------------------------
        DIMENSION BD(5),SAND(5),CLAY(5),FCI(5),OMA(5),THETAO(5),THETAS(5)
        DIMENSION X(11,11)
        DATA TS/0.0/
C       -----------------------------------------------------------------
C       *** READ IN THE DATA ***
C
        READ(5,*) D,NLP,LYRSOL
        DO 1003 I9 = 1,LYRSOL
T.01    READ(5,*) BD(I9),SAND(I9),CLAY(I9),FCI(I9),OMA(I9),THETAO(I9),
       &          THETAS(I9)
C
C       *** CALCULATE SOIL CHARACTERISTICS. ***
C
T.02        VSAND=BD(I9)*(1-CLAY(I9)/2.65
            VCLAY=BD(I9)*CLAY(I9)/2.65
          X201 = 1.122E-5/NLP/D
T.03        VMUCK=OMA(I9)*X201/1.30
        TOPORS=1.0-VSAND-VCLAY-VMUCK
        VH2OC = THETAO(I9)
C
        DO 142 I = 1,10
        X(I,11) = TS
        DO 141  J = 1,10
        X(11,J) = VH2OC
C
C       *** CALCULATE HEAT CAPACITY OF SOIL IN EACH CELL ***
C
T.05        HCS=(0.46*(VSAND+VCLAY))+(0.6*VMUCK) + VH2OC
C
C       *** CALCULATE THE THERMAL CONDUCTIVITY OF AIR SATURATED WITH WATER
C           VAPOUR AND WATER AT CELL TEMPERATURE ***
C
T.06        TCH2O=1.33+(4.4E-3*TS)
            TCAIR=0.058+(1.7E-4*TS)
            TCVAP=0.052*EXP(0.058*TS)
T.07        TCSAT=TCAIR+(TCVAP *VH2OC/FCI(I9))
            IF(VH2OC.GE.FCI(I9))TCSAT = TCAIR + TCVAP
C
C       *** CALCULATE AN INTERMEDIATE PARAMETER PERTAINING TO THE THERMAL
C           CONDUCTIVITY OF EACH SOIL CONSTITUENT ***
C
T.08        XSAND=((2/(1+(((20.4/TCH2O)-1)*0.125)))+(1/(1+(((20.4/TCH2O) -
       &          1)*0.75))))/3
            XCLAY=((2/(1+(((7/TCH2O)-1)*0.125)))+(1/(1+(((7/TCH2O)-1) *
       &          0.75))))/3
            XMUCK=((2/(1+(((0.6/TCH2O)-1)*0.5)))+1)/3
T.09        XGAIR=0.0+(0.3333*VH2OC/FCI(I9))
            IF(XGAIR.GE.0.3333)XGAIR = 0.3333
            XAIR=((2/(1+(((TCSAT/TCH2O)-1)*XGAIR)))+(1/(1+(((TCSAT /
       &         TCH2O)-1)*(1-(2*XGAIR))))))/3
C
C       *** CALCULATE THE THERMAL CONDUCTIVITY OF SOIL IN THE CELL ***
C
T.10        TCS=((VH2OC*TCH2O)+(XSAND*VSAND*20.4) +
       &            (XCLAY*VCLAY*7.0)+(XMUCK*VMUCK*0.6) +
       &            (XAIR*(TOPORS -
       &            VH2OC)*TCSAT))/(VH2OC+(XSAND*VSAND) +
       &            (XCLAY*VCLAY)+(XMUCK*VMUCK)+(XAIR *
       &            (TOPORS-VH2OC))) /1000
C
C       *** CALCULATE THE THERMAL DIFFUSIVITY OF THE SOIL IN EACH CELL ***
```

```
      C
T.11        TDS = TCS/HCS
            X(I,J) = TDS
T.12        VH2OC2 = VH2OC*VH2OC
            AINT1   = VH2OC * TS
      C
      C     *** RECORD THE RESULTS ***
      C
T.13        WRITE(6,11) TDS,TS,VH2OC,VH2OC2,AINT1
        11  FORMAT(5F15.8)
            VH2OC = VH2OC + ((THETAS(I9)-THETAO(I9))/9.)
       141   CONTINUE
            TS = TS+5.0
            VH2OC    =THETAO(I9)
       142   CONTINUE
             WRITE(7,22) (X(11,I),I=1,10)
       22    FORMAT(10X,10F10.4)
             DO 12 I = 1,10
             WRITE(7,23)(X(I,11)),(X(I,K),K=1,10)
       23    FORMAT(11F10.4)
      12    CONTINUE
            WRITE(7,25)
      25    FORMAT(1H1)
            TS=0.0
      1003  CONTINUE
            STOP
            END
```

Notes for THERMS

The relationships used in the program nearly all come from Van Wijk's book "Physics of Plant Environment" (1966) and especially chapter 7 by De Vries. In what follows, this reference will be abbreviated as V.W.

T.01 Various characteristics of the soil layer under consideration are read in. SAND includes the fractions of sand and silt in the soil.

T.02 The volume fraction of various soil constituents is calculated assuming a particle density of 2.65. See V.W. at the bottom of his page 211.

T.03 The volume fraction of organic matter is calculated from the weight of organic matter plowed under and the number of soil layers plowed.

T.04 The calculations are made for a range of soil temperatures and soil moisture contents.

T.05 This equation is from V.W., his page 211, equation (7.2).

T.06 The equations were fitted to the curves shown in V.W. his page 211, Figure 7.3.

T.07 V.W. his page 219, equation (7.10).

T.08 The general form of this equation comes from V.W. his page 215, equation (7.5). The parameter values used in the equation for various soil constituents come from V.W., his page 217, the caption to Figure 7.2.

T.09 The shape of the "particles" of air in the soil is assumed to go from spherical in a saturated soil to elongated ellipsoid in dry soil. See V.W. at the bottom of his page 225.

T.10 V.W. his page 216, equation (7.8).

T.11 V.W. his page 107, just below equation (4.5).

T.12 Some terms used in the equation to fit the data on thermal diffusivity are generated here.

T.13 The thermal diffusivity and other terms are written to file 6 ready to fit the equation in TMPSOL on line 84. The fitting routine is a separate program.

FINDQ

FINDQ is a program which is separate from but ancillary to GLYCIM. It is written in standard FORTRAN 77. It was used to generate estimates of the atmospheric transmission coefficient for various latitudes and times of year given data on radiation received on the earth's surface on cloudless days at those places and times. It is included in this publication so that readers who want to know how these estimates were generated can read the equations for themselves. The estimates of atmospheric transmission coefficient are used in CLYMAT.

```
            PROGRAM FINDQ
      C     --------------------------( FINDQ )-------------------------------
      C     THIS PROGRAM DETERMINES VALUES OF ATMOSPHERIC TRANSMISSION
      C     COEFFICIENT Q
      C     -----------------------------------------------------------------
            IMPLICIT DOUBLE PRECISION (A-H,O-Z)
            REAL LATUDE
            DIMENSION C1(9), Q(6,12),X1(6,12),WATATM(6,12),WATPOT(6,12),
           &    RATIO(6,12),D1(48)
F.01        DATA C1/ 0.3964D-0,0.3631D 1, 0.3838D-1, 0.7659D-1, 0.0000D 0,
           1          -0.2297D 2, -0.3885D 0, -0.1587D-0, -0.1021D-1/
F.02        DATA D1/ 1.9,11.6,36.5,84.5,162.9,279.2,440.5,653.9,926.2,1263.8,
           &         1673.2,2160.1,2730.3,3389.0,4141.1,4991.1,5943.0,7000.6,
           &         8166.8,9444.6,10836.1,12343.0,13966.7,15707.8,17566.7,
           &         19543.0,21636.1,23844.6,26166.8,28600.6,31143.0,33791.1,
           &         36541.1,39389.0,42330.3,45360.1,48473.2,51663.8,54926.2,
           &         58253.9,61640.5,65079.2,68562.9,72084.5,75636.5,79211.6,
           &         82801.9,86400.0/
      C     ----------------------( FORMATS )--------------------------------
         34 FORMAT(3X,5HJDAY=,6X,2H15,6X,2H46,6X,2H74,5X,3H105,5X,3H135,
           & 5X,3H166,5X,3H196,5X,3H227,5X,3H258,5X,3H288,5X,3H319,5X,3H349)
         36 FORMAT (3X,F4.1,3X,12(F8.3))
      C     -----------------------------------------------------------------
            DO 4 M=1,12
F.03        READ(3,*) JDAY,(X1(L,M),L=1,6)
            DO 8 L=1,6
            LATUDE=L*10.0
            RI = X1(L,M) * 1376974.0
      C
      C     *** CALCULATE SOLAR DECLINATION.***
      C
F.04        DEGRAD = 3.1416/180.0
            XLAT = LATUDE * DEGRAD
            DEC = C1(1)
            DO 20 I = 2,5
               N = I - 1
               J = I + 4
               PHI = N * 0.01721 * JDAY
         20 DEC = DEC + C1(I) *DSIN(PHI) + C1(J) * DCOS(PHI)
            DEC = DEC * DEGRAD
      C
      C     *** CALCULATE DAYLENGTH ***
      C
F.05        DAYLNG = DACOS((-0.014544-DSIN(XLAT) * DSIN(DEC))/(DCOS(XLAT) *
           1 DCOS(DEC))) * 7.6394
      C
      C     *** CALCULATE POTENTIAL SOLAR RADIATION ON CROP AT NOON ***
      C
F.06        X4=(24.0-DAYLNG)*1800.0
      C     1800=3600/2
            X5=1.0+SIN(X4*7.272E-5+4.7124)
      C     7.272E-5=3.1416*2/24/3600  &  4.7124=3.1416*3/2
            I=INT(X4/900.0)
            X6=1.0+SIN(I*900.0*7.272E-5+4.7124)
            X8=D1(I)+((X5+X6)*(X4-(900.0*I)))
            X9=(X8+(X5*DAYLNG*3600.0))/86400.0
      C     86400=24*3600
            WATPOT(L,M)=(RI/(1.0-X9))/86400.0*(2.0-X5)
      C
      C     *** CALCULATE SOLAR RADIATION INCIDENT ON TOP OF EARTH'S
      C         ATMOSPHERE AT SOLAR NOON.***
      C
F.07        R = 1 + (0.01674 * DSIN((JDAY - 93.5)* 0.9863 *DEGRAD))
```

```
              X7 = DSIN(XLAT) * DSIN(DEC)
              X2 = DCOS(XLAT) * DCOS(DEC)
              X3 = X7 + X2
F.08      WATATM(L,M) = 1325.4 * X3 / R**2
C
C         *** CALCULATE AND WRITE OUT Q ***
C
          RATIO(L,M)=WATPOT(L,M)/WATATM(L,M)
          DUM=(RATIO(L,M)*2.0)-0.93+(0.02/X3)
          IF(DUM.LE.0.0)THEN
          Q(L,M)=0.0
          ELSE
F.09      Q(L,M) = DUM**X3
          END IF
    8     CONTINUE
    4     CONTINUE
          WRITE(2,34)
          DO 7 L=1,6
          LATUDE=L*10.0
    7     WRITE (2,36) LATUDE,(Q(L,M),M=1,12)
          STOP
          END
```

Notes for FINDQ

Most of the equations used in this program are also used in the CLYMAT subroutine and the reader is referred there for their derivation.

F.01 See note 3.01.

F.02 See note 3.02

F.03 Data on the total radiation incident on the earth under a cloudless sky for various latitudes and times of year is read in here. The data come from Budyko (1974) his pages 46 and 47, Table 3.

F.04 See note 3.06

F.05 See note 3.07

F.06 See note 3.12. In this application of the equations, the radiation integral used is the potential maximum for a cloudless day. Therefore the noon value of radiation flux density will also be the potential.

F.07 See note 3.08.

F.08 See note 3.09.

F.09 In note 3.16, it is shown that direct + diffuse radiation flux at the earth's surface = (WATATM/2*(U + Q**(1/SINALT)))
At noon this is equal to WATPOT so we can solve for Q.

------------(DICTIONARY FOR GLYCIM, THERMS AND FINDQ)---------------

PERIOD REFERS TO THE PART OF A DAY USED AS A TIME INTERVAL IN THE CALCULATIONS. THIS IS NORMALLY EITHER ONE TENTH OF THE PHOTOPERIOD OR THE ENTIRE NIGHT PERIOD .

SOIL SLAB REFERS TO A REPRESENTATIVE VOLUME OF SOIL IN WHICH PHYSICAL CONDITIONS,ROOT GROWTH,ETC ARE FOLLOWED. THIS SOIL SLAB IS ONE CENTIMETER THICK IN THE DIRECTION OF THE ROWS,STRETCHES FROM ONE ROW TO THE NEXT AND IS ANY DEPTH DESIRED. MOST CALCULATIONS ARE ONLY DONE ON A HALF SOIL SLAB STRETCHING FROM ROW TO MID-ROW. THE SOIL SLAB IS DIVIDED INTO NK COLUMNS AND NL LAYERS OF SOIL CELLS EACH W CM WIDE AND D CM DEEP.

PHOTOSYNTHESIS,RESPIRATION,TRANSPIRATION AND EVAPORATION RATES ARE ALL EXPRESSED ON A PER UNIT GROUND AREA.

ORGAN DRY WEIGHTS ARE FOR STRUCTURAL MATERIAL ONLY, I.E. THEY EXCLUDE STARCH AND SUGAR, UNLESS OTHERWISE STATED.

A	FACTOR FOR CHANGING SOLAR RADIATION UNITS.
AA,AB,...AR	CHARACTER ARRAY WHICH HOLDS TITLE GIVEN TO RUN BY USER
ABSCIS	SWITCH TO INDICATE IF A LEAF WITH A BRANCH IN ITS AXIL SHOULD BE DROPPED (=1 IF IT SHOULD).
ABSEED(J)	NUMBER OF SEEDS NOT GROWN FOR J CALCULATION PERIODS.
ABSLPN	NITROGEN LOST BY LEAF AND PETIOLE ABSCISSION (GMS).
ABSPOD	NUMBER OF ABSCISED PODS.
ABSPON	NITROGEN LOST BY POD ABSCISSION(GMS).
ABSWT(J)	DRY WEIGHT OF EACH SEED NOT GROWN FOR J CALCULATION PERIODS.
ADBW	ACTUAL CHANGE IN DRY WEIGHT OF BRANCH STEM UNDER CONSIDERATION IN THIS PERIOD (GMS).
ADLA	ACTUAL CHANGE IN AREA OF MAINSTEM LEAF UNDER CONSIDERATION IN THIS PERIOD (CM**2).
ADLW	ACTUAL CHANGE IN DRY WEIGHT OF MAINSTEM LEAF UNDER CONSIDERATION IN THIS PERIOD (GMS).
ADMH	ACTUAL CHANGE IN MAINSTEM HEIGHT IN THIS PERIOD (CM).
ADMW	ACTUAL CHANGE IN MAINSTEM DRY WEIGHT IN THIS PERIOD (GM).
ADPL	ACTUAL CHANGE IN LENGTH OF PETIOLE UNDER CONSIDERATION IN THIS PERIOD (CM).
ADPW	ACTUAL CHANGE IN DRY WEIGHT OF PETIOLE UNDER CONSIDERATION IN THIS PERIOD (GMS).
ADRL(L,K)	ACTUAL CHANGE IN ROOT LENGTH IN SOIL CELL L,K FOR PAST PERIOD (CM).
ADWR(L,K)	ACTUAL RATE OF INCREASE IN ROOT DRY WEIGHT IN SOIL CELL L,K (GMS/HR).
AFWUP(L,K)	PREDICTION OF AMOUNT OF WATER TO BE EXTRACTED FROM SOIL CELL L,K BY ROOTS OVER THE NEXT CALCULATION PERIOD ADJUSTED FOR UNDER OR OVER-ESTIMATE FROM THE LAST PERIOD (GMS).
AFWUPS	AFWUP(L,K) ACCUMULATED OVER ALL CELLS IN SOIL SLAB AND OVER TIME (GMS).
AINT1	VH2OC * TS
AIRDR(J)	VOLUMETRIC WATER CONTENT OF AIR DRY SOIL IN HORIZON J (CC/CC).
AL	DISTANCE TO A GAS-IMPERMEABLE LAYER FROM THE SOIL CELL UNDER CONSIDERATION (CM).
ALP(I)	ARRAY OF SYMBOLS REPRESENTING LEAVES AND PETIOLES AND

	USED IN PLOTTING A DIAGRAM OF THE PLANT.
AMAX1	FORTRAN FUNCTION TO SELECT MAXIMUM VALUE.
ASCRAT(I)	VALUE OF SCRATO (= FACTOR TO ADJUST CARBON SUPPLY TO SHOOT ORGANS DEPENDING ON CARBON AVAILABILITY)AT TIME I.
ASGT(I)	VALUE OF SGT (=PROPORTION OF TIME FOR WHICH SHOOT GROWS: LIMITED BY SHOOT TURGOR) AT TIME I.
AVNRAT	NITROGEN SUPPLY/DEMAND FOR VEGETATIVE PARTS AVERAGED OVER A DAY
AVP	ACTUAL WATER VAPOR PRESSURE FOR DAY (ASSUMED CONSTANT)(KPA
AWR(L,K)	ACTUAL INCREASE IN ROOT WEIGHT IN SOIL CELL L,K FOR PAST PERIOD (GMS).
AWUP(L,K)	RATE OF WATER EXTRACTION FROM SOIL CELL L,K BY ROOTS (GMS/HR).
AWUPS	SUM OF AWUP(L,K) OVER CELLS IN A HALF SOIL SLAB (GMS/HR).
AWUPSS	ACCUMULATED AMOUNT OF WATER EXTRACTED BY ROOTS FROM A HALF SOIL SLAB (GMS).
B	FACTOR FOR CHANGING TEMPERATURE UNITS.
BD(J)	BULK DENSITY OF SOIL IN HORIZON J (GM/CC)
BDC(L,K)	BULK DENSITY OF SOIL IN CELL L,K (GM/CC).
BETA(J)	SLOPE OF GRAPH OF LOG (HYDRAULIC CONDUCTIVITY) VS. VOLUMETRIC WATER CONTENT FOR HORIZON J OF SOIL.
BETA0	INTERMEDIATE STAGE IN CALCULATION.
BLAREA(I,J)	AREA OF TRIFOLIOLATE I ON BRANCH J(CM**2).
BLEAFW(I,J)	DRY WEIGHT OF TRIFOLIOLATE I ON BRANCH J (GM)
BMAIN	MAINTENANCE RESPIRATION RATE (MG CO2/M**2/S).
BMAINN	CARBON USED IN MAINTENENCE RESPIRATION OVERNIGHT (GM/PLANT/HR).
BPETL(I,J)	LENGTH OF PETIOLE BEARING TRIFOLIOLATE I ON BRANCH J (CM).
BPETW(I,J)	DRY WEIGHT OF PETIOLE TO TRIFOLIATE I ON BRANCH J (GM).
BRASEN	MINIMUM NITRATE REDUCTASE ACTIVITY FOR ALL LEAVES ON PLANT (MICROMOLE NO2/GM LEAF FRESH WT./HR).
BRNCH	REAL FORM OF NBRNCH(=NUMBER OF BRANCHES ON PLANT).
BRNCHH(J)	LENGTH OF BRANCH J (CM).
BRNCHW(J)	DRY WEIGHT OF BRANCH J (GMS).
BSITES	NUMBER OF POSSIBLE SITES FOR BRANCHES ON PLANT.
BTPL	LOWEST VALUE OF LEAF TURGOR PRESSURE REACHED SO FAR TODAY (BARS)
C	FACTOR FOR CHANGING TEMPERATURE UNITS.
C1(I)	FILE OF DATA FOR CALCULATING SOLAR DECLINATION.
CASG	CARBON AVAILABLE FOR GROWTH OF SHOOT ORGANS OTHER THAN SEEDS (GMS).
CB(K)	CONCENTRATION OF OXYGEN AT THE UPPER BOUNDARY OF THE SOIL CELL IN COLUMN K OF THE SOIL LAYER UNDER CONSIDERATION (GM/CC).
CCCT	CANOPY CONDUCTANCE TO CO2 TRANSFER (M/S).
CCTLF	CONDUCTANCE TO CO2 TRANSFER FOR THE LEAF LAYER UNDER CONSIDERATION (M/S).
CEC	CANOPY EXTINCTION COEFFICIENT.
CEXPOD	CARBON EXTRACTABLE FROM THE PODS (GMS).
CLAY(J)	CLAY CONTENT OF HORIZON J AS A PROPORTION OF SOIL DRY WEIGHT.
CLDFAC	CLOUD COVER FACTOR FOR THIS LATITUDE.

CLIMAT(I)	ARRAY OF CLIMATIC VARIABLES FOR THE NEXT DAY.
CLOUD	PROPORTION OF SKY COVERED WITH CLOUD (1 = FULL COVER).
CLUE	CANOPY LIGHT UTILIZATION EFFICIENCY (MG CO2/JOULE).
CMAXSG	MAXIMUM AMOUNT OF CARBON THAT CAN BE USED BY THE SHOOT GROWING AT THE POTENTIAL RATE LIMITED ONLY BY TEMPERATURE AND SHOOT TURGOR (GMS).
CMINSG	MINIMUM AMOUNT OF CARBON REQUIRED BY THE SHOOT TO GROW AT THE POTENTIAL RATE LIMITED ONLY BY TEMPER- ATURE AND SHOOT TURGOR (GMS).
CNFIX	CARBON USED FOR NITROGEN FIXATION DURING THIS PERIOD IN THE SOIL CELL UNDER CONSIDERATION (GMS).
CNFIXS	SUM OF CNFIX OVER SOIL CELLS IN HALF SOIL SLAB (GMS).
CNINT	CARBON AVAILABLE TO INITIATE NEW NODULES IN A HALF SOIL SLAB (GMS).
CO2	AMBIENT CO2 CONCENTRATION (PPM BY VOLUME).
COND(L,K)	UNSATURATED HYDRAULIC CONDUCTIVITY OF SOIL CELL L,K ((CC H2O)/(CM MOVED)/(CM SOIL WATER POTENTIAL GRADIENT)/DAY).
CONRES	CO2 RESPIRED DURING SYNTHESIS OF 1 GM OF ROOT (GM).
CONVL	AMOUNT OF CARBON NEEDED TO MAKE UNIT LEAF DRY WEIGHT (GM/GM).
CONVN	AMOUNT OF CARBON NEEDED TO MAKE UNIT NODULE DRY WEIGHT (GM/GM).
CONVPE	AMOUNT OF CARBON NEEDED TO MAKE UNIT PETIOLE DRY WEIGHT (GM/GM).
CONVPO	AMOUNT OF CARBON NEEDED TO MAKE UNIT POD DRY WEIGHT (GM/GM).
CONVR	AMOUNT OF CARBON NEEDED TO MAKE UNIT ROOT DRY WEIGHT (GM/GM).
CONVSE	AMOUNT OF CARBON NEEDED TO MAKE UNIT SEED DRY WEIGHT (GM/GM).
CONVST	AMOUNT OF CARBON NEEDED TO MAKE UNIT STEM DRY WEIGHT (GM/GM).
COVER	PROPORTION OF SOIL COVERED BY CROP.
CXT(L,K)	OXYGEN CONCENTRATION IN CELL L,K (VOLUME FRACTION OF AIR)
D	DEPTH OF EACH SOIL CELL (CM).
DI(I)	FILE OF DATA USED IN CALCULATING INSTANTANEOUS RADIATION FLUX DENSITY FROM RADIATION INTEGRAL.
DABS	FORTRAN FUNCTION FOR DOUBLE PRECISION ABSOLUTE VALUE.
DATA	FLAG USED TO DESIGNATE A PARTICULAR DATA SET FOR SPECIAL TREATMENT (USUALLY OF SOIL WATER CONTENT)
DAY	FLOATING POINT FORM OF IDAY+1.
DAYBR(J)	NUMBER OF DAYS SINCE BRANCH J WAS INITIATED.
DAYFRC	LOCAL VARIABLE USED TO CALCULATE HRANG FOR EACH ITIME.
DAYLNG	DAYLENGTH (HOURS).
DEADRN	NITROGEN LOST BY ROOT DEATH (GMS).
DEC	SOLAR DECLINATION (RADIANS) (FIRST CALCULATED IN DE- GREES).
DEFICN	SEED NITROGEN DEFICIENCY (GM NITROGEN/PLANT).
DEFNOD	AMOUNT OF CARBON NEEDED TO INCREASE NODULE WEIGHT FROM THAT AT R1 TO THAT REQUIRED TO FIX NITROGEN FOR AN AVERAGE LOAD OF SEEDS (GM/PLANT)
DEGRAD	DEGREE TO RADIAN CONVERSION FACTOR (PI/180).
DEL	SLOPE OF SATURATION VAPOUR PRESSURE CURVE AT AIR TEM-

	PERATURE (KPA/DEG. C).
DET	SWITCH TO INDICATE IF PLANT IS DETERMINATE (=0 FOR DETERMINATE AND =1 FOR INDETERMINATE).
DIA	REAL FORM OF IDAY (=DAYS AFTER COTYLEDON STAGE).
DIFFT	HYDRAULIC DIFFUSIVITY OF SOIL IN SURFACE CELL UNDER CONSIDERATION (CM**2/DAY).
DIFF(L,K)	HYDRAULIC DIFFUSIVITY OF SOIL IN CELL L,K (CM**2/DAY).
DIFFO(J)	HYDRAULIC DIFFUSIVITY OF SOIL IN HORIZON J AT A SOIL WATER POTENTIAL OF -15 BARS (CM**2/DAY).
DIFFIN	PROPORTION OF DIFFUSE RADIATION INTERCEPTED BY "SOLID" ROWS.
DIFFM	HYDRAULIC DIFFUSIVITY OF SOIL IN SURFACE CELL UNDER CONSIDERATION ACCOUNTING FOR LAYER OF AIR DRY SOIL (CM**2/DAY).
DIFIMG	MAXIMUM DAYLENGTH FOR IMMEDIATE FLORAL INDUCTION(HRS).
DIFINT(I)	PROPORTION OF SKY OBSCURED BY "SOLID" ROWS FROM A GIVEN POINT AT SOIL LEVEL.
DIFL(L,K)	MEAN HYDRAULIC DIFFUSIVITY OF SOIL BETWEEN CELL L,K AND THE CELL TO ITS LEFT (CM**2/DAY).
DIFMAX	MAXIMUM HYDRAULIC DIFFUSIVITY OF SOIL BETWEEN ANY 2 CELLS IN THE SOIL PROFILE (CM**2/DAY).
DIFU(L,K)	MEAN HYDRAULIC DIFFUSIVITY OF SOIL BETWEEN CELL L,K AND THE CELL ABOVE (CM**2/DAY).
DIFWAT(I)	PROPORTION OF TOTAL RADIATION THAT IS DIFFUSE AT TIME(I).
DIMPL	DEPTH TO A GAS-IMPERMEABLE LAYER FROM THE SOILSURFACE(CM)
DIRINT(I)	PROPORTION OF DIRECT RADIATION INTERCEPTED BY "SOLID" ROWS AT TIME(I).
DLSUM1,ETC.	INTERMEDIATE STAGES IN CALCULATIONS.
DNIT	NITRIFICATION OF NH4 (MG/CC).
DNMIN	AMOUNT OF MINERALIZED ORGANIC NITROGEN (MG/CM***3)
DPN	INTERMEDIATE STEP IN CALCULATION OF FWU (L,K)
DPR(L,K)	APPARENT DIFFUSION RATE OF OXYGEN IN SOIL CELL L,K (CM**2/SEC).
DPSIO2	CHANGE IN LEAF WATER POTENTIAL CORRESPONDING TO A CHANGE OF 2 BARS IN LEAF TURGOR PRESSURE (BARS)
DR	INCREMENT IN RSTAGE FOR THIS PERIOD.
DRAIN	ACCUMULATED WATER DRAINING OUT OF THE BOTTOM OF THE PROFILE (MM).
DRASEN	NITRATE REDUCTASE ACTIVITY FOR GROWING LEAF UNDER CONSIDERATION (MICROMOLE NO2/GM LEAF FRESH WT./HR).
DRL(L,K)	POTENTIAL CHANGE IN ROOT LENGTH IN SOIL CELL L,K FOR PAST PERIOD (CM).
DROP	SWITCH TO INDICATE IF LOWEST LEAVES ARE SELF SUPPORTING (=0) OR SHOULD BE DROPPED (=1).
DSHF	HEAT FLUX DOWN INTO SOIL (CAL/CM**2/SEC).
DUM	INTERMEDIATE STAGE IN CALCULATION.
DUMB	PARAMETER USED TO CHANGE THE BOTTOM BOUNDARY CONDITION FOR WATER MOVEMENT IN THE SOIL PROFILE FOR SOME SITES AND DATA SETS (CC/CC).
DUMC	PARAMETER USED TO CHANGE THE BOTTOM BOUNDARY CONDITION FOR WATER MOVEMENT IN THE SOIL PROFILE FOR SOME SITES AND DATA SETS.
DV	INCREMENT IN VSTAGE FOR THIS PERIOD.
DW	D * W

E	FACTOR FOR CHANGING RAIN UNITS.
EAVES	REAL FORM OF LEAVES (=NUMBER OF LEAVES ON MAINSTEM).
EO(I)	POTENTIAL TRANSPIRATION (=LEAF WATER EVAPORATION) RATE FOR A MEAN PLANT AT TIME I (GM/PLANT/HR).
EOR	POTENTIAL TRANSPIRATION RATE PER HALF CM OF ROW(GM/HR)
EORS	ACCUMULATED POTENTIAL TRANSPIRATION FROM LEAVES PER HALF CM OF ROW (GMS).
EORSCF	EOR*SCF=POTENTIAL TRANSPIRATION RATE FROM LEAVES PER HALF CM OF ROW AS LIMITED BY STOMATAL CLOSURE(GMS/DAY)
EORSCS	ACCUMULATED POTENTIAL TRANSPIRATION FROM LEAVES PER HALF CM OF ROW AS LIMITED BY STOMATAL CLOSURE (GMS).
EPO	POTENTIAL EVAPORATION RATE FROM THE CROP (GM/M**2/HR).
EQSC	POTENTIAL RATE OF CARBON USE IN SYNTHESIZING SEEDS TO FILL THE PODS CURRENTLY ON THE PLANT (GM/PLANT/HR).
ESO	POTENTIAL EVAPORATION RATE FROM SOIL SURFACE (GM/M**2/HR).
ESP	RATE AT WHICH SOIL CAN SUPPLY WATER FOR EVAPORATION (GM/M**2/HR).
ET	ACTUAL OVERNIGHT TRANSPIRATION(= LEAF EVAPORATION) RATE FOR A MEAN PLANT (GRAM/HOUR).
ETA(J)	A SOIL CHARACTERISTIC PARAMETER RELATING VOLUMETRIC WATER CONTENT TO WATER POTENTIAL FOR SOIL IN HORIZON J (DIMENSIONLESS).
EWU(K)	WATER EVAPORATED FROM EXPOSED SURFACE CELL K OVER A GIVEN TIME INTERVAL (GRAMS).
EWUS	ACCUMULATED WATER EVAPORATED FROM EXPOSED SOIL SURFACE (UNTIL CANOPY CLOSES) (GMS).
EXTRA	DAYLENGTH/20 (HOUR).
F	FACTOR FOR CHANGING WIND UNITS.
FC(L)	VOLUMETRIC WATER CONTENT OF LAYER L AT FIELD CAPACITY.
FCI(J)	VOLUMETRIC WATER CONTENT OF HORIZON J AT FIELD CAPACITY.
FCSH	FACTOR FOR CALCULATING SENSIBLE HEAT CONVECTED FROM SOIL SURFACE WHEN IT IS HOTTER THAN THE AIR (CAL / CM**2/MIN/DEG. C).
FELWR	FACTOR FOR CALCULATING EXTRA LONG-WAVE RADIATION FROM SOIL SURFACE WHEN IT IS HOTTER THAN THE AIR (CAL/CM**2/MIN/DEG. C).
FERN	FERTILIZER NITROGEN APPLIED (LBS/ACRE).
FHL(L,K)	FLOW OF HEAT TO THE LEFT OUT OF CELL L,K (DEG.C).
FHU(L,K)	FLOW OF HEAT UPWARD OUT OF CELL L,K (DEG.C).
FILDAY	NO. OF DAYS SINCE SEED FILL STARTED, I.E.,SINCE R4.
FILL	SEED FILL RATE AT 24 DEG. C (MG/SEED/DAY).
FINT	FACTOR FOR INTEGRATING GROWTH, ETC. OVER A PERIOD GIVEN A RATE ASSUMING RATE VARIES SINUSOIDALLY DURING DAY (HR).
FIXC	NET CARBON FIXATION RATE ALLOWING FOR STOMATAL CLOSURE (G/PLANT/HOUR).
FIXCS	ACCUMULATED NET CARBON FIXED (G/PLANT)
FIXDAY	NET CARBON FIXED,ACCUMULATED OVER ONE DAY (G/PLANT)
FIXNO3	NITROGEN REDUCED FROM NO3 (GM N/PLANT).
FLWRWT	DRY WEIGHT OF FLOWERS (GMS/PLANT).
FMIN	MINERALIZATION FRACTION OF SOIL ORGANIC NITROGEN.
FNH4	FRACTION OF FERTILIZER NITROGEN WHICH IS AMMONIUM.
FNICN	NET FLOW OF WATER INTO CELL DURING CURRENT PERIOD (CC).

FNIT	NITRIFICATION OF NH4.
FNL(L,K)	FLOW OF NITROGEN TO THE LEFT OUT OF CELL L,K DURING THE CURRENT PERIOD (MG).
FNO3	FRACTION OF FERTILIZER NITROGEN WHICH IS NITRATE.
FNU(L,K)	FLOW OF NITROGEN UPWARDS OUT OF CELL L,K DURING THE CURRENT PERIOD (MG).
FPODS	NUMBER OF NEW PODS INITIATED
FRTWT	DRY WEIGHT OF PODS PLUS SEEDS (GMS/PLANT).
FRUITC	RATE OF CARBON SUPPLY TO PODS AND SEEDS (GM/PLANT/HR)
FSEEDS	NUMBER OF NEW SEEDS INITIATED
FWICN	NET FLOW OF WATER INTO CELL DURING CURRENT PERIOD (CC).
FWL(L,K)	FLOW OF WATER TO THE LEFT OUT OF CELL L,K DURING THE CURRENT PERIOD (CC).
FWU(L,K)	FLOW OF WATER UPWARDS OUT OF CELL L,K DURING THE CURRENT PERIOD (CC).
FWUP(L,K)	AMOUNT OF WATER EXTRACTED FROM SOIL CELL L,K BY ROOTS (GMS).
G	FACTOR USED TO PREVENT INSTABILITY BUT REDUCE THE NUMBER OF ITERATIONS TO A MINIMUM IN CAPFLO.
GAMMA	PSYCHROMETRIC CONSTANT (KPA/DEG. C).
H	FACTOR USED TO PREVENT INSTABILITY BUT REDUCE THE NUMBER OF ITERATIONS TO A MINIMUM IN TMPSOL.
H2O	TEMPORARY VOLUME OF WATER IN SOIL CELL (CC/CM**2).
HCS(L,K)	HEAT CAPACITY OF SOIL IN CELL L,K (CAL/CC/DEG. C).
HRANG(I)	HOUR ANGLE BETWEEN SOLAR NOON AND TIME I (RADIANS).
HUNLUX	SOLAR ELEVATION WHEN LIGHT INTENSITY IS 100 LUX.
IDAY	DAYS AFTER COTYLEDON STAGE, I.E., VC.
IDAYX	IDAY + 1.
IHPERD	IPERD/2.
ILOW	NUMBER OF LOWEST TRIFOLIOLATE REMAINING ON THE MAIN STEM.
ILOWB(J)	NUMBER OF LOWEST TRIFOLIOLATE REMAINING ON BRANCH J
INITBR(J)	NUMBER OF TOP LEAF ON MAINSTEM WHEN BRANCH J WAS INTIATED
IP	INTEGRAL FORM OF P
IPERD	NUMBER OF CALCULATION PERIODS IN THE PHOTOPERIOD.
ITB	NUMBER OF NODES ON THE BRANCH UNDER CONSIDERATION (= NBTPLF(J)).
ITIME	TIME OF DAY COUNTING IN ONE TENTH OF A DAY INCREMENTS FROM DAWN. THE MID-POINT OF THE INCREMENT.
ITRIF	TRIFOLIOLATE LEAF NUMBER ON AXIS (0=UNIFOLIOLATES).
JDAY	DAY OF YEAR.
JDAYX	JDAY + 1.
JDFRST	DAY OF YEAR ON WHICH MODEL RUN STARTS.
JDLAST	DAY OF YEAR ON WHICH MODEL RUN STOPS.
JHPERD	IHPERD + 1.
JPERD	IPERD + 1.
K	COLUMN IN SOIL PROFILE.
KLJ	NUMBER OF FIRST EXPOSED SOIL COLUMN COUNTING FROM BASE OF STEM.
KLJM1	KLJ - 1.
KM1	K - 1.
KOUNT	A LOCAL COUNTER.
KP1	K + 1.
L	LAYER IN SOIL PROFILE.

LA	FLAG TO INDICATE IF PRINTOUT OF INDIVIDUAL LEAF AREAS AND PETIOLE LENGTHS IS NEEDED (=1 IF IT IS).
LAMDAC	ALBEDO OF CROP (DIMENSIONLESS).
LAMDAS	ALBEDO OF SOIL (DIMENSIONLESS).
LAREAB	AREA OF BRANCH LEAVES (CM**2/PLANT).
LAREAM	AREA OF MAINSTEM LEAVES(CM**2/PLANT).
LAREAT	TOTAL LEAF AREA PER PLANT (CM**2).
LATUDE	LATITUDE (DEGREES).
LCAI	LEAF AREA PER UNIT AREA COVERED BY CROP CANOPY.
LEAFWT	DRY WEIGHT OF LEAVES (GMS/PLANT).
LEAVES	NUMBER OF LEAVES ON MAINSTEM.
LFWT	DRY WEIGHT OF LEAVES INCLUDING STARCH AND SUGAR THEY CONTAIN (GMS/PLANT).
LINE	APPROXIMATE LINE NUMBER IN CODE USED TO IDENTIFY WRITE STATEMENTS.
LKH	NK * NL * 0.5
LKR	NLR * NKR
LKRP1	NLRP1 * NKRP1
LLAREA	AREA OF LOWEST LAYER OF LEAVES ON THE PLANT (CM**2).
LLFWT	DRY WEIGHT OF LOWEST LAYER OF LEAVES ON THE PLANT(GMS)
LLUE	LEAF LIGHT UTILIZATION EFFICIENCY (MG CO2/JOULE).
LM1	L-1
LONDAY	DAYS OF MILD NITROGEN STRESS SINCE LAST LEAF ABSCISED.
LOXT	FLAG TO INDICATE IF PRINTOUT OF OXYGEN CONCENTRATION DISTRIBUTION IN THE SOIL PROFILE IS NEEDED (=1 IF IT IS).
LP1	L+1
LPNDT	FLAG TO INDICATE IF PRINTOUT OF DATA ON CARBON FIXATION AND ALLOCATION IS NEEDED (=1 IF IT IS).
LRGF	FLAG TO INDICATE IF PRINTOUT OF ROOT GROWTH FACTOR, RGCF, DISTRIBUTION IN THE SOIL PROFILE IS NEEDED (=1 IF IT IS).
LRTS	FLAG TO INDICATE IF PRINTOUT OF ROOT DRY WEIGHT DISTRIBUTION IN THE SOIL PROFILE IS NEEDED (=1 IF IT IS).
LSTP	FLAG TO INDICATE IF PRINTOUT OF TEMPERATURE DISTRIBUTION IN THE SOIL PROFILE IS NEEDED (=1 IF IT IS).
LTC	LEAF TRANSMISSION COEFFICIENT.
LTOPS	FLAG TO INDICATE IF PRINTOUT OF SYMBOLIC PLANT DIAGRAM IS NEEDED (=1 IF IT IS).
LVHC	FLAG TO INDICATE IF PRINTOUT OF WATER CONTENT DISTRIBUTION IN THE SOIL PROFILE IS NEEDED (=1 IF IT IS).
LW	FLAG TO INDICATE IF PRINTOUT OF INDIVIDUAL LEAF AND PETIOLE DRY WEIGHTS IS NEEDED (=1 IF IT IS).
LYRSOL	NUMBER OF SOIL HORIZONS WITH DIFFERENT CHARACTERISTICS
LYTRES	LIGHT RESPIRATION RATE (MG CO2/M**2/S).
M	INTEGER PART OF LCAI+1.
MATURE	SWITCH TO INDICATE THAT ALL LEAVES HAVE ABSCISED AND DRYWEIGHT GAIN HAS STOPPED.
MAXPOD	MAXIMUM NUMBER OF PODS PLANT CAN SUPPORT WITH PRESENT NITROGEN CONTENT.
MECHR	SOIL MECHANICAL RESISTANCE TO ROOT GROWTH (BARS).
MG	MATURITY GROUP NUMBER (GROUP 00 = -1).
MLAREA(J)	AREA OF MAINSTEM TRIFOLIOLATE J (CM**2).
MLEAFW(J)	DRY WEIGHT OF MAINSTEM TRIFOLIOLATE J (GMS).
MPETL(I)	LENGTH OF MAIN STEM PETIOLE BEARING TRIFOLIOLATE I (CM).
MPETW(I)	DRY WEIGHT OF PETIOLE TO MAINSTEM TRIFOLIOLATE I (GM).

MSTEMH	MAINSTEM HEIGHT ABOVE COTYLEDONARY NODE (CM).
MSTEMW	MAINSTEM DRY WEIGHT ABOVE COTYLEDONARY NODE (GMS).
MSW1	SWITCH TO INDICATE IF DAILY WET BULB TEMPERATURES ARE AVAILABLE (THEY ARE IF MSW1=1).
MSW2	SWITCH TO INDICATE IF DAILY WIND IS AVAILABLE (IT IS IF MSW2 = 1).
MSW3	SWITCH TO INDICATE IF INITIAL SOIL PROFILE WATER CONTENTS ARE AVAILABLE (THEY ARE IF MSW3 = 1).
NBP1	NBRNCH + 1
NBRNCH	NUMBER OF BRANCHES ON PLANT
NBTPLF(J)	NUMBER OF THE TOP LEAF ON BRANCH J.
NCD	NUMBER OF CLIMATIC VARIBLES AVAILABLE FOR EACH DAY
NCPM	MAXIMUM AMOUNT OF CARBON THAT CAN BE STORED IN THE NODULES IN A HALF SOIL SLAB (GMS).
NCPOOL	CARBON STORED IN THE NODULES IN A HALF SOIL SLAB (GMS).
NEXLP	NITROGEN EXTRACTED FROM DYING LEAVES AND PETIOLES (GMS).
NEXP	NUMBER OF SOIL CELLS EXPOSED BETWEEN EDGE OF CROP AND HALFWAY BETWEEN ROWS.
NEXT	NEXT DAY WHEN OUTPUT IS REQUIRED.
NFCF(L,K)	PROPORTIONAL REDUCTION OF NODULE NITROGEN FIXATION IN SOIL CELL L,K FROM ALL PHYSICAL CAUSES.
NFCF1	PROPORTIONAL REDUCTION OF NODULE NITROGEN FIXATION CAUSED BY SOIL TEMPERATURE.
NFCF2	PROPORTIONAL REDUCTION OF NODULE NITROGEN FIXATION CAUSED BY SOIL OXYGEN CONCENTRATION.
NFIXNU(L,K)	NUMBER OF NODULES CAPABLE OF FIXING NITROGEN IN SOIL CELL L,K.
NFIXWT(L,K)	DRY WEIGHT OF NODULES CAPABLE OF FIXING NITROGEN IN SOIL CELL L,K (GMS).
NFLWRS	MEAN NUMBER OF FLOWERS PER PLANT.
NFRQ	FREQUENCY OF OUTPUT REQUIRED (DAYS).
NFX	FLAG TO INDICATE IF PRINTOUT OF NODULE DRY WEIGHT DISTRIBUTION IN THE SOIL PROFILE IS NEEDED (=1 IF IT IS).
NJ(L)	NUMBER OF SOIL HORIZON IN LAYER L.
NK	NUMBER OF COLUMNS IN SOIL PROFILE BETWEEN ROWS. MUST BE AN EVEN NUMBER SO THAT NKH IS A WHOLE NUMBER.
NKH	NK * 0.5
NKN	NUMBER OF LAST SOIL COLUMN CONTAINING NODULES.
NKNP	VARIABLE USED TO SET NKN.
NKR	NUMBER OF LAST SOIL COLUMN CONTAINING ROOTS.
NKRP	VARIABLE USED TO SET NKR.
NKRP1	NUMBER OF LAST SOIL COLUMN CONTAINING ROOTS + 1.
NL	NUMBER OF LAYERS IN SOIL PROFILE.
NLN	NUMBER OF DEEPEST SOIL LAYER CONTAINING NODULES.
NLNP	VARIABLE USED TO SET NLN.
NLP	NUMBER OF LAYERS PLOWED.
NLR	NUMBER OF LOWEST SOIL LAYER CONTAINING ROOTS.
NLRP	VARIABLE USED TO SET NLR.
NLRP1	NUMBER OF LOWEST SOIL LAYER CONTAINING ROOTS + 1.
NNEWWT(I,L,K)	DRY WEIGHT OF NODULES INITIATED IN SOIL CELL L,K ON I DAYS AGO (GMS).
NODB	SPECIFIES THE LINE ON WHICH SYMBOLS WILL BE PRINTED FOR THE BRANCH NODE UNDER CONSIDERATION.
NODC	RATE OF CARBON SUPPLY TO NODULES (GMS/PLANT/HR).
NODE	SPECIFIES THE LINE ON WHICH SYMBOLS WILL BE PRINTED FOR

	THE MAINSTEM NODE UNDER CONSIDERATION. AT THE END OF SOYPLT IT IS THE LINE NUMBER OF THE HIGHEST NODE.
NODES	NUMBER OF NODES PER PLANT.
NODIE(L,K)	NUMBER OF DAYS SINCE NODULES IN SOIL CELL L,K FIXED ANY NITROGEN.
NODN	NITROGEN ACQUIRED FROM NODULES (GMS).
NODNEW(I,L,K)	NUMBER OF NODULES INITIATED IN SOIL CELL L,K ON I DAYS AGO.
NODNUM(L,K)	NUMBER OF NODULES IN SOIL CELL L,K.
NODSIT	NUMBER OF SITES AVAILABLE FOR NODULE INITIATION IN SOIL CELL UNDER CONSIDERATION.
NODWAR(J)	ELEMENT OF AN ARRAY IN WHICH SOIL CELL COORDINATES ARE HELD IN ORDER OF DECREASING VALUES OF NFCF(L,K) I.E., DECREASING POTENTIAL FOR NITROGEN FIXATION.
NODWT	DRY WEIGHT OF NODULES IN A HALF SOIL SLAB (GMS).
NPODS	NUMBER OF PODS PLANT CAN SUPPORT IF NITROGEN IS UNLIMITED.
NPRM	NUMBER OF ELEMENTS IN USE IN THE PRM (PARAMETER) ARRAY.
NRATIO	NITROGEN SUPPLY/DEMAND FOR VEGETATIVE PARTS.
NRATS	ACCUMULATOR FOR SUMMING (NRATIO*PERIOD) OVER A DAY
NSTG	SPECIFIES THE POSITION OF SYMBOLS ALONG THE LINE ALLOCATED TO THE NODE UNDER CONSIDERATION.
NTOPLF	NUMBER OF THE TOP TRIFOLIOLATE ON THE MAINSTEM.
NYTE	SWITCH TO CONTROL OVERNIGHT CALCULATIONS (= 0 DURING DAY AND = 1 AT NIGHT)
OIL	PROPORTION OF OIL IN THE SEED.
OMA	ORGANIC MATTER ADDED TO THE PLOW ZONE AT BEGINNING OF SEASON (LB/ACRE).
OSMFAC	FACTOR DESCRIBING ABILITY OF PLANT TO OSMOREGULATE WHEN WATER STRESSED (=CHANGE IN OSMOTIC POTENTIAL/CHANGE IN WATER POTENTIAL).
OSMREG	SWITCH TO INDICATE THAT OSMOREGULATION SHOULD OCCUR. (POSITIVE VALUE DECREASES LEAF OSMOTIC POTENTIAL).
PADWR(L,K)	ACTUAL RATE OF INCREASE IN ROOT DRY WEIGHT IN SOIL CELL L,K DURING PAST CALCULATION PERIOD (GMS/HR).
PAR(I)	LIGHT INCIDENT ON THE CROP AT TIME I (WATTS/M**2).
PARINT(I)	FRACTION OF LIGHT INTERCEPTED BY THE CROP AT TIME I.
PCRL	RATE AT WHICH CARBON WOULD BE SUPPLIED TO GROWING ROOTS IN A HALF SOIL SLAB IF ALL POTENTIAL SHOOT GROWTH HAD BEEN SATISFIED (GMS/HR).
PCRPAR(J)	ELEMENT OF AN ARRAY IN WHICH SOIL CELL COORDINATES ARE HELD IN ORDER OF DECREASING VALUES OF PDWR(L,K) (I.E., DECREASING ROOT GROWTH POTENTIAL).
PCRQ	RATE AT WHICH CARBON WOULD BE SUPPLIED TO GROWING ROOTS IN A HALF SOIL SLAB IF ALL TRANSLOCATED CARBON WENT TO THE ROOTS (GMS/HR).
PCRS	ACTUAL RATE AT WHICH CARBON IS SUPPLIED TO ROOTS IN A HALF SOIL SLAB (GMS/HR).
PCRTS	SUM OF POTENTIAL RATES OF CARBON USE BY ROOTS IN SELECTED SOIL CELLS IN A HALF SOIL SLAB (GMS/HR).
PCST	MAXIMUM RATE OF CARBON USE BY SHOOT EXPANDING AT POTENTIAL RATE (GM/PLANT/HR).
PDBC	MINIMUM RATE OF CARBON USE IN SYNTHESIZING BRANCH STEMS (GM/PLANT/HR).
PDBLA	POTENTIAL RATE OF CHANGE IN BRANCH LEAF AREA (CM**2/PLANT/HR).

PDBLC	MINIMUM RATE OF CARBON USE IN SYNTHESIZING BRANCH LEAVES (GM/PLANT/HR).
PDBLV	POTENTIAL RATE OF CHANGE IN BRANCH LEAF VOLUME (CM**3/PLANT/HR).
PDBLW	POTENTIAL RATE OF CHANGE IN BRANCH LEAF DRY WEIGHT (GM/PLANT/HR).
PDBPC	MINIMUM RATE OF CARBON USE IN SYNTHESIZING BRANCH PETIOLES (GM/PLANT/HR).
PDBPW	POTENTIAL RATE OF CHANGE IN BRANCH PETIOLE DRY WEIGHT (GM/PLANT/HR).
PDBRW(J)	POTENTIAL RATE OF CHANGE IN DRY WEIGHT OF BRANCH STEM J (GMS/HR).
PDBW	POTENTIAL RATE OF CHANGE IN BRANCH STEM DRY WEIGHT (GM/PLANT/HR).
PDFC	MINIMUM RATE OF CARBON USE IN SYNTHESIZING FLOWERS (GM/PLANT/HR).
PDFW	POTENTIAL RATE OF CHANGE IN FLOWER DRY WEIGHT (GM / PLANT/HR).
PDLAB(I,J)	POTENTIAL RATE OF CHANGE IN AREA OF LEAF I ON BRANCH J (CM**2/HR).
PDLAM(I)	POTENTIAL RATE OF CHANGE IN AREA OF MAINSTEM LEAF I (CM**2/HR).
PDLAT	POTENTIAL RATE OF CHANGE IN AREA OF LEAF UNDER CON-SIDERATION LIMITED ONLY BY TEMPERATURE (CM**2/HR).
PDMC	MINIMUM RATE OF CARBON USE IN SYNTHESIZING MAINSTEMS (GM/PLANT/HR).
PDMH	POTENTIAL RATE OF CHANGE IN MAINSTEM HEIGHT (CM).
PDMLA	POTENTIAL RATE OF CHANGE IN MAINSTEM LEAF AREA (CM**2/PLANT/HR).
PDMLC	MINIMUM RATE OF CARBON USE IN SYNTHESIZING MAINSTEM LEAVES (GM/PLANT/HR).
PDMLV	POTENTIAL RATE OF CHANGE IN MAINSTEM LEAF VOLUME (CM**3/PLANT/HR).
PDMLW	POTENTIAL RATE OF CHANGE IN MAINSTEM LEAF DRY WEIGHT (GM/PLANT/HR).
PDMPC	MINIMUM RATE OF CARBON USE IN SYNTHESIZING MAINSTEM PETIOLES (GM/PLANT/HR).
PDMPW	POTENTIAL RATE OF CHANGE IN MAINSTEM PETIOLE DRY WEIGHT (GM/PLANT/HR).
PDMW	POTENTIAL RATE OF CHANGE IN MAINSTEM DRY WEIGHT (GM/PLANT/HR).
PDNCS	CARBON NEEDED TO GROW EXISTING NODULES IN A HALF SOIL SLAB AT THE POTENTIAL RATE (GMS).
PDNFW	POTENTIAL CHANGE IN DRY WEIGHT OF NODULES CAPABLE OF FIXING NITROGEN IN SOIL CELL UNDER CONSIDERATION (GMS).
PDNODC	POTENTIAL RATE OF SUPPLY OF CARBON TO NODULES (GM/PLANT/HR)
PDNW(L,K)	POTENTIAL CHANGE IN WEIGHT OF EXISTING NODULES IN SOIL CELL L,K (GMS).
PDNWM	MAXIMUM POTENTIAL CHANGE IN DRY WEIGHT OF EACH NODULE DURING THIS PERIOD (GMS).
PDNWS	POTENTIAL CHANGE IN WEIGHT OF EXISTING NODULES IN A HALF SOIL SLAB (GMS).
PDPLB(I,J)	POTENTIAL RATE OF CHANGE IN LENGTH OF PETIOLE TO LEAF I ON BRANCH J (CM/HR).

PDPLM(I)	POTENTIAL RATE OF CHANGE IN LENGTH OF MAINSTEM PETIOLE TO LEAF I (CM/HR).
PDPLT	POTENTIAL RATE OF CHANGE IN LENGTH OF PETIOLE UNDER CONSIDERATION LIMITED ONLY BY TEMPERATURE (CM/HR).
PDPODC	MINIMUM RATE OF CARBON USE IN SYNTHESIZING PODS (GM/PLANT/HR).
PDPODW	POTENTIAL RATE OF CHANGE IN POD DRY WEIGHT (GMS / PLANT/HR).
PDPWB(I,J)	POTENTIAL RATE OF CHANGE IN DRY WEIGHT OF PETIOLE TO LEAF I ON BRANCH J (GM/HR).
PDPWM(I)	POTENTIAL RATE OF CHANGE IN DRY WEIGHT OF MAINSTEM PETIOLE TO LEAF I (GMS/HR).
PDR	POTENTIAL RATE OF CHANGE IN RSTAGE (REPRODUCTIVE STAGE OF GROWTH) (RSTAGES/HOUR).
PDRL(L,K)	POTENTIAL RATE OF CHANGE IN ROOT LENGTH IN SOIL CELL L,K IF CARBON IS NOT LIMITING (CM/HR).
PDSC	POTENTIAL RATE OF CARBON USE IN SYNTHESIZING SEEDS (GMS/PLANT/HR).
PDSCS	POTENTIAL AMOUNT OF CARBON USED IN SYNTHESIZING SEEDS SINCE SOME SEEDS WERE LOST MARKED FOR ABORTION (GMS/PLANT).
PDSW	POTENTIAL RATE OF CHANGE IN SEED DRY WEIGHT (GM/PLANT/HR).
PDV	POTENTIAL RATE OF CHANGE IN VSTAGE (VEGETATIVE STAGE OF GROWTH) (VSTAGES/HR).
PDWR(L,K)	POTENTIAL RATE OF INCREASE OF ROOT DRY WEIGHT IN SOIL CELL L,K (GMS/HR).
PDWRS	POTENTIAL RATE OF CHANGE IN ROOT DRY WEIGHT IN HALF SOIL SLAB (GMS/HR).
PERIOD	LENGTH OF CALCULATION PERIOD UNDER CONSIDERATION (HRS)
PERTWT(L,K)	PERMANENT DRY WEIGHT OF ROOT IN SOIL CELL L,K I.E. ROOT NOT KILLED BY DROUGHT OR LOW OXYGEN (GM).
PETWT	PETIOLE DRY WEIGHT (GMS/PLANT).
PG	CANOPY GROSS PHOTOSYNTHETIC RATE ASSUMING TEMPERATURE NOT LIMITING (MG CO2/M**2/S).
PGCOR	CORRECTION TO CANOPY GROSS PHOTOSYNTHETIC RATE CAUSED BY TEMPERATURE LIMITATION (MG CO2/M**2/S).
PGLF	GROSS PHOTOSYNTHETIC RATE FOR THE LEAF LAYER UNDER CONSIDERATION ASSUMING TEMPERATURE NOT LIMITING (MG CO2/M**2/S).
PGLIM	UPPER LIMIT OF LEAF GROSS PHOTOSYNTHESIS DETERMINED BY AIR TEMPERATURE (MG CO2/M**2/S).
PGROSS	CANOPY GROSS PHOTOSYNTHESIS RATE (MG CO2/M**2/S).
PHI	INTERMEDIATE STEP IN CALCULATION.
PHOTO	EFFECTIVE PHOTOPERIOD FOR SOYBEANS (HOURS).
PI	3.1416
PILD	LEAF OSMOTIC POTENTIAL AT DAWN (BARS).
PILDMX	MAXIMUM LEAF OSMOTIC POTENTIAL (BARS).
PILOSM	LEAF OSMOTIC POTENTIAL AT DAWN ADJUSTED FOR OSMO-REGULATION CAUSED BY WATER STRESS (BARS).
PINDEX	NET PHOTOSYNTHESIS RATE UNDER "WELL WATERED" CONDI-TIONS (MG CO2/M**2/S).
PLANTN	TOTAL NITROGEN IN REDUCED FORM IN PLANT (GMS).
PMAP(I,J,K)	ARRAY USED TO PLOT A SYMBOLIC DIAGRAM OF THE PLANT WHERE I = NUMBER OF LINE ON WHICH A GIVEN NODE WILL BE PRINTED,

	J SPECIFIES THE POSITION ON THAT LINE FOR EACH SYMBOL REPRESENTING ORGANS AT THAT NODE AND K = NUMBER OF THE AXIS (MAINSTEM OR BRANCH) UNDER CONSIDERATION.
PN	NET PHOTOSYNTHESIS RATE ALLOWING FOR STOMATAL CLOSURE (MG CO2/M**2/S).
PNLLFS	ACCUMULATED NET PHOTOSYNTHETIC RATE FOR LOWEST LEAVES ON PLANT (MG CO2/M**2/S)
PNNWWT	POTENTIAL CHANGE IN DRY WEIGHT OF NODULES IN AGE CLASS AND SOIL CELL UNDER CONSIDERATION (GMS).
PNONEW	NUMBER OF NEW NODULES WHICH COULD BE INITIATED WITH THE CARBON AVAILABLE.
PODN	TOTAL NITROGEN CONTENT OF PODS(GM/PLANT)
PODS	MEAN NUMBER OF PODS PER PLANT.
PODWT	DRY WEIGHT OF PODS (GMS/PLANT).
POPROW	PLANT POPULATION PER METER OF ROW.
PPDRL(L,K)	PDRL(L,K) AT PREVIOUS CALCULATION TIME (CM/HR).
PPSIL	PSIL AT PREVIOUS CALCULATION TIME (BARS).
PRENOD	RATE AT WHICH CARBON IS SUPPLIED TO NODULES DURING FLOWERING TO GROW THEM IN PREPARATION FOR SEEDFILL (GM/PLANT/HR)
PRM(I)	PARAMETER I.
PROTIN	PROPORTION OF PROTEIN IN THE SEED.
PSIFAC(J)	FACTOR USED IN CALCULATING PSIS FOR HORIZON J: THAT PART OF THE EQUATION THAT IS INDEPENDENT OF WATER CONTENT.
PSIL(I)	LEAF WATER POTENTIAL AT TIME I (BARS).
PSILO	LEAF WATER POTENTIAL AT ZERO TURGOR (BARS).
PSILD	LEAF WATER POTENTIAL AT DAWN (BARS).
PSILT	LEAF WATER POTENTIAL WHICH JUST PREVENTS ALL SHOOT GROWTH AT TIME UNDER CONSIDERATION (BARS).
PSIRD(L,K)	WATER POTENTIAL IN SOIL CELL L,K AT DAWN (BARS).
PSIS(L,K)	WATER POTENTIAL OF SOIL CELL L,K (BARS).
PSISAT(J)	WATER POTENTIAL OF SOIL IN HORIZON J AT WHICH AIR STARTS TO ENTER THE SATURATED SOIL (BARS).
PSISI	INITIAL SOIL WATER POTENTIAL (BARS).
PSISM	SOIL WATER POTENTIAL AVERAGED OVER CELLS FROM WHICH WATER IS EXTRACTED WHEN POTENTIAL SHOOT GROWTH IS SATISFIED; WEIGHTED FOR THE AMOUNT OF WATER EXTRACTED (BARS).
PSIST	SOIL WATER POTENTIAL AVERAGED OVER CELLS FROM WHICH WATER IS EXTRACTED WHEN ALL CARBON TRANSLOCATED GOES TO THE ROOTS; WEIGHTED FOR THE AMOUNT OF WATER EXTRACTED (BARS).
PSWT	POTENTIAL SEED DRY WEIGHT IF ALL SEEDS WERE GROWING (GM/PLANT).
PTPL	LEAF TURGOR PRESSURE AT PREVIOUS CALCULATION TIME (BARS).
PTPLD	LEAF TURGOR PRESSURE AT DAWN YESTERDAY (BARS).
PWUP(L,K)	RATE OF WATER EXTRACTION FROM SOIL CELL L,K BY ROOTS AT PREVIOUS CALCULATION TIME (GMS/HR).
Q	ATMOSPHERIC TRANSMISSION COEFFICIENT (DIMENSIONLESS).
QCST	MINIMUM RATE OF CARBON USE BY SHOOT EXPANDING AT POTENTIAL RATE (GM/PLANT/HR).
QZ	BALANCE OF THE CARBON ACCOUNT IN THE MODEL. IT IS POSITIVE IF THE MODEL IS LEAKING CARBON (GM/PLANT).
R	RADIUS VECTOR OF THE EARTH.
RADINT(I)	FRACTION OF SOLAR RADIATION INTERCEPTED BY THE CROP

	AT TIME I.
RAIN	RAINFALL (MM/DAY).
RASEN	CUMULATIVE NITRATE REDUCTASE ACTIVITY FOR ALL LEAVES ON PLANT (GMS N/PLANT/HOUR).
RATIO(L,M)	EQUALS WATPOT/WATATM FOR LATITUDE L*10 AND MONTH M.
RCPMAX	MAXIMUM AMOUNT OF CARBON THAT CAN BE STORED IN THE ROOT (GMS/PLANT).
RCPAV	RATE AT WHICH CARBON BECOMES AVAILABLE FROM RCPOOL FOR ROOT GROWTH (GM/PLANT/HR)
RCPOOL	CARBON STORED IN THE ROOT (GMS/PLANT).
REDN	PROPORTION OF GROSS PHOTOSYNTHETIC RATE POSSIBLE WITH CURRENT LEAF NITROGEN CONTENT.
REDSEN	REDN OR SEN , WHICHEVER IS SMALLEST
REMC	REMAINING CARBON AVAILABLE FOR SHOOT GROWTH(GM/PLANT)
REMNO3	NITROGEN REMAINING AS NO3 IN THE PLANT (GM N/PLANT).
REQN	TOTAL NITROGEN REQUIRED BY PLANT AT MAXIMUM NITROGEN CONTENT (GM)
REQSC	REMAINING REQUIREMENT FOR CARBON TO GROW SEEDS AT POTENTIAL RATE (GMS).
RESPS(L,K)	CO2 RESPIRED IN SOIL CELL L,K IN THIS PERIOD (GM).
REW	RADIATION USED TO EVAPORATE WATER FROM THE SOIL SURFACE (WATTS/M**2).
RGCF(L,K)	PROPORTIONAL REDUCTION OF ROOT GROWTH FROM ALL PHYSICAL CAUSES IN SOIL CELL L,K.
RGCF1	PROPORTIONAL REDUCTION OF ROOT GROWTH CAUSED BY ME-CHANICAL SOIL RESISTANCE AND SOIL WATER POTENTIAL.
RGCF2	PROPORTIONAL REDUCTION OF ROOT GROWTH CAUSED BY SOIL TEMPERATURE.
RGCF3	PROPORTIONAL REDUCTION OF ROOT GROWTH CAUSED BY DIS-TANCE FROM STEM BASE.
RHS(K)	RADIATION FALLING ON TOP SOIL CELL IN COLUMN K BUT NOT USED TO EVAPORATE WATER (WATTS/M**2).
RI	DAILY SOLAR RADIATION INTEGRAL (JOULE/M**2).
RIADJ	FACTOR USED TO ADJUST RI SO THAT RADIATION FLUX DENSITY CAN BE CALCULATED FROM THE EQUATION FOR A FULL SINE WAVE (DIMENSIONLESS).
RNC	NET RADIATION ON THE CROP ASSUMING COMPLETE COVER (WATTS/M**2).
RNLU	NET UPWARD LONG WAVE RADIATION (WATTS/M**2).
RNNH4	RESIDUAL NITROGEN AS AMMONIUM IN SOIL AT BEGINNING OF SEASON (LB/ACRE).
RNNO3	RESIDUAL NITROGEN AS NITRATE IN SOIL AT BEGINNING OF SEASON (LB/ACRE).
RNODC	REMAINING CARBON AVAILABLE FOR NODULES IN A HALF SOIL SLAB (GMS).
RNS	NET RADIATION ON THE SOIL SURFACE ASSUMING BARE SOIL (WATTS/M**2).
ROOTWT	DRY WEIGHT OF ROOTS (GMS/PLANT).
ROUGH	A CROP SURFACE ROUGHNESS PARAMETER.
ROWANG	ROW ORIENTATION MEASURED EASTWARD FROM NORTH(DEG.).
ROWIDE	EFFECTIVE WIDTH OF ROW (I.E., SHADOW WIDTH) (CM).
ROWINC(I)	DISTANCE BETWEEN A ROW AND THE MIDPOINT OF AN INCRE-MENT OF ROWSPACING (CM).
ROWSP	ROW SPACING (CM).
RPERD	FLOATING POINT FORM OF IPERD.

RRRM	RADIAL RESISTANCE OF OLD ROOTS PER CM OF ROOT IN SOIL CELL (BAR.HR/GM).
RRRY	RADIAL RESISTANCE OF YOUNG ROOTS PER CM OF ROOT IN SOIL CELL (BAR.HR/GM).
RSTAGE	REPRODUCTIVE GROWTH STAGE.
RSTPC	FROM R3 TO R4,THE DIFFERENCE BETWEEN THE RATE OF CARBON SUPPLY NEEDED TO GROW EXISTING PODS AND THE RATE OF CARBON SUPPLY THAT WOULD BE NEEDED TO FILL THOSE PODS (GM/PLANT/HR).
RTMINW	MINIMUM DRY WEIGHT OF ROOT THAT MUST BE PRESENT IN A CELL BEFORE IT CAN GROW INTO ADJACENT CELLS (GMS).
RTWL	AVERAGE ROOT DRY WEIGHT PER UNIT LENGTH (GM/CM).
RTWT(L,K)	DRY WEIGHT OF ROOT IN SOIL CELL L,K (GMS).
RUTDEN(L,K)	ROOT DENSITY IN CELL L,K (CM/CC).
RVR(L,K)	ROOT VASCULAR RESISTANCE BETWEEN BASE OF STEM AND SOIL CELL L,K (BAR.HR/GM).
RVRL	ROOT VASCULAR RESISTANCE PER CM OF ROOT (BAR.HR/GM).
S(L,K)	AERATION POROSITY OF THE SOIL IN CELL L,K (CC/CC).
SAND(J)	SAND PLUS SILT CONTENT OF HORIZON J AS A PROPORTION OF SOIL DRY WEIGHT.
SARANG(I)	ANGLE BETWEEN ROW ORIENTATION AND SOLAR AZIMUTH AT TIME(I) (RADIANS).
SC	RADIUS OF SOIL CYLINDER THROUGH WHICH WATER TRAVELS TO REACH ROOTS (CM).
SCF	STOMATAL CLOSURE FACTOR FOR REDUCING H2O AND CO2 FLUX (=1 WHEN STOMATA FULLY OPEN).
SCOND(J)	SATURATED HYDRAULIC CONDUCTIVITY OF SOIL IN HORIZON J ((CC H2O)/(CM MOVED)/(CM SOIL WATER POTENTIAL GRADIENT)/DAY).
SCPMAX	MAXIMUM AMOUNT OF CARBON THAT CAN BE STORED IN THE SHOOT (GM/PLANT).
SCPOOL	CARBON STORED IN THE SHOOT (GMS/PLANT).
SCRATO	FACTOR TO ADJUST CARBON SUPPLY TO SHOOT ORGANS DEPENDING ON CARBON AVAILABILITY (SEE GROWTH ROUTINE).
SCRATS	ACCUMULATOR FOR SUMMING (SCRATO*PERIOD) OVER A DAY
SDEPTH(J)	MAXIMUM DEPTH OF SOIL IN HORIZON J (CM).
SDWLIM	FACTOR FOR LIMITING SEED DRY WT. GAIN AT END OF POD FILL.
SDWMAX	MAXIMUM SEED DRY WEIGHT (GM/SEED).
SEEDN	TOTAL NITROGEN CONTENT OF SEEDS (GMS/PLANT).
SEEDS	NUMBER OF SEEDS PER PLANT.
SEEDWT	SEED DRY WEIGHT (GMS/PLANT).
SEN	PHOTOSYNTHETIC MULTIPLIER ACCOUNTING FOR SENESCENCE AND BASED ON THE AGE OF THE YOUNGEST LEAF.
SEQSC	(POTENTIAL RATE OF CARBON USE IN SYNTHESIZING SEEDS TO FILL THE PODS CURRENTLY ON THE PLANT PLUS ANOTHER POD FOR EVERY TWO NODES) * A FACTOR (GM/PLANT/HR)
SGT	PROPORTION OF TIME FOR WHICH SHOOT GROWS: LIMITED BY SHOOT TURGOR.
SGTLI	PROPORTION OF SHOOT GROWING TIME LOST IRRETRIEVABLY BECAUSE OF LOW TURGOR
SGTLT	PROPORTION OF SHOOT GROWING TIME LOST TEMPORARILY WHILE TURGOR IS DECREASING. IT IS REGAINED WHEN TURGOR INCREASES.
SGTR	PROPORTION OF TIME FOR WHICH SHOOT GROWS: LIMITED BY SHOOT TURGOR OR CARBON AVAILABILITY.

SHTWT	DRY WT OF LEAVES + STEMS + PETIOLES (GM/PLANT).
SINALT(I)	SINE OF SOLAR ALTITUDE AT TIME(I).
SINAZI(I)	SINE OF SOLAR AZIMUTH AT TIME(I).
SINK	INDICATOR OF SOURCE/SINK BALANCE FOR CARBON (POSITIVE IF SINK IS SMALLER THAN SOURCE).
SITE	FLAG USED TO DESIGNATE A PARTICULAR SITE FOR SPECIAL TREATMENT (USUALLY OF SOIL WATER CONTENT)
SLA	SPECIFIC LEAF AREA (=LEAF AREA/LEAF DRYWEIGHT)(CM**2/GM).
SLOW	VARIABLE USED TO REDUCE GROWTH OF VEGETATIVE ORGANS AFTER FLOWERING STARTS.
SOAKN(L)	NITROGEN SOAKING INTO SOIL LAYER L (MG NITROGEN /CM**2).
SOAKW(L)	WATER SOAKING INTO SOIL LAYER L (CC/CM**2).
SOLALT(I)	SOLAR ALTITUDE AT TIME(I) (RADIANS).
SOLALV	SOLAR ALTITUDE WHEN LIGHT INTENSITY IS 10 LUX (RADIANS).
SOLALW	SOLAR ALTITUDE WHEN LIGHT INTENSITY IS 100 LUX (RADIANS).
SOLAZI(I)	SOLAR AZIMUTH AT TIME(I) (RADIANS).
SORES(L,K)	SOIL AND ROOT RESPIRATION RATE IN SOIL CELL L,K (GM/CC/SEC).
SR(L,K)	SOIL RESISTANCE TO WATER FLOW IN CELL L,K (BAR.HR/GM).
STEMWT	DRY WEIGHT OF STEMS (GMS/PLANT).
STG	STRING OF SYMBOLS REPRESENTING FLOWERS PODS AND SEEDS USED IN PLOTTING A DIAGRAM OF THE PLANT.
STPOOL	CARBON STORED IRRETRIEVABLY IN STEMS AND PETIOLES (GM/PLANT)
SUM1, ETC.	INTERMEDIATE STAGES IN CALCULATIONS.
SUMLW	SUM OF DRY WEIGHTS OF CERTAIN GROWING LEAVES (GMS).
SUPNO3	PREDICTED NO3 UPTAKE FOR THE NEXT CALCULATION PERIOD (GMS N/PLANT).
SVPA	WATER SATURATION VAPOUR PRESSURE AT THE DRY BULB TEMPERATURE (KPA).
SVPW	WATER SATURATION VAPOUR PRESSURE AT THE WET BULB TEMPERATURE (KPA).
SW10,ETC.	SWITCHES TO INDICATE COMPLETION OF CERTAIN OPERATIONS.
SWR3	SWITCH TO INDICATE R3 HAS BEEN REACHED (= 1 WHEN RSTAGE.GT.3).
SWR4	SWITCH TO INDICATE R4 HAS BEEN REACHED (= 1 WHEN RSTAGE.GT.4).
SWV1	SWITCH TO INDICATE V1 HAS BEEN REACHED (=1 WHEN VSTAGE.GT.1)
SWV2	SWITCH TO INDICATE V2 HAS BEEN REACHED (=1 WHEN VSTAGE.GT.2)
SX	EFFECTIVE SOIL CELL DIMENSION IN THE DIRECTION OF DIFFUSION (CM).
T	PERIOD COVERED BY CALCULATION (SECS).
T0	TIME AFTER VC WHEN THE BOTTOM BOUNDARY CONDITION FOR WATER MOVEMENT IN THE SOIL PROFILE STARTS CHANGING FOR SOME SITES AND DATA SETS (DAYS).
TA	INITIAL SOIL TEMPERATURE IN THE SURFACE LAYER (DEG.C).
TAIR(I)	AIR TEMPERATURE AT TIME(I) (DEG. C).
TAIRL	EFFECTIVE AIR TEMPERATURE (= TAIR(I) WITHIN PRESCRIBED LIMITS)(DEG.C).
TAIRSS	AIR TEMPERATURE AT SUNSET (DEG C).
TAUM	PARAMETER USED TO CALCULATE LEAF CONDUCTANCE TO CO2

	(M**3/JOULE).
TAUN	PARAMETER USED TO CALCULATE LEAF CONDUCTANCE TO CO2 (SEC.M**2/JOULE).
TAUNIN	MINIMUM VALUE OF TAUN (SEC.M**2/JOULE).
TB(J)	NUMBER OF LEAVES ON BRANCH J (=VSTAGE FOR BRANCH).
TBMAH(J)	TBMAX(J) + 0.5
TBMAX(J)	MAXIMUM NUMBER OF NODES ON BRANCH J OF DETERMINATE PLANTS.
TBS	NUMBER OF LEAVES ON BRANCHES.
TC	TEMPERATURE OF THE SOIL BELOW THE TEST VOLUME (DEG. C)
TCADS	THERMAL CONDUCTIVITY OF AIR-DRY SURFACE SOIL (CAL/CM/SEC/DEG.C).
TCAIR	THERMAL CONDUCTIVITY OF DRY AIR AT CELL TEMPERATURE (MILLICAL/CM/SEC/DEG. C).
TCH2O	THERMAL CONDUCTIVITY OF WATER AT CELL TEMPERATURE (MILLICAL/CM/SEC/DEG. C).
TCS(L,K)	THERMAL CONDUCTIVITY OF SOIL IN CELL L,K (CAL/CM / SEC/DEG. C).
TCSAT	THERMAL CONDUCTIVITY OF AIR SATURATED WITH WATER VAPOUR AT CELL TEMPERATURE (MILLICAL/CM/SEC/DEG. C).
TCSM	THERMAL CONDUCTIVITY OF SOIL IN TOP HALF OF SURFACE CELL (CAL/CM/SEC/DEG.C).
TCVAP	THERMAL CONDUCTIVITY ATTRIBUTABLE TO WATER VAPOUR MOVEMENT IN SATURATED AIR AT CELL TEMPERATURE (MILLICAL/CM/SEC/DEG. C).
TDAY	AVERAGE DAYTIME AIR TEMPERATURE (DEG C).
TDL(L,K)	MEAN THERMAL DIFFUSIVITY OF SOIL BETWEEN CELL L,K AND THE CELL TO IT'S LEFT (CM**2/SEC).
TDMAX	MAXIMUM THERMAL DIFFUSIVITY OF SOIL BETWEEN ANY 2 CELLS IN THE PRIFILE (CM**2/SEC).
TDRY	DRY BULB TEMPERATURE CORRESPONDING TO TWET (DEG C).
TDS(L,K)	THERMAL DIFFUSIVITY OF SOIL IN CELL L,K(CM**2/SEC)
TDSX(I,J)	PARAMETER I IN THE EQUATION TO CALCULATE THERMAL DIFFUSIVITY FOR THE SOIL IN HORIZON J. EMPIRICAL EQUATION DERIVED FROM MECHANISTIC MODEL THERMS AND USED AT THE START OF TMPSOL.
TDU(L,K)	MEAN THERMAL DIFFUSIVITY OF SOIL BETWEEN CELL L,K AND THE CELL ABOVE (CM**2/SEC).
TENLUX	SOLAR ELEVATION WHEN LIGHT INTENSITY IS 10 LUX.
TERM2,ETC	INTERMEDIATE STEPS IN CALCULATION.
TEST1,ETC	INTERMEDIATE STEPS IN CALCULATION.
THETAO(J)	VOLUMETRIC WATER CONTENT OF SOIL IN HORIZON J AT A SOIL WATER POTENTIAL OF -15 BARS (CC/CC).
THETAI	INITIAL WATER CONTENT OF BOTTOM LAYER IN SOIL PROFILE FOR SOME SITES AND DATA SETS WHERE THIS IS KNOWN (CC/CC).
THETAR(J)	THETAO(J) SLIGHTLY REDUCED.
THETAS(J)	VOLUMETRIC WATER CONTENT OF SOIL IN HORIZON J AT SATURATION.
TLAGE	TIME SINCE LEAF EXPANSION STOPPED (DAYS).
TLIFE	FUNCTIONAL LIFE OF LEAF AFTER STOPPING EXPANSION (DAYS).
TMAX	MAXIMUM AIR TEMPERATURE DURING THE DAY (DEG C).
TMAXHR	TIME OF MAXIMUM AIR TEMPERATURE MEASURED FROM DAWN (HOURS).
TMIN	MINIMUM AIR TEMPERATURE DURING THE DAY (DEG C).
TMINT	MINIMUM AIR TEMPERATURE DURING THE NEXT DAY (DEG C).

TMX5WA	INTERMEDIATE STEP IN CALCULATION.
TNAIRD(K)	DEPTH OF AIR DRY SOIL ON SURFACE CELL K AS A PROPORTION OF HALF CELL DEPTH.
TNL	THICKNESS OF NEW LEAVES (CM).
TNYT	AVERAGE NIGHT-TIME AIR TEMPERATURE (DEG. C).
TOPLF	REAL FORM OF NTOPLF(NUMBER OF THE TOP TRIFOLIOLATE ON THE MAINSTEM).
TOPORS(L,K)	TOTAL (DRY) SOIL POROSITY IN CELL L,K (CC/CC).
TOPWT	DRY WEIGHT OF STEMS PLUS PETIOLES INCLUDING STARCH AND SUGAR THEY CONTAIN (GMS/PLANT).
TOTNO3	TOTAL NITROGEN IN PLANT AS NO3 AT START OF CALCULATION PERIOD (GMS NITROGEN/PLANT).
TPI	2 * PI = 6.2832
TPL	LEAF TURGOR PRESSURE (BARS).
TPLD	LEAF TURGOR PRESSURE AT DAWN (BARS).
TPLT	LEAF TURGOR PRESSURE WHICH JUST PREVENTS ALL SHOOT GROWTH AT TIME UNDER CONSIDERATION (BARS).
TPRD(L,K)	ROOT TURGOR PRESSURE IN CELL L,K AT DAWN MINUS THE THRESHOLD TURGOR FOR GROWTH (BARS).
TRANSC	POTENTIAL RATE OF CARBON TRANSLOCATION FROM THE SHOOT TO GROWING ORGANS (GM/PLANT/HR).
TRIFMG	INTERCEPT IN EQUATION RELATING TRIFOLILOLATE EXPAN-SION RATE TO TEMPERATURE.
TRNDAY	AMOUNT OF CARBON TRANSLOCATED FROM THE SHOOT TO GROWING ORGANS SO FAR TODAY (GM/PLANT).
TS(L,K)	TEMPERATURE OF SOIL IN CELL L,K (DEG. C).
TSI	INITIAL SOIL TEMPERATURE FOR THE LAYER UNDER CONSIDERATION (DEG.C).
TWET	WET BULB TEMPERATURE (DEG C).
ULAREA	AREA OF UNIFOLIOLATES (CM**2).
ULEAFW	DRY WEIGHT OF UNIFOLIOLATES (GMS).
UPETL	LENGTH OF PETIOLES TO UNIFOLIOLATES (CM).
UPETW	DRY WEIGHT OF PETIOLES TO UNIFOLIOLATES (GMS).
UPH2ON	WATER EXTRACTED FROM A HALF SOIL SLAB BY ROOTS OVERNIGHT (GMS).
UPH2OS	WATER EXTRACTED FROM A HALF SOIL SLAB BY ROOTS DURING PAST PERIOD (GMS).
UPTH2O(L,K)	WATER EXTRACTED FROM SOIL CELL L,K BY ROOTS DURING PAST PERIOD (GMS).
USEDCS	AMOUNT OF CARBON USED IN GROWING PLANT SO FOR (GMS)
VCLAY(L,K)	VOLUME FRACTION OF CLAY IN CELL L,K.
VCLAYI	VOLUME FRACTION OF CLAY IN LAYER UNDER CONSIDERATION
VEGSRC	RATE OF CARBON SUPPLY TO THE VEGETATIVE PARTS OF THE SHOOT AND ROOT(GM/PLANT/HR).
VH2OC(L,K)	VOLUMETRIC WATER CONTENT OF SOIL CELL L,K (CC/CC).
VH2OC2	(VH2OC)**2
VH2OCM	MEAN VOLUMETRIC WATER CONTENT OF EXPOSED SURFACE SOIL CELLS.
VH2OCS	ACCUMULATED VOLUMETRIC WATER CONTENT OF EXPOSED SURFACE SOIL CELLS.
VH2OEQ	USED TO CALCULATE WATER FLUX ACROSS A SOIL HORIZON BOUNDARY. SOIL WATER POTENTIAL IN HORIZON B IS USED TO CALCULATE AN EQUIVALENT WATER CONTENT VH2OEQ USING THE PF CURVE FOR HORIZON A (CC/CC).
VH2OI(L)	INITIAL VOLUMETRIC WATER CONTENT OF SOIL IN LAYER L

VLOST	NUMBER OF VEGETATIVE STAGES LOST BECAUSE OF WATER STRESS.
VMUCK(L,K)	VOLUME FRACTION OF ORGANIC MATTER IN CELL L,K.
VNC(L,K)	AMOUNT OF NITROGEN PER UNIT OF SOIL VOLUME IN CELL L,K (MG/CC).
VNH4C(L,K)	AMOUNT OF NITROGEN AS NH4 PER UNIT SOIL VOLUME IN CELL L,K (MG/CC).
VNO3C(L,K)	AMOUNT OF NITROGEN AS NO3 PER UNIT SOIL VOLUME IN CELL L,K (MG/CC).
VPD	WATER VAPOUR PRESSURE DEFICIT (KPA).
VSAND(L,K)	VOLUME FRACTION OF SAND PLUS SILT IN CELL(L,K).
VSANDI	VOLUME FRACTION OF SAND IN LAYER UNDER CONSIDERATION.
VSORES	SOIL AND ROOT RESPIRATION RATE IN SOIL CELL UNDER CONSIDERATION (ATMOSPHERIC PARTIAL PRESSURE).
VSTAGE	VEGETATIVE GROWTH STAGE.
VSTMAH	VSTMAX+0.5
VSTMAX	MAXIMUM NUMBER OF TRIFOLIOLATE NODES ON MAINSTEM OF DETERMINATE PLANTS.
W	WIDTH OF EACH SOIL CELL (CM).
WATACT	ACTUAL RADIATION INCIDENT AT EARTH'S SURFACE AT NOON (WATTS/M**2).
WATADJ	FACTOR USED TO ADJUST RADIATION FLUX DENSITY AFTER CALCULATING IT FROM THE EQUATION FOR A FULL SINE WAVE (DIMENSIONLESS).
WATATM	RADIATION INCIDENT AT THE TOP OF THE ATMOSPHERE AT NOON (WATTS/M**2).
WATLF	LIGHT (PAR) INTERCEPTED BY THE LEAVES IN THE LAYER UNDER CONSIDERATION (WATTS/M**2).
WATPL	LIGHT (PAR) INCIDENT ON THE CROP CANOPY AS MEASURED WITH THE SENSOR FACING THE SUN (WATTS/M**2).
WATPOT	POTENTIAL RADIATION INCIDENT AT EARTH'S SURFACE AT NOON (WATTS/M**2).
WATRAT	PROPORTION OF RADIATION THAT CAN PENETRATE THE CLOUD COVER (=WATACT/WATPOT).
WATTSM(I)	ACTUAL RADIATION INCIDENT AT EARTH'S SURFACE AT TIME(I) (WATTS/M**2).
WATWK	AVERAGE LIGHT FLUX DENSITY ON CROP FOR PAST WEEK (WATTS/M**2).
WFMIN	ACTUAL SOIL MOISTURE CONTENT EXPRESSED AS A FRACTION OF FIELD CAPACITY.
WFNIT	MOISTURE FACTOR AFFECTING NITRIFICATION.
WIND	WINDSPEED AT 2 METRES (KM/HOUR).
WPN	INTERMEDIATE STEP IN THE CALCULATION OF FWL(L,K)
WUPOS	RATE OF WATER UPTAKE FROM A HALF SOIL SLAB WHEN LEAF TURGOR PRESSURE = 0 BARS. (GMS/HR).
WUP2S	RATE OF WATER UPTAKE FROM A HALF SOIL SLAB WHEN LEAF TURGOR PRESSURE = 2 BARS. (GMS/HR).
WUPDS	RATE OF WATER UPTAKE FROM A HALF SOIL SLAB IF LEAF WATER POTENTIAL HAS NOT CHANGED IN PAST PERIOD (GM/HR)
WUPGS	RATE OF WATER UPTAKE FROM A HALF SOIL SLAB FOR VARIOUS VALUES OF LEAF WATER POTENTIAL. USED TO SELECT A VALUE OF LEAF WATER POTENTIAL ITERATIVELY (GMS/HR).
WUPM(L,K)	RATE OF WATER UPTAKE FROM SOIL CELL(L,K) BY ROOTS MORE THAN 0.2 DAYS OLD, WHEN LEAF WATER POTENTIAL IS AT THE THRESHOLD (I.E., PREVENTS SHOOT GROWTH) (GMS/HR).
WUPMS	SUM OF WUPM(L,K) AND WUPN(L,K) OVER ALL SOIL CELLS IN

	A HALF SOIL SLAB (GMS/HR).
WUPN(L,K)	RATE OF WATER UPTAKE FROM SOIL CELL(L,K) BY ROOTS BETWEEN 0.1 AND 0.2 DAYS OLD, WHEN LEAF WATER POTENTIAL IS AT THE THRESHOLD (I.E., PREVENTS SHOOT GROWTH) (GMS.HR).
WUPRS	RATE OF WATER UPTAKE FROM A HALF SOIL SLAB BY NEW ROOTS GROWN AFTER SHOOT GROWTH POTENTIAL HAS BEEN SATISFIED (GMS/HR).
WUPSI	SUM OF (SOIL WATER POTENTIAL)*(RATE OF WATER EXTRACTION) OVER CELLS IN A HALF SOIL SLAB FROM WHICH WATER IS EXTRACTED (BAR.GM/HR).
WUPT(L,K)	RATE OF WATER UPTAKE FROM SOIL CELL(L,K) WHEN LEAF WATER POTENTIAL IS AT THE THRESHOLD (I.E., PREVENTS SHOOT GROWTH) (GMS/HR).
WUPTS	RATE OF WATER UPTAKE FROM A HALF SOIL SLAB BY NEW ROOTS GROWN WHEN ALL TRANSLOCATED CARBON GOES TO THE ROOTS (GMS/HR).
WWKLF	AVERAGE LIGHT FLUX DENSITY FOR PAST WEEK ON THE LEAF LAYER UNDER CONSIDERATION (WATTS/M**2).
X(I,K)	ARRAY USED IN THERMS TO HOLD VALUES OF TDS FOR VARIOUS COMBINATIONS OF VH2OC AND TS.
X0	DUMP FOR UNWANTED NUMBERS.
X1..X999	INTERMEDIATE STAGES IN CALCULATIONS.
XAIR	INTERMEDIATE STAGE IN CALCULATION OF SOIL THERMAL CONDUCTIVITY.
XCLAY	INTERMEDIATE STAGE IN CALCULATION OF SOIL THERMAL CONDUCTIVITY.
XDUM	INTERMEDIATE STAGE IN CALCULATION.
XGAIR	INTERMEDIATE STAGE IN CALCULATION OF SOIL THERMAL CONDUCTIVITY.
XLAT	LATITUDE (RADIANS).
XMUCK	INTERMEDIATE STAGES IN CALCULATION OF SOIL THERMAL CONDUCTIVITY
XSAND	INTERMEDIATE STAGE IN CALCULATION OF SOIL THERMAL CONDUCTIVITY
XTRAC	CARBON LOST BECAUSE IT CANNOT BE USED OR STORED (GM/PLANT).
XTRACS	XTRAC ACCUMULATED OVER TIME (GM/PLANT).
YRL(L,K)	LENGTH OF YOUNG ROOT IN SOIL CELL L,K (CM).
Z	HEIGHT OF TOP LEAVES ABOVE SOIL(CM).
ZNEW	NEW HEIGHT OF TOP LEAVES ABOVE SOIL (CM).

```
C     *** COMMON BLOCK ***
       REAL LATUDE,LLUE,LTC,LAREAT,MSTEMH,MPETL,MLEAFW,MPETW,LEAFWT,
      &      NSTRES,NRATIO,NODN,MLAREA,MSTEMW,NFIXWT,NNEWWT,LAREAM,
      &      LAREAB,LAMDAC,LAMDAS,LCAI,LLFWT,LLAREA,LYTRES,MECHR,NCPM,
      &      NCPOOL,NFCF1,NFCF2,NFCF,NODC,NODWAR,NODWT,NODIE,NFIXNU,
      &      NODNEW,NODNUM,NODSIT,LFWT,NEXLP,NRATS
       INTEGER DET,PCRPAR,VSTMAZ,FILDAY,SWR3,SWR4,SW20,DROP,SW10,
      &       SW30,VSTMAX,TBMAX,SWV1,SWV2,BSITES,SITE,DATA,ABSCIS,DAYBR
       COMMON/ A /A,ABSCIS,ABSLPN,ABSPON,ADRL(20,10),ADWR(20,10),
      &           AFWUP(20,10),AFWUPS,AIRDR(5),ASCRAT(11),ASGT(11),
      &           AVP,AWR(20,10),AWUP(20,10),AWUPS,AWUPSS
       COMMON/ B /B,BD(5),BDC(20,10),BETA(5),BLAREA(20,20),BLEAFW(20,20),
      &           BPETL(20,20),BPETW(20,20),BRNCHH(20),BRNCHW(20),BTPL
       COMMON/ C /C,CEC,CLAY(5),CLDFAC,CLIMAT(7),CLOUD,COND(20,10),
      &           CONRES,CONVN,CONVPO,CONVSE,COVER,CO2,CXT(20,10)
       COMMON/ D /D,DATA,DAYBR(20),DEADRN,DET,DIFFO(5),DIFIMG,DIMPL,
      &           DRAIN,DROP,DUMB,DUMC
       COMMON/ E /E,EO(11),EORS,EORSCS,EQSC,ETA(5),EWU(11),EWUS
       COMMON/ F /F,FC(20),FCI(5),FERN,FILDAY,FILL,FIXC,FIXCS,
      &           FIXDAY,FLWRWT,FNH4,FNO3,FRUITC,FWUP(20,10)
       COMMON/ G /G,GAMMA
       COMMON/ H /H
       COMMON/ I /IDAY,ILOW,ILOWB(20),INITBR(20),ITIME,IPERD
       COMMON/ J /JDAY,JDFRST,JDLAST
       COMMON/ K /KLJ
       COMMON/ L /LAREAB,LAREAM,LAREAT,LATUDE,LCAI,LEAFWT,LKH,LKR,
      &           LKRP1,LLUE,LPNDT,LTC,LYRSOL
       COMMON/ M /MATURE,MG,MLAREA(25),MLEAFW(25),MPETL(25),
      &           MPETW(25),MSTEMH,MSTEMW,MSW1,MSW2
       COMMON/ N /NBRNCH,NBTPLF(20),NCD,NCPOOL,NEXLP,NFCF(20,10),
      &           NFIXNU(20,10),
      &           NFIXWT(20,10),NFLWRS,NJ(20),NK,NKH,NKN,NKR,NKRP1,
      &           NL,NLN,NLP,NLR,NLRP1,NNEWWT(10,20,10),NODC,NODES,
      &           NODIE(20,10),NODN,NODNEW(10,20,10),NODNUM(20,10),
      &           NODWAR(200),NODWT,NPRM,NRATIO,NTOPLF,NYTE
       COMMON/ O /OMA,OSMREG
       COMMON/ P /PADWR(20,10),PAR(10),PARINT(10),PCRPAR(200),PCRS,
      &           PCST,PDBRW(20),PDFW,PDLAB(20,20),PDLAM(3),PDMH,PDMW,
      &           PDNODC,PDNW(20,10),PDPLB(20,20),PDPLM(3),PDPODC,PDPODW,
      &           PDPWB(20,20),PDPWM(3),PDR,PDSC,PDSW,PDV,PDWR(20,10),
      &           PERIOD,PERTWT(20,10),PETWT,PHOTO,PILD,PILOSM,PLANTN,
      &           PNLLFS,PODS,PODWT,POPROW,
      &           PPDRL(20,10),PRM(99),PSIL(10),
      &           PSILD,PSIS(20,10),PSISAT(5),PSISM,PWUP(20,10)
       COMMON/ Q /Q,QCST
       COMMON/ R /RADINT(10),RAIN,RCPOOL,REMNO3,RESPS(20,10),
      &           RGCF(20,10),RHS(10),RI,RNNO3,RNNH4,
      &           ROOTWT,ROUGH,ROWANG,ROWSP,RRRM,RRRY,RSTAGE,
      &           RSTPC,RTMINW,RTWL,RTWT(20,10),RUTDEN(20,10),
      &           RVR(20,10),RVRL
       COMMON/ S /SAND(5),SCF,SCOND(5),SCPOOL,SCRATO,SCRATS,
      &           SDEPTH(5),SDWMAX,SEEDS,SEEDWT,SGT,SGTLI,SGTLT,SGTR,
      &           SHTWT,SINK,SITE,SLOW,STEMWT,STPOOL,SUPNO3,SWR3,SWR4,SWV1,SWV2
       COMMON/ T /TA,TAIR(11),TAUN,TB(20),TBMAH(20),TC,TCADS,TDSX(5,5),
      &           THETAI,THETAS(5),THETAO(5),TLAGE,TLIFE,
      &           TNAIRD(10),TNL,TNYT,TO,TOPORS(20,10),TPL,TPLD,
      &           TRANSC,TRIFMG,TRNDAY,TS(20,10)
       COMMON/ U /ULAREA,ULEAFW,UPETL,UPETW,USEDCS
       COMMON/ V /VCLAY(20,10),VEGSRC,VH2OC(20,10),VH2OI(20),VLOST,
      &           VMUCK(20,10),VNC(20,10),VNH4C(20,10),VNO3C(20,10),
      &           VSAND(20,10),VSTAGE,VSTMAH
       COMMON/ W /W,WATRAT,WATTSM(10),WATWK,WIND
       COMMON/ X /XTRACS
       COMMON/ Y /YRL(20,10)
       COMMON/ Z /Z
```

Order of Variables in Input Files for GLYCIM

Lines of data sometimes occupy more than one line in a file or on the printed page. Therefore, the symbol # has been used below to indicate an obligatory new line in the files.

File 1 variables: weather data

MSW1, MSW2, A,B,C,E,F.,

WIND (only if daily wind is unavailable, in which case MSW2 = 0.)

JDAY, RI, TMAX, TMIN, RAIN, WIND, TWET, TDRY

(Each day's weather data starts on a new line. WIND and TWET, TDRY are optional. They may be omitted if unavailable.

N. B. The weather data can be in any units. The factors A, B, C, E and F are used in CLYMAT, lines 94 through 109 to convert the data to the units given in the dictionary.)

File 3 variables: soil data

LYRSOL, SITE, DATA, NL, NK, D, NLP, DIMPL, TC, MSW3

SDEPTH(1), BD(1), SAND(1), CLAY(1), PSISAT(1), THETAS(1), FCI(1), THETAO(1), AIRDR(1), SCOND(1), DIFFO(1), BETA(1), ETA(1), TDSX(1,1) TDSX(2,1) TDSX(3,1), TDSX(4,1), TDSX(5,1)

SDEPTH(2), BD(2), (etc. for each soil horizon.)

VH2OI(1), VH2OI(2) (etc. for each soil layer, if available.)

NLR,NKR,RTWT(1,1),RTWT(2,1) (etc. for block of soil containing roots.)

File 4 variables: crop location, timing, spacing etc.

MG, DET, LATUDE, JDFRST, JDLAST, ROWSP, POPROW, ROWANG, NFRQ, FILL, CO2, SDWMAX

RNNO3, RNNH4, FERN, FNO3, FNH4, OMA

File 5, variables: miscellaneous parameters

NPRM, PRM (1), PRM (2).......PRM(NPRM)

Example of Input to and Output from GLYCIM

This example is for a crop of soybeans grown on the Plant Science Research Farm, MAFES, Mississippi State University in 1979. First the input data files are listed. These are followed by the standard output for several days of the season as simulated by the code in this publications.

Crop Culture

Soybean, cv Forrest seed were sown on 19 June, 1979 in rows 67 cm apart on the Plant Science Research Farm, Mississippi State. No fertilizer was applied. The plants reached the cotyledon stage, VC on 25 June and on 27 June they were thinned to an average of 14.3 plants per metre of row. Starting on 9 July, when the plants had reached V3, and every week thereafter, 4 blocks of plants each consisting of 0.5 m of row were selected at random and harvested. The number, size and weight of various organs on the shoot were recorded but the roots were not examined. The final harvest occurred on 15 October.

Input Files

File 1, the weather file, is the longest because each day's data occupies a separate line. On the first line MSW1 and MSW2 are both given the value of 0 because no daily wind or wet and dry bulb temperatures are available. A, the multiplier for solar radiation, is set at 41860 because the data in the file are in langleys and must be converted to kJ m^{-2} in CLYMAT before they are used. The temperature data are already in ^{o}C, the units required by the model, so the values of B and C are chosen so that the temperature values are not altered in CLYMAT. In this case the equation used is: temperature in new units = (temperature in old units-C)/B. Similarly, E, the multiplier for rainfall and F, the multiplier for wind speed are set at 1 because rain and wind are already in the correct units of km hr^{-1} and mm day^{-1} respectively. On the second line is given the average seasonal wind speed for the site in km hr^{-1}. This line is only used when daily wind speed is unavailable and the value must be given in the correct units because the F multiplier does not act on this value. On the third and subsequent lines are listed the day of the year, total daily solar radiation, maximum and minimum temperature and rainfall.

(File 2 is an output file.)

File 3 is the soil file and File 4 is the crop file. The order of the variables in these files is as listed earlier and the values must be given in the units specified in the dictionary. In File 3, LYRSOL is the number of horizons for which there are data. MSW3 is set equal to zero if there are no data on the initial water contents (VH2OI) of each layer of the soil. The distribution of root weight (RTWT) at the start of a run (VC stage) is arbitrary and should be changed to conform to any root restricting layers in the soil.

File 5 contains the PRM parameter array. These are parameters, to which the model has proved sensitive, or parameters that differ between cultivars. Great sensitivity to certain inputs often indicates an area of weakness in a model, so these parameters can also be viewed as markers for potential trouble spots. They mostly occur in PNET and SHTGRO.

Output

The printed output always starts with a banner identifying the model, followed by a line of up to 72 characters to identify the run. This line of characters is supplied by the user in response to the prompt "ENTER RUN IDENTIFICATION" received at the terminal. Next comes a listing of important input variables. Everything else is optional apart from the final report on the status of the plant at maturity. The variable NFRQ in File 4 is used to control the interval in days between reports on plant status during the season. Thus if NFRQ = 7, the reports occur weekly. Other options are selected interactively. In response to the prompt,

"ENTER 1 UNDER THE FIRST LETTER OF EACH PLOT YOU WISH TO SEE
'ROOTS RGCF VH2OC TS CXT NFIXWT TOPS PNDT LW LA'"

the user makes his selection. In the text that follows we will identify optional blocks of output and the codeword that controls them with statements like "this block of output is optional (Select ROOTS)". This indicates that if the user wants to see that particular block of output he should put a 1 under ROOTS when the prompt appears at the start of the run.

All the variable names used in the output are the same as those used in the code and the values are given in the units listed in the dictionary.

Immediately after the run identification are two lines giving the sizes and weights of various organs on the plant at the start of the run (VC stage). These and the other plant characteristics listed on these lines are input from the BLK:D subroutine. They are followed by 2 lines listing some File 4 variables starting with maturity group number, MG. Next come 2 more lines giving the initial sizes and weights of various organs, mostly taken from BLK:D. The line starting RNNO3 lists some more input data from File 4 and this is followed by the number of soil horizons for which data are available. The next two blocks of data list various soil characteristics by horizon starting at the top of the profile. Another block shows the initial distribution of root dry weight in the soil cells by layers and columns. The data in these 3 blocks all come from File 3. The last of the input data to be listed are the parameters from File 5. These are written out in free format.

The first block of model output data consists of certain variables associated with the acquisition and utilization of carbon. One complete line of data is written to the output file each time the model goes through PNET, ten times a day. A heading identifies each variable in the line. The first number on each line refers to the place in the code where the WRITE statement is to be found. This block of output is optional (select PNDT) and its selection results in about 1000 lines of output. It is useful for tracing faults in the model.

A standard report of plant status for one day in the season starts on the second page of output. The first line identifies the day. This is followed by 5 lines showing the current status of some plant variables. Most of the variable names are sufficiently close to full names to be recognizable and they are all listed with their units in the dictionary. The next line lists the current status of the shoot and root carbon pools and the accumulated amounts of carbon fixed by the plant and used in growth. This is followed by a block of 5 lines in which are listed the values taken by 5 variables in each calculation period of the previous day. In this run there were ten calculation periods during the photoperiod and one overnight period. Thus, there are 10 values listed for RADINT, solar radiation interception, which is clearly only applicable during the photoperiod and 11 values listed for EO, potential transpiration rate, which is also applicable at night. At the end of the line for leaf water potential, PSIL are listed some associated plant and soil variables. Below this block of 5 lines is a single line starting with EORS which lists the accumulated amounts of water potentially and actually transpired, taken up from the soil by roots, evaporated from the soil surface and drained out of the bottom of the soil profile.

The next block of data is optional (select LW). It shows the dry weights of the petiole, PETW and the leaf, LEAFW at each node on the main stem and the first 8 branches. Trifoliolate nodes are numbered on the left and U denotes the unifoliolate node. Cotyledons are included with the unifoliolates. The last line of the block shows the dry weight of each of the stems, STEMW.

The next block of data is similar and is also optional (select LA). It shows petiole lengths, PETL, leaf areas, LAREA and stem lengths, HEIGHT.

Following this are six blocks of data relating to soil and root conditions and placed in pairs across the printed page. Each block shows conditions in soil cells from the plant row, which is on the left hand edge of the block, to the mid-row, which is on the right hand edge. This can be seen most easily in the first block of data on root dry weight, RTWT where the greatest accummulation of roots is in the left hand column of cells. The cells go down the full depth of the profile and that depth is selected by the user in File 3. All of these blocks of data are optional. To see RTWT select ROOTS. For all the other blocks of data select the variable name that appears above that block.

The final option (select TOPS) is a simple diagram of the plant showing the location of existing organs on the main stem and first 8 branches. The symbols used are as follows: I = node, 0 = cotyledon, -0 = unifoliolate, --000 = trifoliolate, * = flowers, D = green pods and $ = dry pods. The model makes no attempt to predict the location of flowers and pods on the plant so the diagram shows them on all trifoliolate nodes when they are present.

On the last page of sample output reproduced here, near the botton of the page there is the message "DAY 85 LEAF 7 PARASITIC". A similar message occurs each time a mainstem trifoliolate leaf is abscised. It gives the day of the season, the leaf number counting from the base of the stem and the reason the leaf was abscised. PARASITIC indicates that the leaf was no longer self-supporting. BRANCH PUSH indicates that a branch growing in the leaf axil had reached a certain size. MILD N STRESS indicates that the plant was experiencing a nitrogen stress.

FILE 1

0 0 41860 1 0 1 1
7.5

176	600.0	25.6	16.7	0.0
177	600.0	29.0	16.0	0.0
178	606.0	29.4	15.9	0.0
179	606.0	29.8	17.6	0.0
180	486.0	32.6	21.2	3.0
181	636.0	32.6	22.2	0.0
182	578.0	32.0	16.2	0.0
183	633.0	35.6	20.7	0.0
184	578.0	35.4	20.7	0.0
185	624.0	35.3	22.7	0.0
186	594.0	36.0	25.2	12.7
187	396.0	31.8	24.8	0.0
188	225.0	26.3	22.5	0.0
189	304.0	28.5	21.6	0.0
190	391.0	30.0	21.1	0.0
191	402.0	29.0	17.8	0.0
192	80.0	24.3	18.4	25.0
193	576.0	30.7	21.9	0.0
194	398.0	31.8	24.1	25.0
195	225.0	26.3	22.5	0.0
196	400.0	30.0	23.0	0.0
197	596.0	33.8	23.7	0.0
198	572.0	32.5	23.2	0.0
199	456.0	32.0	22.5	0.0
200	402.0	29.8	21.8	21.6
201	281.0	29.4	22.7	0.0
202	437.0	29.3	23.0	0.0
203	570.0	31.5	22.5	0.0
204	431.0	28.0	23.7	6.4
205	400.0	30.0	24.0	25.0
206	200.0	31.1	23.3	6.1
207	400.0	30.1	22.0	0.5
208	538.0	29.4	21.6	25.0
209	512.0	31.7	23.6	0.0
210	386.0	30.5	23.0	0.0
211	616.0	30.9	21.2	3.0
212	562.0	33.2	22.7	0.0
213	545.0	31.7	23.1	0.0
214	431.0	31.6	21.9	0.3
215	517.0	31.9	19.1	2.2
216	579.0	33.0	20.8	0.0
217	549.0	32.2	22.0	0.0
218	635.0	33.0	21.6	0.0
219	612.0	33.7	21.5	0.0
220	516.0	34.8	23.2	22.1
221	519.0	33.3	20.7	0.0
222	584.0	32.9	21.8	0.0
223	580.0	32.0	23.7	0.0
224	540.0	31.0	20.0	0.0
225	500.0	30.0	17.0	0.0
226	500.0	30.3	15.2	0.0

227	597.0	30.4	17.5	0.0
228	496.0	28.7	20.7	0.0
229	579.0	29.1	15.1	0.0
230	532.0	31.1	18.5	0.0
231	513.0	32.4	19.8	0.0
232	490.0	33.3	20.5	0.0
233	492.0	33.2	21.6	0.6
234	332.0	28.2	19.0	3.0
235	333.0	28.7	18.6	0.0
236	262.0	27.5	18.9	0.0
237	357.0	28.7	20.0	16.0
238	404.0	29.2	20.0	8.1
239	484.0	29.7	18.9	0.0
240	486.0	30.5	18.2	0.0
241	406.0	30.9	20.5	0.0
242	519.0	31.5	19.8	0.0
243	583.0	28.8	17.9	0.0
244	369.0	28.4	19.3	0.0
245	352.0	26.3	19.1	10.2
246	410.0	29.3	20.5	0.0
247	552.0	30.4	18.1	0.0
248	561.0	29.8	20.0	0.0
249	468.0	30.1	19.2	0.0
250	470.0	29.6	19.2	3.8
251	480.0	28.0	15.0	0.0
252	494.0	26.4	12.8	0.0
253	504.0	25.2	15.1	0.0
254	434.0	26.8	13.0	0.0
255	376.0	26.8	14.4	0.0
256	100.0	20.0	19.0	25.0
257	489.0	22.3	15.6	0.0
258	500.0	21.6	17.0	0.0
259	153.0	21.2	18.0	0.0
260	350.0	21.2	19.0	25.0
261	92.0	23.9	19.6	25.0
262	190.0	25.3	20.0	0.0
263	57.0	22.0	19.8	2.3
264	184.0	24.2	22.4	25.0
265	505.0	25.9	16.0	0.0
266	534.0	25.0	13.6	0.0
267	441.0	26.0	13.7	0.0
268	470.0	26.5	15.0	0.0
269	360.0	26.5	14.8	0.0
270	32.6	26.1	18.0	0.0
271	429.0	27.6	16.1	0.0
272	453.0	27.8	14.9	0.0
273	424.0	28.3	15.5	0.0
274	455.0	29.9	15.0	0.0
275	423.0	29.9	14.1	0.0
276	459.0	25.7	6.5	0.0
277	436.0	19.4	12.6	25.0
278	440.0	19.5	5.0	0.0
279	450.0	24.0	7.6	0.0
280	455.0	25.6	6.8	0.0
281	460.0	30.1	10.4	0.0

282	277.0	28.0	14.7	0.0
283	426.0	16.9	6.8	3.8
284	292.0	22.4	6.1	0.0
285	336.0	30.3	10.3	0.0
286	360.0	18.0	15.0	0.0
287	459.0	20.2	3.2	0.0
288	412.0	22.7	4.6	0.0

FILE 3

```
2 1 1 10 10 10 2 200 25 0
 15 1.5 .53 .47 -0.1 .33 .33 .18 .03 3.0 6.34E-5 43.9 3.97
0.8159E-2 0.5248E-4 -0.9263E-4  0.0       -0.7424E-2
100 1.5 .53 .47 -0.1 .42 .42 .25 .03 3.0 1.00E-4 74.7 3.97
0.8314E-2 0.4964E-4 -0.8780E-4 -0.3034E-2 0.0
6 2 2.2E-3 1.8E-3 1.5E-3 1.2E-3 .9E-3 .3E-3
1.1E-3 .9E-3 .6E-3 .3E-3 0 0
```

FILE 4

```
5 0 33.5 176 288 67 14.3 140 7 7.5 330 0.136
10 50 0 0 0 5000
```

FILE 5

```
27 0.99 0.98 0.5 0.6 0.6 0.99 0.005 0.6 0.045 0.5 0.0024 1.0 3.3 4.0
0.009 0.017 0.045 0.015 2.0 0.05 14 0.8 0.015 60.0 1.0 0.6 1.0
```

```
********************* GLYCIM *********************
*     A DYNAMIC SIMULATOR FOR SOYBEAN CROPS CREATED*
*BY MEMBERS OF THE USDA CROP SIMULATION RESEARCH   *
*UNIT, THE MISSISSIPPI AGRICULTURAL AND FORESTRY   *
*EXPERIMENT STATION AND THE AGRONOMY DEPARTMENT AT *
*MISSISSIPPI STATE UNIVERSITY, WITH COLLEAGUES IN  *
*THE S107 REGIONAL SOYBEAN MODELING PROJECT.       *
*     FUNDED BY THE AGRISTARS YIELD MODELING       *
*PROJECT AND THE USDA/DOE PROJECT ON CROP RESPONSE *
*TO INCREASED ATMOSPHERIC CARBON DIOXIDE.          *
*     ADDRESS CORRESPONDENCE TO BASIL ACOCK,       *
*P.O.BOX 5367, MISS.STATE,MISSISSIPPI 39762.       *
****************************************************

MS 1979 DATA

 ULEAFW=0.010 UPETW=0.001 ULAREA=3.3300UPETL=,0.170 MSTEMH=0.500 Z=5.5000 LEAFWT=0.03 PETWT=0.0010 STEMWT=0.0020  ROOTWT=0.035
LAREAT= 7.330 CEC=0.550 LLUE=0.012 LTC=0.170 SCPOOL=0.0124 RRRM=8000.0 RRRY=1500.0 RVRL=17.50 TLIFE=30.0 RTWL= 0.9E-04

 MG= 5 DET=0 LATUDE=33.50 JDFRST=176 JDLAST=288 ROWSP= 67.0 POPROW=14.3
ROWANG=140.0 NFRQ=42 FILL=7.5 CO2= 330. SDWMAX=0.136
 LEAFWT= 0.030    PETWT= 0.001    STEMWT= 0.002    PODWT= 0.000    SEEDWT= 0.000    ROOTWT= 0.035 NODWT=0.0000
 PODS=  0.00       SEEDS=  0.00       LAREAM=    7.33       LAREAB=    0.00       LAREAT=    7.33

 RNNO3= 10.000 RNNH4=50.0000 FERN= 0.0000 FNO3=  0.000 FNH4=  0.000 OMA=  5000.0
 LYRSOL=2

HORIZON SDEPTH  BD    SAND  CLAY  PSISAT THETAS FCI    THETAO AIRDR  SCOND DIFFO     BETA  ETA
  TOP    15.0   1.50  0.53  0.47  -0.10  0.330  0.330  0.180  0.030   3.0  0.63E-04  43.9  3.97
   2    100.0   1.50  0.53  0.47  -0.10  0.420  0.420  0.250  0.030   3.0  0.10E-03  74.7  3.97

HORIZON   TDSX(1)   TDSX(2)   TDSX(3)   TDSX(4)   TDSX(5)
  TOP    0.82E-02  0.52E-04 -0.93E-04  0.00E+00 -0.74E-02
   2     0.83E-02  0.50E-04 -0.88E-04 -0.30E-02  0.00E+00


LAYER 1 RTWT(1,K)=0.0022  0.0011
LAYER 2 RTWT(1,K)=0.0018  0.0009
LAYER 3 RTWT(1,K)=0.0015  0.0006
LAYER 4 RTWT(1,K)=0.0012  0.0003
LAYER 5 RTWT(1,K)=0.0009  0.0000
LAYER 6 RTWT(1,K)=0.0003  0.0000


         27  0.9900000     0.9800000      0.5000000      0.6000000
 0.6000000      0.9700000     4.9999999E-03  0.6000000     4.5000002E-02
 0.5000000      2.4000001E-03   1.000000      3.300000      4.000000
 8.9999996E-03  1.7000001E-02  4.5000002E-02  1.5000000E-02   2.000000
 5.0000001E-02   14.00000      0.8000000     1.5000000E-02   60.00000
  1.000000      0.6000000      1.000000

LINE TRANSC    FIXC      SCPOOL    SEN    REDN  SCF  RCPOOL    QZ   SCRATO SGTR XTRACS    USEDCS    FIXCS     TAUN      SINK


1449 0.162E-02 0.327E-03 0.105E-01 1.000 0.998 1.00 0.000E+00 0.000 0.000 0.000 0.350E-01 0.000E+00 0.471E-03 0.450E-01  0.00
1449 0.165E-02 0.406E-03 0.106E-01 1.000 1.172 1.00 0.000E+00 0.000 1.000 1.000 0.350E-01 0.485E-03 0.105E-02 0.450E-01  0.00
1449 0.166E-02 0.423E-03 0.106E-01 1.000 1.170 1.00 0.000E+00 0.000 1.000 1.000 0.350E-01 0.103E-02 0.166E-02 0.450E-01  0.00
1449 0.166E-02 0.424E-03 0.107E-01 1.000 1.169 1.00 0.000E+00 0.000 1.000 1.000 0.350E-01 0.162E-02 0.227E-02 0.450E-01  0.00
1449 0.166E-02 0.422E-03 0.106E-01 1.000 1.168 1.00 0.000E+00 0.000 1.000 1.000 0.350E-01 0.227E-02 0.288E-02 0.450E-01  0.00
1449 0.164E-02 0.419E-03 0.106E-01 1.000 1.166 1.00 0.000E+00 0.000 1.000 1.000 0.350E-01 0.297E-02 0.348E-02 0.450E-01  0.00
1449 0.162E-02 0.414E-03 0.104E-01 1.000 1.164 1.00 0.000E+00 0.000 1.000 1.000 0.350E-01 0.372E-02 0.408E-02 0.450E-01  0.00
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 SITUATION AT DAWN ON IDAY= 42       JDAY=217
LEAFWT= 4.805   PETWT= 1.428   STEMWT= 3.619   PODWT= 0.000   SEEDWT= 0.000   ROOTWT= 2.327 NODWT=0.0574
PODS=  0.00      SEEDS=  0.00      LAREAM= 953.28      LAREAB= 513.55      LAREAT=1466.83
VSTAGE=10.4     NTOPLF=10       RSTAGE= 1.0      VLOST= 0.0
PDNODC= 0.01        NRATIO= 0.99
 ILOW= 3      Z= 60.8       NBRNCH= 6
 SCPOOL= 0.169   RCPOOL= 0.005   XTRACS= 0.000 FIXCS= 6.510 USEDCS= 6.349
RADINT(ITIME=1,10)  0.97  0.99  0.99  0.92  0.88  0.92  1.00  0.99  0.99  0.97
    EO(ITIME=1,11)   0.1   6.3  14.0  19.2  22.7  24.8  24.3  18.9  11.1   3.2   1.9
  PSIL(ITIME=1,10)  -2.0  -3.7  -7.3  -9.2 -10.8 -11.6 -11.2  -9.4  -6.2  -2.6  PSILD=  -2.0  PILD=  -9.3 PSISM=  -1.1
   SGT(ITIME=1,11) 1.000 0.438 0.000 0.000 0.000 0.000 1.148 1.767 2.333 2.507 1.246
SCRATO(ITIME=1,11) 1.000 1.000 0.000 0.000 0.000 0.000 1.000 1.000 0.382 0.004 0.000
 EORS= 186.6   EORSCS= 186.1   AWUPSS= 185.3   AFWUPS= 184.4   EWUS=   1.9   DRAIN= 113.1
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	MAIN STEM		BRANCH 1		BRANCH 2		BRANCH 3		BRANCH 4		BRANCH 5		BRANCH 6		BRANCH 7		BRANCH 8	
	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW
10	0.037	0.166	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
9	0.099	0.352	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
8	0.176	0.531	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
7	0.176	0.548	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
6	0.160	0.543	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5	0.122	0.522	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
4	0.087	0.455	0.024	0.069	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3	0.051	0.329	0.081	0.183	0.034	0.092	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	0.000	0.000	0.122	0.266	0.072	0.181	0.027	0.083	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1	0.000	0.000	0.062	0.176	0.055	0.161	0.031	0.101	0.013	0.047	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
U	0.000	0.000																
STEMW	1.4		0.7		0.6		0.5		0.3		0.1		0.0		0.0		0.0	

	MAIN STEM		BRANCH 1		BRANCH 2		BRANCH 3		BRANCH 4		BRANCH 5		BRANCH 6		BRANCH 7		BRANCH 8	
	PETL	LAREA	PETL	LAREA	PETL	LAREA	PETL	LAREA	PETL	LAREA	PETL	LAREA	PETL	LAREA	PETL	LAREA	PETL	LAREA
10	8.4	47.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9	17.6	99.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8	25.8	147.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7	25.0	147.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6	23.0	145.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5	19.3	142.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	15.4	128.1	6.1	27.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	10.5	95.0	15.4	69.7	7.9	35.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	20.2	99.3	14.0	68.9	6.6	32.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	0.0	0.0	11.7	63.2	11.0	59.8	7.1	38.7	3.4	18.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
U	0.0	0.0																
HEIGHT	47.3		32.8		28.6		22.8		16.6		7.3		1.9		0.0		0.0	

RTWT

0.009	0.006	0.003	0.001	0.000
0.016	0.011	0.004	0.001	0.000
0.021	0.014	0.004	0.000	0.000
0.020	0.013	0.003	0.000	0.000
0.014	0.009	0.002	0.000	0.000
0.009	0.004	0.001	0.000	0.000
0.004	0.001	0.000	0.000	0.000
0.001	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000

RGCF

0.512	0.526	0.534	0.341	0.341
0.298	0.589	0.596	0.485	0.485
0.283	0.594	0.599	0.542	0.542
0.215	0.583	0.586	0.558	0.558
0.571	0.573	0.574	0.560	0.560
0.565	0.566	0.567	0.560	0.560
0.562	0.562	0.563	0.559	0.559
0.560	0.560	0.560	0.559	0.559
0.559	0.559	0.559	0.558	0.558
0.557	0.557	0.557	0.557	0.557

VH2OC

0.23	0.30	0.33	0.33	0.33
0.20	0.24	0.32	0.33	0.33
0.27	0.32	0.41	0.42	0.42
0.27	0.34	0.41	0.42	0.42
0.34	0.39	0.41	0.42	0.42
0.39	0.41	0.41	0.42	0.42
0.41	0.41	0.42	0.42	0.42
0.42	0.42	0.42	0.42	0.42
0.42	0.42	0.42	0.42	0.42
0.42	0.42	0.42	0.42	0.42

TS

24.45	24.63	24.72	21.87	21.87
25.28	25.40	25.49	24.09	24.09
25.38	25.45	25.51	24.82	24.82
25.28	25.33	25.36	25.02	25.02
25.18	25.20	25.22	25.06	25.06
25.11	25.13	25.14	25.05	25.05
25.08	25.08	25.09	25.04	25.04
25.05	25.05	25.06	25.04	25.04
25.03	25.03	25.04	25.03	25.03
25.02	25.02	25.02	25.01	25.01

CXT

0.210	0.210	0.210	0.210	0.210
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209

NFIXWT

0.005	0.002	0.000	0.000	0.000
0.007	0.004	0.001	0.000	0.000
0.009	0.006	0.001	0.000	0.000
0.006	0.004	0.000	0.000	0.000
0.004	0.001	0.000	0.000	0.000
0.003	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000

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      I--000
000--I
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000--I        000--I        000--I
      I             I--000        I--000        I--000
      I       000--I        000--I        000--I        000--I
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LINE	TRANSC	FIXC	SCPOOL	SEN	REDN	SCF	RCPOOL	QZ	SCRATO	SGTR	XTRACS	USEDCS	FIXCS	TAUN	SINK
1449	0.273E-01	0.191E-01	0.157E+00	1.000	0.993	1.00	0.498E-02	0.000	0.000	0.799	0.233E+01	0.635E+01	0.654E+01	0.450E-01	-1.00
1449	0.344E-01	0.357E-01	0.181E+00	1.000	1.001	1.00	0.336E-02	0.000	1.000	1.000	0.233E+01	0.637E+01	0.659E+01	0.450E-01	-1.00
1449	0.387E-01	0.427E-01	0.200E+00	1.000	1.002	1.00	0.223E-01	0.000	1.000	0.437	0.234E+01	0.638E+01	0.664E+01	0.450E-01	-1.00
1449	0.401E-01	0.448E-01	0.206E+00	1.000	1.003	1.00	0.673E-01	0.000	0.000	0.000	0.235E+01	0.639E+01	0.671E+01	0.450E-01	-1.00
1449	0.409E-01	0.448E-01	0.212E+00	1.000	1.005	1.00	0.114E+00	0.000	0.000	0.000	0.237E+01	0.640E+01	0.677E+01	0.450E-01	-1.00
1449	0.414E-01	0.443E-01	0.216E+00	1.000	1.007	1.00	0.161E+00	0.000	0.000	0.000	0.239E+01	0.641E+01	0.683E+01	0.450E-01	-1.00
1449	0.399E-01	0.389E-01	0.214E+00	1.000	1.009	0.91	0.209E+00	0.000	0.000	0.000	0.240E+01	0.642E+01	0.688E+01	0.450E-01	-1.00
1449	0.416E-01	0.392E-01	0.225E+00	1.000	1.009	1.00	0.201E+00	0.000	1.000	1.330	0.242E+01	0.646E+01	0.693E+01	0.450E-01	-1.00
1449	0.393E-01	0.325E-01	0.220E+00	1.000	1.007	1.00	0.193E+00	0.000	1.000	1.736	0.243E+01	0.652E+01	0.698E+01	0.450E-01	-1.00
1449	0.328E-01	0.171E-01	0.198E+00	1.000	1.004	1.00	0.185E+00	0.000	0.424	2.252	0.245E+01	0.659E+01	0.700E+01	0.450E-01	-1.00
1449	0.330E-01	0.223E-01	0.192E+00	1.000	0.997	1.00	0.613E-02	0.000	1.000	1.262	0.252E+01	0.680E+01	0.703E+01	0.450E-01	1.00
1449	0.406E-01	0.400E-01	0.217E+00	1.000	0.992	1.00	0.414E-02	0.000	1.000	1.000	0.253E+01	0.682E+01	0.709E+01	0.450E-01	1.00
1449	0.445E-01	0.473E-01	0.233E+00	1.000	0.993	1.00	0.287E-01	0.000	1.000	0.313	0.254E+01	0.684E+01	0.715E+01	0.450E-01	1.00
1449	0.459E-01	0.500E-01	0.239E+00	1.000	0.995	1.00	0.759E-01	0.000	0.000	0.000	0.255E+01	0.686E+01	0.722E+01	0.450E-01	1.00
1449	0.441E-01	0.429E-01	0.237E+00	1.000	0.997	0.86	0.128E+00	0.000	0.000	0.000	0.257E+01	0.687E+01	0.728E+01	0.450E-01	1.00
1449	0.430E-01	0.402E-01	0.233E+00	1.000	0.999	0.82	0.179E+00	0.000	0.000	0.000	0.259E+01	0.688E+01	0.733E+01	0.450E-01	1.00
1449	0.417E-01	0.379E-01	0.228E+00	1.000	1.001	0.81	0.228E+00	0.000	0.000	0.000	0.260E+01	0.689E+01	0.739E+01	0.450E-01	1.00

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SITUATION AT DAWN ON IDAY= 84       JDAY=259
LEAFWT= 6.350   PETWT= 2.278   STEMWT= 6.016   PODWT= 6.322   SEEDWT=13.989   ROOTWT= 9.446 NODWT=0.1956
PODS= 88.17        SEEDS=220.42        LAREAM= 696.39       LAREAB=1326.29        LAREAT=2022.68
VSTAGE=13.5       NTOPLF=13        RSTAGE= 5.0       VLOST= 0.6
PDNODC= 0.01          NRATIO= 0.99
ILOW= 7        Z= 67.3         NBRNCH= 6
SCPOOL= 0.127   RCPOOL= 0.000    XTRACS= 0.000 FIXCS=33.519 USEDCS=28.616
RADINT(ITIME=1,10)  1.00  1.00  1.00  1.00  1.00  1.00  1.00  1.00  1.00  1.00
    EO(ITIME=1,11)   0.0   5.3  11.7  16.8  19.9  20.5  18.5  14.1   7.6   1.7   1.1
  PSIL(ITIME=1,10)  -2.0  -3.4  -7.1  -9.8 -11.2 -11.5 -10.7  -8.5  -5.2  -2.0  PSILD=  -2.0  PILD= -11.3 PSISM=  -1.1
   SGT(ITIME=1,11) 1.000 0.687 0.062 0.062 0.301 0.740 1.086 1.219 1.329 1.324 1.001
SCRATO(ITIME=1,11) 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
EORS= 710.6   EORSCS= 669.2   AWUPSS= 662.9   AFWUPS= 658.8   EWUS=   2.0   DRAIN= 120.2
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	MAIN STEM		BRANCH 1		BRANCH 2		BRANCH 3		BRANCH 4		BRANCH 5		BRANCH 6		BRANCH 7		BRANCH 8	
	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW	PETW	LEAFW
13	0.020	0.101	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
12	0.046	0.205	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
11	0.087	0.332	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
10	0.116	0.399	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
9	0.154	0.482	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
8	0.177	0.533	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
7	0.176	0.548	0.017	0.054	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
6	0.000	0.000	0.043	0.115	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5	0.000	0.000	0.077	0.184	0.020	0.062	0.018	0.057	0.002	0.008	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
4	0.000	0.000	0.106	0.224	0.052	0.134	0.044	0.119	0.022	0.065	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3	0.000	0.000	0.144	0.276	0.113	0.236	0.082	0.191	0.047	0.125	0.015	0.048	0.001	0.003	0.000	0.000	0.000	0.000
2	0.000	0.000	0.137	0.288	0.113	0.253	0.087	0.211	0.066	0.175	0.033	0.100	0.017	0.057	0.000	0.000	0.000	0.000
1	0.000	0.000	0.062	0.176	0.057	0.165	0.046	0.141	0.037	0.119	0.027	0.095	0.018	0.065	0.000	0.000	0.000	0.000
U	0.000	0.000																
STEMW	1.8		1.1		0.8		0.7		0.7		0.5		0.4		0.0		0.0	

	MAIN STEM		BRANCH 1		BRANCH 2		BRANCH 3		BRANCH 4		BRANCH 5		BRANCH 6		BRANCH 7		BRANCH 8	
	PETL	LAREA	PETL	LAREA	PETL	LAREA	PETL	LAREA	PETL	LAREA	PETL	LAREA	PETL	LAREA	PETL	LAREA	PETL	LAREA
13	4.5	25.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
12	9.1	51.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
11	14.8	84.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
10	18.9	107.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9	23.5	133.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8	25.9	147.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7	25.0	147.6	4.1	18.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6	0.0	0.0	8.7	39.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5	0.0	0.0	13.7	62.3	4.6	20.7	4.2	19.0	0.8	3.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	17.7	80.2	10.0	45.4	8.8	39.7	5.0	22.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	0.0	0.0	22.4	101.6	18.6	84.5	14.2	64.4	9.4	42.5	3.7	16.8	0.4	1.6	0.0	0.0	0.0	0.0
2	0.0	0.0	21.7	107.1	19.0	93.5	15.4	75.6	12.0	58.9	7.0	34.2	4.0	19.8	0.0	0.0	0.0	0.0
1	0.0	0.0	11.7	63.2	11.3	61.3	9.7	52.4	7.9	42.6	5.9	32.2	4.1	22.3	0.0	0.0	0.0	0.0
U	0.0	0.0																
HEIGHT	57.0		43.7		34.0		32.8		31.9		23.4		18.0		0.0		0.0	

RTWT

0.017	0.014	0.007	0.002	0.000
0.031	0.026	0.014	0.004	0.000
0.044	0.038	0.019	0.005	0.001
0.050	0.041	0.022	0.006	0.001
0.047	0.038	0.020	0.006	0.001
0.038	0.030	0.015	0.005	0.001
0.027	0.020	0.011	0.003	0.000
0.018	0.015	0.005	0.001	0.000
0.011	0.008	0.003	0.000	0.000
0.008	0.006	0.002	0.000	0.000

RCCF

0.229	0.231	0.232	0.232	0.232
0.260	0.260	0.261	0.261	0.261
0.287	0.287	0.287	0.287	0.287
0.314	0.314	0.314	0.314	0.314
0.343	0.343	0.343	0.343	0.343
0.374	0.374	0.374	0.374	0.374
0.407	0.407	0.407	0.407	0.407
0.441	0.441	0.441	0.441	0.441
0.471	0.478	0.478	0.478	0.478
0.516	0.516	0.516	0.516	0.516

VH2OC

0.28	0.30	0.32	0.33	0.33
0.21	0.23	0.31	0.32	0.33
0.26	0.26	0.26	0.41	0.41
0.25	0.26	0.26	0.41	0.41
0.26	0.26	0.26	0.27	0.42
0.26	0.26	0.26	0.30	0.42
0.26	0.26	0.26	0.38	0.41
0.26	0.26	0.32	0.40	0.41
0.26	0.27	0.37	0.40	0.41
0.27	0.31	0.38	0.40	0.41

TS

19.63	19.66	19.68	19.69	19.69
20.31	20.32	20.33	20.33	20.33
20.86	20.87	20.87	20.87	20.87
21.39	21.39	21.39	21.39	21.39
21.91	21.92	21.92	21.92	21.92
22.44	22.44	22.44	22.44	22.44
22.96	22.96	22.96	22.96	22.96
23.47	23.47	23.47	23.47	23.47
23.98	23.98	23.99	23.99	23.99
24.49	24.49	24.49	24.49	24.49

CXT

0.210	0.210	0.210	0.210	0.210
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209
0.209	0.209	0.209	0.209	0.209

NFIXWT

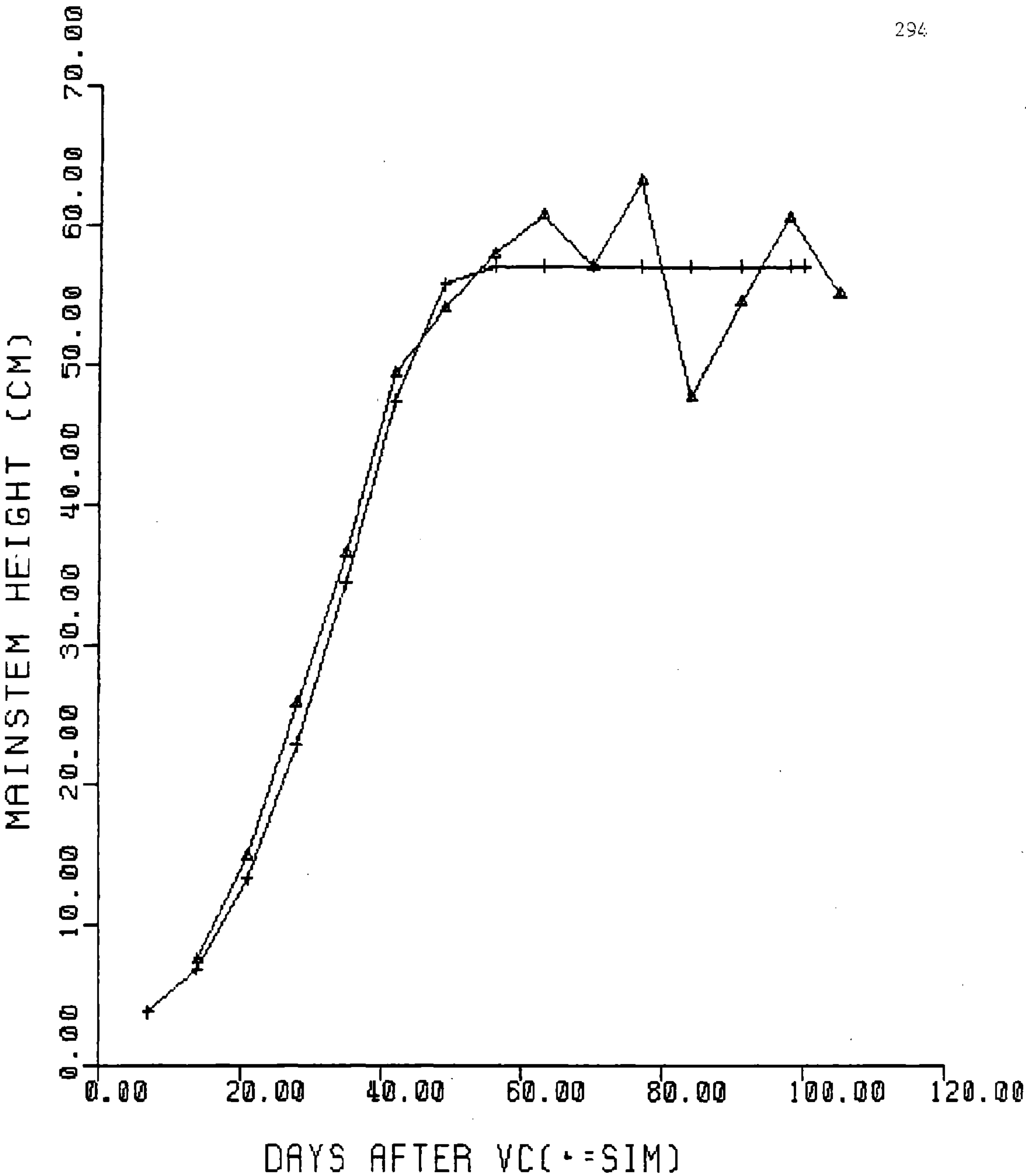
0.009	0.004	0.001	0.000	0.000
0.014	0.009	0.002	0.000	0.000
0.020	0.014	0.003	0.001	0.000
0.014	0.011	0.004	0.002	0.001
0.011	0.005	0.005	0.002	0.001
0.009	0.004	0.004	0.002	0.000
0.004	0.003	0.005	0.001	0.000
0.005	0.006	0.002	0.001	0.000
0.003	0.002	0.001	0.000	0.000
0.002	0.001	0.000	0.000	0.000

```
000--I$
    $I--000
000--I$
    $I--000
000--I$
    $I--000
000--I$       000--I$
    $I            $I--000
     I$       000--I$       000--I$       000--I$       000--I$          I$            I$
    $I            $I--000       $I--000       $I--000       $I--000       $I            $I
     I$       000--I$       000--I$       000--I$       000--I$       000--I$       000--I$
    $I            $I--000       $I--000       $I--000       $I--000       $I--000       $I--000
     I$       000--I$       000--I$       000--I$       000--I$       000--I$       000--I$
     I
   . I
```

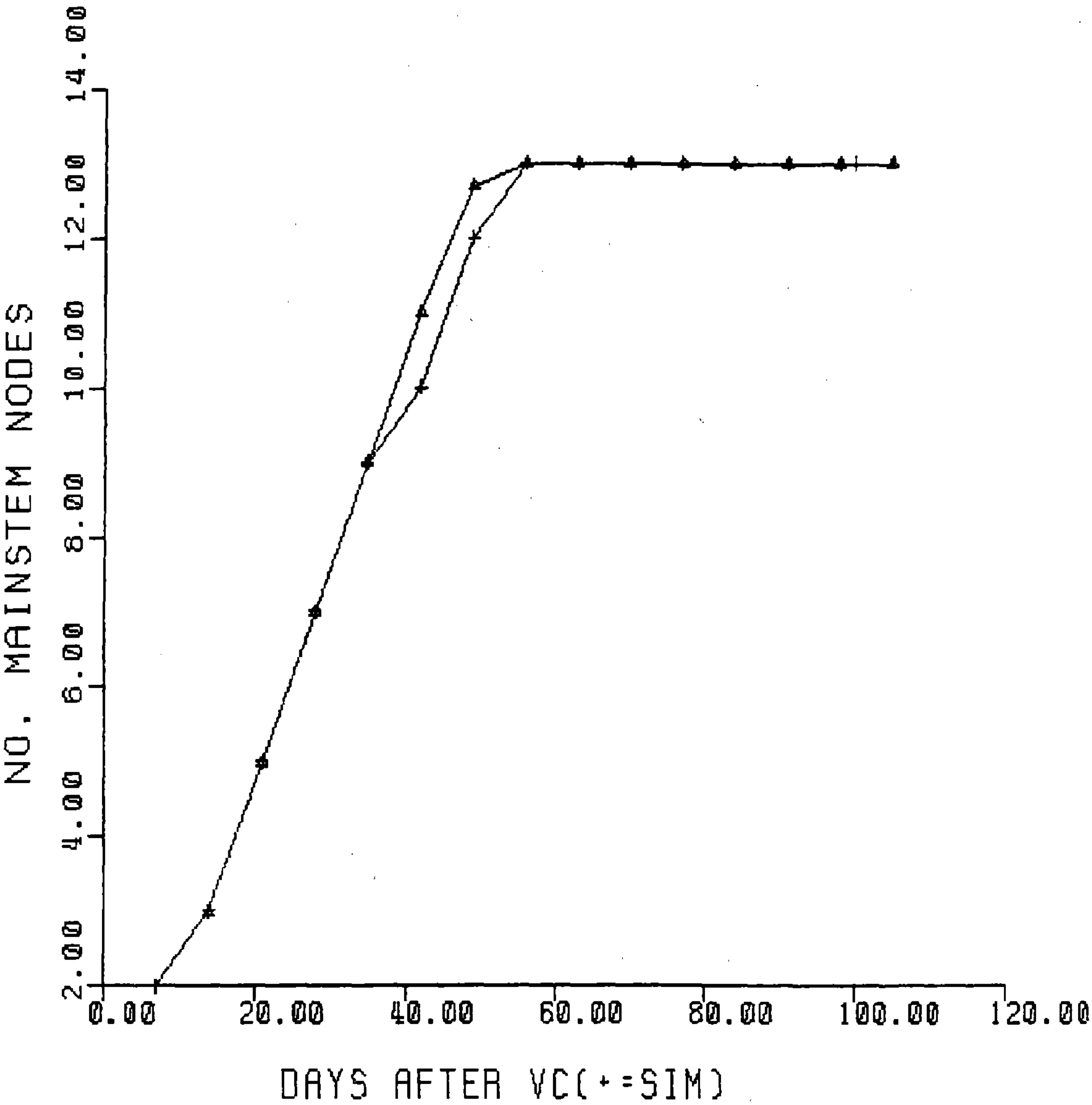
LINE	TRANSC	FIXC	SCPOOL	SEN	REDN	SCF	RCPOOL	GZ	SCRATO	SGTR	XTRACS	USEDCS	FIXCS	TAUN	SINK
1449	0.179E-01	0.638E-02	0.113E+00	0.809	0.995	1.00	0.953E-04	4.788	0.000	0.000	0.945E+01	0.286E+02	0.335E+02	0.170E-01	-1.00
1449	0.206E-01	0.193E-01	0.111E+00	0.809	1.000	1.00	0.667E-04	4.788	0.000	0.000	0.945E+01	0.286E+02	0.336E+02	0.170E-01	-1.00
1449	0.237E-01	0.286E-01	0.117E+00	0.809	0.999	1.00	0.467E-04	4.788	0.000	0.000	0.945E+01	0.287E+02	0.336E+02	0.170E-01	-1.00
1449	0.265E-01	0.346E-01	0.127E+00	0.809	0.998	1.00	0.327E-04	4.788	0.000	0.000	0.945E+01	0.287E+02	0.336E+02	0.170E-01	-1.00
1449	0.288E-01	0.375E-01	0.138E+00	0.809	0.997	1.00	0.229E-04	4.788	0.000	0.000	0.945E+01	0.287E+02	0.337E+02	0.170E-01	-1.00
1449	0.301E-01	0.375E-01	0.147E+00	0.809	0.996	1.00	0.160E-04	4.788	0.000	0.000	0.945E+01	0.288E+02	0.337E+02	0.170E-01	-1.00
1449	0.302E-01	0.346E-01	0.153E+00	0.809	0.996	1.00	0.112E-04	4.788	0.000	0.000	0.945E+01	0.288E+02	0.338E+02	0.170E-01	-1.00
1449	0.288E-01	0.286E-01	0.152E+00	0.809	0.996	1.00	0.785E-05	4.788	0.000	0.000	0.945E+01	0.288E+02	0.338E+02	0.170E-01	-1.00
1449	0.255E-01	0.193E-01	0.145E+00	0.809	0.996	1.00	0.550E-05	4.788	0.000	0.000	0.945E+01	0.289E+02	0.338E+02	0.170E-01	-1.00
1449	0.200E-01	0.638E-02	0.128E+00	0.809	0.996	1.00	0.385E-05	4.788	0.000	0.000	0.945E+01	0.289E+02	0.338E+02	0.170E-01	-1.00
1449	0.840E-02	0.145E-01	0.346E-01	0.760	0.997	1.00	0.957E-07	4.788	0.000	0.000	0.945E+01	0.290E+02	0.338E+02	0.170E-01	-1.00
DAY 85 LEAF 7 PARASITIC															
1449	0.164E-01	0.348E-01	0.573E-01	0.760	1.003	1.00	0.670E-07	4.788	0.000	0.000	0.945E+01	0.290E+02	0.339E+02	0.170E-01	-1.00
1449	0.237E-01	0.476E-01	0.868E-01	0.760	1.014	1.00	0.470E-07	4.788	0.000	0.000	0.945E+01	0.291E+02	0.340E+02	0.170E-01	-1.00

Example of a Validation Run with GLYCIM

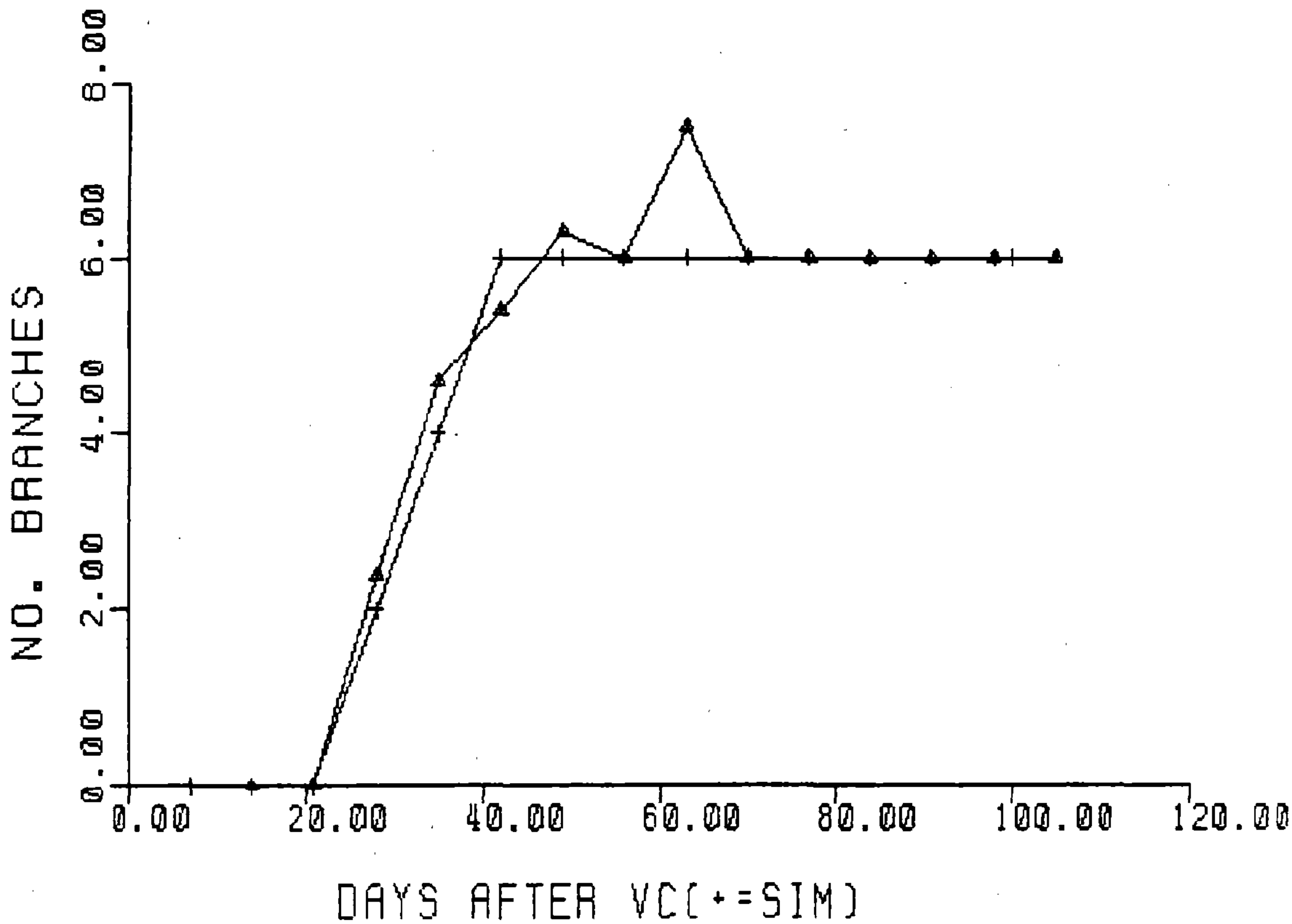
The following figures show a comparison of real and simulated data for the 1979 Mississippi soybean crop that was used in the previous example of input and output. The plotting routine used to produce the figures is not a part of GLYCIM. The observed data are represented by triangles and the simulated data by crosses.



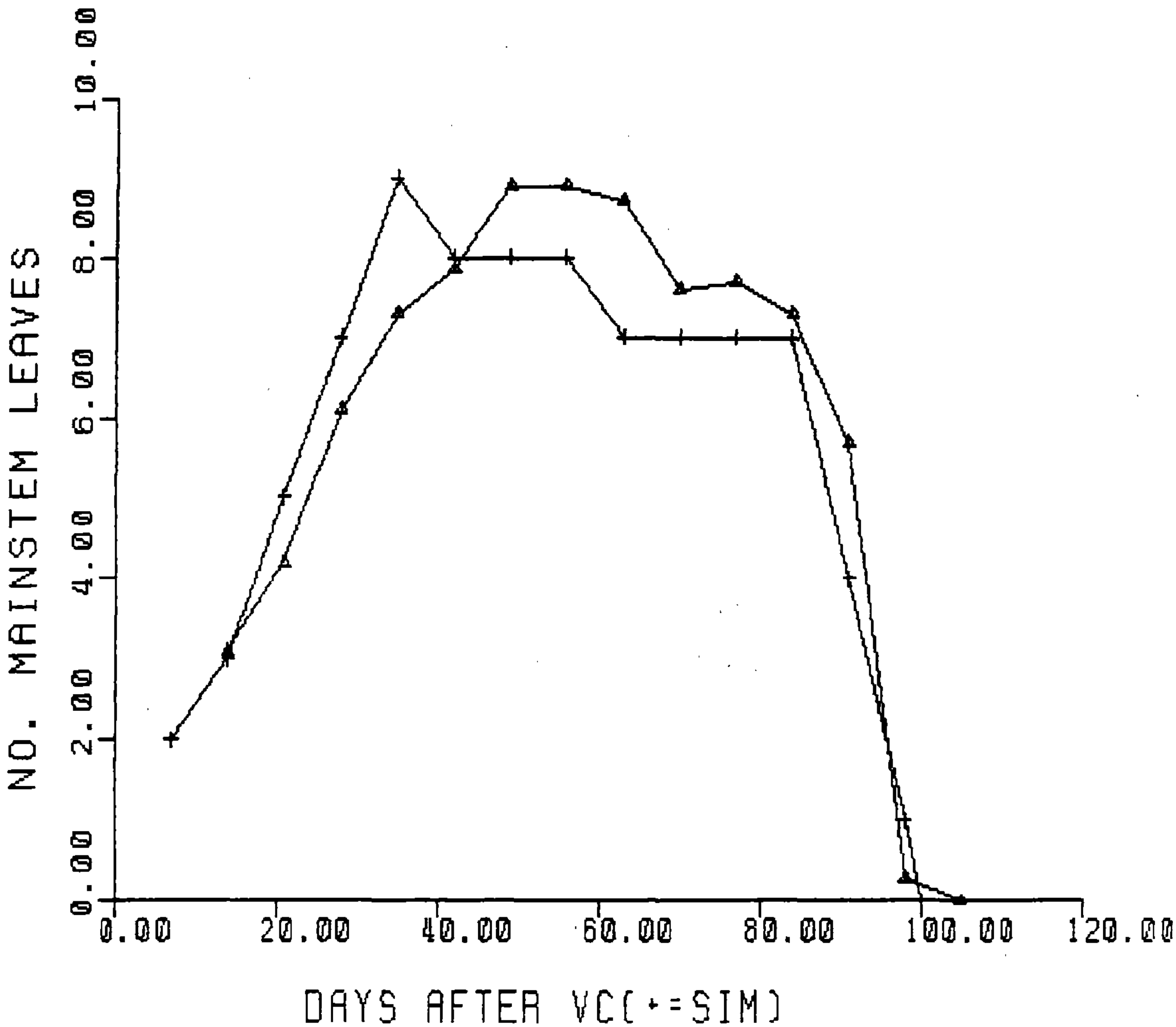
Mainstem height above the cotyledonary node



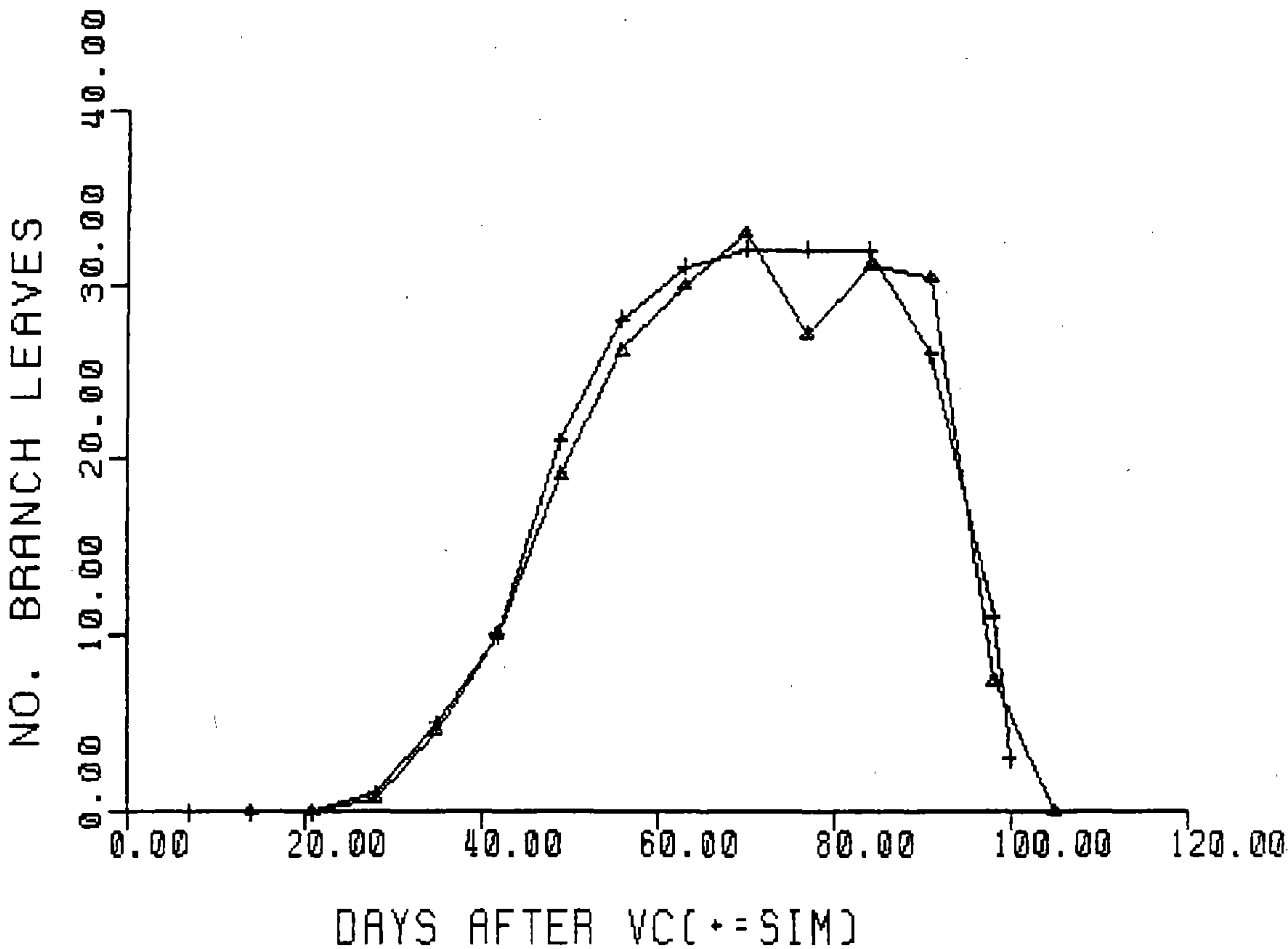
Number of mainstem nodes



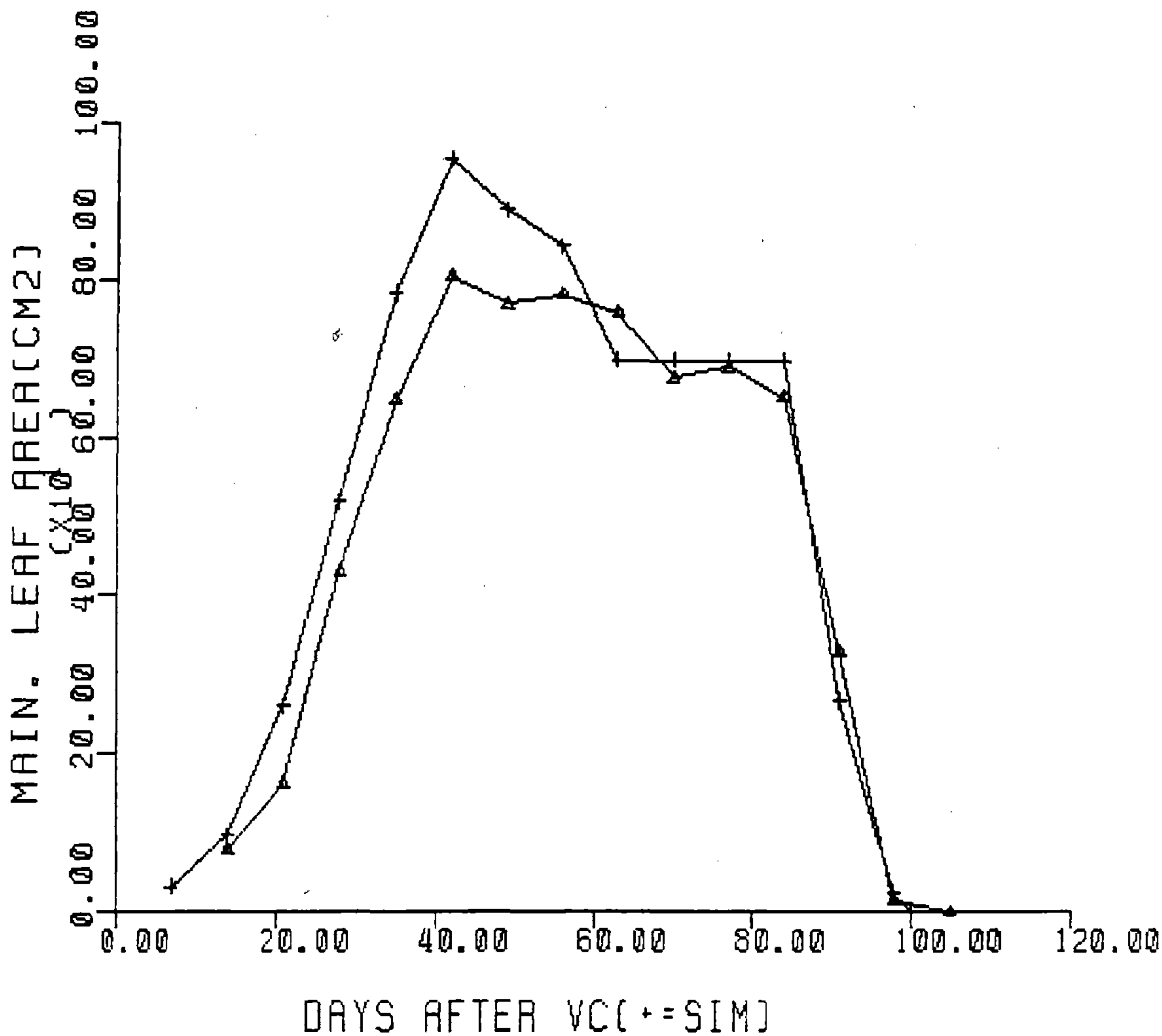
Number of vegetative branches



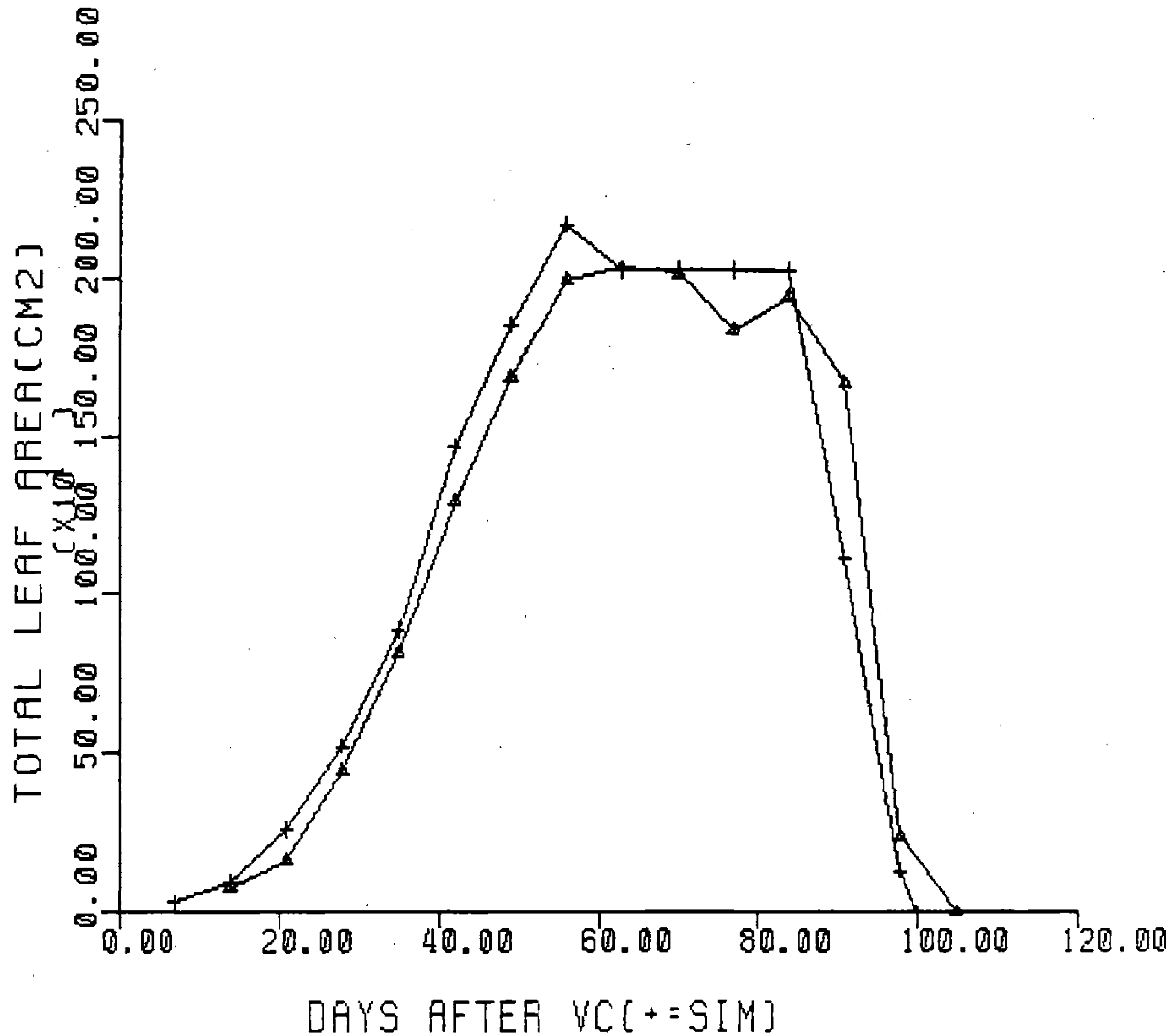
Number of trifoliolate leaves on the mainstem



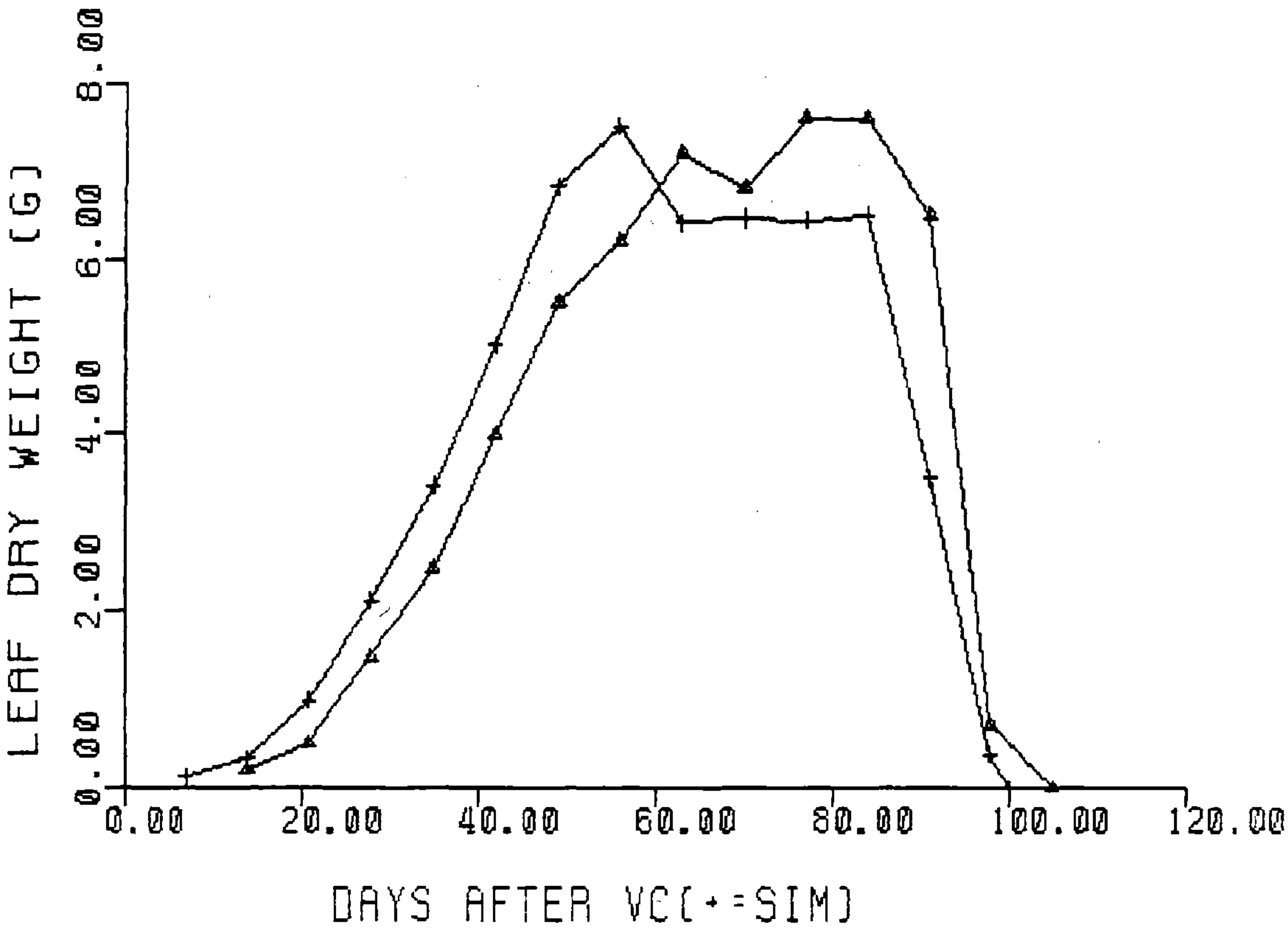
Number of trifoliolate leaves on the branches



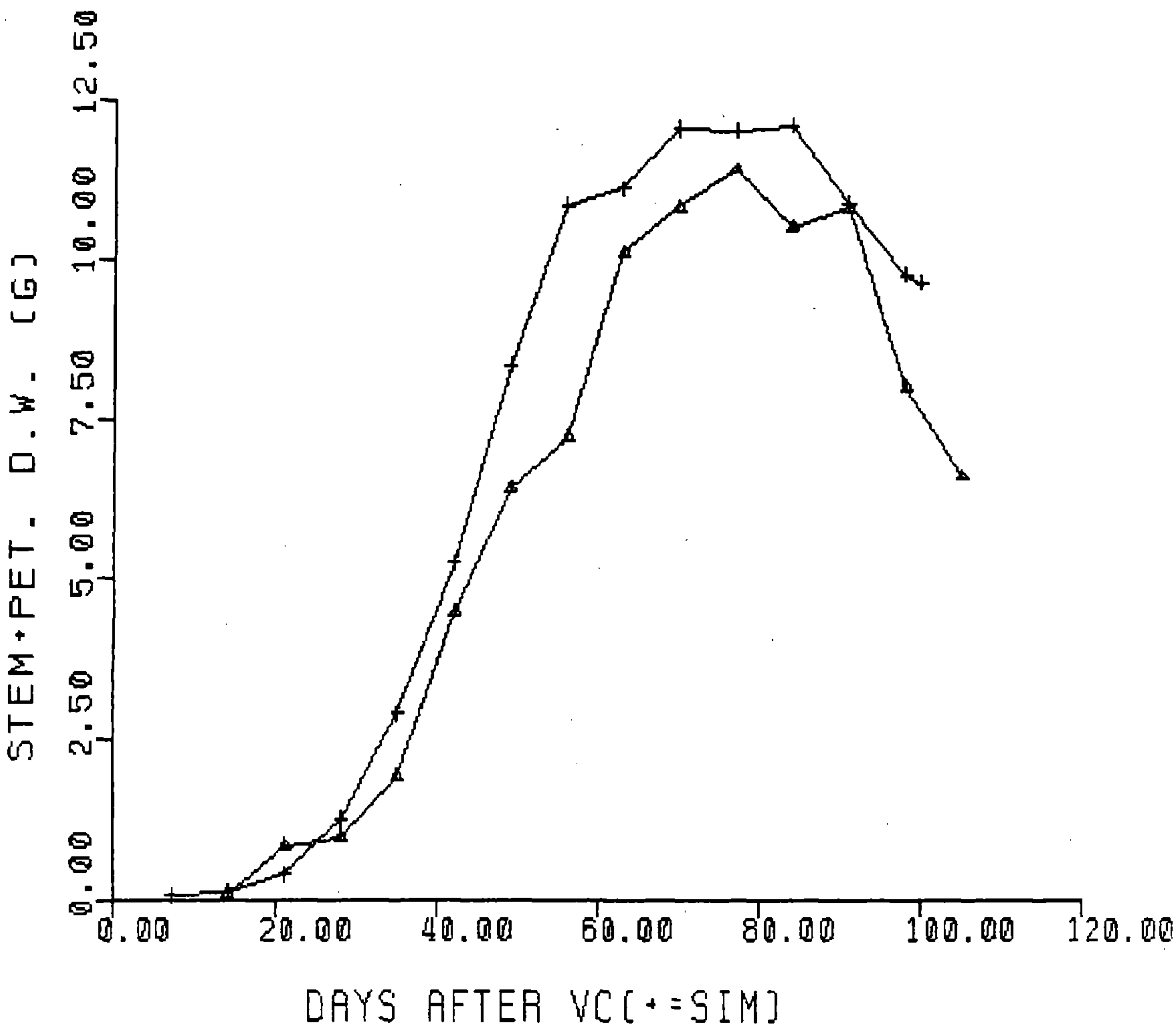
Area of leaves on the mainstem



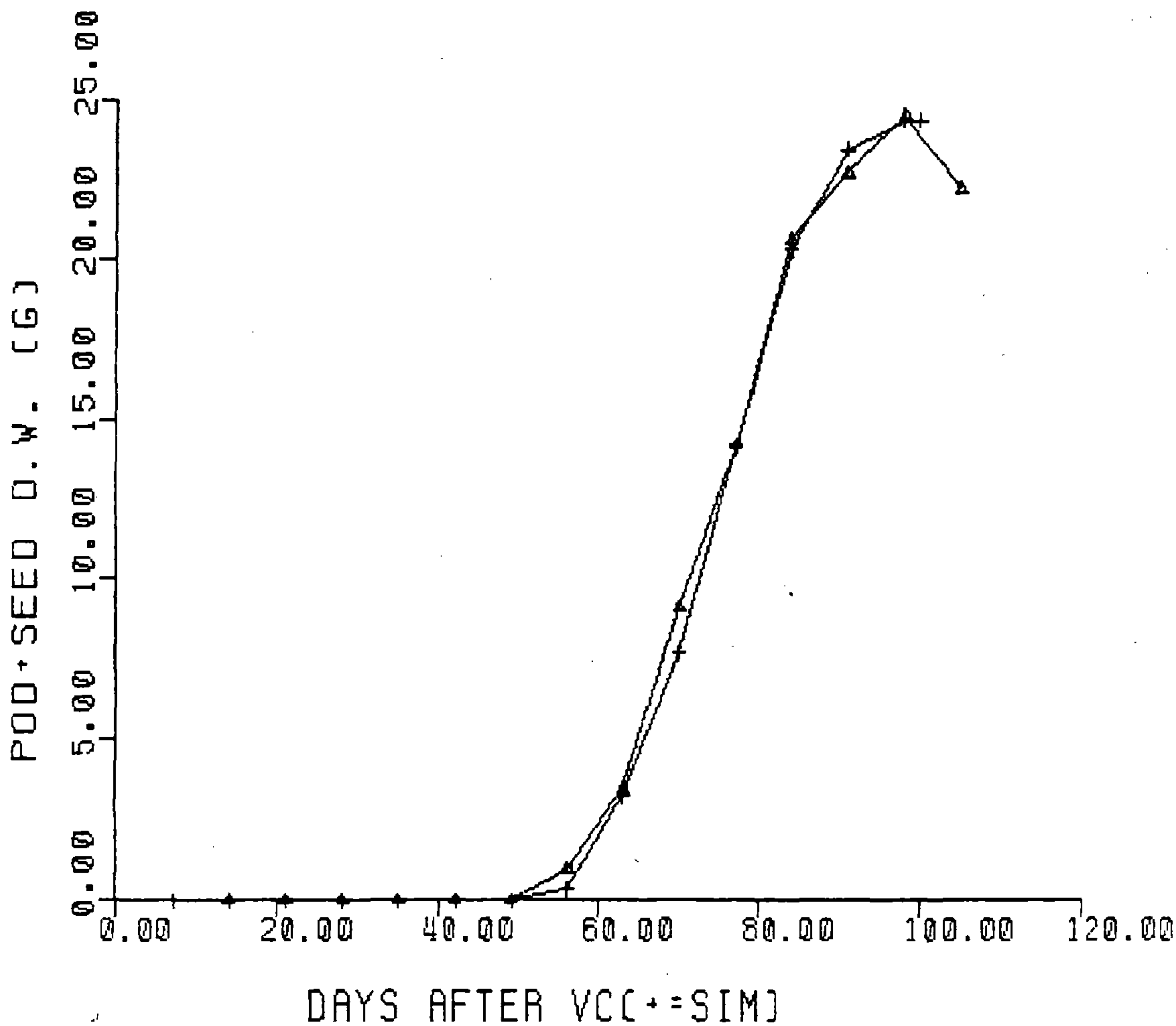
Total leaf area on the plant



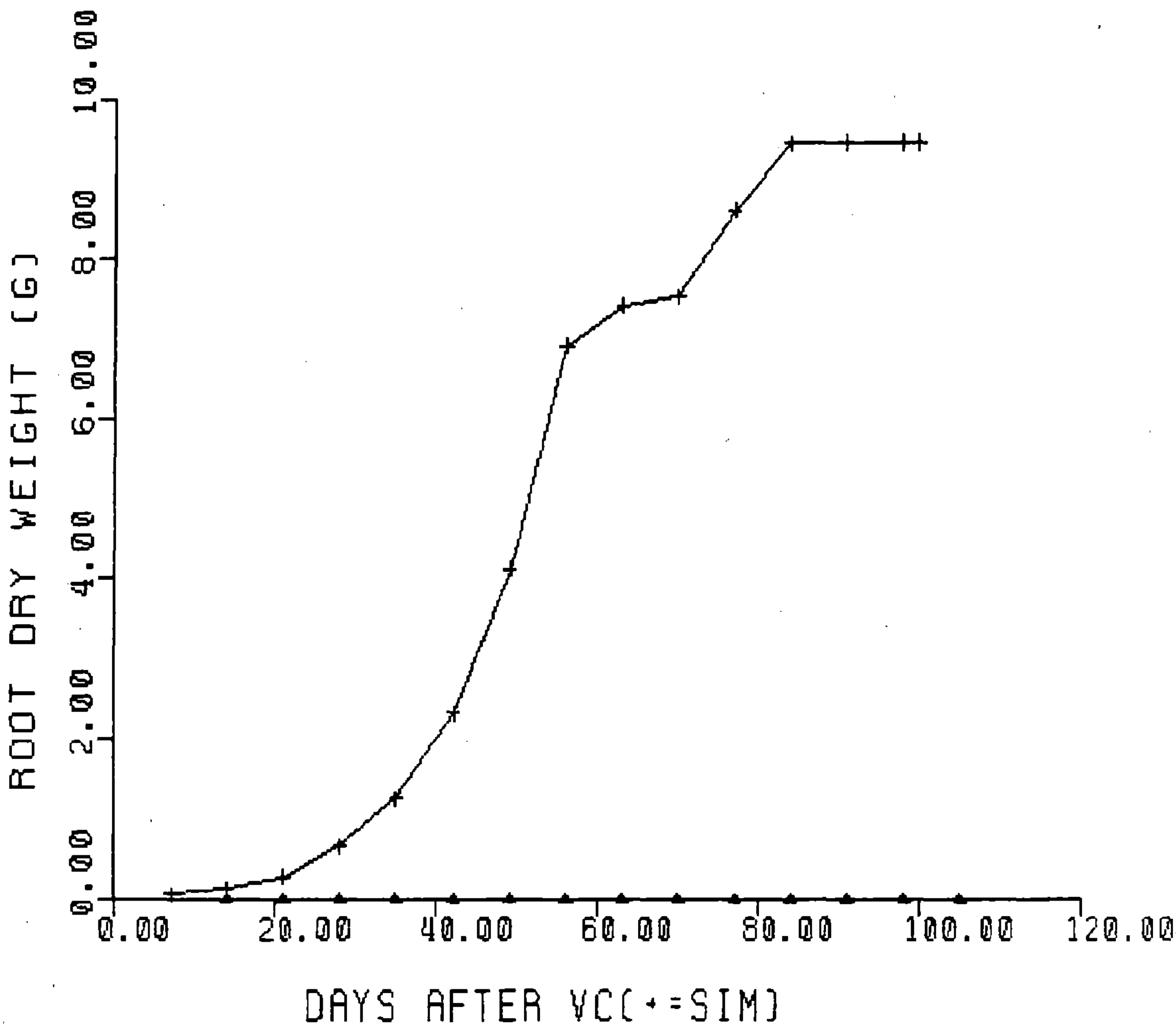
Total leaf dry weight



Dry weight of stems and petioles



Dry weight of podwalls and seeds.



Dry weight of roots (no observations available)

Constraints on the Use of GLYCIM

The model was written for use in the northern hemisphere between latitudes 10 and 55°. There is no reason why it should not be extended to cover a wider range of latitudes including those in the southern hemisphere.

Although there is a determinate/indeterminate switch in the model, it has never been validated against indeterminate varieties and is unlikely to give a good simulation. The model has been parameterized for "Forrest" soybean. When it was validated with field data for "Bragg", we had to adjust the parameters (a) relating air temperature to stem extension and leaf expansion and (b) controlling the length of the flowering period. Similar parameter changes are likely to be necessary with other varieties. Maturity group number takes care of a lot of the difference between varieties. However, we know that there are at least two physiological processes operating to determine the time between sowing and harvest, because occasionally plants from different maturity groups ripen in the "wrong" order. Hence, the need to adjust some parameters between varieties is to be expected.

Finally, the values that may be taken by some variables are limited by the size of arrays in the model. NK, the number of columns in the full soil profile must be an even number so that NKH (=NK/2) is a whole number. Neither NK nor NL, the number of layers in the soil profile, can exceed 20. Characteristics for no more than 5 soil horizons can be put into the model. IPERD, the number of calculation periods in the photoperiod, cannot exceed 10. The maximum permissible numbers of organs on the shoot are as follows: (a) Number of trifoliolates on the mainstem = 25. (b) Number of branches = 20. (c) Number of trifoliolates on each branch = 20.

Strengths and Weaknesses of GLYCIM 82A

The great strength of GLYCIM is that it is almost completely mechanistic at the level of the processes involved in the transfer of materials around the soil, plant and atmosphere. It summarizes current knowledge about these processes and is structured so that new knowledge can be assimilated. It is an expert system that makes the knowledge available to non-experts. It is also a vehicle for physiologists, who specialize in certain organs or processes, to test their sub-models in a whole-crop model.

The members of the team that built GLYCIM have considerable expertise in light interception, leaf and crop photosynthesis and respiration, leaf water relations and soil physical processes. These processes in the model have a solid foundation, except where the scientific community as a whole has to plead ignorance. The team members have many years experience as whole-crop physiologists experimenting in controlled environments. They are, therefore, aware of the danger of naively supposing that yield is a simple function of photosynthesis or transpiration or any other single process. As a result, all processes are dealt with at a similar level of detail in the model.

For many of the processes concerned with partitioning carbon and initiating and aborting organs, we have little idea what mechanisms are involved. It is a strength of the model that mechanisms for these processes have been proposed; mechanisms that are plausible and are compatible with available data. It is a weakness that they have not been more thoroughly tested.

The greatest weakness of GLYCIM 82A is that senescence is not being simulated correctly. The leaves either fall prematurely or stay on too long. The mechanism controlling leaf abscission is obviously wrong. Many possible mechanisms have been tested but none has proved workable so far.

Another weakness, that has much less effect on the performance of the model, is the way in which rainfall is handled. In GRAFLO, it is assumed that all rain enters the soil and wets it layer by layer to field capacity. This is unrealistic. Depending on the infiltration rate, some rain runs off the soil surface. The rain that enters the soil, first wets it to saturation and then drains to field capacity. Getting this water movement correct would also improve our simulation of soil oxygen flux and soil heat flux.

Finally, it is a weakness that the model has not received more validation. But how much is enough and when, in the long development process, should a model be published? In the authors' opinion, GLYCIM will benefit more from public criticism at this stage than it will from further in-house development. We hope to hear from you!

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